# ON THE INTEGRABLE THEORY OF CHIRAL YANGIANS AND RELATED CHIRAL QUANTUM GROUPS

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Abstract. For a chiral algebra $\mathcal{A}$ on a smooth algebraic curve $X$, we define the *ordered chiral homology* $\int_X^{\mathrm{ord}} \mathcal{A}$ as the derived pushforward of the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$, the moduli of ordered tuples of distinct points. The Beilinson–Drinfeld Ran space $\mathrm{Ran}(X)$ is the $\Sigma_\bullet$-coinvariant quotient, and the averaging map $\mathrm{av}\colon R\Gamma(\mathrm{Ran}^{\mathrm{ord}}(X), \mathcal{F}^{\mathrm{ord}}) \to R\Gamma(\mathrm{Ran}(X), \mathcal{F}^{\Sigma})$ recovers the Beilinson–Drinfeld chiral homology. For $\mathsf{E}_\infty$-chiral algebras (all standard vertex algebras), the Kontsevich–Tamarkin formality of the little-disks operad gives a quasi-isomorphism between the ordered and symmetric theories; for genuinely $\mathsf{E}_1$-chiral algebras (Yangians, Etingof–Kazhdan quantum vertex algebras), this formality fails and the descent is lossy, depending on the choice of a Drinfeld associator.

The main structural result is the chiral quantum group equivalence: on the Koszul locus, three structures on an $\mathsf{E}_1$-chiral algebra (the vertex $R$-matrix, the chiral $A_\infty$-structure, and the chiral coproduct) determine each other, forming an equivalence triangle mediated by the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ and its deconcatenation coproduct.

Five computations span the $G/L/C/M$ shadow classification: Heisenberg (class $G$, regular KZ, ordered = symmetric), $V_k(\mathfrak{sl}_2)$ (class $L$, KZ conformal blocks), $\beta\gamma$ (class $C$, nilpotent $R$-matrix), Virasoro (class $M$, irregular KZ connection with Stokes data), and the Yangian $Y_\hbar(\mathfrak{sl}_2)$, which is the first example where ordered chiral homology is strictly richer than symmetric: the kernel $\ker(\mathrm{av})$ is non-trivial at every degree $n \geq 2$, with $\dim \ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ on the $d$-dimensional representation (Proposition 2.6).

For $\mathsf{E}_\infty$-chiral algebras, the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ carries an $\mathsf{E}_2$-coalgebra structure from the curve geometry, and the Higher Deligne Conjecture (Lurie, Francis) equips the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ with an $\mathsf{E}_3$-algebra structure ($\mathsf{E}_2 + 1 = \mathsf{E}_3$). This $\mathsf{E}_3$ structure does not extend to genuinely $\mathsf{E}_1$-chiral algebras, where the symmetric bar does not exist and the ordered bar yields only $\mathsf{E}_2$ on the derived centre. For $V_k(\mathfrak{g})$ at non-critical level, we construct a filtered $\mathsf{E}_3$-algebra $\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g})$ in $\mathcal{D}$-modules on $\mathrm{Ran}(X)$ whose $\mathsf{E}_3$-deformation space is $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$. The same space classifies the Costello–Francis–Gwilliam perturbative Chern–Simons $\mathsf{E}_3$-algebra; formal disk restriction recovers the CFG algebra at the associated-graded and $\mathsf{P}_3$-bracket level. We conjecture that the two formal deformation *families* are isomorphic as $\mathsf{E}_3$-algebras; the matching of deformation spaces is proved, but the chain-level identification is open.

The construction extends to genus 1 and genus 2: the ordered chiral homology $\int_{E_\tau}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2)$ on an elliptic curve $E_\tau$ is computed degree by degree, with the KZ connection replaced by the Knizhnik–Zamolodchikov–Bernard connection (Bernard [5], Felder [32]) and the $B$-cycle monodromy producing the quantum group parameter $q = e^{2\pi i\hbar}$. At genus 2, the period matrix is a $2 \times 2$ symmetric matrix in the Siegel upper half-space, the bar propagator is $d \log E(z, w)$ where $E$ is the prime form, and the degree-2 Euler characteristic jumps to $-12$ (from $-4$ at genus 1). The kernel $\ker(\mathrm{av})$ grows strictly with the genus.

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## 1. Introduction

This paper constructs the *ordered chiral homology* $\int_X^{\mathrm{ord}} \mathcal{A}$ of a chiral algebra $\mathcal{A}$ on an algebraic curve $X$: the derived pushforward of an ordered factorisation $\mathcal{D}$-module on the moduli of *labelled* tuples of distinct points, with no symmetric-group quotient imposed.

Beilinson and Drinfeld's chiral homology [4] requires that $\mathcal{A}$ be a *commutative* factorisation algebra: the factorisation product, which merges local observables at disjoint collections of points, must be symmetric under permutation of the inputs.

Every vertex algebra in the standard landscape (Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}$-algebras, $\beta\gamma$ systems, lattice algebras) satisfies this commutativity hypothesis. The OPE $Y(a,z)Y(b,w)$ and $Y(b,w)Y(a,z)$ agree (after clearing poles) by the locality axiom, and the factorisation product descends to the symmetric quotient $X^{(n)} = X^n/\Sigma_n$.

The hypothesis fails for the class of algebras that the first volume of this series [34] identifies as the *genuinely* $\mathsf{E}_1$*-chiral* objects: the Yangians $Y_\hbar(\mathfrak{g})$, the Etingof–Kazhdan quantum vertex algebras [18], and more broadly any chiral algebra whose operator product is governed by a non-trivial vertex $R$-matrix $S(z)$ satisfying the quantum Yang–Baxter equation. For these algebras, the factorisation product is *braided*, not commutative: swapping two groups of inputs introduces a twist by $S(z)$.

The deficiency is not in the algebras but in the available integration theory. There is no $\mathcal{D}$-module obstruction to working on ordered spaces: $\mathcal{D}$-modules on $X^n$, on the ordered Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(X)$, and on Mok's log-smooth compactifications are perfectly standard objects, studied since Kashiwara and Malgrange. The correct foundational object is the *ordered Ran space* $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$, the moduli of ordered tuples of

distinct points. The Beilinson–Drinfeld Ran space is the $\Sigma_\bullet$-coinvariant quotient $\mathrm{Ran}(X) = \mathrm{Ran}^{\mathrm{ord}}(X)/\Sigma_\bullet$, and the commutative factorisation structure on $\mathrm{Ran}(X)$ is the equivariant shadow of the ordered factorisation structure on $\mathrm{Ran}^{\mathrm{ord}}(X)$ that exists for any chiral algebra, commutative or not.

This paper constructs the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$, defines ordered chiral homology as the derived pushforward of this single object, and establishes the basic properties of the resulting invariant.

1.1. **Main results.** Three theorems and a conjecture emerge from the ordered theory. The first constructs the $\mathcal{D}$-module and defines ordered chiral homology; the second separates it from topological Hochschild; the third identifies the $\mathsf{E}_3$ structure that only the ordered picture sees. At the representation-theoretic level, Proposition 2.6 proves that $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ on $V^{\otimes n}$ for any $d$-dimensional representation $V$ of any simple $\mathfrak{g}$, with generating function $\Delta P_d(t) = (1-dt)^{-1} - (1-t)^{-d}$; Conjecture 2.4 assembles the expected global structure of the ordered derived chiral centre of the chiral Yangian, including the $\mathsf{E}_2$-algebra structure and the recovery of $\mathfrak{g}$ from the Gerstenhaber bracket.

**Theorem 1.1** (Ordered chiral homology)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve $X$ of genus $g$, and let $r(z)$ denote the classical $r$-matrix (the degree-2 projection of the Maurer–Cartan element $\Theta_{\mathcal{A}}$, the universal element of the $\mathsf{E}_1$ convolution $L_\infty$-algebra $\mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1}$ that packages the full bar differential of $\mathcal{A}$; see §6).*

(i) (Ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$.) *The OPE of $\mathcal{A}$, extended meromorphically across the diagonals, defines a factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_n U_n(X)$. Its restriction to the $n$-th stratum is the holonomic $\mathcal{D}$-module*

$$\mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big|_{U_n} = \mathcal{A}^{\boxtimes n}\big|_{U_n}, \qquad \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A}) := j_* j^*(\mathcal{A}^{\boxtimes n}) \in \mathcal{D}\text{-mod}(X^n), \tag{1.1}$$

*where $j\colon U_n = X^n \setminus \bigcup_{1\le a<b\le n} \Delta_{ab} \hookrightarrow X^n$ is the inclusion of the ordered configuration space. No $\Sigma_n$-equivariance is imposed. The factorisation product, the KZ connection, and the chiral differential are encoded in the single object $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$.*

(ii) (Ordered chiral homology.) *The* ordered chiral homology *of $\mathcal{A}$ over $X$ is the derived pushforward*

$$\int_X^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big). \tag{1.2}$$

*The degree-by-degree decomposition $\bigoplus_{n\ge 0} R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A}))$ computes this pushforward stratum by stratum.*

(iii) (Symmetric descent.) *The quotient* $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ *induces the averaging map*

$$\int_X^{\mathrm{ord}} \mathcal{A} = R\Gamma\big(\mathrm{Ran}^{\mathrm{ord}}(X),\, \mathcal{F}^{\mathrm{ord}}\big) \xrightarrow{\mathrm{av}} R\Gamma\big(\mathrm{Ran}(X),\, \mathcal{F}^{\Sigma}\big) = \int_X \mathcal{A}, \tag{1.3}$$

*which is a quasi-isomorphism for* $\mathsf{E}_\infty$*-chiral* $\mathcal{A}$*. For* $\mathsf{E}_1$*-chiral* $\mathcal{A}$*, the descent is lossy: the kernel of* $\mathrm{av}\colon R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}}) \twoheadrightarrow R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}})_{\Sigma_n}$ *is non-trivial at* $n \geq 2$*: at* $n = 2$ *the* $R$*-matrix twist produces* $\ker(\mathrm{av}_2) \neq 0$ *(Example 2.3), and at general degree* $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ *on the* $d$*-dimensional representation (Proposition 2.6). At* $n \geq 3$ *the Drinfeld associator contributes* $\mathrm{GRT}_1$*-dependent data.*

**Theorem 1.2** (Topological Hochschild vs. ordered chiral homology of $D^\times$)**.** *Let* $\mathcal{A}$ *be a chiral algebra on a curve* $X$*, and let* $D^\times = \mathrm{Spec}\ \mathbb{C}((z))$ *be the punctured formal disk.*

(i) (Topological Hochschild.) *Viewing* $\mathcal{A}$ *as an* $\mathsf{E}_1$*-topological algebra by forgetting the holomorphic structure, the factorisation homology* $\int_{S^1} \mathcal{A}$ *is the Hochschild homology* $\mathrm{HH}_*(\mathcal{A})$*, computed by the cyclic bar complex with the Hochschild differential using the associative product* $m_2\colon \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ *and the higher* $A_\infty$ *operations* $m_k$ *(the chiral* $A_\infty$*-structure maps; defined formally in Definition 6.2).*

(ii) (Ordered chiral homology of $D^\times$.) *The ordered chiral chain complex at degree* $n$ *on* $D^\times$ *is*

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{total}}\Big) \tag{1.4}$$

*where* $\Omega^*_{\log}$ *denotes logarithmic differential forms on the Fulton–MacPherson compactification,* $s^{-1}$ *is the desuspension* ($|s^{-1}v| = |v| - 1$)*,* $\bar{\mathcal{A}} = \ker(\varepsilon)$ *is the augmentation ideal, and*

$$d_{\mathrm{total}} = d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZ}} \tag{1.5}$$

*with* $d_{\mathrm{dR}}$ *the de Rham differential,* $d_{\mathrm{bar}}$ *the bar differential encoding the OPE, and*

$$\nabla_{\mathrm{KZ}} = \sum_{i<j} r_{ij}(z_i - z_j)\, dz_{ij}, \qquad dz_{ij} = d(z_i - z_j), \tag{1.6}$$

*the Knizhnik–Zamolodchikov connection. The Arnold forms* $\omega_{ij} = d\log(z_i - z_j)$ *are generators of the cohomology algebra* $H^*(\mathrm{Conf}_n(\mathbb{C}))$ *and appear as coefficients in the bar differential* $d_{\mathrm{bar}}$*; they are* not *the connection* 1*-form of* $\nabla_{\mathrm{KZ}}$*.*

(iii) (Formality bridge.) *For* $\mathsf{E}_\infty$*-chiral* $\mathcal{A}$*, the formality bridge (Theorem 7.5) gives a quasi-isomorphism* $\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow{\sim} \mathrm{HH}_*(\mathcal{A})$*: ordered and symmetric chiral homologies agree as chain complexes, though the transferred* $A_\infty$*-structure may carry non-trivial higher operations for class* $L/C/M$*. For genuinely* $\mathsf{E}_1$*-chiral* $\mathcal{A}$ *(where* $S(z) \neq \mathrm{id}$*), this quasi-isomorphism does not exist.*

**Theorem 1.3** (The $\mathsf{E}_3$-algebra and Chern–Simons)**.** *Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra and $V_k(\mathfrak{g})$ the affine Kac–Moody vertex algebra at level $k$.*

(i) (The $\mathsf{E}_3$ structure.) *The curve geometry provides the $\mathsf{E}_2$ structure on $\overline{B}^{\Sigma}(V_k(\mathfrak{g}))$ (Warning 3.7). By the Higher Deligne Conjecture (Lurie [24], Francis [21]), the Hochschild cohomology of this $\mathsf{E}_2$-coalgebra*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) = \mathrm{HH}^*\big(\overline{B}^{\Sigma}(V_k(\mathfrak{g})),\ \overline{B}^{\Sigma}(V_k(\mathfrak{g}))\big) \tag{1.7}$$

*carries a natural $\mathsf{E}_3$-algebra structure ($\mathsf{E}_2 + 1 = \mathsf{E}_3$).*

(ii) (Deformation classification.) *The space of $\mathsf{E}_3$-deformations of $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ is non-canonically isomorphic to $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$. For simple $\mathfrak{g}$, $H^3(\mathfrak{g})$ is one-dimensional, so there is a single deformation parameter at each order in $(k+h^\vee)$.*

(iii) (Comparison.) *The same $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$ classifies, by Costello–Francis–Gwilliam [12]: perturbative Chern–Simons quantisations, filtered $\mathsf{E}_3$-algebras deforming $C^*(\mathfrak{g})$, braided monoidal deformations of $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$, and quasi-triangular quasi-Hopf deformations of $U(\mathfrak{g})$; and by Etingof–Kazhdan [17, 18]: quantum vertex $R$-matrices on $V(\mathfrak{g}, K)$, forming a $\mathrm{GRT}_1$-torsor. All six classification spaces are in natural bijection.*

(iv) (Topological enhancement.) *By Khan–Zeng [28], the Sugawara Virasoro element at non-critical level upgrades the holomorphic-topological symmetry of the $3d$ gauge theory to fully topological symmetry. Khan–Zeng [28] prove this at the classical (PVA) level; the quantum extension holds because the Sugawara element exists as an exact operator in $V_k(\mathfrak{g})$ at every non-critical level, satisfying the full Virasoro commutation relations without quantum corrections to the BRST-exactness argument (see Proposition 8.23). At the critical level $k = -h^\vee$, the Sugawara element is undefined and the topological enhancement does not apply; the topological nature of the perturbative $\mathsf{E}_3$ in the formal neighbourhood of critical level holds order-by-order in $(k+h^\vee)$ (since the Sugawara element exists at any nonzero departure from critical level).*

1.2. **Architecture of the paper.** The paper opens with the motivating example (§2): the Yangian $Y_\hbar(\mathfrak{sl}_2)$ exhibits all three structures of the chiral quantum group equivalence and the non-trivial kernel ker(av) that ordered chiral homology captures but the Beilinson–Drinfeld theory cannot see. The motivating example culminates in Conjecture 2.4, which predicts the full structure of the ordered derived chiral centre (infinite-dimensional, $\mathsf{E}_2$ but not $\mathsf{E}_3$, recovering $\mathfrak{g}$ from the Gerstenhaber bracket), and Proposition 2.6, which proves the kernel dimension formula $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ for any $d$-dimensional representation of any simple Lie algebra.

The general theory follows. Section 3 recalls the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy and the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X)$. Section 4 builds the ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$; Section 5 defines ordered chiral homology as the derived pushforward and establishes symmetric descent to the Beilinson–Drinfeld theory.

The main structural result is the chiral quantum group equivalence (§6): on the Koszul locus, three structures on an $\mathsf{E}_1$-chiral algebra (the vertex $R$-matrix, the chiral $A_\infty$-structure, and the chiral coproduct) determine each other, forming an equivalence triangle mediated by the ordered bar complex. The chiral Yangian datum (Definition 6.7) packages this equivalence into a single four-component object $(\mathcal{A}, S(z), \Delta^{\mathrm{ch}}, \{m_k^{\mathrm{ch}}\})$ satisfying four axioms (Koszul self-recovery, Verdier $R$-inversion, braided module category, modular MC tower).

Section 7 separates ordered chiral homology from topological Hochschild through four levels of formality. Section 8 identifies the $\mathsf{E}_3$ structure via Chern–Simons theory. The examples in §9 cover all four shadow classes $G/L/C/M$. Section 10 extends the construction to genus 1: the ordered chiral homology $\int_{E_\tau}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2)$ is computed degree by degree on an elliptic curve, where the KZB connection replaces the KZ connection and the $B$-cycle monodromy produces the quantum group parameter $q = e^{2\pi i\hbar}$. Section 11 extends the construction further to genus 2, where the $2 \times 2$ period matrix, the prime form, and the genus-2 surface braid group produce a richer theory with non-vanishing Euler characteristics at every degree. Section 12 places the construction against Ben-Zvi–Brochier–Jordan and Latyntsev. Section 13 collects open problems.

## 2. The motivating example: $Y_\hbar(\mathfrak{sl}_2)$

The Yangian $Y_\hbar(\mathfrak{sl}_2)$ is the simplest genuinely $\mathsf{E}_1$-chiral algebra: it is the first example where ordered chiral homology is strictly richer than the Beilinson–Drinfeld (symmetric) theory. Before developing the general machinery, we present this example in full to exhibit the three structures (vertex $R$-matrix, chiral $A_\infty$-structure, chiral coproduct) and the non-trivial kernel ker(av) that motivate the entire construction.

**Proposition 2.1** (Yangian presentations)**.** *The following are isomorphic as filtered associative algebras:*

1. *The Yangian $Y_\hbar(\mathfrak{g})$ in the RTT presentation: generated by $T_{ij}(u)$ with $R_{12}(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R_{12}(u-v)$, where $R(u) = 1 + \hbar P/u$ and $\hbar = 1/(k+h^\vee)$.*
2. *The Yangian $Y(\mathfrak{g})$ in the Drinfeld first presentation: generated by $\{x_i, \xi_i : i \in I\}$ with Serre-type relations (Drinfeld [10]).*
3. *The $\mathsf{E}_1$-chiral Koszul dual $\mathcal{A}^!_{\mathrm{ord}}$ of $V_k(\mathfrak{g})$: the algebra obtained from the ordered bar coalgebra $B^{\mathrm{ord}}(V_k(\mathfrak{g}))$ by spectral strictification, with RTT relations recovered from collision residues on $\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{C})$.*

*The isomorphism* (i) $\cong$ (ii) *is classical* [10]. *The isomorphism* (ii) $\cong$ (iii) *is the spectral Drinfeld strictification* ([34, §III]*; proved for all simple* $\mathfrak{g}$).

*Remark* 2.2 (Spectral parameter locus). The spectral parameter $z$ lives on the structures (the coproduct $\Delta_z$, the $R$-matrix $R(z)$, the transfer matrix $T(z)$, the evaluation functor $\mathrm{ev}_z$), not on the algebra symbol. The algebra $Y_\hbar(\mathfrak{g})$ is a single filtered algebra; $z$ parametrises representations and coalgebra maps.

**Example 2.3** (Yangian $Y_\hbar(\mathfrak{sl}_2)$ as a chiral quantum group: three structures and the ordered chiral center)**.** The Yangian $Y_\hbar(\mathfrak{sl}_2)$ is the first fully worked-out $\mathsf{E}_1$-chiral quantum group: an $\mathsf{E}_1$-chiral algebra equipped with chiral coproduct $\Delta_z$, spectral $R$-matrix $R(z)$, and quasi-triangularity, making $\mathrm{Mod}_{Y_\hbar(\mathfrak{sl}_2)}$ a braided monoidal category (equivalently, $\mathsf{E}_2$ in Cat). It is the first example in which ordered chiral homology differs from Beilinson–Drinfeld (symmetric) chiral homology. The $R$-matrix is independent input, not derived from any $\mathsf{E}_\infty$-chiral OPE. The formality bridge (Theorem 7.5) does not apply, because the factorisation $\mathcal{D}$-module is not $\Sigma_n$-equivariant. We exhibit all three structures of the chiral quantum group equivalence (Theorem 6.5) concretely, then compute the ordered chiral center at low degrees and identify $\ker(\mathrm{av}) \neq 0$.

**Data.** Let $\mathfrak{g} = \mathfrak{sl}_2$ with standard basis $\{e, f, h\}$, brackets $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$, and normalised Killing form $(e, f) = 1$, $(h, h) = 2$. The dual Coxeter number is $h^\vee = 2$. The deformation parameter is $\hbar = 1/(k + h^\vee) = 1/(k + 2)$ (Remark 8.7). The inverse Casimir is

$$\Omega = e \otimes f + f \otimes e + \tfrac{1}{2}\, h \otimes h \;\in\; \mathfrak{g} \otimes \mathfrak{g}. \tag{2.1}$$

In the fundamental representation $V = \mathbb{C}^2$: $(\rho \otimes \rho)(\Omega) = P - \frac{1}{2}\, I$, where $P$ is the permutation operator on $V \otimes V$.

**Structure (I): the $R$-matrix.** The Yang $R$-matrix on $V \otimes V$ is

$$R(u) = u\, I + \hbar\, P \;\in\; \mathrm{End}(V \otimes V)[u], \tag{2.2}$$

where $u$ is the spectral parameter (the difference of evaluation points on $\mathbb{C}$). This satisfies the quantum Yang–Baxter equation

$$R_{12}(u - v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u - v) \tag{2.3}$$

in $\mathrm{End}(V^{\otimes 3})[u, v]$, and the unitarity condition $R_{12}(u)\, R_{21}(-u) = (\hbar^2 - u^2)\, I$ (scalar: invertible for $u \neq \pm\hbar$).

At the classical level, $R(u) = u\, I + \hbar\, P = u(I + \hbar\, \Omega/u + O(\hbar^2))$ after restoring the Casimir from $P = \Omega + \frac{1}{2} I$ on $V \otimes V$. The classical $r$-matrix $r(z) = \hbar\, \Omega/z$ is in KZ normalisation; at $\hbar = 0$ (i.e. $k \to \infty$), $r = 0$; at the critical level $k = -2$, $\hbar$ diverges.

The eigenvalues of $R(u)$ on $V \otimes V$ are $u + \hbar$ on $\mathrm{Sym}^2 V$ (dimension 3) and $u - \hbar$ on $\bigwedge^2 V$ (dimension 1); the $R$-matrix has a zero on $\bigwedge^2 V$ at $u = \hbar$, selecting the antisymmetric channel.

**Structure (II): the $A_\infty$ structure.** The chiral $A_\infty$-structure maps $m_k^{\mathrm{ch}}\colon Y_\hbar^{\otimes k} \to Y_\hbar$ are the operations extracted from the ordered bar complex via the bar-cobar correspondence. The transfer matrix

$$T(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix} \in \mathrm{End}(V) \otimes Y_\hbar(\mathfrak{sl}_2)[\![u^{-1}]\!] \tag{2.4}$$

encodes the generating functions of the algebra.

The binary operation $m_2^{\mathrm{ch}}$ is the RTT relation: for any two entries $T_{ij}(u)$ and $T_{kl}(v)$ of the transfer matrix,

$$R_{12}(u-v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R_{12}(u-v), \tag{2.5}$$

where subscripts indicate the auxiliary space factor. Expanding this $4 \times 4$ matrix equation produces the Yangian relations in RTT form. The ten independent component relations include:

- diagonal commutativity: $[A(u), A(v)] = 0$, $[D(u), D(v)] = 0$;
- off-diagonal commutativity: $[B(u), B(v)] = 0$, $[C(u), C(v)] = 0$;
- the cross-relations, e.g. $(u-v)[B(u), C(v)] = \hbar\big(D(u)A(v) - D(v)A(u)\big)$.

The higher operations $m_k^{\mathrm{ch}}$ for $k \geq 3$ involve the Drinfeld associator $\Phi_{\mathrm{KZ}}$. The binary product $m_2^{\mathrm{ch}}$ is strictly associative only up to the holonomy of the KZ connection on $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$: the correction is $m_3^{\mathrm{ch}}(a,b,c) = \Phi_{\mathrm{KZ}} \cdot m_2(m_2(a,b),c) - m_2(a, m_2(b,c))$, where $\Phi_{\mathrm{KZ}}$ is the regularised holonomy between the two maximal boundary strata of $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The Gauss decomposition $T(u) = F(u)\, K(u)\, E(u)$ (lower-triangular, diagonal, upper-triangular) translates the RTT relations into the Drinfeld presentation: generators $\{E_r, F_r, H_r\}_{r \geq 0}$ with relations

$$[H_r, H_s] = 0, \tag{2.6}$$

$$[H_0, E_s] = 2E_s, \quad [H_0, F_s] = -2F_s, \tag{2.7}$$

$$[E_r, F_s] = H_{r+s}, \tag{2.8}$$

$$[H_{r+1}, E_s] - [H_r, E_{s+1}] = \hbar(H_r E_s + E_s H_r), \tag{2.9}$$

and the Serre relation $[E_{r+1}, [E_r, E_s]] - [E_r, [E_{r+1}, E_s]] = 0$ (and similarly for $F$). The leading modes $E_0 = e$, $F_0 = f$, $H_0 = h$ generate a copy of $U(\mathfrak{sl}_2)$ inside $Y_\hbar(\mathfrak{sl}_2)$; the PBW filtration has $\mathrm{gr}\, Y_\hbar(\mathfrak{sl}_2) \cong U(\mathfrak{sl}_2[t])$.

**Structure (III): the chiral coproduct.** The Drinfeld coproduct on $Y_\hbar(\mathfrak{sl}_2)$ is

$$\Delta_z(T(u)) = T(u) \,\dot{\otimes}\, T(u-z), \tag{2.10}$$

where $\dot{\otimes}$ denotes matrix multiplication in the auxiliary space $\mathrm{End}(V)$ and ordinary (completed) tensor product in the $Y_\hbar$ factor. The spectral parameter $z$ is the difference coordinate $z_1 - z_2$ on $\mathrm{Conf}_2(\mathbb{C})$. On Drinfeld generators:

$$\Delta_z(E_0) = E_0 \otimes 1 + 1 \otimes E_0 + \hbar\, H_0 \otimes E_0 \cdot z^{-1} + O(z^{-2}), \tag{2.11}$$

$$\Delta_z(H_0) = H_0 \otimes 1 + 1 \otimes H_0 + 2\hbar\, (E_0 \otimes F_0 - F_0 \otimes E_0) \cdot z^{-1} + O(z^{-2}). \tag{2.12}$$

The coproduct is coassociative up to the Drinfeld associator $\Phi$:

$$(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_2} = \Phi_{123}(z_1, z_2) \cdot (\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1} \cdot \Phi_{123}(z_1, z_2)^{-1},$$

where $\Phi_{123}(z_1, z_2) \in \exp(\widehat{\mathrm{Lie}}(\Omega_{12}, \Omega_{23}))$ is the KZ associator with spectral parameters.

The $R$-matrix is recovered from the coproduct: $R(u) = \sigma \circ \Delta_z \circ (\Delta_z^{\mathrm{op}})^{-1}$ evaluated on $V \otimes V$, where $\sigma$ is the flip. This is the content of the equivalence (I)$\Leftrightarrow$(III) in Theorem 6.5.

**The three structures are equivalent.** The ordered bar complex $\overline{B}^{\mathrm{ord}}(Y_\hbar(\mathfrak{sl}_2)) = T^c(s^{-1}\overline{Y}_\hbar)$ (desuspended augmentation ideal, deconcatenation coproduct) is the universal datum from which all three structures are recovered:

(a) The bar differential encodes the $A_\infty$ maps (coderivations of $T^c$ biject with multilinear operations): the RTT relation (2.5) is the degree-2 bar differential, and the Serre relations are the degree-3 component.
(b) The deconcatenation coproduct on $T^c$ induces the Drinfeld coproduct (2.10) via cobar dualisation.
(c) The $R$-matrix (2.2) is the degree-2 KZ monodromy: the parallel transport of the connection $\nabla = d + \hbar\,\Omega/(z_1 - z_2) \cdot d(z_1 - z_2)$ around the diagonal.

The passage $r(z) = \hbar\,\Omega/z \xrightarrow{\text{exponentiation}} R(u) = u\,I + \hbar\,P \xrightarrow{\text{RTT}}$ Yangian relations $\xrightarrow{\text{Gauss}}$ Drinfeld presentation is the ordered bar construction applied to $V_k(\mathfrak{sl}_2)$, producing $Y_\hbar(\mathfrak{sl}_2)$ as the $\mathsf{E}_1$-chiral Koszul dual.

**Part B: the ordered chiral center.** We compute $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(Y_\hbar(\mathfrak{sl}_2), Y_\hbar(\mathfrak{sl}_2))$ at low degrees on the formal disk $D$, and identify the kernel of the averaging map av.

*Degree* 0*: the center.* The degree-0 cohomology is the chiral center $Z(Y_\hbar(\mathfrak{sl}_2))$: those elements $a$ with $a_{(n)}b = 0$ for all $b$ and $n \geq 0$. The quantum determinant $\mathrm{qdet}\ T(u) = A(u)\,D(u - \hbar) - B(u)\,C(u - \hbar)$ is central in $Y_\hbar(\mathfrak{sl}_2)$ [37]. The center is the polynomial algebra generated by the coefficients of $\mathrm{qdet}\ T(u)$:

$$\mathrm{HH}^{0,\mathrm{ord}}_{\mathrm{ch}} = Z(Y_\hbar(\mathfrak{sl}_2)) = \mathbb{C}[\mathrm{qdet}\ T(u)]. \tag{2.13}$$

The averaging map $\mathrm{av}_0$ is an isomorphism at degree 0 (there is nothing to symmetrise), so $\ker(\mathrm{av}_0) = 0$.

*Degree* 2*: the KZ local system and* $\ker(\mathrm{av}_2) \neq 0$*.* This is the degree where the ordered theory diverges from the symmetric. The ordered chiral Hochschild complex at degree 2 involves the de Rham cohomology of $\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})$ with coefficients in the KZ local system. At degree 2, the leading-order KZ connection $\nabla = d + \hbar\,\Omega\,d\log(z_1 - z_2)$ captures the classical $r$-matrix data $r(z) = \hbar\,\Omega/z$. The full quantum $R$-matrix $R(u) = u\,I + \hbar\,P$ contributes higher-order corrections visible at degree $\geq 3$ through the Drinfeld associator; at degree 2 the connection is:

$$\nabla_{\mathrm{KZ}} = d + \frac{\hbar\,\Omega}{z_1 - z_2}\,d(z_1 - z_2) = d + \hbar\,\Omega\,d\log(z_1 - z_2), \tag{2.14}$$

with coefficients in $\mathrm{End}(Y_\hbar^{\otimes 2})$. Setting $w = z_1 - z_2$, the flat sections on $\mathbb{C}^\times$ satisfy $\partial\Phi/\partial w = (\hbar\,\Omega/w)\,\Phi$.

For the Yangian on the fundamental representation $V \otimes V$, the Casimir decomposes along the irreducible $\mathfrak{sl}_2$-components:

$$V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V, \qquad \Omega\big|_{\mathrm{Sym}^2 V} = +\tfrac{1}{2}, \quad \Omega\big|_{\bigwedge^2 V} = -\tfrac{3}{2}.$$

The monodromy of $\nabla_{\mathrm{KZ}}$ around $w = 0$ on each isospin component is

$$M_{\mathrm{sym}} = e^{2\pi i\hbar/2} = e^{\pi i/(k+2)}, \qquad M_{\mathrm{alt}} = e^{-3\pi i/(k+2)}. \tag{2.15}$$

The averaging map at degree 2 is $\mathrm{av}_2\colon H^*_{\mathrm{dR}}(\mathrm{Conf}_2^{\mathrm{ord}}, \nabla_{\mathrm{KZ}}) \to H^*_{\mathrm{dR}}(\mathrm{Conf}_2, \nabla_{\mathrm{KZ}})^{\Sigma_2}$, where $\Sigma_2$ acts by the flip $w \mapsto -w$ composed with the $R$-matrix.

For an $\mathsf{E}_\infty$-chiral algebra, $R = \mathrm{id}$ and the flip acts by the Koszul sign; $\mathrm{av}_2$ is an isomorphism and $\ker(\mathrm{av}_2) = 0$. For the Yangian, $R(u) \neq \mathrm{id}$: the $R$-matrix mixes the $\mathrm{Sym}^2 V$ and $\bigwedge^2 V$ channels non-trivially, and the $\Sigma_2$-action on the ordered de Rham cohomology is the composition of the geometric flip with $R$.

The kernel is spanned by those flat sections that are antisymmetric under this twisted $\Sigma_2$-action:

$$\ker(\mathrm{av}_2) \cong \bigwedge^2 V = \mathbb{C}. \tag{2.16}$$

The generator is the flat section $\Phi_{\mathrm{alt}}(w) = w^{-3\hbar/2}$ on the antisymmetric channel of $V \otimes V$. The full $R$-matrix $R(u) = u\,I + \hbar\,P$ evaluates on the antisymmetric component as $(u - \hbar)\,\mathrm{id}$; the $\Sigma_2$-flip sends $\Phi_{\mathrm{alt}}(w)$ to $R_{21}(-w)\,\Phi_{\mathrm{alt}}(-w) = -\Phi_{\mathrm{alt}}(w)$ (the $R$-matrix eigenvalue $-1$ on $\bigwedge^2 V$ at the leading order cancels the geometric sign). This class is in the ordered cohomology but maps to zero under $\mathrm{av}_2$: it is invisible to the symmetric chiral homology.

The scalar shadow surviving $\mathrm{av}_2$ is $\kappa = 3(k+2)/4$ (equation (9.9)).

*Degree* 3*: the Drinfeld associator in* $\ker(\mathrm{av}_3)$*.* At degree 3, the KZ system on $\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})$ is

$$d\Phi = \hbar\left(\frac{\Omega_{12}}{z_1 - z_2}\,d(z_1 - z_2) + \frac{\Omega_{13}}{z_1 - z_3}\,d(z_1 - z_3) + \frac{\Omega_{23}}{z_2 - z_3}\,d(z_2 - z_3)\right)\Phi.$$

The fundamental group $\pi_1(\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})) = PB_3$ (pure braid group on three strands). The KZ monodromy gives a representation $\rho_{\mathrm{KZ}}\colon PB_3 \to \mathrm{GL}(V^{\otimes 3})$.

The Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ is the regularised holonomy of this connection along the real path from the boundary stratum $|z_1 - z_2| \ll |z_2 - z_3|$ to the stratum $|z_2 - z_3| \ll |z_1 - z_2|$ in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The averaging map $\mathrm{av}_3$ sends ordered data to $\Sigma_3$-coinvariants. At degree 3, the relevant representation is the adjoint $\mathfrak{g} = \mathfrak{sl}_2$ (dimension 3), not the fundamental $V = \mathbb{C}^2$ (dimension 2) used at degree 2: the convolution algebra acts on $\mathfrak{g}^{\otimes n}$ via the adjoint action (the Casimir operators $\Omega_{ij}$ act through

the adjoint representation on each tensor factor), and the Drinfeld associator lives in $\Lambda^3(\mathfrak{g})$. The kernel of $\mathrm{av}_3$ contains the associator:

$$\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g}) = \mathbb{C}, \tag{2.17}$$

one-dimensional, carrying the associator class. This computation follows from the structure of the cohomology of $\mathrm{Conf}_3^{\mathrm{ord}}(\mathbb{C})$: the top-degree Arnold class $\omega_{12} \wedge \omega_{23} \in H^2(\mathrm{Conf}_3^{\mathrm{ord}})$ maps to the alternating representation of $\Sigma_3$, and the KZ-twisted version (with coefficients in $\mathfrak{g}^{\otimes 3}$) gives $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g})$. For $\mathfrak{g} = \mathfrak{sl}_2$: $\dim \Lambda^3(\mathfrak{g}) = \binom{3}{3} = 1$.

For $\mathsf{E}_\infty$-chiral algebras, the formality bridge forces $\ker(\mathrm{av}_3) = 0$: the associator is in the image of a $\Sigma_3$-equivariant quasi-isomorphism and carries no independent ordered data. For the Yangian, the $\mathcal{D}$-module is not $\Sigma_3$-equivariant (because $R \neq \mathrm{id}$), and the associator survives as a non-trivial element of the ordered chiral center invisible to any symmetric theory.

*Explicit computation in* $\ker(\mathrm{av}_3)$. On the fundamental representation $V = \mathbb{C}^2$ of $\mathfrak{sl}_2$, the KZ system at three points reduces to a $2 \times 2$ Fuchsian system on the multiplicity space $M = \mathrm{Hom}_{\mathfrak{sl}_2}(V_1, V^{\otimes 3})$, where $V_j$ is the spin-$j/2$ irreducible (so $V_1 = \mathbb{C}^2$ is the fundamental). Since $V^{\otimes 3} \cong V_3 \oplus V_1^{\oplus 2}$, $M$ is 2-dimensional: its two basis vectors correspond to the (12)-pair coupled to spin 0 (the $\bigwedge^2 V$ channel) and spin 1 (the $\mathrm{Sym}^2 V$ channel). The Casimir $\Omega_{12}$ acts on $M$ as $A = \mathrm{diag}(-\frac{3}{2}, \frac{1}{2})$ (eigenvalue $-\frac{3}{2}$ on the spin-0 channel, $\frac{1}{2}$ on spin-1). The representation of $\Sigma_3$ on $M$ has character $(2, 0, 0, 0, -1, -1)$: it is the standard representation with no trivial summand. The Reynolds operator $\frac{1}{6}\sum_{\sigma \in \Sigma_3} \sigma$ therefore vanishes identically on $M$, so the *entire* 2-dimensional multiplicity space lies in $\ker(\mathrm{av}_3)$. The Drinfeld associator acts on $M$ as a pure $\mathsf{E}_1$ invariant with no symmetric shadow.

At level $k = 2$ ($\kappa = 3(k+2)/4 = 3$), direct numerical integration of the Fuchsian system gives

$$\Phi_{\mathrm{KZ}}\big|_M \;=\; \begin{pmatrix} 0.884 & -0.204 \\ 0.306 & 1.061 \end{pmatrix}, \qquad \det \Phi_{\mathrm{KZ}}\big|_M = 1,$$

with eigenvalues on the unit circle (the connection is unitary on $M$). The off-diagonal entry $|\Phi_{12}| = 0.204$ measures the channel-mixing amplitude between the $j = 0$ and $j = 1$ sectors; it decays as $2.849/\kappa^2$ at large $\kappa$, consistent with the semiclassical expansion

$$\Phi_{\mathrm{KZ}}\big|_M \;=\; I \;-\; \frac{\zeta(2)}{\kappa^2}\,[t_{12},\, t_{23}] \;+\; O(1/\kappa^3),$$

where $t_{ij} = \Omega_{ij}/(k + h^\vee)$ and $[t_{12}, t_{23}] \neq 0$ on $M$ (reflecting the non-abelian Lie bracket). This is an explicit non-trivial class in $\ker(\mathrm{av}_3)$: an ordered invariant of the Yangian that is invisible to every symmetric chiral-homological observable.

*Degree $n$: braid group monodromy.* At degree $n$, the ordered chiral center carries the full pure braid group monodromy of the KZ system with the

quantum $R$-matrix:

$$\rho_{\mathrm{KZ}}^{(n)}\colon PB_n \;\to\; \mathrm{GL}\big(V^{\otimes n}\big).$$

The symmetric shadow at degree $n$ retains only the scalar invariants: the traces of the monodromy matrices and the Markov-trace data. The kernel $\ker(\mathrm{av}_n)$ carries the full matrix-valued monodromy, from which knot invariants (the Jones polynomial and its coloured generalisations) are extracted.

The Poincaré series of the ordered chiral centre, evaluated on the fundamental representation $V = \mathbb{C}^2$, admits a closed-form generating function. At degree $n$, the ordered datum is $V^{\otimes n}$ (dimension $2^n$) and the symmetric shadow is $\mathrm{Sym}^n(V)$ (dimension $n+1$), so

$$P_V^{\mathrm{ord}}(t) = \sum_{n\geq 0} 2^n\, t^n = \frac{1}{1-2t}, \qquad P_V^{\Sigma}(t) = \sum_{n\geq 0}(n+1)\, t^n = \frac{1}{(1-t)^2}.$$

The kernel series recording the genuinely $\mathsf{E}_1$ data at each degree is

$$\Delta P_V(t) \;=\; P_V^{\mathrm{ord}}(t) - P_V^{\Sigma}(t) \;=\; \frac{1}{1-2t} - \frac{1}{(1-t)^2} \;=\; \frac{t^2}{(1-2t)(1-t)^2}, \tag{2.18}$$

with coefficients $\dim\ker(\mathrm{av}_n) = 2^n - (n+1)$ for $n \geq 0$. The first ten values are $0, 0, 1, 4, 11, 26, 57, 120, 247, 502$: the kernel vanishes at degrees 0 and 1, starts at degree 2 with the antisymmetric $R$-matrix block $\bigwedge^2 V = \mathbb{C}$, and grows exponentially thereafter. The symmetric shadow retains $O(n)$ dimensions out of $O(2^n)$; almost all ordered data is invisible to the Beilinson–Drinfeld theory at large degree.

**Why this is new.** This is the first instance where the formality bridge (Theorem 7.5) fails: the $\mathcal{D}$-module is not $\Sigma_n$-equivariant because $R(u) \neq \mathrm{id}$, so the ordered chiral homology carries strictly more data than the symmetric theory.

| $n$ | $\dim\ker(\mathrm{av}_n)$ | Ordered datum ($V = \mathbb{C}^2$) | Symmetric shadow |
|---|---|---|---|
| 0 | 0 | $Z(Y_\hbar) = \mathbb{C}[\mathrm{qdet}]$ | $Z(Y_\hbar)$ |
| 1 | 0 | $V = \mathbb{C}^2$ | $V$ |
| 2 | 1 | $V^{\otimes 2}$, dim 4; $R$-matrix | $\mathrm{Sym}^2 V$, $\kappa = 3(k{+}2)/4$ |
| 3 | 4 | $V^{\otimes 3}$, dim 8; $\supseteq$ associator | $\mathrm{Sym}^3 V$, dim 4 |
| 4 | 11 | $V^{\otimes 4}$, dim 16 | $\mathrm{Sym}^4 V$, dim 5 |
| $n$ | $2^n{-}(n{+}1)$ | $V^{\otimes n}$, dim $2^n$; $PB_n$-monodromy | $\mathrm{Sym}^n V$, dim $n{+}1$ |

At degree 3, the 4-dimensional kernel contains the Drinfeld associator class $\Lambda^3(\mathfrak{g}) = \mathbb{C}$ as a distinguished one-dimensional subspace; the remaining 3 dimensions carry the mixed-symmetry Schur–Weyl components of $V^{\otimes 3}$ under $\Sigma_3$.

The computations in Example 2.3 determine the ordered chiral centre at low degrees. We now state the expected global structure.

**Conjecture 2.4** (Ordered derived chiral centre of the chiral Yangian)**.** *Let* $\mathfrak{g} = \mathfrak{sl}_2$ *and let* $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ *be the chiral Yangian on a smooth curve* $X$*.*

(i) (Infinite-dimensionality.) *The ordered derived chiral centre*

$$Z_{\mathrm{ch}}^{\mathrm{der,ord}}\big(Y_\hbar(\mathfrak{sl}_2)^{\mathrm{ch}}\big)$$

*is infinite-dimensional as a graded vector space. This stands in contrast to the derived chiral centre of any* $\mathsf{E}_\infty$*-chiral algebra, which has finite-dimensional cohomology concentrated in degrees* $\{0,1,2\}$ (Theorem H)*.*

(ii) (Poincaré series – fundamental representation.) *On the fundamental representation* $V = \mathbb{C}^2$*, the ordered and symmetric Poincaré series are*

$$P_V^{\mathrm{ord}}(t) = \frac{1}{1-2t}, \qquad P_V^{\Sigma}(t) = \frac{1}{(1-t)^2},$$

*and the kernel generating function is*

$$\Delta P_V(t) \;=\; \frac{1}{1-2t} - \frac{1}{(1-t)^2} \;=\; \frac{t^2}{(1-2t)(1-t)^2}, \tag{2.19}$$

*with* $n$*-th coefficient* $\dim\ker(\mathrm{av}_n) = 2^n - (n+1)$ $(= 0,0,1,4,11,26,57,120,247,502,\ldots)$*. At degree* $2$*:* $\dim\ker(\mathrm{av}_2) = 1$*, the antisymmetric* $R$*-matrix block* $\bigwedge^2 V$ (Example 2.3)*. At degree* $3$*:* $\dim\ker(\mathrm{av}_3) = 4$*, containing the Drinfeld associator class* $\Lambda^3(\mathfrak{g}) = \mathbb{C}$ *and three mixed-symmetry Schur–Weyl dimensions.*

(iii) (Poincaré series – adjoint representation.) *On the adjoint representation* $\mathfrak{g} = \mathfrak{sl}_2$ $(\dim\mathfrak{g} = 3)$*, the kernel generating function is*

$$\Delta P_{\mathfrak{g}}(t) \;=\; \frac{1}{1-3t} - \frac{1}{(1-t)^3} \;=\; \frac{t^2(3-t)}{(1-3t)(1-t)^3}, \tag{2.20}$$

*with* $n$*-th coefficient* $\dim\ker(\mathrm{av}_n) = 3^n - \binom{n+2}{2}$ $(= 0,0,3,17,66,222,701,\ldots)$*. For a representation of dimension* $d$*, the general formula is*

$$\Delta P_d(t) \;=\; \frac{1}{1-dt} - \frac{1}{(1-t)^d}, \qquad \dim\ker(\mathrm{av}_n) = d^n - \tbinom{n+d-1}{n}. \tag{2.21}$$

*The ordered data grows exponentially* ($d^n$) *while the symmetric shadow grows polynomially* ($O(n^{d-1})$)*; at large degree, the kernel carries asymptotically all of the ordered content. At each degree* $n$*, the* $PB_n$*-monodromy of the Knizhnik–Zamolodchikov system furnishes* $\ker(\mathrm{av}_n)$ *with a faithful representation of the pure braid group.*

(iv) ($\mathsf{E}_2$-algebra structure.) *The ordered derived chiral centre carries an* $\mathsf{E}_2$*-algebra structure via the classical Deligne conjecture applied to the ordered bar coalgebra* $B^{\mathrm{ord}}(Y_\hbar^{\mathrm{ch}})$*. It does* not *carry an* $\mathsf{E}_3$*-structure: the* $\mathcal{D}$*-module underlying* $Y_\hbar^{\mathrm{ch}}$ *does not descend to the symmetric power* $X^{(n)}$*, so the symmetric bar* $\overline{B}^{\Sigma}$ *does not exist, and the higher Deligne conjecture* (*which produces* $\mathsf{E}_3$ *from the* $\mathsf{E}_2$*-coalgebra* $\overline{B}^{\Sigma}$) *is*

*inapplicable. The* $\mathsf{E}_2$*-structure is the maximal algebraic structure visible to the ordered theory.*

(v) (Gerstenhaber bracket recovers $\mathfrak{g}$.) *The degree-*1 *component of the ordered derived chiral centre, equipped with the* $\mathsf{E}_2$ *Gerstenhaber bracket, satisfies*

$$\mathrm{HH}^1_{\mathrm{ord}}\big(Y_\hbar(\mathfrak{sl}_2)^{\mathrm{ch}}\big) \;\cong\; \mathfrak{sl}_2$$

*as a Lie algebra. The bracket arises from the classical* $r$*-matrix* $r(z) = \hbar\,\Omega/z$ *(equation (2.1); on* $V \otimes V$*,* $\Omega = P - \frac{1}{2}I$*): at first order in* $\hbar$*, the collision residue of* $r(z)$ *recovers the Lie bracket* $[X, Y]$*.*

(vi) (Symmetric comparison.) *The averaging map* $\mathrm{av}\colon Z^{\mathrm{der,ord}}_{\mathrm{ch}} \to Z^{\mathrm{der},\Sigma}_{\mathrm{ch}}$ *is surjective. Its kernel*

$$\ker(\mathrm{av}) \;=\; \bigoplus_{n\geq 2} \ker(\mathrm{av}_n)$$

*carries the full pure braid group monodromy data of the KZ system; this is the genuinely* $\mathsf{E}_1$ *content of the ordered theory, invisible to every symmetric chiral-homological observable.*

*Remark* 2.5 (Categorical Drinfeld centre and the passage to $\mathsf{E}_3$). The conjecture above describes the *algebraic* ordered derived centre, which is an $\mathsf{E}_2$-algebra. To recover an $\mathsf{E}_3$-structure, one must pass through the categorical Drinfeld centre: replacing the chiral Yangian $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ by its *category of modules* $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$-mod and computing the Drinfeld centre $\mathcal{Z}(Y_\hbar(\mathfrak{g})^{\mathrm{ch}}\text{-mod})$ produces a braided monoidal category whose deformation-class is an $\mathsf{E}_3$-algebra. This is the content of the categorified averaging step in the $\mathsf{E}_n$ operadic circle ($\mathsf{E}_1 \to \mathsf{E}_2 \to \mathsf{E}_3$): the Drinfeld centre categorifies the Reynolds averaging map, and the passage from $\mathsf{E}_2$ to $\mathsf{E}_3$ requires working at the categorical rather than algebraic level. Concretely, the Drinfeld centre $\mathcal{Z}(Y_\hbar(\mathfrak{sl}_2)^{\mathrm{ch}}\text{-mod})$ should recover the quantum group category $\mathrm{Rep}_q(\mathfrak{sl}_2)$ at $q = e^{2\pi i/(k+2)}$. This identification is the subject of the DK-5 programme; see [34, §6] for the current status in type $A$.

The computations in Example 2.3 and Conjecture 2.4(v) concern the kernel of the averaging map on the ordered chiral centre, a cohomological object. The following proposition establishes the underlying representation-theoretic input: the dimension of $\ker(\mathrm{av}_n)$ on the bare tensor power $V^{\otimes n}$, before passing to bar-complex cohomology.

**Proposition 2.6** (Kernel of the Reynolds projector: general simple Lie algebras)**.** *Let* $\mathfrak{g}$ *be any simple Lie algebra over* $\mathbb{C}$*, let* $V$ *be any finite-dimensional representation of* $\mathfrak{g}$ *with* $d = \dim V$*, and let* $\mathrm{av}_n\colon V^{\otimes n} \to (V^{\otimes n})_{\Sigma_n}$ *denote the Reynolds averaging map (projection onto the* $\Sigma_n$*-coinvariant quotient). Then*

$$\dim\ker(\mathrm{av}_n\big|_{V^{\otimes n}}) \;=\; d^n - \binom{n+d-1}{d-1}. \tag{2.22}$$

*The formula depends only on $d = \dim V$, not on $\mathfrak{g}$ or on the $\mathfrak{g}$-module structure of $V$.*

*Special cases. For $\mathfrak{g} = \mathfrak{sl}_2$ with the fundamental representation $V = \mathbb{C}^2$: $\dim\ker(\mathrm{av}_n) = 2^n - (n+1)$. For any $\mathfrak{g}$ with $d$-dimensional representation $V$, at $n = 2$: $\dim\ker(\mathrm{av}_2) = d^2 - d(d{+}1)/2 = d(d{-}1)/2 = \dim\bigwedge^2 V$.*

*Proof.* The Reynolds projector $\mathrm{av}_n = \frac{1}{n!}\sum_{\sigma\in\Sigma_n}\sigma$ acts on $V^{\otimes n}$ by permuting tensor factors. Its image is the $\Sigma_n$-invariant subspace $\mathrm{Sym}^n(V)$, regardless of any Lie algebra structure on $V$. The dimension $\dim\mathrm{Sym}^n(V) = \binom{n+d-1}{d-1}$ (the number of degree-$n$ monomials in $d$ variables) depends only on $d = \dim V$, as the stars-and-bars argument uses no algebraic structure beyond the vector space. Since $\dim V^{\otimes n} = d^n$, the kernel has dimension $d^n - \binom{n+d-1}{d-1}$.

When $\mathfrak{g} = \mathfrak{gl}_r$ with $V = \mathbb{C}^r$, Schur–Weyl duality refines this to an isotypic decomposition. The tensor power decomposes into $(\Sigma_n, \mathfrak{gl}_r)$-bimodules:

$$V^{\otimes n} \;\cong\; \bigoplus_{\lambda\vdash n,\ \ell(\lambda)\le r} S_\lambda \otimes V_\lambda(\mathfrak{gl}_r), \tag{2.23}$$

where $\lambda$ runs over partitions of $n$ with at most $r$ rows, $S_\lambda$ is the corresponding Specht module of $\Sigma_n$, and $V_\lambda(\mathfrak{gl}_r)$ is the irreducible $\mathfrak{gl}_r$-representation of highest weight $\lambda$. The projector $\mathrm{av}_n$ selects the trivial isotypic component $S_{(n)} = \mathbb{C}$ and annihilates every $S_\lambda$ with $\lambda \neq (n)$. The image is

$$\mathrm{im}(\mathrm{av}_n) \;\cong\; S_{(n)} \otimes V_{(n)}(\mathfrak{gl}_r) \;=\; \mathrm{Sym}^n(\mathbb{C}^r),$$

and the kernel decomposes by isotypic components:

$$\ker(\mathrm{av}_n) \;\cong\; \bigoplus_{\substack{\lambda\vdash n\\ \ell(\lambda)\le r,\ \lambda\neq(n)}} S_\lambda \otimes V_\lambda(\mathfrak{gl}_r).$$

For a general simple $\mathfrak{g}$ acting on $V$, the Schur–Weyl decomposition involves the commutant algebra $\mathrm{End}_{\mathfrak{g}}(V^{\otimes n})$ rather than the group algebra of $(V)$, and the individual summands carry different $\mathfrak{g}$-representations with different dimensions. The total kernel dimension, however, depends only on $\dim\mathrm{Sym}^n(V) = \binom{n+d-1}{d-1}$, which is determined by $d$ alone. □

*Remark* 2.7 (Explicit values and Conjecture 2.4). Table 1 records the kernel dimensions at low degrees for the fundamental representations of $\mathfrak{sl}_2$ through $\mathfrak{sl}_5$ (of dimensions $d = 2, 3, 4, 5$), together with the fundamental representations of $\mathfrak{so}_5$ ($d = 5$) and $G_2$ ($d = 7$). Since the formula depends only on $d = \dim V$, the rows for $\mathfrak{sl}_5$ and $\mathfrak{so}_5$ are identical. These are representation-theoretic upper bounds on the dimensions of the corresponding components of the ordered chiral centre in Conjecture 2.4(v): $\ker(\mathrm{av}_n)$ on the ordered chiral centre is a subquotient of $\ker(\mathrm{av}_n\big|_{V^{\otimes n}})$, obtained by passing to bar-complex cohomology (only the cocycles in the non-trivial $\Sigma_n$-isotypics survive). At degree 2 for $\mathfrak{sl}_2$: the full 1-dimensional kernel $\bigwedge^2 V = \mathbb{C}$ survives (Example 2.3, equation (2.16)). At degree $n \ge 3$, the kernel on the chiral centre is in general strictly smaller than the representation-theoretic bound $d^n - \binom{n+d-1}{d-1}$.

| | $n=2$ | $n=3$ | $n=4$ | $n=5$ | $n=6$ |
|---|---|---|---|---|---|
| $\mathfrak{sl}_2$ ($d$=2) | 1 | 4 | 11 | 26 | 57 |
| $\mathfrak{sl}_3$ ($d$=3) | 3 | 17 | 66 | 222 | 701 |
| $\mathfrak{sl}_4$ ($d$=4) | 6 | 44 | 221 | 968 | 4,012 |
| $\mathfrak{sl}_5$, $\mathfrak{so}_5$ ($d$=5) | 10 | 90 | 555 | 2,999 | 15,415 |
| $G_2$ *fund.* ($d$=7) | 21 | 259 | 2,191 | 16,345 | 116,725 |
| *Formula* | $d^n - \binom{n+d-1}{d-1}$ | | | | |

TABLE 1. Dimension of $\ker(\mathrm{av}_n\big|_{V^{\otimes n}})$ for a $d$-dimensional representation $V$ at degrees $n = 2, \ldots, 6$ (Proposition 2.6). The formula depends only on $d = \dim V$.

## 3. Background: chiral algebras and the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy

The ordered theory rests on a single structural observation: every chiral algebra has two independent invariants (chirality hierarchy and topological ladder), and the standard Beilinson–Drinfeld theory sees only their $\mathsf{E}_\infty$-equivariant shadow. This section recalls the two hierarchies and the bridge theorem that identifies $\mathsf{E}_1$-chiral algebras with $\mathsf{E}_2$-topological data.

3.1. **Two orthogonal axes.** A chiral algebra carries two independent pieces of structure: how commutative its OPE is (the chirality hierarchy, $\mathsf{E}_\infty$-chiral $\subset$ $\mathsf{E}_1$-chiral) and how many real dimensions its configuration spaces see (the topological ladder, $\mathsf{E}_1 \subset \mathsf{E}_2 \subset \cdots \subset \mathsf{E}_\infty$). Conflating the two is the source of most errors in the subject; separating them is the starting point of the ordered theory.

**Definition 3.1** (Chirality hierarchy)**.** For chiral algebras on a fixed algebraic curve $X$, the *chirality hierarchy* $\mathsf{E}_\infty$-chiral $\subset$ $\mathsf{E}_1$-chiral records the commutativity of the OPE:

(i) An $\mathsf{E}_\infty$*-chiral algebra* has symmetric OPE: the vertex operators $Y(a,z)$ and $Y(b,w)$ commute after clearing poles. Every standard vertex algebra (Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}$-algebras, lattice algebras) is $\mathsf{E}_\infty$-chiral. The symmetry means the factorisation product descends to $X^{(n)} = X^n/\Sigma_n$.

(ii) An $\mathsf{E}_1$*-chiral algebra* has braided OPE: the vertex operators satisfy $S$-locality

$$(z-w)^N Y(a,z)Y(b,w) = (z-w)^N Y(b,w)Y(a,z) \cdot S(z-w) \tag{3.1}$$

where $S(z)$ is a vertex $R$-matrix satisfying the quantum Yang–Baxter equation, unitarity, the shift condition, and the hexagon axiom (Etingof–Kazhdan [18]). The factorisation product lives on ordered configurations $X^n$, not on $X^{(n)}$.

**Warning 3.2** (Multiple notions of $\mathsf{E}_1$-chiral)**.** At least five notions coexist in the literature: (A) strict ChirAss-algebra (ChirAss is the chiral associative operad, governing $A_\infty$-chiral algebras), (B) $A_\infty$-algebra in the chiral endomorphism operad (structure maps meromorphic in spectral parameters), (C) Etingof–Kazhdan quantum vertex algebra (five axioms including $S$-locality), (D) $A_\infty$-algebra object in $\mathsf{E}_1$-chiral algebras (double $A_\infty$), and (E) factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$. On the Koszul locus, (B) and (C) determine each other via a Drinfeld associator $\Phi$ (Theorem 6.5); (A) is a special case of (B); (E) is the $\mathcal{D}$-module realisation of (B). Each notion admits its own derived centre construction, and the output operad depends on which construction is used (see [34, Warning 2.21]). In this paper, Definition 3.1 uses Notion (C) for the algebraic definition and Notion (E) for the geometric realisation.

**Definition 3.3** (Koszul locus)**.** An $\mathsf{E}_1$-chiral algebra $\mathcal{A}$ lies on the *Koszul locus* if the bar-cobar adjunction is an equivalence: $\Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$. Equivalently, the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is a Koszul resolution of $\mathcal{A}$. Standard vertex algebras at generic level ($V_k(\mathfrak{g})$ for $k$ not a non-negative integer) and Yangians $Y_\hbar(\mathfrak{g})$ lie on the Koszul locus; the Koszul property can fail at special levels where the bar complex acquires extra cohomology.

**Definition 3.4** (The ordered Ran space)**.** For a smooth algebraic curve $X$, the *ordered Ran space* is

$$\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_{n\geq 1} U_n(X), \qquad U_n(X) = X^n \setminus \bigcup_{i<j} \Delta_{ij},$$

the moduli of ordered finite subsets of $X$: a point of $\mathrm{Ran}^{\mathrm{ord}}(X)$ is an ordered tuple $(x_1, \ldots, x_n)$ of distinct points of $X$. The *ordered factorisation structure* on $\mathrm{Ran}^{\mathrm{ord}}(X)$ encodes the merging of ordered subsets: for $(x_1, \ldots, x_k)$ and $(x_{k+1}, \ldots, x_n)$ with disjoint underlying sets, the concatenation $(x_1, \ldots, x_n)$ defines a map

$$\mathrm{union}_{k,n-k} \colon \mathrm{Ran}^{\mathrm{ord}}(X)_k \times_{\mathrm{disj}} \mathrm{Ran}^{\mathrm{ord}}(X)_{n-k} \to \mathrm{Ran}^{\mathrm{ord}}(X)_n.$$

**Definition 3.5** (The Ran space as coinvariant quotient)**.** The Beilinson–Drinfeld *Ran space* $\mathrm{Ran}(X) = \coprod_{n\geq 1} X^{(n)}$, where $X^{(n)} = X^n/\Sigma_n$ is the $n$-th symmetric product, is the $\Sigma_\bullet$-coinvariant quotient:

$$\mathrm{Ran}(X) = \mathrm{Ran}^{\mathrm{ord}}(X)/\Sigma_\bullet.$$

The factorisation structure on $\mathrm{Ran}(X)$ is the $\Sigma_\bullet$-equivariant shadow of the ordered factorisation structure on $\mathrm{Ran}^{\mathrm{ord}}(X)$: for disjoint finite subsets $I, J \subset X$, the merging map $\mathrm{Ran}(X)_I \times \mathrm{Ran}(X)_J \to \mathrm{Ran}(X)_{I\cup J}$ is the descent of $\mathrm{union}_{k,n-k}$ through the symmetric group quotient.

*Remark* 3.6 (Chiral algebras, $\mathsf{E}_2$, and the role of the curve)*.* An $\mathsf{E}_\infty$-chiral algebra $\mathcal{A}$ on a complex curve $X$ is a factorisation algebra on $\mathrm{Ran}(X)$; for an

$\mathsf{E}_1$-chiral algebra, the factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ lives on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 3.4), and descends to $\mathrm{Ran}(X)$ only when the braiding is trivial. In either case, $\mathcal{A}$ is *not* an $\mathsf{E}_2$-algebra in the operadic sense. The $\mathsf{E}_2$ structure lives on the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$, provided by the curve geometry (Warning 3.7). The holomorphic data (the spectral parameter $z$, the $r$-matrix $r(z)$, the KZ connection) refines this topological $\mathsf{E}_2$ structure; forgetting the complex structure of $X$ recovers the bare $\mathsf{E}_2$-coalgebra.

**Warning 3.7** (Bar-cobar gives $\mathsf{E}_1$; the curve gives $\mathsf{E}_2$)**.** The abstract bar-cobar adjunction for $\mathsf{E}_1$-algebras is:

$$\mathrm{Bar}\colon \mathsf{E}_1\text{-Alg} \rightleftarrows \mathsf{E}_1\text{-CoAlg}\colon \mathrm{Cobar}.$$

The output is an $\mathsf{E}_1$-coalgebra with the *deconcatenation* coproduct: $\Delta(a_1 \otimes \cdots \otimes a_n) = \sum_{k=0}^{n}(a_1 \otimes \cdots \otimes a_k) \otimes (a_{k+1} \otimes \cdots \otimes a_n)$. This is $\mathsf{E}_1$ (it respects the linear ordering of tensor factors).

The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ of a chiral algebra on a curve $X$ carries the *coshuffle* coproduct: $\Delta(a_1 \cdots a_n) = \sum_{I \sqcup J = \{1,\ldots,n\}} a_I \otimes a_J$ (sum over all unordered bipartitions). This is $\mathsf{E}_2/\mathsf{E}_\infty$: it uses the $\Sigma_n$-symmetry from unordered configuration spaces $\mathrm{Conf}_k(X)$ of the curve, whose real $2k$-dimensional topology carries the $\mathrm{FM}_2 \simeq \mathsf{E}_2$ operad.

The passage from $\overline{B}^{\mathrm{ord}}$ ($\mathsf{E}_1$, deconcatenation) to $\overline{B}^{\Sigma}$ ($\mathsf{E}_2$, coshuffle) is the $\Sigma_n$-coinvariant quotient. This quotient *introduces* the curve geometry: it replaces ordered configurations with unordered ones, and the deconcatenation with the coshuffle.

Without the curve, bar-cobar gives $\mathsf{E}_1$, and Higher Deligne gives the centre $\mathsf{E}_2$ ($\mathsf{E}_1 + 1$; this is the classical Deligne conjecture for associative algebras). With the curve, the bar is $\mathsf{E}_2$, and Higher Deligne gives the centre $\mathsf{E}_3$ ($\mathsf{E}_2 + 1$). The extra operadic level is the curve.

**Definition 3.8** (Topological ladder)**.** The *topological ladder* $\mathsf{E}_1 \subset \mathsf{E}_2 \subset \mathsf{E}_3 \subset \cdots \subset \mathsf{E}_\infty$ is indexed by real manifold dimension:

(i) $n = 1$: operations on $\mathrm{Conf}_k(\mathbb{R})$. $\mathsf{E}_1$-algebras $= A_\infty$-algebras $=$ homotopy associative algebras.
(ii) $n = 2$: operations on $\mathrm{Conf}_k(\mathbb{R}^2) \cong \mathrm{Conf}_k(\mathbb{C})$. $\mathsf{E}_2$-algebras govern braided monoidal categories.
(iii) $n = 3$: operations on $\mathrm{Conf}_k(\mathbb{R}^3)$. $\mathsf{E}_3$-algebras arise from perturbative Chern–Simons theory (Costello–Francis–Gwilliam [12]).

**Definition 3.9** (Shadow classification $G/L/C/M$)**.** The *shadow classification* ([34, §II]) partitions $\mathsf{E}_\infty$-chiral algebras by the pole structure of the classical $r$-matrix $r(z)$:

(i) Class $G$ (Gaussian): $r$-matrix pole order $\leq 1$, no structure constants (Heisenberg). Shadow depth $r = 2$.
(ii) Class $L$ (Lie): $r$-matrix pole order 1, non-abelian (affine Kac–Moody). Shadow depth $r = 3$.
(iii) Class $C$ (Clifford): nilpotent $r$-matrix ($\beta\gamma$, $bc$). Shadow depth $r = 4$.

(iv) Class $M$ (modular): $r$-matrix pole order $\geq 2$ (Virasoro, $\mathcal{W}$-algebras). Shadow depth $r = \infty$.

Here $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} = \mathrm{Hom}_{\Sigma}(B(\mathcal{A}), \mathcal{A})$ is the symmetric convolution algebra (the $\Sigma_n$-coinvariant quotient of the $\mathsf{E}_1$ convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1}$), equipped with convolution $L_\infty$ brackets $\ell_k$ induced by the bar coproduct; and the *shadow depth* $r$ is the smallest integer such that $\ell_k = 0$ for $k \geq r+1$ in $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$.

**Theorem 3.10** (Bridge; [34, Theorem 1.1(ii)])**.** *The chiral bar-cobar adjunction on a complex curve $X$ is the holomorphic refinement of the $n = 2$ chapter of $\mathsf{E}_n$ Koszul duality. The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ of a chiral algebra $\mathcal{A}$ on $X$ is an $\mathsf{E}_2$-coalgebra (from the coshuffle coproduct on $\mathrm{Conf}_k(X) \cong \mathrm{Conf}_k(\mathbb{R}^2)$), not merely an $\mathsf{E}_1$-coalgebra (which is what the abstract bar-cobar adjunction would give). For an $\mathsf{E}_1$-chiral algebra, the braid-group action on $\mathrm{Conf}_k(\mathbb{C})$ is non-trivial (the $R$-matrix braiding); for an $\mathsf{E}_\infty$-chiral algebra, it factors through the symmetric group. In both cases, the $\mathsf{E}_2$ structure on $\overline{B}^{\Sigma}$ comes from the curve, not from bar-cobar.*

*Remark* 3.11 (The bridge at the level of operads)*.* The real Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R})$ is isomorphic to the Stasheff associahedron $K_n$ (Sinha [42]). The inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ induces an embedding

$$K_n = \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R}) \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C}) \tag{3.2}$$

of the $A_\infty$ operad into the ordered Fulton–MacPherson operad. An $\mathsf{E}_1$-chiral algebra structure is an operadic map $A_\infty \to \mathrm{End}_{\mathcal{A}}^{\mathrm{ch,ord}}$ factoring through (3.2). The image of $K_n$ in $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ is real-codimension $n$: the $A_\infty$ operations are a thin slice of the full chiral operations.

## 4. $\mathcal{D}$-modules on ordered configuration spaces

The integration theory for $\mathsf{E}_\infty$-chiral algebras lives on symmetric products $X^{(n)} = X^n/\Sigma_n$, but the $\mathsf{E}_1$ theory requires ordered configurations $X^n$. What are the $\mathcal{D}$-modules on $X^n$ that encode the OPE, and what happens to them at the diagonals? The answer is the ordered factorisation $\mathcal{D}$-module: a holonomic $\mathcal{D}$-module on $X^n$ whose singularities along the diagonals are precisely the OPE poles.

### 4.1. Ordered configuration spaces and their compactifications.

The configuration space $U_n(X) = X^n \setminus \bigcup \Delta_{ij}$ and its Fulton–MacPherson compactification are the geometric substrate for ordered chiral homology. The compactification resolves the diagonals by blowup, producing boundary strata indexed by ordered collision patterns: rooted planar trees with $n$ labelled leaves.

**Definition 4.1** (Ordered configuration space)**.** Let $X$ be a smooth algebraic curve over $\mathbb{C}$. For $n \geq 0$, the *ordered configuration space* is the smooth

quasi-projective variety

$$U_n = U_n(X) := X^n \setminus \bigcup_{1\le i<j\le n} \Delta_{ij} \tag{4.1}$$

where $\Delta_{ij} = \{(x_1, \dots, x_n) \in X^n : x_i = x_j\}$ is the $(i, j)$-diagonal. For $n = 0$, $U_0 = \{\varnothing\} = \operatorname{Spec} \mathbb{C}$. For $n = 1$, $U_1 = X$.

**Definition 4.2** (Ordered FM compactification)**.** The *ordered Fulton–MacPherson compactification* $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(X)$ is the real oriented blowup of $X^n$ along all diagonals $\Delta_{ij}$ $(i < j)$, iterated in order of increasing codimension. It is a smooth manifold with corners whose interior is $U_n(X)$ and whose boundary strata are indexed by *ordered collision patterns*: rooted planar trees $T$ with $n$ labelled leaves, where each internal vertex records a group of points colliding simultaneously.

For each ordered collision pattern $T$ with internal vertices $v_1, \dots, v_r$ of valences $k_1, \dots, k_r$, the corresponding boundary stratum is

$$\partial_T \overline{\mathrm{FM}}_n^{\mathrm{ord}}(X) \cong \prod_{i=1}^{r} \overline{\mathrm{FM}}_{k_i}^{\mathrm{ord}}(T_{x_i}X) \ \times\ \overline{\mathrm{FM}}_{n'}^{\mathrm{ord}}(X) \tag{4.2}$$

where $T_{x_i}X \cong \mathbb{C}$ is the tangent space at the collision point $x_i$, and $n' = n - \sum_i (k_i - 1)$ is the number of surviving points after collisions.

*Remark* 4.3 ($\mathcal{D}$-modules on ordered spaces). The category $\mathcal{D}\text{-mod}(X^n)$ of $\mathcal{D}$-modules on $X^n$ is the standard abelian category of quasi-coherent sheaves with flat connection (Kashiwara [27], Hotta–Takeuchi–Tanisaki [23]). The diagonals $\Delta_{ij} \subset X^n$ are smooth divisors; the $*$-extension $j_*$ and the $!$-extension $j_!$ along $j\colon U_n \hookrightarrow X^n$ are the standard meromorphic and distributional extensions. The classical Riemann–Hilbert correspondence applies to regular holonomic $\mathcal{D}$-modules on $X^n$; for irregular holonomic $\mathcal{D}$-modules, one needs the irregular Riemann–Hilbert correspondence of D'Agnolo–Kashiwara [16]. Both cases arise in the chiral setting (see Proposition 4.6 and Warning 4.7).

4.2. **The ordered factorisation $\mathcal{D}$-module.** Given the configuration spaces and their compactifications, one extends the external product $\mathcal{A}^{\boxtimes n}$ meromorphically across the diagonals. The resulting $*$-extension $j_*j^*(\mathcal{A}^{\boxtimes n})$ is the ordered factorisation $\mathcal{D}$-module: a holonomic $\mathcal{D}$-module on $X^n$ whose singularity type along each diagonal $\Delta_{ij}$ is dictated by the OPE pole order of $\mathcal{A}$.

**Construction 4.4** (Ordered factorisation $\mathcal{D}$-module)**.** Let $\mathcal{A}$ be a chiral algebra on $X$. The *external product* $\mathcal{A}^{\boxtimes n}$ is the $\mathcal{D}$-module on $X^n$ defined by

$$\mathcal{A}^{\boxtimes n} := \mathcal{A} \boxtimes \mathcal{A} \boxtimes \cdots \boxtimes \mathcal{A} = \mathrm{pr}_1^*\mathcal{A} \otimes_{\mathcal{O}_{X^n}} \mathrm{pr}_2^*\mathcal{A} \otimes_{\mathcal{O}_{X^n}} \cdots \otimes_{\mathcal{O}_{X^n}} \mathrm{pr}_n^*\mathcal{A} \tag{4.3}$$

where $\mathrm{pr}_i\colon X^n \to X$ is the $i$-th projection.

The *ordered factorisation $\mathcal{D}$-module* at degree $n$ is the $*$-extension:

$$\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A}) := j_*j^*(\mathcal{A}^{\boxtimes n}) \in \mathcal{D}\text{-mod}(X^n). \tag{4.4}$$

Concretely, sections of $\mathcal{F}_n^{\mathrm{ord}}$ over an open $V \subset X^n$ are sections of $\mathcal{A}^{\boxtimes n}$ over $V \cap U_n$ that extend meromorphically across the diagonals. The poles along $\Delta_{ij}$ are the OPE poles: if the OPE $Y(a,z)b = \sum_{k \geq -N} a_{(k)} b\, z^{-k-1}$ has poles of order at most $N$ for all $a, b \in \mathcal{A}$, then sections of $\mathcal{F}_n^{\mathrm{ord}}$ have poles of order at most $N$ along each $\Delta_{ij}$.

**Definition 4.5** (Ordered factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$)**.** For a chiral algebra $\mathcal{A}$ on $X$, the *ordered factorisation $\mathcal{D}$-module* is the $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 3.4) whose restriction to the $n$-th stratum $U_n(X)$ is

$$\mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big|_{U_n} = \mathcal{A}^{\boxtimes n}\big|_{U_n}$$

with the factorisation product: for the merging map $\mathrm{union}_{k,n-k}$, the factorisation isomorphism

$$\mathrm{union}^*_{k,n-k}\mathcal{F}^{\mathrm{ord}}\big|_{U_n} \;\simeq\; \mathcal{F}^{\mathrm{ord}}\big|_{U_k} \boxtimes \mathcal{F}^{\mathrm{ord}}\big|_{U_{n-k}}$$

is the OPE (the meromorphic extension across the diagonals). The KZ connection endows $\mathcal{F}^{\mathrm{ord}}$ with a flat connection on each stratum, compatible with the factorisation maps. The degree-by-degree $\mathcal{D}$-modules $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ of Construction 4.4 are the restrictions of this single object to the strata of $\mathrm{Ran}^{\mathrm{ord}}(X)$.

**Proposition 4.6** (Properties of the ordered factorisation $\mathcal{D}$-module)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve $X$.*

(i) (Holonomicity and singularity type.) *$\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ is a holonomic $\mathcal{D}$-module on $X^n$. The singularity type along $\bigcup \Delta_{ij}$ depends on the pole order of the $r$-matrix:*
- Class $G$ (Heisenberg): $r(z) = k/z$, *pole order* 1. *Regular singularities.*
- Class $L$ (affine KM): $r(z) = k\Omega/z$ (*trace-form*), *pole order* 1. *Regular singularities.*
- Class $C$ ($\beta\gamma$): *$r$ is a constant nilpotent endomorphism* (*$d$-log absorbs the simple OPE pole*). *Regular singularities.*
- Class $M$ (Virasoro, $\mathcal{W}_N$): $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, *pole order* 3. Irregular *singularities of Poincaré rank* 2.

(ii) (The KZ connection.) *The flat connection on $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})$ restricts on $U_n$ to the standard product connection on $\mathcal{A}^{\boxtimes n}|_{U_n}$ plus the Knizhnik–Zamolodchikov correction:*

$$\nabla_n = \nabla_{\mathcal{A}}^{\boxtimes n} + \sum_{i<j} r_{ij}(z_i - z_j)\, dz_{ij} \tag{4.5}$$

*where $r_{ij}(z) \in \mathrm{End}(\mathcal{A} \otimes \mathcal{A})((z))$ is the classical $r$-matrix* (*degree-2 projection of $\Theta_{\mathcal{A}}$*) *acting on the $(i,j)$ tensor factors, and $dz_{ij} = d(z_i - z_j)$ is the connection 1-form.*

*For affine Kac–Moody at level $k$, two equivalent conventions coexist. In the trace-form convention, with $\Omega_{\mathrm{tr}}$ the Casimir built from the*

*normalised trace form* $(X,Y)_{\mathrm{tr}} = \mathrm{tr}(\mathrm{ad}_X \circ \mathrm{ad}_Y)/(2h^\vee)$*:*

$$\nabla_n^{\mathrm{KM}} = \nabla^{\boxtimes n} + \sum_{i<j} \frac{k\,(\Omega_{\mathrm{tr}})_{ij}}{z_i - z_j}\, d(z_i - z_j). \tag{4.6}$$

*In the KZ normalisation, with* $\Omega$ *the inverse Killing form Casimir:*

$$\nabla_n^{\mathrm{KM}} = \nabla^{\boxtimes n} + \sum_{i<j} \frac{\Omega_{ij}}{(k+h^\vee)(z_i - z_j)}\, d(z_i - z_j). \tag{4.7}$$

*The bridge identity* $k\,(\Omega_{\mathrm{tr}})_{ij} = \Omega_{ij}/(k+h^\vee)$ *equates the two at generic* $k$ *(see* [34, §III.2]*). Both have a* simple pole *along* $\Delta_{ij}$*: the connection is Fuchsian.*

(iii) (Flatness.) $\nabla_n^2 = 0$ *on* $U_n$*, following from the classical Yang–Baxter equation for* $r(z)$*.*

(iv) (Equivariant descent.) *For* $\mathsf{E}_\infty$*-chiral* $\mathcal{A}$ *(where* $S(z) = \mathrm{id}$ *up to Koszul signs): ${\mathcal{F}}_n^{\mathrm{ord}}$ is* $\Sigma_n$*-equivariant, and* $({\mathcal{F}}_n^{\mathrm{ord}})^{\Sigma_n} \simeq {\mathcal{F}}_n^{\Sigma}$ *is the Beilinson–Drinfeld factorisation* $\mathcal{D}$*-module on* $X^{(n)}$*.*

(v) (Non-equivariance for $\mathsf{E}_1$.) *For* $\mathsf{E}_1$*-chiral* $\mathcal{A}$*: ${\mathcal{F}}_n^{\mathrm{ord}}$ is* not $\Sigma_n$*-equivariant. The braiding* $S(z)$ *on the* $(i,j)$ *factor introduces monodromy as the* $i$*-th and* $j$*-th points circle each other.*

*Proof.* Part (i): holonomicity follows from $j_*$ preserving the holonomic property (Kashiwara's theorem [27]). The external product $\mathcal{A}^{\boxtimes n}$ is holonomic on $X^n$ (being an external tensor of holonomic $\mathcal{D}$-modules on curves). The singularity type is determined by the pole structure of the connection. For class $G$ and $L$, the $r$-matrix $r(z)$ has a simple pole: writing the connection locally as $d + r(z)\,dz$ with $r(z) = A/z + O(1)$, the singularity is regular (Fuchsian). For class $C$, the $d$-log absorption of the OPE simple pole produces a connection with at worst logarithmic singularity: regular. For class $M$, the Virasoro $r$-matrix has a cubic pole $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$; the connection $d + r(z)\,dz$ has a pole of order 3 in $dz$, giving an irregular singularity of Poincaré rank $3-1=2$. Holonomic $\mathcal{D}$-modules on proper varieties have finite-dimensional de Rham cohomology regardless of the regularity of singularities (Mebkhout [40]).

Part (ii): the KZ connection arises from the Maurer–Cartan element. In the convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1} = \bigoplus_n \mathrm{End}((s^{-1}\bar{\mathcal{A}})^{\otimes n})$, the MC element $\Theta_{\mathcal{A}}$ at degree 2 gives $r(z) = \Theta_{\mathcal{A}}|_{\text{degree } 2} \in \mathrm{End}(\mathcal{A}\otimes\mathcal{A})((z))$. The connection 1-form is $r_{ij}(z_i - z_j)\,dz_{ij}$ where $dz_{ij} = d(z_i - z_j)$. Note: the Arnold forms $\omega_{ij} = d\log(z_i - z_j) = dz_{ij}/(z_i - z_j)$ are generators of $H^*(\mathrm{Conf}_n(\mathbb{C}))$ and appear as coefficients in the bar differential $d_{\mathrm{bar}}$; they are distinct from the connection 1-form.

Part (iii): the flatness condition $\nabla_n^2 = 0$ splits into three contributions: $(\nabla^{\boxtimes n})^2 = 0$ (individual connections); the cross-term $[\nabla^{\boxtimes n}, \sum r_{ij}\,dz_{ij}] = 0$ (from the shift condition $[\partial_z, r(z)] = -r'(z)$ paired with $d(dz_{ij}) = 0$); and $(\sum r_{ij}\,dz_{ij})^2 = 0$, which expands as $\sum_{i<j<k}([r_{12}, r_{13}] + [r_{12}, r_{23}] +$

$[r_{13}, r_{23}])\, dz_{ij} \wedge dz_{jk} = 0$, where the coefficient vanishes by the classical Yang–Baxter equation.

Part (iv): for $\mathsf{E}_\infty$-chiral $\mathcal{A}$, the locality axiom gives $S(z) = \mathrm{id}$ (up to signs from the Koszul convention), so the KZ monodromy is trivial and $\mathcal{F}_n^{\mathrm{ord}}$ descends to $X^{(n)}$.

Part (v): for $\mathsf{E}_1$-chiral $\mathcal{A}$, the KZ monodromy around $\Delta_{ij}$ is $\exp(2\pi i \cdot r_{ij})$, which acts by the quantum $R$-matrix $S_{ij}$ (Drinfeld–Kohno theorem). Since $S \neq \mathrm{id}$, $\mathcal{F}_n^{\mathrm{ord}}$ does not descend. □

**Warning 4.7** (Regularity dichotomy: gauge theory vs. gravity)**.** The singularity type of the factorisation $\mathcal{D}$-module correlates with a physical distinction (heuristic; the 3d interpretation requires constructing the bulk theory, which is not done in this paper).

- Regular singularities (class $G/L/C$): the connection is Fuchsian; the classical Riemann–Hilbert correspondence of Deligne applies. In the physics literature, the corresponding 3d theories are gauge theories (Chern–Simons for class $L$; abelian for class $G$; free matter for class $C$).
- Irregular singularities (class $M$): the connection has a higher-order pole; the classical Riemann–Hilbert correspondence does not apply, and one needs the irregular Riemann–Hilbert correspondence of D'Agnolo–Kashiwara [16]. The expected 3d theory involves gravity (holomorphic-topological gravity in the Beltrami parametrisation for Virasoro; see [13]). The irregular singularity carries Stokes data and wall-crossing phenomena (for the Virasoro algebra, at $c = -22/5$ the Stokes multipliers exhibit a wall-crossing).

The regularity type depends on the pole order of $r(z)$, which is the OPE pole order minus one ($d$-log absorption): class $G/L/C$ have $r$-matrix pole order $\leq 1$, class $M$ has pole order $\geq 2$.

## 5. Ordered chiral homology

The ordered factorisation $\mathcal{D}$-module $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ is a holonomic $\mathcal{D}$-module with flat connection. Its de Rham cohomology is a finite-dimensional graded vector space at each degree. Assembling these across degrees, with the chiral differential from FM boundary strata, produces the ordered chiral homology of $\mathcal{A}$ over $X$.

5.1. **Definition.** The de Rham cohomology of a holonomic $\mathcal{D}$-module on a proper variety is finite-dimensional (Mebkhout). Applied to $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}^{\mathrm{ord}}(X)$, this gives a graded vector space at each degree; the cross-degree chiral differential assembles these into a chain complex.

**Definition 5.1** (Ordered chiral homology via $\mathrm{Ran}^{\mathrm{ord}}(X)$)**.** Let $\mathcal{A}$ be a chiral algebra on a smooth proper curve $X$, and let $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ be the ordered

factorisation $\mathcal{D}$-module on $\mathrm{Ran}^{\mathrm{ord}}(X)$ (Definition 4.5). The *ordered chiral homology* is the derived pushforward to the point:

$$\int_X^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(X),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big) \;=\; \pi_*\mathcal{F}^{\mathrm{ord}}(\mathcal{A}) \tag{5.1}$$

where $\pi\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{pt}$ is the projection. The degree-by-degree decomposition

$$\int_X^{\mathrm{ord}} \mathcal{A} \;=\; \Big(\bigoplus_{n\geq 0} R\Gamma_{\mathrm{dR}}\big(X^n,\ \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})\big),\ \ d_{\mathrm{ch}}\Big) \tag{5.2}$$

is the computation of this pushforward stratum by stratum, where $\mathrm{dR}(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{D}} \Omega^{\bullet}_{X^n}$ is the de Rham complex of the $\mathcal{D}$-module $\mathcal{F}$ and $d_{\mathrm{ch}}$ is the *chiral differential*: the alternating sum of the restriction maps to the codimension-1 boundary strata of $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(X)$, corresponding to pairwise collisions of adjacent points. At each degree, the *ordered chiral homology at degree $n$* is

$$\int_{X,n}^{\mathrm{ord}} \mathcal{A} \;:=\; R\Gamma_{\mathrm{dR}}\big(X^n,\ \mathcal{F}_n^{\mathrm{ord}}(\mathcal{A})\big). \tag{5.3}$$

5.2. **Symmetric descent.** The quotient map $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ (Definition 3.5) induces the natural map from ordered to symmetric chiral homology.

**Proposition 5.2** (Symmetric descent recovers Beilinson–Drinfeld)**.** *Let $\mathcal{A}$ be an $\mathsf{E}_\infty$-chiral algebra on a smooth proper curve $X$. The quotient $q\colon \mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$ induces the averaging map*

$$\int_X^{\mathrm{ord}} \mathcal{A} = R\Gamma\big(\mathrm{Ran}^{\mathrm{ord}}(X),\, \mathcal{F}^{\mathrm{ord}}\big) \xrightarrow{\mathrm{av}} R\Gamma\big(\mathrm{Ran}(X),\, \mathcal{F}^{\Sigma}\big) = \int_X \mathcal{A}, \tag{5.4}$$

*which is a quasi-isomorphism. At each degree, the $\Sigma_n$-coinvariants of the ordered chiral homology recover the Beilinson–Drinfeld chiral homology:*

$$\Big(\int_{X,n}^{\mathrm{ord}} \mathcal{A}\Big)_{\Sigma_n} \;\simeq\; R\Gamma_{\mathrm{dR}}\big(X^{(n)},\ \mathcal{F}_n^{\Sigma}(\mathcal{A})\big) \;=:\; \int_{X,n} \mathcal{A}. \tag{5.5}$$

*Proof.* For $\mathsf{E}_\infty$-chiral $\mathcal{A}$, $\mathcal{F}_n^{\mathrm{ord}}$ is $\Sigma_n$-equivariant (Proposition 4.6(iv)). The equivariant descent for $\mathcal{D}$-modules works through the quotient stack $[X^n/\Sigma_n]$: the category of $\Sigma_n$-equivariant $\mathcal{D}$-modules on $X^n$ is equivalent to the category of $\mathcal{D}$-modules on $[X^n/\Sigma_n]$. Since $\Sigma_n$ is a finite group acting in characteristic zero, Maschke's theorem ensures that representations of $\Sigma_n$ are semisimple; the quotient stack $[X^n/\Sigma_n]$ and the coarse moduli space $X^{(n)} = X^n/\Sigma_n$ have equivalent $\mathcal{D}$-module categories. The quotient $\pi\colon X^n \to X^{(n)}$ is ramified along the diagonals, but equivariant descent is not affected by ramification in characteristic zero.

The de Rham functor commutes with $\Sigma_n$-coinvariants (which are exact for the symmetric group acting on vector spaces over $\mathbb{C}$ of characteristic zero), giving: $R\Gamma_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}})_{\Sigma_n} \simeq R\Gamma_{\mathrm{dR}}(X^{(n)}, (\mathcal{F}_n^{\mathrm{ord}})^{\Sigma_n}) \simeq R\Gamma_{\mathrm{dR}}(X^{(n)}, \mathcal{F}_n^{\Sigma})$. The right side is the Beilinson–Drinfeld chiral homology at degree $n$.

The cross-degree chiral differential $d_{\mathrm{ch}}$ is $\Sigma$-equivariant for $\mathsf{E}_\infty$-chiral algebras because the factorisation maps (collision/OPE) are symmetric: the OPE $Y(a,z)b$ and $Y(b,z)a$ agree up to signs. This ensures that the total complex, not just the fixed-degree pieces, descends to the symmetric quotient. □

**Proposition 5.3** (Lossy descent for $\mathsf{E}_1$-chiral algebras)**.** *For an $\mathsf{E}_1$-chiral algebra $\mathcal{A}$, the natural map*

$$\mathrm{av}_n\colon \int_{X,n}^{\mathrm{ord}} \mathcal{A} \longrightarrow \Big(\int_{X,n}^{\mathrm{ord}} \mathcal{A}\Big)_{\Sigma_n} \tag{5.6}$$

*is surjective but not injective for $n \geq 2$. The kernel $\ker(\mathrm{av}_n)$ carries ordered data invisible to the Beilinson–Drinfeld theory:*

(i) *At $n = 2$: the full spectral $r$-matrix $r(z)$ versus the scalar $\kappa(\mathcal{A})$ (for abelian algebras, $\kappa = \mathrm{av}(r(z))$; for non-abelian KM, $\kappa = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$, the Sugawara shift).*

(ii) *At $n = 3$: the Drinfeld associator $\Phi$ appears in $\ker(\mathrm{av}_3)$. The space of sections of $\mathrm{av}_3$ compatible with the Maurer–Cartan equation is a $\mathrm{GRT}_1$-torsor.*

(iii) *At $n \geq 4$: the kernel carries higher-degree ordered shadow data from the $\mathsf{E}_1$ obstruction tower.*

*Proof.* By [34, the non-splitting proposition], the short exact sequence $0 \to \ker(\mathrm{av}) \to \mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \to 0$ does not split as dg Lie algebras. At $n = 2$: for a genuinely $\mathsf{E}_1$-chiral algebra, $R(z) \neq \mathrm{id}$ and the $\Sigma_2$-action on the ordered de Rham cohomology is twisted by $R$; the antisymmetric eigenspace of this twisted action gives $\ker(\mathrm{av}_2) \neq 0$ (see Example 2.3 for the Yangian case). At $n \geq 3$: the cross-degree obstruction (the bar differential is $\Sigma_{n-2}$-equivariant but not $\Sigma_n$-equivariant) forces non-trivial elements of $\ker(\mathrm{av}_n)$. The $\mathrm{GRT}_1$-torsor structure at $n = 3$ follows from Drinfeld's theorem on associators combined with the Etingof–Kazhdan quantisation theorem. □

5.3. **Genus $g \geq 1$ curvature.** At genus zero the bar differential squares to zero and ordered chiral homology is an ordinary chain complex. At genus $g \geq 1$, this fails: the fiberwise bar differential acquires curvature proportional to the Koszul invariant $\kappa(\mathcal{A})$ (the scalar $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$, the degree-2 average of the classical $r$-matrix; computed explicitly in §9) times the Hodge class $\omega_g = c_1(\lambda)$ on $\overline{\mathcal{M}}_g$. The curvature is a *base* class (on moduli), not a *fiber* form (on the curve).

*Remark* 5.4 (Curvature at higher genus). At genus $g \geq 1$, the fiberwise bar differential acquires curvature:

$$d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g \qquad \text{(UNIFORM-WEIGHT)} \tag{5.7}$$

where $\kappa(\mathcal{A})$ is the Koszul invariant of $\mathcal{A}$ and $\omega_g = c_1(\lambda)$ is the first Chern class of the Hodge line bundle on $\overline{\mathcal{M}}_g$; $\omega_g$ is a class on the *moduli space* $\overline{\mathcal{M}}_g$, not a form on the curve. The scalar form of the curvature holds in

the uniform-weight sector; at multi-weight, a cross-channel correction $\delta d^2_{\mathrm{cross}}$ must be included.

Three points require emphasis.

(a) The $\mathcal{D}$-module connection at *fixed* degree $n$ remains flat: $\nabla_n^2 = 0$ on $U_n$. The curvature (5.7) appears in the *cross-degree* chiral differential $d_{\mathrm{ch}}$, which connects different degrees.
(b) The curvature $\kappa(\mathcal{A}) \cdot \omega_g$ is a *scalar* (it acts on all tensor factors uniformly), so it is $\Sigma_n$-invariant: it survives descent to the symmetric quotient unchanged.
(c) The period-corrected differential $D^{(g)}$ of [34, §II.3] restores $d^2 = 0$ at the total level by incorporating the Hodge correction.

## 6. Chiral quantum groups: $R$-matrices, $A_\infty$-structures, and chiral coproducts

The classical theory of quantum groups rests on an equivalence between three descriptions of the same structure: a quasi-triangular Hopf algebra $(H, \Delta, R)$ can be specified by its $R$-matrix, by its coproduct, or by an operadic $A_\infty$-algebra map. The goal of this section is to establish the chiral analogue: an equivalence between three descriptions of an $\mathsf{E}_1$-chiral algebra equipped with quantum-group-type structure. The primitive datum is the $\mathsf{E}_1$ ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with its deconcatenation coproduct, coassociative over $(\mathrm{ChirAss})^!$; the $R$-matrix and the chiral coproduct are both recoverable from this datum.

6.1. **Three structures on an $\mathsf{E}_1$-chiral algebra.** An $\mathsf{E}_1$-chiral algebra admits three equivalent descriptions: by a vertex $R$-matrix $S(z)$ (braiding data), by chiral $A_\infty$ structure maps $m_k^{\mathrm{ch}}$ (higher associativity data), and by a chiral coproduct $\Delta^{\mathrm{ch}}$ (coalgebra data). Each description captures a different projection of the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$.

**Definition 6.1** ($\mathsf{E}_1$-chiral algebra with vertex $R$-matrix)**.** An $\mathsf{E}_1$-chiral algebra with vertex $R$-matrix is a pair $(\mathcal{A}, S(z))$ where $\mathcal{A}$ is a graded vertex algebra and $S(z) \in \mathrm{End}(\mathcal{A} \otimes \mathcal{A})[[z, z^{-1}]]$ is a vertex $R$-matrix satisfying:

(i) ($S$-locality.) For all $a, b \in \mathcal{A}$ and sufficiently large $N$:

$$(z-w)^N\, Y(a,z)\, Y(b,w) \;=\; (z-w)^N\, Y(b,w)\, Y(a,z) \cdot S(z-w). \tag{6.1}$$

(ii) (Quantum Yang–Baxter equation.) In $\mathrm{End}(\mathcal{A}^{\otimes 3})$:

$$S_{12}(z_1-z_2)\, S_{13}(z_1-z_3)\, S_{23}(z_2-z_3) \;=\; S_{23}(z_2-z_3)\, S_{13}(z_1-z_3)\, S_{12}(z_1-z_2). \tag{6.2}$$

(iii) (Unitarity.) $S_{12}(z)\, S_{21}(-z) = \mathrm{id}$.
(iv) (Shift condition.) The $R$-matrix intertwines the translation operator $T$: $[T \otimes \mathrm{id} + \mathrm{id} \otimes T,\, S(z)] = \partial_z S(z)$.

(v) (Hexagon axiom.) On $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, the braiding and associativity satisfy the coherence: for the two standard paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$ connecting the same boundary strata,

$$S_{12}(z_{12})\,\Phi_{123}^{-1}\,S_{13}(z_{13})\,\Phi_{132} \;=\; \Phi_{213}^{-1}\,S_{23}(z_{23})\,\Phi_{123} \tag{6.3}$$

where $\Phi_{ijk}$ is the Drinfeld associator evaluated on the three-point KZ connection.

These are the axioms of Etingof–Kazhdan [18] for a quantum vertex operator algebra.

**Definition 6.2** ($A_\infty$-algebra in the chiral endomorphism operad)**.** An $A_\infty$-algebra structure on $\mathcal{A}$ in the chiral endomorphism operad $\mathrm{End}_{\mathcal{A}}^{\mathrm{ch}}$ is an operadic map

$$A_\infty \;\longrightarrow\; \mathrm{End}_{\mathcal{A}}^{\mathrm{ch}} \tag{6.4}$$

where $\mathrm{End}_{\mathcal{A}}^{\mathrm{ch}}(n) = \mathrm{Hom}(\mathcal{A}^{\otimes n},\ \mathcal{A}((\lambda_1))\cdots((\lambda_{n-1})))$ is the spectral substitution operad ([34, §3]). Concretely, the data consist of chiral structure maps

$$m_k^{\mathrm{ch}}\colon \mathcal{A}^{\otimes k} \longrightarrow \mathcal{A}((\lambda_1))\cdots((\lambda_{k-1})), \qquad k \geq 1, \tag{6.5}$$

satisfying the Stasheff $A_\infty$-relations in the chiral endomorphism operad: for each $n \geq 1$,

$$\sum_{\substack{i+j=n+1\\ 1\leq p\leq i}} (-1)^{\epsilon}\, m_i^{\mathrm{ch}} \circ_p m_j^{\mathrm{ch}} \;=\; 0, \tag{6.6}$$

where $\circ_p$ is the partial composition via spectral substitution ( $\lambda$-variable nesting), and $\epsilon$ carries the standard Koszul sign from the suspended degree $\|m_k\| = k-2$. The structure maps $m_k^{\mathrm{ch}}$ are meromorphic in the spectral parameters $\lambda_1, \ldots, \lambda_{k-1}$. At $k=2$, the map $m_2^{\mathrm{ch}}$ is the OPE vertex operation $Y(-,z)$. At $k=1$, $m_1^{\mathrm{ch}} = d$ is the differential. The operations $m_k^{\mathrm{ch}}$ for $k \geq 3$ record the obstruction to strict associativity of the chiral product.

**Definition 6.3** ($\mathsf{E}_1$-chiral algebra with compatible chiral coproduct)**.** An $\mathsf{E}_1$-chiral algebra with compatible chiral coproduct is a triple $(\mathcal{A}, Y, \Delta^{\mathrm{ch}})$ where $(\mathcal{A}, Y)$ is a chiral algebra and

$$\Delta^{\mathrm{ch}}\colon \mathcal{A} \longrightarrow (\mathcal{A}\,\hat{\otimes}\,\mathcal{A})((z)) \tag{6.7}$$

is a chiral coproduct (a map to the completed tensor product with spectral parameter) satisfying:

(i) (Coassociativity up to associator.) The two iterated coproducts agree up to conjugation by the Drinfeld associator $\Phi$:

$$(\Delta^{\mathrm{ch}} \otimes \mathrm{id}) \circ \Delta^{\mathrm{ch}} \;=\; \Phi \cdot (\mathrm{id} \otimes \Delta^{\mathrm{ch}}) \circ \Delta^{\mathrm{ch}} \cdot \Phi^{-1}. \tag{6.8}$$

(ii) (OPE compatibility.) The coproduct intertwines the vertex operation: for $a, b \in \mathcal{A}$,

$$\Delta^{\mathrm{ch}}(Y(a,z)\,b) \;=\; Y^{(2)}(\Delta^{\mathrm{ch}}(a), z) \cdot \Delta^{\mathrm{ch}}(b) \tag{6.9}$$

where $Y^{(2)}$ denotes the tensorwise vertex operation on $\mathcal{A}\,\hat{\otimes}\,\mathcal{A}$.

(iii) (Counit.) There exists a counit $\varepsilon\colon \mathcal{A} \to \mathbb{C}$ with $(\varepsilon \otimes \mathrm{id}) \circ \Delta^{\mathrm{ch}} = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta^{\mathrm{ch}}$, recovering the augmentation of $\mathcal{A}$.

The chiral coproduct determines a vertex $R$-matrix by

$$S(z) \;:=\; \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1} \tag{6.10}$$

where $\sigma$ is the graded flip and $\Delta^{\mathrm{ch,op}} = \sigma \circ \Delta^{\mathrm{ch}}$ is the opposite coproduct. This is the chiral analogue of the classical formula $R = \Delta^{\mathrm{op}}(\Delta^{-1}(-))$ for a quasi-triangular Hopf algebra.

*Remark* 6.4 (The role of the Drinfeld associator)*.* The Drinfeld associator $\Phi \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ controls the failure of strict coassociativity: the two parenthesizations traverse different paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, and $\Phi$ is the KZ parallel transport between them. It also appears in $\ker(\mathrm{av}_3)$ of Proposition 5.3, invisible to the symmetric shadow. For $\mathsf{E}_\infty$-chiral algebras, $\Phi$ acts trivially on coinvariants; for genuinely $\mathsf{E}_1$-chiral algebras, it is essential structural data.

6.2. **The equivalence theorem.** The three descriptions of the preceding subsection are not independent: a vertex $R$-matrix determines the chiral $A_\infty$ operations via the bar complex, the $A_\infty$ operations determine the chiral coproduct via cobar dualization, and the coproduct recovers the $R$-matrix. The equivalences close into a triangle, each arrow depending on the choice of a Drinfeld associator $\Phi$.

**Theorem 6.5** (Chiral quantum group equivalence)**.** *The following three structures on an $\mathsf{E}_1$-chiral algebra $\mathcal{A}$ on the Koszul locus determine each other, up to the choice of a Drinfeld associator $\Phi$:*

(I) $\mathsf{E}_1$*-chiral algebra with vertex $R$-matrix $S(z)$* (*Definition* 6.1)

(II) $A_\infty$*-algebra in the chiral endomorphism operad* $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ (*Definition* 6.2)

(III) $\mathsf{E}_1$*-chiral algebra with compatible chiral coproduct* $\Delta^{\mathrm{ch}}$ (*Definition* 6.3)

*The equivalences form a triangle:*

$$\begin{tikzcd}
\text{(I)} \arrow[rr, shift left, "\alpha"] \arrow[dr, shift left, "\gamma"] & & \text{(II)} \arrow[ll, shift left, "\alpha^{-1}"] \arrow[dl, shift left, "\beta"] \\
& \text{(III)} \arrow[ul, shift left, "\bar{\gamma}^{-1}"] \arrow[ur, shift left, "\bar{\beta}^{-1}"] &
\end{tikzcd} \tag{6.11}$$

*where each arrow is an equivalence of categories.*

*Proof.* We prove each direction of the triangle (6.11).

**(I) → (II): $R$-matrix gives $A_\infty$ structure.** The Stasheff associahedron $K_n$ is isomorphic to the real Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R})$ (Sinha [42]); the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ gives the embedding $K_n \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ of (3.2). The chiral $A_\infty$-maps $m_k^{\mathrm{ch}}$ are defined by restricting the OPE to this real locus:

$$m_k^{\mathrm{ch}}(a_1, \dots, a_k) \;=\; \int_{K_k} Y(a_1, z_1) \cdots Y(a_k, z_k)\, \omega_k, \tag{6.12}$$

where $\omega_k$ is the canonical volume form on $K_k$ induced by the bar propagator.

The Stasheff identity at degree $n$ (equation (6.6)) follows from Stokes' theorem on $K_{n+1} = \overline{\mathrm{FM}}_{n+1}^{\mathrm{ord}}(\mathbb{R})$. The codimension-1 boundary $\partial K_{n+1}$ has $\binom{n+1}{2} - 1$ facets, one for each consecutive cluster $(z_i, z_{i+1}, \dots, z_{i+j})$ of length $j+1 \geq 2$ (with $i + j \leq n+1$ and $j+1 \leq n$). Each facet is isomorphic to $K_{r+1} \times K_{s+1}$ with $r + s = n$, recording the cluster and the residual configuration. The integral of $d\omega_{n+1}$ over $K_{n+1}$ vanishes, and the boundary contributions give $\sum (-1)^\epsilon\, m_i^{\mathrm{ch}} \circ_p m_j^{\mathrm{ch}} = 0$: the ordered AOS reduction ([34, §5]).

At codimension 2, where two simultaneous consecutive collisions ($z_i \to z_{i+1}, z_j \to z_{j+1}$) with $|i - j| \geq 2$ occur, the compatibility condition is the hexagon axiom (6.3): the two codimension-2 strata correspond to the two standard paths in $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$, and the hexagon ensures that the iterated boundary contributions match. At codimension 3 (triple simultaneous collision in $\overline{\mathrm{FM}}_4^{\mathrm{ord}}(\mathbb{C})$), the compatibility is the pentagon relation for $\Phi$, which is equivalent to the QYBE (6.2) for $S(z)$.

$S$-locality (6.1) ensures that the integrals (6.12) converge: the factor $(z-w)^N$ in $S$-locality bounds the pole orders of the OPE along each boundary divisor.

This construction defines the functor $\alpha$. Its inverse $\alpha^{-1}$ recovers the $R$-matrix from the $A_\infty$-structure as the degree-2 holonomy: $S(z) = \mathcal{P} \exp\bigl(\int_{\gamma(z)} m_2^{\mathrm{ch}}\bigr)$, where $\gamma(z)$ is the KZ parallel-transport path in $\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})$.

**(II) → (III): $A_\infty$ structure gives chiral coproduct.** The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ (where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal) carries the deconcatenation coproduct

$$\Delta[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n]. \tag{6.13}$$

The $A_\infty$-maps $m_k^{\mathrm{ch}}$ determine the bar differential $d_{\overline{B}^{\mathrm{ord}}}$ by the standard formula $d_{\overline{B}^{\mathrm{ord}}} = \sum_{k \geq 1} \hat{m}_k^{\mathrm{ch}}$, where $\hat{m}_k^{\mathrm{ch}}$ is the coderivation extension of $m_k^{\mathrm{ch}}$ to $T^c(s^{-1}\bar{\mathcal{A}})$. The relation $d_{\overline{B}^{\mathrm{ord}}}^2 = 0$ is equivalent to the Stasheff identities (6.6), so the $A_\infty$ data make $(\overline{B}^{\mathrm{ord}}(\mathcal{A}), d_{\overline{B}^{\mathrm{ord}}}, \Delta)$ into a $\mathsf{E}_1$-chiral coassociative dg coalgebra over $(\mathsf{ChirAss})^!$ ([34, Theorem 1.2]): the differential $d_{\overline{B}^{\mathrm{ord}}}$ comes

from the chiral product (OPE collision residues) and the deconcatenation $\Delta$ comes from the cofree tensor coalgebra structure.

The chiral coproduct on $\mathcal{A}$ is constructed as follows. On the Koszul locus (where the bar-cobar adjunction is an equivalence), the cobar functor $\Omega$ applied to $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ recovers $\mathcal{A}$: $\mathcal{A} \simeq \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A}))$ ([34, Theorem 1.1(i)]). The deconcatenation $\Delta$ on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is a morphism of dg coalgebras $\Delta\colon \overline{B}^{\mathrm{ord}}(\mathcal{A}) \to \overline{B}^{\mathrm{ord}}(\mathcal{A}) \otimes \overline{B}^{\mathrm{ord}}(\mathcal{A})$; applying $\Omega$ produces

$$\Delta^{\mathrm{ch}}\colon \mathcal{A} \simeq \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\Omega(\Delta)} \Omega(\overline{B}^{\mathrm{ord}}(\mathcal{A}) \otimes \overline{B}^{\mathrm{ord}}(\mathcal{A})) \to (\mathcal{A} \hat{\otimes} \mathcal{A})((z)), \tag{6.14}$$

where the last map uses the Künneth quasi-isomorphism $\Omega(C \otimes D) \xrightarrow{\sim} \Omega(C) \hat{\otimes} \Omega(D)$ for conilpotent coalgebras.

The deconcatenation coproduct is strictly coassociative on $T^c(s^{-1}\bar{\mathcal{A}})$: the two iterated coproducts $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$ by direct computation (the $(n+1)(n+2)/2$ terms on each side match by the unique decomposition of a word into three consecutive subwords). The associator $\Phi$ enters when passing through the cobar functor: the formality quasi-isomorphism underlying $\Omega$ depends on $\Phi$, so $\Delta^{\mathrm{ch}}$ is coassociative only up to conjugation by $\Phi$, as required by Definition 6.3(i).

The OPE compatibility (6.9) follows from the coderivation property of $d_{\overline{B}^{\mathrm{ord}}}$ with respect to $\Delta$: the bar differential commutes with the coproduct, which dualizes under $\Omega$ to the intertwining of $\Delta^{\mathrm{ch}}$ with the vertex operation $Y$. The counit $\varepsilon\colon \mathcal{A} \to \mathbb{C}$ is the augmentation, and the counit axiom (Definition 6.3(iii)) holds because the bar complex is built on $\bar{\mathcal{A}} = \ker(\varepsilon)$.

This construction defines the functor $\beta$. Its inverse $\beta^{-1}$ recovers the $A_\infty$-maps from a chiral coproduct by reading off the bar differential: the $k$-th structure map $m_k^{\mathrm{ch}}$ is the component of $d_{\overline{B}^{\mathrm{ord}}}$ at tensor degree $k$.

**(III) $\to$ (I): chiral coproduct gives $R$-matrix.** Given a chiral coproduct $\Delta^{\mathrm{ch}}$ satisfying Definition 6.3, the vertex $R$-matrix is defined by (6.10):

$$S(z) \;:=\; \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1}, \tag{6.15}$$

the chiral analogue of the classical Drinfeld formula for quasi-triangular quasi-Hopf algebras [11]. We verify the five axioms of Definition 6.1.

*QYBE.* Expand the left side of (6.2) using the definition (6.15). Each factor $S_{ij}(z_{ij})$ introduces one coproduct and one inverse opposite coproduct; composing the three factors yields an expression involving three applications of the coassociativity relation (6.8). In each application, the two iterated coproducts differ by conjugation by $\Phi$. The pentagon identity for $\Phi$ (Drinfeld [11]; see also Etingof–Kazhdan [17], §3) ensures that the three associator insertions cancel pairwise, and the remaining coproduct factors rearrange to give the right side of (6.2).

*Unitarity.* By definition, $S_{21}(-z) = \sigma \circ S_{12}(-z) \circ \sigma$, and $\Delta^{\mathrm{ch,op,op}} = \Delta^{\mathrm{ch}}$ (applying the flip twice is the identity). Composing:

$$S_{12}(z)\, S_{21}(-z) = \sigma \circ \Delta^{\mathrm{ch}} \circ (\Delta^{\mathrm{ch,op}})^{-1} \cdot \sigma \circ \Delta^{\mathrm{ch,op}} \circ (\Delta^{\mathrm{ch}})^{-1} = \mathrm{id}.$$

*Hexagon.* The hexagon (6.3) encodes the compatibility of $S(z)$ with $\Phi$ on three-point configurations. Substituting (6.15) into the left side, each $S_{ij}$ contributes a coproduct and an inverse opposite coproduct, and each $\Phi_{ijk}^{\pm 1}$ reparenthesizes. The OPE compatibility (6.9) (the chiral Leibniz rule: $\Delta^{\mathrm{ch}}$ intertwines $Y$) ensures that the product-coproduct rearrangement through $\Phi$ matches the right side. This computation is the vertex-algebraic analogue of Drinfeld's proof that a quasi-Hopf coproduct produces a quasi-triangular structure [11]; the adaptation to the chiral setting (spectral-parameter dependence, meromorphic completions) follows Etingof–Kazhdan [18], §4.

*Shift condition.* The translation operator $T = L_{-1}$ satisfies $[T, Y(a,z)] = \partial_z Y(a,z)$. The OPE compatibility (6.9) applied to the vacuum gives $\Delta^{\mathrm{ch}}(T \cdot a) = (T \otimes \mathrm{id} + \mathrm{id} \otimes T) \cdot \Delta^{\mathrm{ch}}(a)$ (the counit axiom ensures the two tensor factors share $T$ additively). Differentiating (6.15) in $z$ and using this intertwining yields $[T \otimes \mathrm{id} + \mathrm{id} \otimes T,\, S(z)] = \partial_z S(z)$.

*$S$-locality.* The OPE compatibility (6.9) maps singular OPE terms to tensor products of singular terms. For $a, b \in \mathcal{A}$ and $N$ sufficiently large, $(z-w)^N Y(a,z)Y(b,w)$ is regular. Applying $\Delta^{\mathrm{ch}}$ to both sides of the OPE and using the definition of $S(z-w)$ via (6.15) gives $(z-w)^N Y(a,z)Y(b,w) = (z-w)^N Y(b,w)Y(a,z) \cdot S(z-w)$, which is (6.1).

This construction defines $\gamma$. Its inverse $\gamma^{-1}$ recovers the coproduct from $(\mathcal{A}, S(z))$ by composing $\alpha$ and $\beta$: $\gamma^{-1} = \beta \circ \alpha$.

**Triangle coherence.** The three functors satisfy $\gamma = \alpha^{-1} \circ \beta^{-1}$. Each functor is an equivalence with explicit inverse as constructed above, and the triangle (6.11) commutes: $\alpha \circ \gamma \circ \beta = \mathrm{id}$. □

*Remark* 6.6 (Independent proof via the cohomological Hall algebra). The equivalence (I) ↔ (II) ↔ (III) admits a second, independent proof for algebras arising as critical CoHAs of quivers with potential, using the framework of Kontsevich–Soibelman [30], Schiffmann–Vasserot [44], Yang–Zhao [48], and Jindal–Kaubrys–Latyntsev [25]. The logic is orthogonal to the bar-cobar adjunction above: rather than constructing the triangle abstractly from the ordered bar complex, one exhibits all three structures simultaneously on a single geometric object and verifies that they determine each other.

**Step 1: simultaneous existence of (I) and (III) on the CoHA.** Let $(Q, W)$ be a quiver with potential. The critical CoHA

$$\mathcal{H}_{Q,W} = \bigoplus_{d \in \mathbb{Z}_{\geq 0}^{Q_0}} H^{G_d}_{\mathrm{BM},*}(\mathcal{M}_d^{\mathrm{nil}}, \varphi_W)$$

carries two structures of independent geometric origin:

(a) *Vertex coproduct* $\Delta_{\mathrm{CoHA}}$, from restriction to Levi subalgebras $G_{d_1} \times G_{d_2} \hookrightarrow G_d$; and
(b) *$R$-matrix* $R(z)$, from the Maulik–Okounkov stable envelope construction [29] on the Nakajima variety associated to $Q$.

Jindal–Kaubrys–Latyntsev [25] prove that these two structures satisfy the vertex bialgebra axiom (their Theorems A and C):

$$\Delta_{\mathrm{CoHA},z}(Y(a,w)\,b) = Y^{(2)}(\Delta_{\mathrm{CoHA},z}(a),w)\,\Delta_{\mathrm{CoHA},z}(b), \tag{6.16}$$

which is the OPE compatibility axiom (6.9). The $R$-matrix satisfies the QYBE and unitarity by the stable envelope formalism [29]; the quasi-coassociativity of $\Delta_{\mathrm{CoHA}}$ follows from the associativity of the Hall product on the Drinfeld double [48]. This establishes the equivalence (I) $\leftrightarrow$ (III) by construction: the $R$-matrix and the coproduct coexist on $\mathcal{H}_{Q,W}$ and satisfy the quasi-triangularity relation $R(z)\,\Delta^{\mathrm{ch}} = \Delta^{\mathrm{ch,op}}\,R(z)$.

**Step 2: extraction of (II) from the CoHA vertex bialgebra.** The CoHA $\mathcal{H}_{Q,W}$, equipped with its $R$-matrix, is an $\mathsf{E}_1$-chiral algebra in the sense of Definition 6.1. The functor $\alpha$ of the equivalence triangle (6.11) extracts the chiral $A_\infty$ operations $\{m_k^{\mathrm{ch}}\}$ by restricting the OPE to the associahedra $K_k \hookrightarrow \overline{\mathrm{FM}}_k^{\mathrm{ord}}(\mathbb{C})$ as in (6.12). The Stasheff identities follow from Stokes' theorem on $K_{k+1}$ as before; the convergence of the integrals is guaranteed by $S$-locality of the $R$-matrix. This gives (I) $\to$ (II) on the CoHA.

The reverse direction (II) $\to$ (I) is the degree-2 holonomy recovery of the $R$-matrix from $m_2^{\mathrm{ch}}$, which depends only on the formal structure of the Stasheff associahedra and not on the CoHA origin.

**Step 3: the triangle closes.** By the JKL vertex bialgebra theorem, the CoHA simultaneously carries (I), (II), and (III). To verify that these three structures *determine* each other (not merely coexist), one checks the uniqueness directions. On the Koszul locus:

(i) *(I) determines (III):* given an $R$-matrix satisfying the QYBE and $S$-locality, the quasi-triangular coproduct is unique up to gauge equivalence by the Drinfeld reconstruction [11] (spectral parameter version: Etingof–Kazhdan [17]). The JKL coproduct on the CoHA is the unique one compatible with the MO $R$-matrix.

(ii) *(III) determines (I):* the Drinfeld formula (6.15) recovers $S(z)$ from $\Delta^{\mathrm{ch}}$, independently of the CoHA construction.

(iii) *(I) $\leftrightarrow$ (II):* the extraction of $A_\infty$ operations from the $R$-matrix is the Stasheff–FM construction (direction $\alpha$), whose inverse recovers the $R$-matrix as holonomy (direction $\alpha^{-1}$). Neither direction uses the CoHA.

The triangle coherence $\alpha \circ \gamma \circ \beta = \mathrm{id}$ then follows from the uniqueness of each vertex.

**Independence from the bar-cobar proof.** The argument above differs from the proof of Theorem 6.5 in two respects. First, the equivalence (I) $\leftrightarrow$ (III) is established by the geometric CoHA construction (stable envelopes for the $R$-matrix, Levi restriction for the coproduct, JKL for the vertex bialgebra axiom), bypassing the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ and the cobar dualization (6.14). Second, the uniqueness of the coproduct

given the $R$-matrix is deduced from the Drinfeld reconstruction theorem for quasi-triangular vertex bialgebras, rather than from the bar-cobar adjunction on the Koszul locus. The functor $\alpha$ (extraction of $A_\infty$ operations from the $R$-matrix via associahedron integrals) is shared between the two proofs, but it enters as an a posteriori observation in the CoHA argument, not as the engine of the construction.

**Scope.** The CoHA proof applies to $\mathsf{E}_1$-chiral algebras arising as critical CoHAs of quivers with potential. For the Jordan quiver, $\mathcal{H}_{\mathrm{Jor}} \cong Y^+(\widehat{\mathfrak{gl}}_1)$ [44], and the equivalence specializes to the chiral quantum group structure on $\mathcal{W}_{1+\infty}[\Psi]$ (Theorem 6.15). For quivers of type $\widehat{A}_{N-1}$, the CoHA is $Y^+(\widehat{\mathfrak{sl}}_N)$, and the JKL coproduct recovers the Drinfeld coproduct on the affine Yangian [25]. Beyond quiver-origin algebras, the CoHA argument does not directly apply; the bar-cobar proof of Theorem 6.5 remains the primary route for general $\mathsf{E}_1$-chiral algebras on the Koszul locus.

The equivalence theorem packages three equivalent descriptions of quantum-group-type structure on an $\mathsf{E}_1$-chiral algebra. The following definition records the datum that the equivalence classifies: the chiral Yangian datum, assembling all four components into a single object.

**Definition 6.7** (Chiral Yangian datum)**.** For simple $\mathfrak{g}$, smooth curve $X$, and formal parameter $\hbar$, the *chiral Yangian datum* $Y_\hbar^{\mathrm{ch}}(\mathfrak{g})$ is the quadruple $(\mathcal{A},\, S(z),\, \Delta^{\mathrm{ch}},\, \{m_k^{\mathrm{ch}}\})$ where:

(i) $\mathcal{A}$ is an $\mathsf{E}_1$-chiral algebra on $\mathrm{Ran}^{\mathrm{ord}}(X)$ with $R$-twisted $S$-locality (Definition 6.1);
(ii) $S(z) = R_{21}(z)\, R_{12}(-z)$ is the spectral $S$-matrix (monodromy on $\mathrm{Conf}_2^{\mathrm{ord}}(X)$);
(iii) $\Delta^{\mathrm{ch}} \colon \mathcal{A} \to \mathcal{A} \hat{\otimes} \mathcal{A}$ is a quasi-coassociative chiral coproduct satisfying quasi-triangularity: $R(z)\, \Delta^{\mathrm{ch}} = \Delta^{\mathrm{ch,op}}\, R(z)$ (Definition 6.3);
(iv) $\{m_k^{\mathrm{ch}}\}_{k\geq 1}$ is the curved $A_\infty$-structure from the bar Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1})$ (Definition 6.2).

These data satisfy four axioms:

(A1) (Koszul self-recovery.) $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$;
(A2) (Verdier by $R$-inversion.) $\mathcal{A}^{!,\mathrm{ch}} \cong Y_{-\hbar}^{\mathrm{ch}}(\mathfrak{g})$;
(A3) (Braided module category.) $\mathrm{Mod}_{\mathcal{A}}$ is $\mathsf{E}_2$ with braiding $\sigma = P \circ R(u)$;
(A4) (Modular MC tower.) $R^{\mathrm{mod}}(z;\hbar) = \sum_{g\geq 0} \hbar^{2g}\, r_g(z) \in \mathrm{MC}(\widehat{L}_{\mathcal{A}}^{\mathrm{mod}})$.

*Remark* 6.8 (Relationship to the $\mathsf{E}_1$-chiral Yangian)*.* The $\mathsf{E}_1$-*chiral Yangian* of [34, §III] requires only component (i): an $\mathsf{E}_1$-factorization algebra on $\mathrm{Ran}^{\mathrm{ord}}(X)$ whose RTT relations reproduce the Yangian $R$-matrix. The chiral Yangian datum (Definition 6.7) is strictly stronger, imposing axioms (A1)–(A4) including the chiral coproduct (iii) and the modular Maurer–Cartan tower (A4). Whether the weaker $\mathsf{E}_1$-chiral Yangian structure determines the full datum is open: the main obstruction is constructing the chiral coproduct from the $\mathsf{E}_1$-factorization data alone.

*Remark* 6.9 (Constructive content of the bar complex). The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ produces three of the four components constructively: the $\mathsf{E}_1$-chiral algebra structure (i) is the bar-cobar adjunction (Theorem A of [34]); the spectral $S$-matrix (ii) is the degree-2 holonomy of the KZ connection on $\mathrm{Conf}_2^{\mathrm{ord}}(X)$; and the $A_\infty$-structure (iv) is the coderivation extension of $\Theta_{\mathcal{A}}$ to $T^c(s^{-1}\bar{\mathcal{A}})$. The chiral coproduct (iii) requires the additional datum of the $\mathsf{E}_1$-chiral coassociative structure: the deconcatenation coproduct on the bar complex is the cofree tensor coalgebra structure, and its passage to $\mathcal{A}$ via the cobar functor $\Omega$ uses the Künneth quasi-isomorphism for conilpotent coalgebras (equation (6.14)).

6.3. **The classical limit.** The chiral quantum group equivalence (Theorem 6.5) is a vertex-algebraic lift of Drinfeld's classical equivalence between quasi-triangular Hopf algebras, $R$-matrices, and coproducts. The lift replaces finite-dimensional vector spaces with modules over $\mathbb{C}((z))$, the symmetric group with the braid group, and strict coassociativity with coassociativity up to the Drinfeld associator $\Phi$.

*Remark* 6.10 (Comparison with the classical Hopf algebra story). The chiral quantum group equivalence (Theorem 6.5) is the vertex-algebraic analogue of the following classical equivalence, due to Drinfeld:

- $(\mathrm{I})_{\mathrm{cl}}$ Quasi-triangular quasi-Hopf algebra $(H, \Delta, R, \Phi)$
- $(\mathrm{II})_{\mathrm{cl}}$ $A_\infty$-algebra in the endomorphism operad $\mathrm{End}_H$
- $(\mathrm{III})_{\mathrm{cl}}$ Bialgebra with $R$-matrix: $R \cdot \Delta(x) = \Delta^{\mathrm{op}}(x) \cdot R$ for all $x \in H$

The passage from classical to chiral introduces two features: the coproduct acquires a spectral parameter ($\Delta^{\mathrm{ch}} \colon \mathcal{A} \to (\mathcal{A} \hat{\otimes} \mathcal{A})((z))$), and strict coassociativity is replaced by coassociativity up to the Drinfeld associator $\Phi$, which appears universally in the chiral setting because the KZ connection on $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$ has non-trivial holonomy.

6.4. **Examples.** The three structures are most transparent for affine Kac–Moody, where the $R$-matrix is the KZ monodromy, the $A_\infty$ operations are the Sugawara OPE restricted to the associahedron, and the chiral coproduct is the Drinfeld coproduct. For the Yangian $Y_\hbar(\mathfrak{g})$, the $R$-matrix is the starting point rather than derived from an $\mathsf{E}_\infty$-chiral OPE. For $\mathsf{E}_\infty$-chiral algebras, all three structures are trivial.

**Example 6.11** (Affine Kac–Moody: Sugawara coproduct)**.** For $V_k(\mathfrak{g})$ at generic level $k \neq -h^\vee$, the three structures are:

(i) ($R$-matrix.) The vertex $R$-matrix is the KZ $R$-matrix: the monodromy of the KZ connection $\nabla = d + k\Omega/(z_1 - z_2) \cdot d(z_1 - z_2)$ around $\Delta_{12}$. The classical $r$-matrix (trace-form) is $r(z) = k\Omega/z$; at $k = 0$, $r = 0$; at $k = -h^\vee$, the critical level, the Sugawara construction degenerates.

(ii) ($A_\infty$ structure.) The chiral $A_\infty$ structure maps are the Sugawara OPE operations restricted to $K_n \hookrightarrow \overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$. At $k = 2$: $m_2^{\mathrm{ch}}(J^a, J^b; \lambda) =$

$[J^a, J^b] + k\,\delta^{ab}/\lambda$ is the affine OPE. The transferred higher operations $m_k^{\mathrm{ch}}$ for $k \geq 3$ encode the associator corrections from the KZ holonomy.

(iii) (Chiral coproduct.) The Drinfeld coproduct on $Y_\hbar(\mathfrak{g})$ is the paradigmatic example. On generators: $\Delta^{\mathrm{ch}}(J^a) = J^a \otimes 1 + 1 \otimes J^a + \hbar\, f^a_{bc}\, J^b \otimes J^c/z + \cdots$, where the spectral parameter $z$ comes from the difference coordinate $z_1 - z_2$ on $\mathrm{Conf}_2(\mathbb{C})$. The deconcatenation coproduct on the ordered bar $\overline{B}^{\mathrm{ord}}(V_k(\mathfrak{g}))$ induces, via the cobar functor, this spectral Drinfeld coproduct.

The Drinfeld–Kohno theorem identifies the KZ monodromy representation (the $R$-matrix of (i)) with the quantum group $R$-matrix on $U_q(\mathfrak{g})$-modules, where $q = \exp(2\pi i/(k + h^\vee))$. The chiral quantum group equivalence (Theorem 6.5) lifts this identification from monodromy representations to the level of the algebra itself.

**Example 6.12** (Yangian: the Drinfeld coproduct as chiral primitive)**.** For the Yangian $Y_\hbar(\mathfrak{g})$, structure (I) is the starting point: the $R$-matrix $R(z) = 1 + \hbar\, r/z + O(\hbar^2)$ is independent input (not derived from any $\mathsf{E}_\infty$-chiral OPE). The Drinfeld coproduct

$$\Delta_z(T^{(n)}) \;=\; \sum_{k=0}^{n} T^{(k)} \otimes T^{(n-k)} \cdot z^{-k} \tag{6.17}$$

(on the generating series $T(u) = 1 + \sum_{n\geq 1} T^{(n)} u^{-n}$) provides structure (III). The ordered bar complex $\overline{B}^{\mathrm{ord}}(Y_\hbar(\mathfrak{g}))$ is the bridge: its deconcatenation coproduct is strictly coassociative and induces, via cobar dualization, the Drinfeld coproduct (6.17) on $Y_\hbar(\mathfrak{g})$. The associator $\Phi$ appears when passing from the strictly coassociative bar level to the algebra level.

The $A_\infty$ structure maps $m_k^{\mathrm{ch}}$ for $k \geq 3$ are the RTT relations of the Yangian: they encode the obstruction to strict associativity of the spectral OPE, governed by the fusion procedure for the $R$-matrix. The bar-cobar inversion $\Omega(\overline{B}^{\mathrm{ord}}(Y_\hbar(\mathfrak{g}))) \xrightarrow{\sim} Y_\hbar(\mathfrak{g})$ identifies these operations with the standard Yangian relations.

**Example 6.13** (Etingof–Kazhdan quantum vertex algebra for $\mathfrak{sl}_2$: full braided VOA structure)**.** The Yangian $Y_\hbar(\mathfrak{sl}_2)$ of Example 2.3 is a quasi-triangular Hopf algebra in the category of topological algebras with a spectral parameter. The Etingof–Kazhdan construction [18] lifts it to a *quantum vertex operator algebra*: an algebra satisfying the full suite of braided VOA axioms ($S$-locality, vertex $R$-matrix, hexagon) of Definition 6.1. The purpose of this example is to exhibit the construction for $\mathfrak{g} = \mathfrak{sl}_2$ concretely, verify the axioms, compute the ordered chiral center with the full quantum $R$-matrix $S(z)$ (not merely its classical leading order), and connect $S$-locality to the chiral $A_\infty$ structure.

**Part A: the EK construction.** The input is the Yang $R$-matrix $R(u) = u\,I + \hbar\,P \in \mathrm{End}(V \otimes V)[u]$ from (2.2), where $V = \mathbb{C}^2$ is the fundamental representation, $P = (\rho \otimes \rho)(\Omega) + \frac{1}{2}\,I$ is the permutation operator, and $\hbar = 1/(k+2)$. The EK quantum VOA is constructed in four steps: quantum function algebra, dual QUE algebra (state space), vertex operator (creation–annihilation decomposition), and vertex $R$-matrix.

*Step 1: the quantum function algebra $F_0(R)$.* The algebra $F_0(R)$ is generated by the matrix entries of a formal power series

$$T(u) = \sum_{n\geq 0} T^{(n)}\,u^n \;\in\; \mathrm{End}(V) \otimes F_0(R)[\![u]\!], \tag{6.18}$$

subject to the RTT relation

$$R_{12}(u-v)\,T_{13}(u)\,T_{23}(v) = T_{23}(v)\,T_{13}(u)\,R_{12}(u-v) \tag{6.19}$$

in $\mathrm{End}(V^{\otimes 2}) \otimes F_0(R)[\![u,v]\!]$. For $R(u) = u\,I + \hbar\,P$, writing $T^{(n)} = \left(\begin{smallmatrix} a_n & b_n \\ c_n & d_n \end{smallmatrix}\right)$, equation (6.19) expands at order $u^m v^n$ to give:

$$[a_m, a_n] = 0, \quad [d_m, d_n] = 0, \quad [b_m, b_n] = 0, \quad [c_m, c_n] = 0, \tag{6.20}$$

$$a_m\,b_n - b_n\,a_m = \hbar\,(b_m\,a_n - b_{m-1}\,a_{n+1}), \tag{6.21}$$

$$[a_m, d_n] = \hbar\,(c_m\,b_n - c_{m-1}\,b_{n+1} - b_m\,c_n + b_{m-1}\,c_{n+1}). \tag{6.22}$$

The algebra $F_0(R)$ carries a coalgebra structure $\Delta_F(T(u)) = T(u) \dot{\otimes} T(u)$ (matrix multiplication in $\mathrm{End}(V)$, tensor product in $F_0(R)$), a counit $\varepsilon_F(T(u)) = I$, and an antipode $S_F(T(u)) = T(u)^{-1}$.

*Step 2: the state space $V_{\mathrm{EK}} = U_0(R)$.* The quantum universal enveloping algebra $U_0(R)$ is the continuous dual of $F_0(R)$:

$$V_{\mathrm{EK}} \;:=\; U_0(R) \;=\; F_0(R)^*_{\mathrm{cont}}, \tag{6.23}$$

a topological Hopf algebra with multiplication dual to the comultiplication on $F_0(R)$. The vacuum vector is the counit:

$$\Omega = \varepsilon_F \in U_0(R), \qquad \langle \Omega, f\rangle = \varepsilon_F(f) \quad \text{for all } f \in F_0(R). \tag{6.24}$$

The translation operator $D\colon U_0(R) \to U_0(R)$ is defined by the shift of spectral parameter:

$$e^{zD}\,T(u_1)\cdots T(u_n)\,\Omega \;=\; T(u_1+z)\cdots T(u_n+z)\,\Omega. \tag{6.25}$$

On the generating modes: $D\,T^{(n)} = n\,T^{(n-1)}$ (with $T^{(-1)} := 0$), reproducing the conformal translation of mode operators.

For $\mathfrak{g} = \mathfrak{sl}_2$: the graded components of $U_0(R)$ are finite-dimensional, with

$$\dim U_0(R)_n \;=\; \dim \mathrm{Sym}^n(\mathfrak{sl}_2[t]) \;=\; \begin{cases} 1 & n=0, \\ 3 & n=1, \\ 9 & n=2, \\ 22 & n=3, \end{cases} \tag{6.26}$$

the Hilbert series of the PBW basis $\{E_r, F_r, H_r : r \geq 0\}$ of $Y_\hbar(\mathfrak{sl}_2)$, confirming the identification $U_0(R) \cong Y_\hbar(\mathfrak{sl}_2)$ as vector spaces.

*Step 3: the vertex operator $Y$.* The vertex operator $Y\colon V_{\mathrm{EK}} \to \mathrm{End}(V_{\mathrm{EK}})[\![z, z^{-1}]\!]$ is constructed from the creation–annihilation decomposition of the quantum current. Define:

- the *creation operator* $\mathbf{T}^{+}(u,z) = T(u+z)$, acting on $V_{\mathrm{EK}}$ by left multiplication;
- the *annihilation operator* $\mathbf{T}^{-}(u,z) = (T^{*}(u+z-K\hbar/2))^{-1}$, where $T^{*}$ is the adjoint transfer matrix (the transpose of $T$ with respect to the Killing form) and $K$ is the level.

The quantum current $\mathbf{T}(u,z)$ is the product

$$\mathbf{T}(u,z) \;=\; \mathbf{T}^{+}(u,z)\cdot\big(\mathbf{T}^{-}(u,z)\big)^{-1} \;\in\; \mathrm{End}(V)\otimes\mathrm{End}(V_{\mathrm{EK}})[\![u^{-1},z,z^{-1}]\!]. \tag{6.27}$$

The vertex operator on generators is

$$Y(T^{(n)}\,\Omega,\,z) \;=\; \mathrm{Res}_{u=0}\; u^{n}\,\mathbf{T}(u,z)\,du, \tag{6.28}$$

and extends to products by the normal-ordering prescription: $Y(a\,b,\,z) = {:}\,Y(a,z)\,Y(b,z)\,{:}$, where the normal ordering places annihilation modes to the right.

On the leading generators ($n=0$): identifying $T^{(0)} = I + \hbar(e\otimes e_{12} + f\otimes e_{21} + \frac{1}{2}h\otimes(e_{11}-e_{22}))$ at leading order, the vertex operators reduce to the standard affine Kac–Moody currents

$$Y(J^{a}\,\Omega,\,z) \;=\; J^{a}(z) = \sum_{n\in\mathbb{Z}} J^{a}_{n}\,z^{-n-1} \tag{6.29}$$

for $a\in\{e,f,h\}$, matching the OPE of $V_k(\mathfrak{sl}_2)$ at leading order in $\hbar$.

*Step 4: the vertex $R$-matrix $S(z)$.* The vertex $R$-matrix is the braiding operator that governs the exchange of vertex operators. It is constructed from the $R$-matrix data via the universal formula of [18]:

$$S(z) \;=\; \mathcal{R}(z)\cdot\sigma \;\in\; \mathrm{End}(V_{\mathrm{EK}}\otimes V_{\mathrm{EK}})[\![z,z^{-1}]\!], \tag{6.30}$$

where $\sigma$ is the graded flip and $\mathcal{R}(z) = \sum_{n\geq 0}\mathcal{R}_n\,z^{-n}$ is the *spectral universal $R$-matrix* of $Y_\hbar(\mathfrak{sl}_2)$, satisfying

$$\mathcal{R}(z)\cdot\Delta_z(x) \;=\; \Delta_z^{\mathrm{op}}(x)\cdot\mathcal{R}(z) \qquad \text{for all } x\in Y_\hbar(\mathfrak{sl}_2). \tag{6.31}$$

On the fundamental representation $V\otimes V$, the spectral universal $R$-matrix evaluates to the Yang $R$-matrix:

$$(\rho\otimes\rho)\big(\mathcal{R}(z)\big) \;=\; R(z) \;=\; z\,I + \hbar\,P \;\in\; \mathrm{End}(V\otimes V)[z]. \tag{6.32}$$

The leading-order expansion of $S(z)$ on $V_{\mathrm{EK}}\otimes V_{\mathrm{EK}}$ is

$$S(z) = \mathrm{id} + \frac{\hbar\,\Omega}{z} + \frac{\hbar^2}{z^2}\Big(\Omega^{(2)} + \tfrac{1}{2}\,\Omega^2\Big) + O(z^{-3}), \tag{6.33}$$

where $\Omega^{(2)}\in U(\mathfrak{g})^{\otimes 2}$ is the second-order Casimir correction (determined by the $R$-matrix unitarity condition) and $\Omega^2 = \Omega\cdot\Omega\in U(\mathfrak{g})^{\otimes 2}$. At leading order ($z^{-1}$ term), the classical $r$-matrix $r(z) = \hbar\,\Omega/z$ is recovered.

**Part A (continued): verification of the braided VOA axioms.** The five axioms of Definition 6.1 are verified for $(V_{\mathrm{EK}},\, Y,\, S(z))$ as follows.

*$S$-locality.* For the generating currents, the $S$-locality

$$(z-w)^N\, Y(a,z)\, Y(b,w) \;=\; (z-w)^N\, Y(b,w)\, Y(a,z)\cdot S(z-w) \tag{6.34}$$

holds at $N=2$: the $\hbar\,\Omega/(z-w)$ correction from $S(z-w)$ accounts for the structure-constant term in the OPE, and the higher-order terms account for the double pole. For descendants, $S$-locality follows by induction on conformal weight via the shift condition.

*QYBE, unitarity, shift, hexagon.* The quantum Yang–Baxter equation on $V_{\mathrm{EK}}^{\otimes 3}$ reduces on $V^{\otimes 3}$ to (2.3), verified in Example 2.3; the extension to $V_{\mathrm{EK}}$ follows from the universal property of $\mathcal{R}(z)$. Unitarity $S_{12}(z)\cdot(\sigma\circ S_{12}(-z)\circ\sigma)=\mathrm{id}$ follows from Drinfeld–Jimbo unitarity. The shift condition

$$[D\otimes\mathrm{id}+\mathrm{id}\otimes D,\; S(z)] \;=\; \partial_z\, S(z) \tag{6.35}$$

follows from the translation-covariance of the Drinfeld coproduct. The hexagon follows from the pentagon relation for $\Phi$ (from the Stasheff boundary of $\overline{\mathrm{FM}}_4^{\mathrm{ord}}(\mathbb{C}) = K_4$) combined with the QYBE for $S$, by the standard argument of Drinfeld [17] lifted to the vertex-algebraic setting.

**Part B: the ordered chiral center of the EK quantum VOA.** We compute the ordered chiral Hochschild cohomology $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(V_{\mathrm{EK}},\, V_{\mathrm{EK}})$ at low degrees on the formal disk $D$, paralleling the computation of Example 2.3 (Part B) but now using the *full* vertex $R$-matrix $S(z)$ rather than the Yang $R$-matrix on $V\otimes V$ alone.

*Degree 2: the KZ connection with the full vertex $R$-matrix.* At degree 2, the KZ connection on $\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})$ with coefficients in $\mathrm{End}(V_{\mathrm{EK}}^{\otimes 2})$ is

$$\nabla^S_{\mathrm{KZ}} = d + \frac{\hbar\,\Omega}{z}\,dz + \frac{\hbar^2}{z^2}\Big(\Omega^{(2)} + \tfrac{1}{2}\,\Omega^2\Big)\,dz + O(z^{-3}), \tag{6.36}$$

where $z = z_1 - z_2$. The connection (6.36) is obtained from $S(z)$ via the general formula $\nabla = d + (\partial_z \log S(z))\,dz$ (logarithmic derivative of the vertex $R$-matrix).

For the classical $V_k(\mathfrak{sl}_2)$ (trivial braiding $S(z)=\mathrm{id}$), the flat sections are $\Phi(z) = z^{\Omega/(k+2)}$ with algebraic monodromy. For the EK quantum VOA ($S(z)\neq\mathrm{id}$), the flat sections acquire quantum corrections:

$$\Phi^S(z) = z^{\hbar\,\Omega}\Big(I + \frac{\hbar^2\,\Omega^{(2)}}{z} + O(z^{-2})\Big). \tag{6.37}$$

*The kernel* $\ker(\mathrm{av})$. The ordered chiral centre computation parallels Example 2.3; the result is identical on cohomology. On $V\otimes V$: $\ker(\mathrm{av}_2)\big|_{V\otimes V} \cong \bigwedge^2 V = \mathbb{C}$, the same one-dimensional space as equation (2.16). The quantum corrections from $S(z)$ shift the subleading terms of the flat sections but do not alter the $\ker(\mathrm{av}_2)$ dimension. The scalar shadow surviving $\mathrm{av}_2$ is $\kappa = 3(k+2)/4$ (equation (9.9)), identical to the classical value. At degree 3,

the kernel $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathfrak{g}) = \mathbb{C}$ carries the quantum-corrected Drinfeld associator

$$\Phi^S_{\mathrm{KZ}} - 1 \;=\; \hbar^2\,\zeta(2)\,[\Omega_{12},\Omega_{23}] + O(\hbar^3), \tag{6.38}$$

whose leading term agrees with the classical KZ associator.

**Part C: $S$-locality as an $A_\infty$ relation.** The braided VOA axioms of the EK quantum VOA are equivalent to the chiral $A_\infty$ relations for $V_{\mathrm{EK}}$ through the chiral quantum group equivalence (Theorem 6.5, (I)⇔(II)). In the ordered bar complex $\overline{B}^{\mathrm{ord}}(V_{\mathrm{EK}}) = T^c(s^{-1}\overline{V}_{\mathrm{EK}})$, the $S$-locality axiom (6.34) governs the reordering of two insertions: the braid group $\pi_1(\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})) = \mathbb{Z}$ acts on degree-2 bar chains by $\sigma_1 \cdot [a|b] = S(z) \cdot [b|a]$. The $S$-locality axiom is the $n = 2$ Stasheff relation, the hexagon is the $n = 3$ relation, and the QYBE is the $n = 4$ relation (from the pentagon for $\Phi$ on $K_4 = \overline{\mathrm{FM}}_4^{\mathrm{ord}}(\mathbb{R})$).

**Summary.** The EK quantum VOA ($V_{\mathrm{EK}}$, $Y$, $S(z)$) provides the first concrete instance of all three structures of the chiral quantum group equivalence (Theorem 6.5) exhibited simultaneously on a single algebra with all axioms verified. The ordered chiral center carries the full quantum $R$-matrix $S(z)$ (not just the classical $r$-matrix $r(z) = \hbar\,\Omega/z$), and the kernel $\ker(\mathrm{av}_2) \neq 0$ demonstrates that the ordered theory is strictly richer than the symmetric Beilinson–Drinfeld chiral homology even at the quantum level.

**Example 6.14** (Heisenberg: trivial case)**.** For the Heisenberg algebra $\mathcal{H}_k$ (class $G$), all three structures are trivial: $S(z) = \mathrm{id}$ (the braiding is symmetric); $m_k^{\mathrm{ch}} = 0$ for $k \geq 3$ (the algebra is strictly associative in the chiral sense); $\Delta^{\mathrm{ch}}(J) = J \otimes 1 + 1 \otimes J$ is cocommutative and strictly coassociative ($\Phi$ acts trivially). The ordered and symmetric data coincide.

**Theorem 6.15** ($\mathcal{W}_{1+\infty}[\Psi]$). *asachiralquantumgroup]Thealgebra $W_{1+\infty}[\Psi]$ carries a chiral quantum group datum in the sense of Theorem 6.5: an $R$-matrix, a chiral coproduct, and an $A_\infty$ structure, all explicit and originating from the cohomological Hall algebra of the Jordan quiver. The precise content is as follows.*

(I) ($R$-matrix.) *The Maulik–Okounkov $R$-matrix* [29] *of quantum toroidal $\mathfrak{gl}_1$, restricted to the CoHA, is* scalar*:*

$$R(z) = g(z) = \frac{(z-h_1)(z-h_2)(z-h_3)}{(z+h_1)(z+h_2)(z+h_3)}, \qquad h_1+h_2+h_3 = 0, \tag{6.39}$$

*where $h_1, h_2, h_3$ are the equivariant parameters of $\mathbb{C}^3$ subject to the Calabi–Yau constraint. The classical limit is $r(z) = \Psi/z$ (class $G$, level prefix $\Psi$).*

(II) (Chiral coproduct at all spins.) *The Drinfeld coproduct on $Y(\widehat{\mathfrak{gl}}_1)$ gives an explicit spectral coproduct $\Delta_z\colon \mathcal{W}_{1+\infty}[\Psi] \to (\mathcal{W}_{1+\infty}[\Psi] \hat{\otimes} \mathcal{W}_{1+\infty}[\Psi])((z))$ satisfying strict coassociativity, the spectral counit, and OPE compatibility at all spins. The transfer matrix $T(u) =$*

*$1+\sum_{n\geq 1}\psi_n\,u^{-n}$ is scalar ($\mathfrak{gl}_1$ has rank 1), and*

$$\Delta_z(T(u)) = T(u)\otimes T(u-z). \tag{6.40}$$

*The general spin-n formula is*

$$\Delta_z(\psi_n) = \psi_n\otimes 1 + \sum_{k=0}^{n-1}\sum_{m=1}^{n-k}\binom{n-k-1}{m-1} z^{n-k-m}\,\psi_k\otimes\psi_m, \tag{6.41}$$

*where $\psi_0 = 1$.*

(III) ($A_\infty$ structure.) *The chiral $A_\infty$-operations $m_k^{\mathrm{ch}}$ are extracted from the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty}) = T^c(s^{-1}\overline{\mathcal{W}}_{1+\infty})$ via the functor $\alpha$ of Theorem 6.5, with input the R-matrix (6.39).*

*The three structures determine each other by Theorem 6.5. The OPE compatibility axiom (6.9) is satisfied at all spins simultaneously by two independent arguments: the coderivation property on the Koszul-locus bar complex, and the JKL vertex bialgebra theorem on the CoHA.*

*Proof.* The argument proceeds in seven steps.

**Step 1** (CoHA identification; Schiffmann–Vasserot [44], Schiffmann–Mellit–Minets–Vasserot [43]). The critical CoHA of the Jordan quiver is

$$\mathcal{H}_{\mathrm{Jor}} = \bigoplus_{n\geq 0} H^{GL_n}_{\mathrm{BM},*}(\mathcal{M}_n^{\mathrm{nil}},\,\varphi_W), \qquad W = \mathrm{Tr}(x[y,z]),$$

where $\mathcal{M}_n^{\mathrm{nil}}$ is the nilpotent locus in the representation variety of the Jordan quiver and $\varphi_W$ denotes the vanishing-cycle twist. Schiffmann–Vasserot [44] proved $\mathcal{H}_{\mathrm{Jor}} \cong Y^+(\widehat{\mathfrak{gl}}_1)$ by matching generators and relations; the critical refinement is [43]. Kontsevich–Soibelman [30] established the general CoHA framework (see also Joyce [26]).

**Step 2** (Bialgebra and Drinfeld double; Yang–Zhao [48]). The CoHA carries a bialgebra coproduct $\Delta_{\mathrm{CoHA}}\colon \mathcal{H}_{\mathrm{Jor}} \to \mathcal{H}_{\mathrm{Jor}}\otimes\mathcal{H}_{\mathrm{Jor}}$ from restriction to Levi subalgebras $GL_{n_1}\times GL_{n_2}\hookrightarrow GL_n$. Yang–Zhao proved that the Drinfeld double is the full affine Yangian: $D(\mathcal{H}_{\mathrm{Jor}}) \cong Y(\widehat{\mathfrak{gl}}_1)$.

**Step 3** (Yangian–$\mathcal{W}$-algebra identification; Procházka–Rapčák [39]). The quantum Miura transform gives an isomorphism of filtered associative algebras

$$Y(\widehat{\mathfrak{gl}}_1) \;\cong\; U(\mathcal{W}_{1+\infty}[\Psi]),$$

intertwining the Yangian filtration with the conformal filtration. The $\psi$-generators map to $\mathcal{W}_{1+\infty}$ fields via a triangular change of basis: $\psi_1 = J$, $\psi_n = W_n + P_n(J, W_2, \ldots, W_{n-1})$ for $n\geq 2$, where $W_2 = T$ (stress tensor), $W_3 = W$ (spin-3 current), and $P_n$ is a nonlinear polynomial in lower-spin fields with coefficients depending on $\Psi$.

**Step 4** (Drinfeld coproduct at all spins). The transfer matrix $T(u) = 1+\sum_{n\geq 1}\psi_n\,u^{-n}$ is a scalar formal power series in $u^{-1}$ ($\mathfrak{gl}_1$ has rank 1).

The Drinfeld coproduct (6.40) is the product of two such series. The expansion (6.41) follows from $(u-z)^{-m} = u^{-m}\sum_{j\geq 0}\binom{m+j-1}{j}(z/u)^j$ applied to $T(u)\otimes T(u-z)$ and collecting the coefficient of $u^{-n}$ (see (2.10) and Example 6.12). Through spin 3:

$$\Delta_z(\psi_1) = \psi_1\otimes 1 + 1\otimes\psi_1, \tag{6.42}$$

$$\Delta_z(\psi_2) = \psi_2\otimes 1 + 1\otimes\psi_2 + \psi_1\otimes\psi_1 + z\,(1\otimes\psi_1), \tag{6.43}$$

$$\begin{aligned}\Delta_z(\psi_3) = {}& \psi_3\otimes 1 + 1\otimes\psi_3 + \psi_1\otimes\psi_2 + \psi_2\otimes\psi_1 \\ & + z\,(\psi_1\otimes\psi_1 + 2\cdot 1\otimes\psi_2) + z^2\,(1\otimes\psi_1).\end{aligned} \tag{6.44}$$

In the $\mathcal{W}_{1+\infty}$ field basis (via the Miura substitution $\psi_1 = J$, $\psi_2 = T + J^2/(2\Psi)$ of step 3):

$$\Delta_z(J) = J\otimes 1 + 1\otimes J, \tag{6.45}$$

$$\Delta_z(T) = T\otimes 1 + 1\otimes T + \tfrac{\Psi-1}{\Psi}\,J\otimes J + z\,(1\otimes J). \tag{6.46}$$

The coefficient $(\Psi-1)/\Psi$ in (6.46) arises from the Miura inversion $T = \psi_2 - J^2/(2\Psi)$: the $\psi_1\otimes\psi_1$ cross-term in $\Delta_z(\psi_2)$ has coefficient 1, and subtracting $(1/(2\Psi))\,\Delta_z(J)^2 = (J^2/(2\Psi))\otimes 1 + (1/\Psi)\,J\otimes J + 1\otimes J^2/(2\Psi)$ replaces the coefficient by $1 - 1/\Psi = (\Psi-1)/\Psi$. At $\Psi = 1$ (free boson) the cross-term vanishes; in the classical limit $\Psi\to\infty$ it tends to 1. The Heisenberg OPE $J(z)\,J(w)\sim\Psi/(z-w)^2$ gives $r(z) = \Psi/z$ (class $G$, level prefix $\Psi$). The general formula (6.41) gives $\Delta_z(\psi_n)$ at every spin in closed form; the coproduct on the $\mathcal{W}_{1+\infty}$ field basis $\{J, T, W, W_4, \ldots\}$ is obtained by inverting the Miura transform, and the computation is triangular and algorithmic at each spin.

**Step 5** ($R$-matrix from Maulik–Okounkov; vertex bialgebra structure; Maulik–Okounkov [29], Jindal–Kaubrys–Latyntsev [25]). The critical CoHA of any quiver with potential carries a vertex coproduct forming a vertex bialgebra ([25], Theorems A and C). The $R$-matrix is

$$R(z) = \frac{\Psi(\sigma^*\mathrm{Ext}^\vee, z)}{\Psi(\mathrm{Ext}, z)},$$

where $\Psi$ on the right denotes the motivic exponential and Ext is the $\mathrm{Ext}^1$-bundle on the stack of quiver representations. The Maulik–Okounkov $R$-matrix [29] restricted to the CoHA gives $R_{\mathrm{CoHA}} = q^{\alpha\cdot\beta}$ ($q = e^{\pi i\Psi}$) with classical limit $r(z) = \Psi/z$.

**Step 6** (OPE compatibility at all spins: two independent arguments). The vertex bialgebra axiom (6.9) requires $\Delta_z(Y(a,w)\,b) = Y^{(2)}(\Delta_z(a), w)\,\Delta_z(b)$ as an identity of vertex operators, not merely of associative algebra elements. We give two independent proofs that this holds at all spins simultaneously.

*Argument A* (*coderivation on the Koszul locus*). The algebra $\mathcal{W}_{1+\infty}[\Psi]$ at generic $\Psi$ lies on the Koszul locus (Definition 3.3: the bar-cobar adjunction $\Omega(\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty}[\Psi])) \xrightarrow{\sim} \mathcal{W}_{1+\infty}[\Psi]$ is an equivalence). By the direction

(II) $\to$ (III) of Theorem 6.5, the bar construction produces a chiral coproduct on $\mathcal{W}_{1+\infty}[\Psi]$ from the deconcatenation coproduct $\Delta$ on $\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty}[\Psi]) = T^c(s^{-1}\overline{\mathcal{W}}_{1+\infty}[\Psi])$ via the cobar dualization (6.14). The bar differential $d_{\overline{B}^{\mathrm{ord}}}$ is a coderivation with respect to $\Delta$:

$$d_{\overline{B}^{\mathrm{ord}}} \circ \Delta = \Delta \circ d_{\overline{B}^{\mathrm{ord}}}. \tag{6.47}$$

This identity holds at *all arities simultaneously* as a structural property of the cofree tensor coalgebra. Under the cobar functor $\Omega$, the coderivation property (6.47) dualizes to the OPE compatibility axiom (6.9): the bar differential encodes the vertex operation $Y(a, w)$, and the deconcatenation encodes the coproduct $\Delta_z$ (see the proof of Theorem 6.5, direction (II) $\to$ (III)).

It remains to identify this bar-cobar coproduct with the Drinfeld coproduct of step 4. The bar-cobar coproduct is determined by the vertex algebra structure of $\mathcal{W}_{1+\infty}[\Psi]$; the Drinfeld coproduct is determined by the generating-series formula (6.40). Both satisfy coassociativity and the counit axiom. On the Koszul locus, the coproduct compatible with a given vertex $R$-matrix is unique (Theorem 6.5: the functors $\alpha$, $\beta$, $\gamma$ form an equivalence triangle). The bar-cobar coproduct is compatible with the Maulik–Okounkov $R$-matrix (by construction via $\alpha$ and $\gamma^{-1}$). The Drinfeld coproduct is compatible with the same $R$-matrix (the Yangian $R$-matrix of step 5 restricts to the MO $R$-matrix). By uniqueness, the two coproducts coincide. Therefore the Drinfeld coproduct satisfies OPE compatibility at all spins.

*Argument B* (*JKL vertex bialgebra on the CoHA*). Jindal–Kaubrys–Latyntsev [25] construct a vertex coproduct on the critical CoHA of any quiver with potential, satisfying the vertex bialgebra axiom (6.9) (their Theorems A and C). For the Jordan quiver, the CoHA is $\mathcal{H}_{\mathrm{Jor}} \cong Y^+(\widehat{\mathfrak{gl}}_1)$ (step 1), and the JKL vertex coproduct recovers the Drinfeld coproduct on the Yangian (their Theorem B). Since the JKL vertex bialgebra axiom is exactly the OPE compatibility axiom (6.9), this gives a second proof at all spins simultaneously.

*RTT triviality* (*corroboration, not proof*). For $\mathfrak{gl}_1$, the Maulik–Okounkov $R$-matrix is scalar: $R(z) = g(z)$ of (6.39). The RTT relation

$$R(u-v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R(u-v) \tag{6.48}$$

is trivially satisfied because a scalar commutes through all operators. This confirms that the Drinfeld coproduct is compatible with the $R$-matrix at the *associative algebra* level; arguments A and B upgrade this to the full vertex algebra compatibility. For $Y(\widehat{\mathfrak{gl}}_N)$ with $N \geq 2$, the $R$-matrix is no longer scalar, and OPE compatibility at each truncation is a separate question (Remark 6.19).

**Step 7** (Full axiom verification). The spectral coproduct (6.40) satisfies all axioms of Definition 6.3:

(a) *Strict coassociativity.* $(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_2}(T(u)) = T(u) \otimes T(u - z_2) \otimes T(u - z_1 - z_2)$ and $(\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1}(T(u)) = T(u) \otimes T(u - z_1) \otimes T(u - z_1 - z_2)$.

For $\mathfrak{gl}_1$ the Drinfeld associator acts trivially (the algebra is abelian), so $\Delta_z$ is *strictly* coassociative.

(b) *Counit.* $\varepsilon(\psi_n) = 0$ for $n \geq 1$, $\varepsilon(1) = 1$ gives $(\varepsilon \otimes \mathrm{id}) \circ \Delta_z(T(u)) = T(u-z)$ and $(\mathrm{id} \otimes \varepsilon) \circ \Delta_z(T(u)) = T(u)$.

(c) *OPE compatibility at all spins.* By step 6: the coderivation property on the bar complex of $\mathcal{W}_{1+\infty}[\Psi]$ (argument A) gives the vertex bialgebra axiom (6.9) at all spins via the Koszul-locus equivalence (Theorem 6.5), and the JKL vertex bialgebra theorem on the CoHA (argument B) provides an independent proof. RTT triviality for $\mathfrak{gl}_1$ corroborates the result at the associative algebra level.

No constant coproduct exists for $c \neq 0$: the conformal anomaly $T(z)\,T(w) \sim (c/2)/(z-w)^4 + \cdots$ obstructs any $z$-independent ansatz $\Delta(T) = T \otimes 1 + 1 \otimes T + X$ because the quartic pole on the left maps to $(c/2 + c/2)/(z-w)^4$ while the right retains $c/2$. The spectral parameter in (6.40) absorbs this obstruction through the shift $u \mapsto u - z$.

The three structures (I)–(III) determine each other by Theorem 6.5: the $R$-matrix is given by step 5, the coproduct by steps 4 and 6, and the $A_\infty$ structure is extracted from $\overline{B}^{\mathrm{ord}}(\mathcal{W}_{1+\infty})$ via the functor $\alpha$ with input the $R$-matrix from step 5. □

*Remark* 6.16 (Structural features of the spectral coproduct). The coproduct of Theorem 6.15 has the following convention-independent features:

(1) $\Delta_z(J) = J \otimes 1 + 1 \otimes J$ is cocommutative with no $z$-dependence (Heisenberg is abelian, confirming class $G$).

(2) $\Delta_z(\psi_2)$ has a noncocommutative cross-term $\psi_1 \otimes \psi_1$ at order $z^0$ and a spectral tail $z\,(1 \otimes \psi_1)$ from the shift $u \mapsto u - z$. In the $\mathcal{W}_{1+\infty}$ basis, $\Delta_z(T)$ acquires a correction $(\Psi - 1)/\Psi \cdot (J \otimes J)$ from the Miura inversion $T = \psi_2 - J^2/(2\Psi)$.

(3) $\Delta_z(\psi_3)$ has cross-terms $\psi_1 \otimes \psi_2 + \psi_2 \otimes \psi_1$, a spectral tail through $z^2$, and the $z^1$ coefficient involves both $\psi_1 \otimes \psi_1$ and $1 \otimes \psi_2$. In the $\mathcal{W}_{1+\infty}$ basis, $\Delta_z(W)$ mixes all three spins with coefficients polynomial in $\Psi$.

(4) The general formula (6.41) gives $\Delta_z(\psi_n)$ at every spin in closed form. The coproduct on the $\mathcal{W}_{1+\infty}$ field basis $\{J, T, W, W_4, \ldots\}$ is obtained by inverting the Miura transform; the computation is triangular and algorithmic at each spin.

*Remark* 6.17 (Effective central charge and intertwining in the Miura basis). The coproduct (6.46) in the $\mathcal{W}_{1+\infty}$ field basis has cross-term coefficient $(\Psi - 1)/\Psi$ (not $1/\Psi$), as derived in step 4. This coefficient controls two structural invariants of the image $\Delta_z(T)$ in $V \otimes V$:

(1) *Effective central charge.* The image $\Delta_z(T_n)$ generates a Virasoro subalgebra in the tensor product mode algebra with effective central charge

$$c_{\mathrm{eff}}(\Psi) = 2 + 2(\Psi - 1)^2. \tag{6.49}$$

At $\Psi = 1$ (free boson): $c_{\text{eff}} = 2$ (two decoupled copies, the cross-term vanishes). At $\Psi = 2$: $c_{\text{eff}} = 4$. The quadratic growth $c_{\text{eff}} \sim 2\Psi^2$ at large $\Psi$ reflects the quadratic Miura nonlinearity $\psi_2 = T + J^2/(2\Psi)$. The formula is independent of the spectral parameter $z$ (the $z$-dependent terms in $\Delta_z(T)$ do not contribute to the vacuum expectation value of the commutator $[\Delta_z(T_2), \Delta_z(T_{-2})]$).

(2) *Intertwining with* $\Delta_z(J)$. The commutator $[\Delta_z(T_n), \Delta_z(J_m)] = -\Psi \cdot m \cdot \Delta_z(J_{n+m})$ has intertwining factor $\Psi$ (not 1). The extra factor $\Psi - 1$ beyond the original algebra relation $[T_n, J_m] = -m\, J_{n+m}$ arises from the cross-term $(\Psi - 1)/\Psi \cdot (J \otimes J)_n$ commuting with $\Delta_z(J_m) = J^L_m + J^R_m$: the left and right Heisenberg commutators each contribute $-((\Psi - 1)/\Psi) \cdot \Psi \cdot m = -(\Psi - 1)m$. At $\Psi = 1$: factor 1 (algebra homomorphism).

Both features are consequences of the Miura nonlinearity: the Drinfeld coproduct is an algebra homomorphism of the Yangian $Y(\widehat{\mathfrak{gl}}_1)$ (preserving the $\psi$-generator relations), but when expressed in the $\mathcal{W}_{1+\infty}$ field basis via the nonlinear Miura inversion $T = \psi_2 - J^2/(2\Psi)$, the image acquires $\Psi$-dependent structural invariants that are absent in the $\psi$-basis.

*Remark* 6.18 (Descent to $\mathcal{W}_N$). Drinfeld–Sokolov reduction gives $\pi_N \colon \mathcal{W}_{1+\infty}[\Psi] \twoheadrightarrow \mathcal{W}_N$ for each $N \geq 2$. Since the spectral coproduct of Theorem 6.15 is defined on the full $\mathcal{W}_{1+\infty}$, it descends to $\mathcal{W}_N$ provided $\pi_N \otimes \pi_N$ intertwines $\Delta_z$ with a coproduct on $\mathcal{W}_N$. For $\mathcal{W}_{1+\infty}$ itself, OPE compatibility holds at all spins by the $GL_1$ triviality argument (step 6 of the proof); for finite $N$, the intertwining $(\pi_N \otimes \pi_N) \circ \Delta_z = \Delta_z^{(N)} \circ \pi_N$ follows from the compatibility of the Miura transform with truncation (Remark 6.19). The tower $\cdots \twoheadrightarrow \mathcal{W}_{N+1} \twoheadrightarrow \mathcal{W}_N \twoheadrightarrow \cdots \twoheadrightarrow \mathcal{W}_2 = \mathrm{Vir}_c$ is a projective system of algebras with spectral coproducts. At $N = 2$: class $M$, $\kappa = c/2$, irregular KZ connection. At $N = 3$: $\kappa(\mathcal{W}_3) = 5c/6$ (where $H_3 - 1 = 5/6$). The Gaiotto–Rapčák $Y$-algebras $Y_{N_1,N_2,N_3}[\Psi]$ [22] are truncations of $\mathcal{W}_{1+\infty}$; the $S_3$-triality is a symmetry of the spectral coproduct, corresponding to three Hecke correspondences on spiked-instanton moduli (Rapčák–Soibelman–Yang–Zhao [38]). The $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}(\mathcal{W}_{1+\infty})$ has irregular singularities of unbounded Poincaré rank (spin-$s$ contributes pole order $2s - 1$ after $d$-log absorption), providing the boundary algebra of higher-spin holographic gravity.

*Remark* 6.19 (Proof structure and remaining questions). Three comments on the proof of Theorem 6.15.

*(1) Why RTT alone does not suffice.* The RTT relation (6.48) is a constraint on the *associative algebra* structure (mode commutators). The vertex bialgebra axiom (6.9) is a constraint on the *vertex algebra* structure (all $n$-products $a_{(n)}b$, including the regular products for $n < 0$). For $\mathfrak{gl}_1$ the RTT relation is trivially satisfied because the $R$-matrix is scalar; this confirms associative-algebra compatibility but does not, by itself, establish the full vertex-algebra compatibility. The two independent arguments of step 6 (the

coderivation property on the Koszul-locus bar complex, and the JKL vertex bialgebra theorem on the CoHA) close this gap.

*(2) Truncation to $\mathcal{W}_N$.* Theorem 6.22 resolves this question: for the truncations $\mathcal{W}_N$ at finite $N \geq 2$, the $R$-matrix of $Y(\widehat{\mathfrak{gl}}_N)$ is Yang's rational $R$-matrix $R(u) = u\,I + \Psi\,P$ (no longer scalar), and the OPE compatibility at each truncation level is established by the same two arguments. The coderivation argument (argument A) applies to $\mathcal{W}_N$ at generic level by the same Koszul-locus reasoning. The JKL argument (argument B) applies to $Y(\widehat{\mathfrak{gl}}_N)$ by their Theorem C for general quivers.

*(3) Antipode and Drinfeld double.* The Schiffmann–Mellit–Minets–Vasserot isomorphism [43] gives the positive half $\mathcal{H}_{\mathrm{Jor}} \cong U(\mathcal{W}_{1+\infty}[\Psi])^+$. The Drinfeld double $D(\mathcal{H}_{\mathrm{Jor}}) \cong Y(\widehat{\mathfrak{gl}}_1)$ (Yang–Zhao [48]) provides the full algebra with antipode $S(T(u)) = T(u)^{-1}$. Whether this lifts to a vertex-algebraic antipode on $\mathcal{W}_{1+\infty}$, yielding a vertex Hopf algebra in the sense of Etingof–Kazhdan, is a natural question beyond the scope of the present theorem.

**Lemma 6.20** (Miura inversion of the spectral coproduct at spin 2)**.** *Let $\psi_1 = J$ and $\psi_2 = T + J^2/(2\Psi)$ be the Miura identification of Procházka–Rapčák [39]. The spectral coproduct $\Delta_z(T) = \Delta_z(\psi_2) - (1/(2\Psi))\,\Delta_z(J)^2$ equals*

$$\Delta_z(T) \;=\; T \otimes 1 + 1 \otimes T + \tfrac{\Psi-1}{\Psi}\, J \otimes J + z\,(1 \otimes J). \tag{6.50}$$

*Proof.* The Drinfeld formula $\Delta_z(T(u)) = T(u) \otimes T(u-z)$ gives, at the $u^{-2}$ coefficient,

$$\Delta_z(\psi_2) \;=\; \psi_2 \otimes 1 + 1 \otimes \psi_2 + \psi_1 \otimes \psi_1 + z\,(1 \otimes \psi_1).$$

Since $\Delta_z$ is an algebra homomorphism and $J = \psi_1$ is primitive,

$$\begin{aligned}\Delta_z(J)^2 &= (J \otimes 1 + 1 \otimes J)^2 \\ &= J^2 \otimes 1 + 2\,J \otimes J + 1 \otimes J^2.\end{aligned}$$

Substituting $T = \psi_2 - J^2/(2\Psi)$:

$$\begin{aligned}\Delta_z(T) &= \Delta_z(\psi_2) - \tfrac{1}{2\Psi}\,\Delta_z(J)^2 \\ &= \Big[\big(T + \tfrac{J^2}{2\Psi}\big) \otimes 1 + 1 \otimes \big(T + \tfrac{J^2}{2\Psi}\big) + J \otimes J + z\,(1 \otimes J)\Big] \\ &\qquad - \tfrac{1}{2\Psi}\Big[J^2 \otimes 1 + 2\,J \otimes J + 1 \otimes J^2\Big] \\ &= T \otimes 1 + 1 \otimes T + \big(1 - \tfrac{1}{\Psi}\big)\,J \otimes J + z\,(1 \otimes J).\end{aligned}$$

Since $1 - 1/\Psi = (\Psi-1)/\Psi$, this is (6.50). The $J^2/(2\Psi)$ terms cancel between the Miura substitution and the $\Delta_z(J)^2$ correction. At $\Psi = 1$ (free boson, $c = 1$) the $J \otimes J$ cross-term vanishes and $\Delta_z(T) = T \otimes 1 + 1 \otimes T + z\,(1 \otimes J)$; in the classical limit $\Psi \to \infty$ the coefficient tends to 1. $\square$

6.5. **The chiral quantum group structure on $\mathcal{W}_N$ for $N \geq 2$.** Theorem 6.15 established the chiral quantum group structure on $\mathcal{W}_{1+\infty}[\Psi]$ via the scalar Maulik–Okounkov $R$-matrix. For the truncations $\mathcal{W}_N = \mathcal{W}_{1+\infty}/_N$ (generators of spins $1, 2, \ldots, N$) the $R$-matrix is no longer scalar: it is Yang's rational $R$-matrix on $\mathbb{C}^N \otimes \mathbb{C}^N$, and the RTT relation is a non-trivial matrix identity.

The general theorem is best understood by first computing everything at $N = 2$, where $\mathcal{W}_2 = \mathrm{Vir}_c$ with $c = 1 - 6(\Psi - 1)^2/\Psi$ and the affine Yangian $Y(\widehat{\mathfrak{gl}}_2)$ has a $2 \times 2$ transfer matrix. At $N = 2$ every matrix is small enough to write in full, and the general pattern becomes visible.

**Example 6.21** (The $N = 2$ chiral quantum group: $Y(\widehat{\mathfrak{gl}}_2)$ and Virasoro)**.** *Transfer matrix.* The $2 \times 2$ transfer matrix is

$$T(u) = I_2 + T^{(1)}u^{-1} + T^{(2)}u^{-2} + \cdots = \begin{pmatrix} t_{11}(u) & t_{12}(u) \\ t_{21}(u) & t_{22}(u) \end{pmatrix}, \tag{6.51}$$

where each entry $t_{ij}(u) = \delta_{ij} + \sum_{r \geq 1} t_{ij}^{(r)}\, u^{-r}$. At degree 1 the four generators are $t_{11}^{(1)}, t_{12}^{(1)}, t_{21}^{(1)}, t_{22}^{(1)}$. The quantum Miura factorisation (6.65) gives

$$T(u) = \begin{pmatrix} u + \Lambda_1^{(1)} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & u + \Lambda_2^{(1)} \end{pmatrix} + \text{lower order},$$

where $\Lambda_1^{(1)}, \Lambda_2^{(1)}$ are the Miura fields of Procházka–Rapčák [39].

*Yang's R-matrix at $N = 2$.* The $R$-matrix is $R(u) = u\, I_4 + \Psi\, P$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$, where $P$ is the permutation $P(e_i \otimes e_j) = e_j \otimes e_i$. In the standard basis $\{e_1 \otimes e_1,\, e_1 \otimes e_2,\, e_2 \otimes e_1,\, e_2 \otimes e_2\}$:

$$R(u) = \begin{pmatrix} u + \Psi & 0 & 0 & 0 \\ 0 & u & \Psi & 0 \\ 0 & \Psi & u & 0 \\ 0 & 0 & 0 & u + \Psi \end{pmatrix}. \tag{6.52}$$

At $\Psi = 0$: $R(u) = u\, I_4$ (no interaction). The eigenvalues of $R(u)$ are $u + \Psi$ (symmetric, multiplicity 3) and $u - \Psi$ (antisymmetric, multiplicity 1). The unitarity relation $R(u)\, R(-u) = (\Psi^2 - u^2)\, I_4$ is verified by direct $4 \times 4$ multiplication. The classical $r$-matrix is $r(z) = \Psi\, P/z$, with $r|_{\Psi=0} = 0$.

*RTT relation: explicit commutation relations.* The RTT relation $R_{12}(u - v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R_{12}(u - v)$ at $N = 2$ encodes the following quadratic relations among the generators. Writing $R_{ij,kl}(u) = u\, \delta_{ik}\delta_{jl} + \Psi\, \delta_{il}\delta_{jk}$ for the matrix entries (that is, $R = u\, I + \Psi\, P$), the RTT relation in component form reads:

$$\sum_{k,l=1}^{2} R_{ab,kl}(u - v)\, t_{ka'}(u)\, t_{lb'}(v) = \sum_{k,l=1}^{2} t_{bk}(v)\, t_{al}(u)\, R_{lk,a'b'}(u - v) \tag{6.53}$$

for all $1 \le a, b, a', b' \le 2$. The key relations extracted at degree 1 are:

$$(u-v)\,[t_{ij}(u),\, t_{kl}(v)] = \Psi\big(t_{kj}(u)\,t_{il}(v) - t_{kj}(v)\,t_{il}(u)\big), \tag{6.54}$$

$$[t_{11}^{(1)},\, t_{12}^{(1)}] = \Psi\, t_{12}^{(1)}, \tag{6.55}$$

$$[t_{11}^{(1)},\, t_{21}^{(1)}] = -\Psi\, t_{21}^{(1)}, \tag{6.56}$$

$$[t_{12}^{(1)},\, t_{21}^{(1)}] = \Psi\,(t_{11}^{(1)} - t_{22}^{(1)}). \tag{6.57}$$

These are the standard $Y(\mathfrak{gl}_2)$ relations (Molev [37], §1.4). For $N = 1$ all four generators collapse to $\psi_1$ and the RTT is trivially satisfied (scalar $R$-matrix).

*Quantum determinant at $N = 2$.* The quantum determinant (6.72) specialises at $N = 2$ to

$$\operatorname{qdet} T(u) = t_{11}(u)\,t_{22}(u-\Psi) - t_{12}(u)\,t_{21}(u-\Psi). \tag{6.58}$$

The $\Psi$-shift $u \mapsto u - \Psi$ in the second column is the quantum correction: at $\Psi = 0$ this reduces to the classical determinant. Centrality: $\operatorname{qdet} T(u)$ commutes with all $t_{ij}(v)$ (Molev [37], Theorem 1.11.2; verified numerically at $\Psi = 1$ by `glN_affine_yangian_chiral_qg_engine.py`).

*Drinfeld coproduct at $N = 2$.* The matrix Drinfeld formula $\Delta_z(T(u)) = T(u) \cdot T(u-z)$ gives explicit $2 \times 2$ matrix multiplication:

$$\Delta_z\begin{pmatrix} t_{11}(u) & t_{12}(u) \\ t_{21}(u) & t_{22}(u) \end{pmatrix} = \begin{pmatrix} t_{11}(u) & t_{12}(u) \\ t_{21}(u) & t_{22}(u) \end{pmatrix} \otimes \begin{pmatrix} t_{11}(u-z) & t_{12}(u-z) \\ t_{21}(u-z) & t_{22}(u-z) \end{pmatrix}, \tag{6.59}$$

where the product is matrix multiplication in $\mathbb{C}^2$ with the tensor product in the algebra. In components:

$$\Delta_z(t_{11}(u)) = t_{11}(u) \otimes t_{11}(u-z) + t_{12}(u) \otimes t_{21}(u-z), \tag{6.60}$$

$$\Delta_z(t_{12}(u)) = t_{11}(u) \otimes t_{12}(u-z) + t_{12}(u) \otimes t_{22}(u-z), \tag{6.61}$$

$$\Delta_z(t_{21}(u)) = t_{21}(u) \otimes t_{11}(u-z) + t_{22}(u) \otimes t_{21}(u-z), \tag{6.62}$$

$$\Delta_z(t_{22}(u)) = t_{21}(u) \otimes t_{12}(u-z) + t_{22}(u) \otimes t_{22}(u-z). \tag{6.63}$$

At $N = 1$: the matrix is $1 \times 1$ and $\Delta_z(T(u)) = T(u) \otimes T(u-z)$ is scalar multiplication, recovering (6.40) of Theorem 6.15. The off-diagonal terms in (6.60)–(6.63) (the $t_{12} \otimes t_{21}$ cross-terms) are the genuinely new feature at $N \ge 2$: they couple different matrix entries through the spectral parameter $z$.

*Identification with Virasoro.* The Procházka–Rapčák isomorphism [39] identifies $\mathcal{W}_2 = \mathrm{Vir}_c$ with $c = 1 - 6(\Psi - 1)^2/\Psi$. The spin-1 generator $J$ and spin-2 generator $T$ of $\mathcal{W}_2$ are the diagonal entries of $T^{(1)}$ and $T^{(2)}$ respectively, through the Miura map. The Sugawara-type relation gives $\kappa(\mathrm{Vir}_c) = c/2$ (census C2). The coproduct $\Delta_z(T)$ at $N = 2$ from the matrix formula above, restricted to the Virasoro subalgebra, recovers Lemma 6.20.

**Theorem 6.22** ($Y(\widehat{\mathfrak{gl}}_N)$ chiral quantum group for $N \ge 2$)**.** *For each $N \ge 2$ and generic $\Psi$, the truncation $\mathcal{W}_N[\Psi]$ carries a chiral quantum group datum in the sense of Theorem 6.5: an $N \times N$ matrix transfer matrix, Yang's*

*rational R-matrix with level prefix $\Psi$, and a chiral coproduct with all OPE compatibility verified. The precise content is as follows.*

(I) ($N \times N$ transfer matrix.) *The affine Yangian $Y(\widehat{\mathfrak{gl}}_N)$ has transfer matrix*

$$T(u) = I_N + \sum_{r\geq 1} T^{(r)}\, u^{-r}, \qquad (T^{(r)})_{ij} = t_{ij}^{(r)}, \quad 1 \leq i,j \leq N, \tag{6.64}$$

*an $N \times N$ matrix of formal power series in $u^{-1}$. The Drinfeld–Sokolov reduction $\pi_N \colon \mathcal{W}_{1+\infty}[\Psi] \twoheadrightarrow \mathcal{W}_N$ sends the scalar transfer matrix $T(u) = 1 + \sum \psi_n\, u^{-n}$ of Theorem 6.15 to the quantum Miura factorisation*

$$T(u) = (u + \Lambda_1^{(1)})(u + \Lambda_2^{(1)}) \cdots (u + \Lambda_N^{(1)}), \tag{6.65}$$

*where the $\Lambda_i^{(1)}$ are the Miura fields of Procházka–Rapčák [39].*

(II) (Yang's rational $R$-matrix.) *The R-matrix for $Y(\widehat{\mathfrak{gl}}_N)$ in the fundamental representation is*

$$R(u) = u\, I_{N^2} + \Psi\, P \quad \in \mathrm{End}(\mathbb{C}^N \otimes \mathbb{C}^N), \tag{6.66}$$

*where $P(e_i \otimes e_j) = e_j \otimes e_i$ is the permutation and $\Psi$ is the level. The classical r-matrix (trace-form, level prefix $\Psi$) is*

$$r(z) = \frac{\Psi\, P}{z}, \tag{6.67}$$

*with $r|_{\Psi=0} = 0$. The unitarity relation is $R(u)\, R(-u) = (\Psi^2 - u^2)\, I$. The Yang–Baxter equation*

$$R_{12}(u-v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u-v) \tag{6.68}$$

*holds for all $N$ and all $\Psi$ (computed numerically at $N = 2, 3$ by* `glN_affine_yangian_chiral_qg_engine.py`*; proved algebraically in Molev [37], Chapter 1). At $N = 1$: $P = 1$ and $R(u) = u + \Psi$ is scalar, recovering the class-G case of Theorem 6.15.*

(III) (Drinfeld coproduct: matrix multiplication.) *The Drinfeld coproduct on $T(u)$ is*

$$\Delta_z(T(u)) = T(u) \cdot T(u-z) \qquad (\text{matrix multiplication in } \mathbb{C}^N), \tag{6.69}$$

*that is, in components:*

$$\Delta_z(t_{ij}(u)) = \sum_{k=1}^{N} t_{ik}(u) \otimes t_{kj}(u-z), \tag{6.70}$$

*where $t_{ij}(u) = \delta_{ij} + \sum_{r\geq 1} t_{ij}^{(r)}\, u^{-r}$. This is coassociative up to the Drinfeld associator $\Phi$: $(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_2}(T(u)) = T(u) \otimes T(u-z_2) \otimes T(u-z_1-z_2)$ and $(\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1}(T(u)) = T(u) \otimes T(u-z_1) \otimes T(u-z_1-z_2)$, strictly coassociative for $\mathfrak{gl}_N$ at generic $\Psi$.*

(IV) (RTT relation: non-trivial for $N \geq 2$.) *The RTT relation*

$$R_{12}(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R_{12}(u-v), \tag{6.71}$$

*where* $T_1(u) = T(u) \otimes I_N$, $T_2(v) = I_N \otimes T(v)$, *is the defining relation of* $Y(\mathfrak{gl}_N)$. *For* $N = 1$, *this is trivially satisfied* (*scalar* $R$). *For* $N \geq 2$ *it is a genuine quadratic constraint on the generators* $t_{ij}^{(r)}$: *the commutator* $[R_{12}, T_1T_2]$ *is generically nonzero* (*verified numerically at* $N = 2, 3$).

(V) (Quantum determinant.) *The quantum determinant*

$$\operatorname{qdet} T(u) = \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma) \prod_{i=1}^{N} T_{i,\sigma(i)}\big(u - (i-1)\Psi\big) \tag{6.72}$$

*is* central *in* $Y(\mathfrak{gl}_N)$: *it commutes with all* $t_{ij}(v)$. *In the evaluation representation,* $\operatorname{qdet} T(u)$ *is proportional to the identity on* $\mathbb{C}^N$ (*verified numerically at* $N = 2, 3$). *The* $\Psi$*-shift* $u \mapsto u - (i-1)\Psi$ *is the quantum correction to the classical determinant* $\det T(u)$.

(VI) ($A_\infty$ structure.) *The chiral* $A_\infty$*-operations* $m_k^{\mathrm{ch}}$ *on* $\mathcal{W}_N[\Psi]$ *are extracted from the ordered bar complex* $\overline{B}^{\mathrm{ord}}(\mathcal{W}_N) = T^c(s^{-1}\overline{\mathcal{W}}_N)$ *via the functor* $\alpha$ *of Theorem 6.5, with input the* $R$*-matrix* (6.66).

*The three structures* ($R$*-matrix, coproduct,* $A_\infty$) *determine each other by Theorem 6.5. The OPE compatibility axiom* (6.9) *is satisfied at all spins* $1, \ldots, N$ *simultaneously by the same two independent arguments as in Theorem 6.15:*

(a) Argument $A$ (coderivation on the Koszul locus). *At generic* $\Psi$, *the algebra* $\mathcal{W}_N[\Psi]$ *lies on the Koszul locus* (*Definition 3.3*). *The bar-cobar adjunction* $\Omega(\overline{B}^{\mathrm{ord}}(\mathcal{W}_N[\Psi])) \xrightarrow{\sim} \mathcal{W}_N[\Psi]$ *is an equivalence, and the coderivation property* $d_{\overline{B}^{\mathrm{ord}}} \circ \Delta = \Delta \circ d_{\overline{B}^{\mathrm{ord}}}$ *on* $\overline{B}^{\mathrm{ord}}(\mathcal{W}_N[\Psi]) = T^c(s^{-1}\overline{\mathcal{W}}_N[\Psi])$ *dualizes to OPE compatibility* (6.9) *at all spins simultaneously* (*Theorem 6.5, direction* (II) $\to$ (III)). *Uniqueness on the Koszul locus identifies this bar-cobar coproduct with the Drinfeld coproduct* (6.69).

(b) Argument $B$ (JKL vertex bialgebra on the CoHA). *The critical CoHA of the Jordan quiver at dimension vector* $n \cdot \delta$ ($\delta$ *the minimal imaginary root of* $\widehat{\mathfrak{gl}}_N$) *satisfies the vertex bialgebra axiom* (6.9) *by Jindal–Kaubrys–Latyntsev* [25], *Theorem C for general quivers. The Schiffmann–Vasserot identification* $\mathcal{H}_{\mathrm{Jor},N} \cong Y^+(\widehat{\mathfrak{gl}}_N)$ *transports this to the Yangian, giving a second proof at all spins simultaneously.*

*Proof.* The proof follows the seven-step structure of Theorem 6.15, with the scalar $R$-matrix replaced by Yang's matrix $R$-matrix.

**Step 1** (CoHA identification; Schiffmann–Vasserot [44]). The critical CoHA of the framed Jordan quiver at rank $N$ satisfies $\mathcal{H}_{\mathrm{Jor},N} \cong Y^+(\widehat{\mathfrak{gl}}_N)$. The $N = 1$ case is Schiffmann–Vasserot [44]; the general case uses the Drinfeld–Sokolov hierarchy for $\widehat{\mathfrak{gl}}_N$.

**Step 2** (Yangian–$\mathcal{W}_N$ identification; Procházka–Rapčák [39]). The quantum Miura transform gives an isomorphism of filtered associative algebras $Y(\widehat{\mathfrak{gl}}_N) \cong U(\mathcal{W}_N[\Psi])$. The transfer matrix (6.64) maps to the Miura factorisation (6.65). At $N = 2$: $\mathcal{W}_2 = \mathrm{Vir}_c$ with $c = 1 - 6(\Psi - 1)^2/\Psi$ and $\kappa(\mathrm{Vir}_c) = c/2$. At $N = 3$: $\mathcal{W}_3$ has generators at spins $2, 3$ and $\kappa(\mathcal{W}_3) = 5c/6$ (where $H_3 - 1 = 5/6$).

**Step 3** ($R$-matrix: Yang's rational $R$-matrix). The $R$-matrix (6.66) in the fundamental representation of $\mathfrak{gl}_N$ is:

$$R(u) = u\, I_{N^2} + \Psi\, P \qquad \text{(additive convention)}.$$

The level prefix $\Psi$ is mandatory: at $\Psi = 0$, $R(u) = u\, I$ (no interaction). The classical $r$-matrix is $r(z) = \Psi\, P/z$, with $r|_{\Psi=0} = 0$. The key properties, verified numerically at $N = 2, 3$ and proved algebraically for all $N$ (Molev [37]):

(i) Permutation involution: $P^2 = I$.
(ii) Unitarity: $R(u)\, R(-u) = (\Psi^2 - u^2)\, I$.
(iii) Yang–Baxter equation (6.68).

At $N = 1$: $P = 1$ on $\mathbb{C}^1 \otimes \mathbb{C}^1$ and $R(u) = u + \Psi$ is scalar, so the RTT relation (6.71) is trivially satisfied. For $N \geq 2$: $P$ is a genuine $N^2 \times N^2$ permutation matrix with eigenvalues $\pm 1$, and the RTT is non-trivial.

**Step 4** (Drinfeld coproduct at all spins). The matrix Drinfeld formula (6.69) gives $\Delta_z(T(u)) = T(u) \cdot T(u - z)$. In components, the coefficient of $u^{-n}$ is:

$$\Delta_z(T^{(n)})_{ij} = \sum_{\substack{a,b\geq 0 \\ a+b+c=n}} \binom{b+c-1}{c} z^c \sum_{k=1}^{N} t_{ik}^{(a)} \otimes t_{kj}^{(b)},$$

where $c = n - a - b \geq 0$ and $\binom{-1}{0} = 1$ by convention. The matrix structure sums over the auxiliary index $k$, distinguishing this from the scalar ($N = 1$) formula (6.41).

**Step 5** (RTT verification). The RTT relation (6.71) is a consequence of the Yang–Baxter equation (6.68) by evaluating the universal $R$-matrix in the fundamental representation. In the evaluation representation $T(u) \mapsto R(u - a)$, the RTT becomes the YBE with shifted spectral parameters. The RTT at $N \geq 2$ encodes genuine quadratic relations among the generators $t_{ij}^{(r)}$: the commutator $[R_{12}(u - v),\, T_1(u)\, T_2(v)]$ is generically nonzero (verified at $N = 2$: commutator norm $= 1.0$; $N = 3$: commutator norm $= 1.0$; $N = 1$: commutator $= 0$).

**Step 6** (OPE compatibility at all spins: two arguments). The two independent proofs of OPE compatibility extend from $\mathcal{W}_{1+\infty}$ to $\mathcal{W}_N$:

*Argument A.* The coderivation property $d_{\overline{B}^{\mathrm{ord}}} \circ \Delta = \Delta \circ d_{\overline{B}^{\mathrm{ord}}}$ on the bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{W}_N[\Psi]) = T^c(s^{-1}\overline{\mathcal{W}}_N[\Psi])$ is a structural property of the cofree tensor coalgebra, valid at each truncation $N$. At generic $\Psi$, $\mathcal{W}_N[\Psi]$ lies on the

Koszul locus, and the bar-cobar adjunction identifies the bar-cobar coproduct with the Drinfeld coproduct by uniqueness (Theorem 6.5).

*Argument B.* The JKL vertex bialgebra theorem [25] applies to the CoHA of the Jordan quiver at arbitrary rank $N$ (their Theorem C holds for general quivers with potential). The vertex coproduct on $\mathcal{H}_{\mathrm{Jor},N} \cong Y^+(\widehat{\mathfrak{gl}}_N)$ satisfies (6.9) and identifies with the Drinfeld coproduct.

**Step 7** (Quantum determinant and full axiom verification). The quantum determinant (6.72) is central in $Y(\mathfrak{gl}_N)$ (Molev [37], Theorem 1.11.2), verified numerically to be proportional to $I_N$ in the evaluation representation at $N = 2, 3$. The coproduct satisfies coassociativity, the counit axiom, and OPE compatibility at all spins $1, \ldots, N$ by step 6. Together with the Yang $R$-matrix and the $A_\infty$ structure from the bar complex, this gives the complete chiral quantum group datum for $\mathcal{W}_N[\Psi]$. □

*Remark* 6.23 (The $N = 1 \to N \geq 2$ transition). The passage from $N = 1$ (Theorem 6.15) to $N \geq 2$ (Theorem 6.22) has the following structural features.

(1) The $R$-matrix transitions from scalar ($R(u) = u + \Psi$, trivialising the RTT) to matrix-valued ($R(u) = u\,I + \Psi\,P$, making the RTT a genuine constraint). This is the non-abelian threshold.
(2) The transfer matrix transitions from the scalar series $T(u) = 1 + \sum \psi_n u^{-n}$ to the $N \times N$ matrix $T(u) = I_N + \sum T^{(r)} u^{-r}$. The Drinfeld coproduct $\Delta_z(T(u)) = T(u) \cdot T(u - z)$ is matrix multiplication in $\mathbb{C}^N$, not scalar multiplication.
(3) The quantum determinant $\operatorname{qdet} T(u)$ is a genuinely new central element for $N \geq 2$ (at $N = 1$ it reduces to $T(u)$ itself).
(4) Both arguments for OPE compatibility (coderivation on the Koszul locus, JKL on the CoHA) apply uniformly for all $N$: argument A uses only the structural properties of the cofree tensor coalgebra $T^c(s^{-1}\overline{\mathcal{W}}_N)$ and the Koszul-locus equivalence; argument B uses JKL for the Jordan quiver at rank $N$.

### 6.6. **Structural consequences.**

**Corollary 6.24** (The ordered bar complex encodes all three structures)**.** *The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, equipped with its bar differential $d_{\overline{B}^{\mathrm{ord}}}$ and deconcatenation coproduct $\Delta$, is the universal object from which all three structures in Theorem 6.5 are recovered:*

(i) *The bar differential $d_{\overline{B}^{\mathrm{ord}}}$ encodes the $A_\infty$ structure maps $m_k^{\mathrm{ch}}$ (by the standard bar-cobar correspondence: coderivations of $T^c(s^{-1}\bar{\mathcal{A}})$ biject with elements of $\prod_k \mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}(k)$).*
(ii) *The deconcatenation coproduct $\Delta$ induces, via cobar dualization, the chiral coproduct $\Delta^{\mathrm{ch}}$.*
(iii) *The vertex $R$-matrix $S(z)$ is recovered as the monodromy of the KZ connection, which is itself the degree-2 projection of the Maurer–Cartan element $\Theta_{\mathcal{A}}$ in $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$.*

*This is the structural content of the* $\mathsf{E}_1$ *primacy principle: the ordered bar complex with deconcatenation is the primitive object; the R-matrix, the chiral coproduct, and the* $A_\infty$ *operations are projections of this single datum.*

*Remark* 6.25 (Relation to factorization quantum groups). When the hexagon holds strictly, the chiral quantum group $(\mathcal{A}, S(z), \Delta^{\mathrm{ch}})$ is a *factorization quantum group* (Latyntsev [31]): its ordered bar complex defines a factorization coalgebra on the curve. When the hexagon holds only up to $\Phi$, one obtains a *quasi-factorization quantum group* with regularized holonomy data $(R, \Phi)$.

## 7. Topological Hochschild versus ordered chiral homology of $D^\times$

Two chain complexes compute invariants of a chiral algebra on a punctured disk: the cyclic bar complex (topological Hochschild, from the circle $S^1 = \partial D$) and the ordered chiral chain complex (from the punctured disk $D^\times$ itself, with its complex structure). For $\mathsf{E}_\infty$-chiral algebras, operad formality identifies the two. For $\mathsf{E}_1$-chiral algebras, they diverge: the ordered complex carries holomorphic data from the spectral parameter that the cyclic bar complex cannot see.

7.1. **Two computations.** Let $\mathcal{A}$ be a chiral algebra on a curve $X$. Fix a point $p \in X$ and let $D = \operatorname{Spec} \mathbb{C}[[z]]$ be the formal disk at $p$, with $D^\times = \operatorname{Spec} \mathbb{C}((z))$ the punctured formal disk. The two chain complexes are the following.

**Definition 7.1** (Topological Hochschild)**.** The *topological Hochschild homology* of $\mathcal{A}$ is the factorisation homology of $\mathcal{A}$, viewed as an $\mathsf{E}_1$-topological algebra, over the circle $S^1 = \partial D$:

$$\mathrm{HH}_*^{\mathrm{top}}(\mathcal{A}) := \int_{S^1} \mathcal{A}. \tag{7.1}$$

This is computed by the cyclic bar complex

$$B_n^{\mathrm{cyc}}(\mathcal{A}) = \mathcal{A}^{\otimes n} \Big/ \big(a_1 \otimes \cdots \otimes a_n \sim (-1)^{|a_n|(|a_1|+\cdots+|a_{n-1}|)} a_n \otimes a_1 \otimes \cdots \otimes a_{n-1}\big) \tag{7.2}$$

with the Hochschild differential $d_{\mathrm{HH}}(a_1, \ldots, a_n) = \sum_{i=1}^{n-1} \pm(a_1, \ldots, m_2(a_i, a_{i+1}), \ldots, a_n) + \pm(m_2(a_n, a_1), a_2, \ldots, a_{n-1})$ and, for algebras with non-trivial $A_\infty$-structure, the higher terms involving $m_k$ for $k \geq 3$.

The input data: the associative product $m_2 \colon \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ and the higher $A_\infty$ operations $m_k$. No dependence on $z$; no meromorphic forms; no Arnold relations; no configuration-space geometry.

**Definition 7.2** (Ordered chiral homology of $D^\times$)**.** The *ordered chiral chain complex on* $D^\times$ *at degree* $n$ is:

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) := \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZ}}\Big). \tag{7.3}$$

The ingredients:

(a) $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)$: the ordered FM compactification of configurations of $n$ distinct points in $D^\times$. This is a smooth manifold with corners.

(b) $\Omega^*_{\log}$: logarithmic differential forms, with logarithmic singularities along the boundary divisors of the FM compactification. The Arnold forms $\omega_{ij} = d\log(z_i - z_j)$ are among the generators; these are coefficients in $d_{\mathrm{bar}}$, not in $\nabla_{\mathrm{KZ}}$.

(c) $d_{\mathrm{bar}}$: the bar differential encoding the full OPE. At the codimension-1 boundary stratum where $z_i \to z_j$ $(i < j)$, the bar differential extracts the residue:

$$d_{\mathrm{bar}}\big(a_1 \otimes \cdots \otimes a_n\big)\big|_{\Delta_{ij}} = \sum_k \pm\, a_1 \otimes \cdots \otimes a_{i\,(k)}\, a_j \otimes \cdots \otimes a_n \;\otimes\; \omega_{ij}^{(k)} \tag{7.4}$$

where $a_{(k)}b$ are the OPE modes and $\omega_{ij}^{(k)}$ are the corresponding forms on the boundary stratum.

(d) $\nabla_{\mathrm{KZ}}$: the KZ connection

$$\nabla_{\mathrm{KZ}} = \sum_{i<j} r_{ij}(z_i - z_j)\, d(z_i - z_j), \tag{7.5}$$

with connection 1-form $r(z)\,dz$, not $r(z)\,d\log z$.

The *ordered chiral homology of* $D^\times$ is the derived pushforward of $\mathcal{F}^{\mathrm{ord}}(\mathcal{A})$ over $\mathrm{Ran}^{\mathrm{ord}}(D^\times)$; the degree-by-degree computation gives:

$$\int_{D^\times}^{\mathrm{ord}} \mathcal{A} = R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(D^\times),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big) = \bigoplus_{n\geq 0} H^*\big(\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A})\big). \tag{7.6}$$

The input data: the full OPE $Y(a,z)b = \sum_k a_{(k)}b\, z^{-k-1}$, the Arnold forms on $\mathrm{Conf}_n^{\mathrm{ord}}(D^\times)$, the FM compactification, and the $r$-matrix. The output depends on the complex structure of $D^\times$ (the coordinate $z$).

*Remark* 7.3 (Stratified perspective). The disk $D$ is a stratified space with closed stratum $\partial D = S^1$ and open stratum $D^\times$. Topological Hochschild $\mathrm{HH}_*^{\mathrm{top}}(\mathcal{A}) = \int_{S^1} \mathcal{A}$ is factorisation homology over the closed stratum; ordered chiral homology $\int_{D^\times}^{\mathrm{ord}} \mathcal{A}$ is factorisation homology over the open stratum with its complex structure. The comparison map $\int_{D^\times}^{\mathrm{ord}} \mathcal{A} \to \mathrm{HH}_*^{\mathrm{top}}(\mathcal{A})$ is induced by the boundary retraction $D^\times \to S^1$, which *forgets the complex structure*. This map is a quasi-isomorphism for $\mathsf{E}_\infty$-chiral algebras (Theorem 7.5 below) and fails for genuinely $\mathsf{E}_1$-chiral algebras (Theorem 7.6): the holomorphic data encoded by the ordered theory (the $r$-matrix, the spectral parameter, the Arnold relations) is invisible to the topological theory on $S^1$.

7.2. **Four levels of formality.** Operad formality (Kontsevich–Tamarkin) guarantees that ordered and symmetric give the same answer for $\mathsf{E}_\infty$-chiral algebras. But this does not guarantee simplicity: the transferred $A_\infty$-structure may carry non-trivial higher operations. Four formality conditions, each

strictly stronger than the last, control the relationship between ordered chiral homology and topological Hochschild.

**Definition 7.4** (Formality hierarchy)**.** (a) (Operad formality.) The chain operad $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ is formal as an operad: it is quasi-isomorphic to its cohomology $H^*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ via a quasi-isomorphism depending on a Drinfeld associator $\Phi$ (Kontsevich–Tamarkin). This is *unconditional*: it depends only on the operad, not on any algebra.

(b) (Algebra formality.) The transferred $A_\infty$-structure on $H^*(\mathcal{A})$ has $m_k^{\mathrm{tr}} = 0$ for $k \geq 3$. This depends on the *specific algebra*: it holds for class $G$ (Heisenberg), and fails for class $L/C/M$.

(c) (Convolution algebra formality.) The modular $L_\infty$ brackets $\ell_k = 0$ for $k \geq 3$ in $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$. This holds *only* for class $G$ ([34, prop:sc-formal-iff-class-g]).

(d) (SC-formality.) The Swiss-cheese type operations $m_k^{\mathrm{SC}} = 0$ for $k \geq 3$. This is equivalent to class $G$.

Operad formality (a) is the weakest condition and the only one that holds unconditionally; it guarantees that ordered and symmetric give the *same answer as chain complexes* for $\mathsf{E}_\infty$-chiral algebras. It does *not* guarantee that the answer is simple: the transferred $A_\infty$-structure may carry non-trivial higher operations (levels (b)–(d) impose progressively stronger conditions on the specific algebra, culminating in SC-formality, which holds only for class $G$).

**Theorem 7.5** (Formality bridge for $\mathsf{E}_\infty$-chiral algebras)**.** *Let $\mathcal{A}$ be an $\mathsf{E}_\infty$-chiral algebra and $\Phi$ a Drinfeld associator. The Kontsevich–Tamarkin operad formality quasi-isomorphism induces a quasi-isomorphism of chain complexes:*

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow[\Phi]{\sim} B_n^{\mathrm{cyc}}(\mathcal{A}). \tag{7.7}$$

*In particular, $\int_{D^\times}^{\mathrm{ord}} \mathcal{A} \simeq \mathrm{HH}_*^{\mathrm{top}}(\mathcal{A})$ as graded vector spaces.*

*The transferred $A_\infty$-structure on the cohomological model $H^*(\mathcal{A})$ may still carry non-trivial higher operations $m_k^{\mathrm{tr}}$ for $k \geq 3$. For class $L$ (affine KM), class $C$ ($\beta\gamma$), and class $M$ (Virasoro): these transferred operations are non-trivial. What operad formality guarantees is that the ordered and symmetric complexes compute the same cohomology.*

*Proof.* The proof proceeds in six steps: the KT formality theorem, the identification of the two chain complexes, the transfer of the bar differential, the vanishing of higher OPE modes, the transferred $A_\infty$-structure, and the failure for genuinely $\mathsf{E}_1$-chiral algebras.

*Step 1: Kontsevich–Tamarkin operad formality.* The chain operad $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$, with structure maps given by the operad composition on Fulton–MacPherson compactifications, is formal: there exists a quasi-isomorphism of operads

$$\varphi_\Phi \colon C_*(\overline{\mathrm{FM}}_n(\mathbb{C})) \xrightarrow{\sim} H^*(\overline{\mathrm{FM}}_n(\mathbb{C})) \tag{7.8}$$

depending on the choice of a Drinfeld associator $\Phi \in \exp(\hat{\mathfrak{t}}_3)$ (Kontsevich [8], Tamarkin [9]). The target $H^*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ is the Arnold algebra: it is generated by classes $\omega_{ij} = [d\log(z_i - z_j)]$ of degree 1 ($1 \le i < j \le n$), subject to the Arnold relation $\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij} = 0$ for all triples $i < j < k$ (Arnold [1]). The quasi-isomorphism $\varphi_\Phi$ is $\Sigma_n$-equivariant and respects the operad composition maps; these properties are inherited from the $\Sigma_n$-equivariance of the FM operad structure.

*Step 2: Identification of the two chain complexes.* The ordered chiral chain complex at degree $n$ (Definition 7.2) is the tensor product

$$\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = \Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(D^\times)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}$$

with the total differential $d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZ}}$. The de Rham factor carries the full algebra of logarithmic differential forms on the FM compactification. The cyclic bar complex $B_n^{\mathrm{cyc}}(\mathcal{A})$ uses only the cohomology $H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$, which is the Arnold algebra. The quasi-isomorphism (7.8) induces a chain map from the de Rham model to the Arnold model:

$$\varphi_\Phi \otimes \mathrm{id}\colon \Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n} \longrightarrow H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}. \tag{7.9}$$

Since $\varphi_\Phi$ is a quasi-isomorphism of differential graded vector spaces, so is $\varphi_\Phi \otimes \mathrm{id}$ (by the Künneth theorem; the algebra factor $(s^{-1}\bar{\mathcal{A}})^{\otimes n}$ is degreewise finite-dimensional).

*Step 3: Transfer of the bar differential.* The bar differential $d_{\mathrm{bar}}$ (equation (7.4)) acts at the codimension-1 boundary stratum $\Delta_{ij}$ where $z_i \to z_j$, extracting the full OPE:

$$d_{\mathrm{bar}}\big(a_1 \otimes \cdots \otimes a_n\big)\big|_{\Delta_{ij}} = \sum_{k\ge 0} \pm\, a_1 \otimes \cdots \otimes a_{i\,(k)}\, a_j \otimes \cdots \otimes a_n \;\otimes\; \omega_{ij}^{(k)}$$

where $\omega_{ij}^{(k)}$ is the differential form on the boundary stratum that pairs with OPE mode $a_{(k)}b$. For $\mathcal{A}$ an $\mathsf{E}_\infty$-chiral algebra, the $\mathcal{D}$-module underlying the OPE is $\Sigma_n$-equivariant: the braiding is trivial ($S(z) = \mathrm{id}$), so $d_{\mathrm{bar}}$ commutes with the $\Sigma_n$-action on $\overline{\mathrm{FM}}_n^{\mathrm{ord}}$. Since $\varphi_\Phi$ is a $\Sigma_n$-equivariant operad map, the transferred differential $d_{\mathrm{bar}}^{\mathrm{tr}} = \pi \circ d_{\mathrm{bar}} \circ \iota$, where $\iota\colon H^* \to \Omega^*_{\log}$ is the inclusion of cohomology representatives and $\pi\colon \Omega^*_{\log} \to H^*$ is the projection from the homotopy transfer data of $\varphi_\Phi$, is well defined on the target complex.

*Step 4: Vanishing of higher OPE modes.* The OPE $Y(a,z)b = \sum_{k\ge 0} a_{(k)}b\, z^{-k-1}$ pairs each mode with a differential form on the FM compactification. The leading mode $a_{(0)}b = m_2(a,b)$ pairs with the 1-form $\omega_{ij} = d\log(z_i - z_j)$: this is the Arnold generator in degree 1, and it represents a non-trivial cohomology class in $H^1(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$.

The higher modes $a_{(k)}b$ for $k \ge 1$ pair with forms $\omega_{ij}^{(k)}$ supported on the boundary strata of the FM compactification. These forms arise from the Laurent expansion of the OPE along the exceptional divisor of the blowup $\overline{\mathrm{FM}}_n \to \mathbb{C}^n$: the mode $a_{(k)}b$ multiplies the residue form $z_{ij}^{-k-1}\, dz_{ij}$ by a

regular form on the remaining coordinates, where $z_{ij} = z_i - z_j$ is the local coordinate on the boundary stratum. For $k \geq 1$, the pole order exceeds 1, so the corresponding form $\omega_{ij}^{(k)}$ involves derivatives of the Green kernel along the exceptional fiber.

Under the projection $\pi\colon \Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \to H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$, these boundary forms vanish:

$$\pi(\omega_{ij}^{(k)}) = 0 \in H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \quad \text{for all } k \geq 1. \tag{7.10}$$

The argument proceeds by degree analysis in the Arnold algebra. The cohomology $H^*(\mathrm{Conf}_n(\mathbb{C}))$ is generated in degree 1 by the classes $[\omega_{ij}]$; every class of degree $\geq 2$ is decomposable, i.e. a product of degree-1 generators subject to the Arnold relation. The forms $\omega_{ij}^{(k)}$ for $k \geq 1$ are localised on the codimension-1 boundary stratum $\Delta_{ij}$ and carry fiber degree $\geq 1$ along the exceptional divisor, in addition to the base degree from $\omega_{ij}$ itself. They are indecomposable in the fiber direction: each involves the local coordinate of a single boundary stratum and cannot be expressed as a product of Arnold generators from distinct boundary strata. Since the Arnold algebra has no indecomposable classes in degree $\geq 2$, these forms must be exact in $\Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}})$. Explicitly, $\omega_{ij}^{(k)}$ for $k \geq 1$ is $d_{\mathrm{dR}}$ of a form of lower degree supported on $\Delta_{ij}$, so it lies in the kernel of $\pi$.

The transferred bar differential on the Arnold model therefore reduces to the contribution from $k = 0$ alone:

$$d^{\mathrm{tr}}_{\mathrm{bar}}\big(a_1 \otimes \cdots \otimes a_n\big) = \sum_{i<j} \pm\, a_1 \otimes \cdots \otimes m_2(a_i, a_j) \otimes \cdots \otimes a_n \ \otimes\ [\omega_{ij}].$$

Here the elements $a_i \in s^{-1}\bar{\mathcal{A}}$ are desuspended augmentation-ideal elements, as in Definition 7.2. The resulting differential on $H^*(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}$ is the bar-construction model of the Hochschild differential $d_{\mathrm{HH}}$: it coincides with the differential on $B_n^{\mathrm{cyc}}(\mathcal{A})$ (Definition 7.1) under the standard identification between the bar complex $T^c(s^{-1}\bar{\mathcal{A}})$ and the Hochschild chain complex $\mathcal{A}^{\otimes n}/\mathrm{cyclic}$.

*Step 5: Transferred $A_\infty$-structure and independence of the chain-level comparison.* The homotopy transfer theorem (Kadeishvili [7]) applied to the quasi-isomorphism $\varphi_\Phi$ produces a transferred $A_\infty$-structure $\{m_k^{\mathrm{tr}}\}_{k\geq 2}$ on $H^*(\mathcal{A})$: $m_2^{\mathrm{tr}} = m_2$, and for $k \geq 3$ the operations $m_k^{\mathrm{tr}}$ are given by sums over planar rooted trees with $k$ leaves, each internal edge decorated by the contracting homotopy $h\colon \Omega^*_{\log} \to \Omega^{*-1}_{\log}$ (satisfying $\mathrm{id} - \iota\pi = [d_{\mathrm{dR}}, h]$) and each vertex by the original multiplication.

These transferred operations encode the interaction between the OPE pole structure and the configuration-space geometry: they are non-trivial precisely when algebra-level formality (Definition 7.4(b)) fails. This occurs for class $L$ (affine Kac–Moody), class $C$ ($\beta\gamma$), and class $M$ (Virasoro, $\mathcal{W}_N$).

The chain-level quasi-isomorphism $\mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \xrightarrow{\sim} B_n^{\mathrm{cyc}}(\mathcal{A})$ does *not* depend on these higher operations. The map $\varphi_\Phi \otimes \mathrm{id}$ is a quasi-isomorphism of

chain complexes by Step 2, and the transferred bar differential equals $d_{\mathrm{HH}}$ by Step 4. The two complexes compute the same cohomology, establishing (7.7), regardless of whether the $A_\infty$-structure is formal. What $m_k^{\mathrm{tr}}$ for $k \geq 3$ controls is the *multiplicative* structure on the cohomology: the $A_\infty$-algebra refinement of $H^*(\mathcal{A})$ as an algebra over the operad $H^*(\overline{\mathrm{FM}})$.

*Step 6: Why the argument fails for genuinely* $\mathsf{E}_1$*-chiral algebras.* For a genuinely $\mathsf{E}_1$-chiral algebra, the $R$-matrix braiding $S(z) \neq \mathrm{id}$ breaks the $\Sigma_n$-equivariance of the bar differential: $d_{\mathrm{bar}}$ no longer commutes with the $\Sigma_n$-action, because the OPE $Y(a,z)b$ and $Y(b,-z)a$ differ by $S(z) \neq \mathrm{id}$. The $\mathcal{D}$-module underlying the OPE does not descend to the symmetric quotient $\mathrm{Conf}_n(\mathbb{C})/\Sigma_n$, so the transfer through $\varphi_\Phi$ cannot reduce the ordered complex to the cyclic bar complex. The ordered chiral complex retains holomorphic data from the spectral-parameter dependence of $S(z)$ that has no counterpart in $B_n^{\mathrm{cyc}}$ (Theorem 7.6). □

**Theorem 7.6** (Formality failure for genuinely $\mathsf{E}_1$-chiral algebras)**.** *Let $\mathcal{A}$ be a genuinely $\mathsf{E}_1$-chiral algebra (tier (c): the $R$-matrix $S(z)$ is not derived from any $\mathsf{E}_\infty$-chiral OPE; $S(z) \neq \mathrm{id}$). The chain-level formality quasi-isomorphism (7.7) does* not *exist.*

(i) (Operad formality holds.) *The Kontsevich–Tamarkin formality of $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ as an operad is unconditional: it depends only on the operad structure, not on the specific algebra.*

(ii) (Equivariance fails.) *The $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}$ is not $\Sigma_n$-equivariant (because $S(z) \neq \mathrm{id}$), so the ordered complex does* not *descend to the symmetric quotient. The ordered chiral homology is strictly richer than topological Hochschild: the ordered complex carries holomorphic data from the spectral parameter dependence of $S(z)$ that has no counterpart in the cyclic bar complex.*

(iii) (Dimension count.) *By Arnold's theorem, the cohomology algebra $H^*(\mathrm{Conf}_n(\mathbb{C}))$ has Poincaré polynomial $\prod_{j=0}^{n-1}(1+jt)$, with total dimension $n!$. The ordered chiral chain complex at degree $n$ therefore has*

$$\dim \mathcal{C}_n^{\mathrm{ord}}(D^\times, \mathcal{A}) = (\dim \bar{\mathcal{A}})^n \cdot n!, \tag{7.11}$$

*where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal (as in Definition 7.2), while the cyclic bar complex has $\dim B_n^{\mathrm{cyc}} = (\dim \mathcal{A})^n$. The factor $n!$ from the Arnold algebra measures the additional holomorphic data carried by the ordered chiral complex beyond the topological Hochschild complex.*

*Proof of the dimension count.* The Poincaré polynomial of $H^*(\mathrm{Conf}_n(\mathbb{C}))$ is $P_n(t) = \prod_{j=0}^{n-1}(1+jt)$ (Arnold [1]). Setting $t = 1$ gives $P_n(1) = \prod_{j=0}^{n-1}(1+j) = n!$. $P_2(1) = 1 \cdot 2 = 2$; at $n = 3$, $P_3(1) = 1 \cdot 2 \cdot 3 = 6$; at $n = 4$, $P_4(1) = 1 \cdot 2 \cdot 3 \cdot 4 = 24$. The ordered chiral complex is $\Omega^*_{\log}(\overline{\mathrm{FM}}_n^{\mathrm{ord}}) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n}$; at the cohomological level, the $\Omega^*_{\log}$ factor contributes the Arnold algebra, whose total dimension is $n!$. □

## 8. The $\mathsf{E}_3$ structure and Chern–Simons theory

The $\mathsf{E}_2$-coalgebra structure on $\overline{B}^{\Sigma}(\mathcal{A})$ (Warning 3.7) and the Higher Deligne Conjecture together equip the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ with a natural $\mathsf{E}_3$-algebra structure ($\mathsf{E}_2+1=\mathsf{E}_3$). The comparison with perturbative Chern–Simons theory (Costello–Francis–Gwilliam) identifies this $\mathsf{E}_3$ structure with the bulk observables of the 3d gauge theory.

8.1. **The derived chiral centre.** The ordered and symmetric chiral Hochschild cochain complexes differ by the presence or absence of $\Sigma_k$-coinvariants. The derived chiral centre is the cohomology of the symmetric version; its ordered refinement carries the full $\mathsf{E}_1$ data.

**Definition 8.1** (Symmetric and ordered chiral Hochschild complexes)**.** Let $\mathcal{A}$ be a chiral algebra on a curve $X$.

(i) The *symmetric chiral Hochschild cochain complex* is:

$$\mathrm{HH}^{*,\Sigma}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) = \bigoplus_{k\geq 0} \Omega^*(\overline{\mathrm{FM}}_k(\mathbb{C})) \otimes_{\Sigma_k} \mathrm{Hom}(\mathcal{A}^{\otimes k},\mathcal{A}), \tag{8.1}$$

with the chiral Hochschild differential. The derived chiral centre is $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) := \mathrm{HH}^{*,\Sigma}_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$.

(ii) The *ordered chiral Hochschild cochain complex* is:

$$\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) = \bigoplus_{k\geq 0} \Omega^*(\overline{\mathrm{FM}}^{\mathrm{ord}}_k(\mathbb{C})) \otimes \mathrm{Hom}(\mathcal{A}^{\otimes k},\mathcal{A}), \tag{8.2}$$

without $\Sigma_k$-coinvariants, with the ordered chiral Hochschild differential.

**Proposition 8.2** (The $\mathsf{E}_3$ structure on the derived centre)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth algebraic curve $X$.*

(i) (Source of the $\mathsf{E}_2$ structure.) *The curve geometry provides the $\mathsf{E}_2$-coalgebra structure on $\overline{B}^{\Sigma}(\mathcal{A})$ (Warning 3.7).*

(ii) (Higher Deligne gives $\mathsf{E}_3$.) *By the Higher Deligne Conjecture (Lurie [24], Francis [21]), the Hochschild cohomology of an $\mathsf{E}_2$-algebra carries a natural $\mathsf{E}_3$-algebra structure. Dualising the $\mathsf{E}_2$-coalgebra $\overline{B}^{\Sigma}(\mathcal{A})$ to an algebra (which requires the finiteness conditions satisfied by standard chiral algebras, see [34, Thm A]):*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{HH}^*(\overline{B}^{\Sigma}(\mathcal{A}),\overline{B}^{\Sigma}(\mathcal{A})) \quad \textit{is an } \mathsf{E}_3\textit{-algebra.} \tag{8.3}$$

*The dimension count: $\mathsf{E}_2$ (curve, $\dim_{\mathbb{R}} X = 2$) $+1$ (centre construction) $=\mathsf{E}_3$.*

(iii) (Scope.) *For $\mathsf{E}_\infty$-chiral algebras, the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ exists as a well-defined $\mathsf{E}_2$-coalgebra and the $\mathsf{E}_3$ structure on the derived centre follows from* (i)–(ii)*. No Virasoro element or conformal symmetry is required. For genuinely $\mathsf{E}_1$-chiral algebras (Yangians, EK*

*quantum vertex algebras), the factorisation $\mathcal{D}$-module does not descend to $X^{(n)}$ (Theorem 7.6), so $\overline{B}^{\Sigma}(\mathcal{A})$ does not exist. The ordered bar $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ carries an $\mathsf{E}_1$-coalgebra structure, giving only $\mathsf{E}_2$ on $\mathrm{HH}^*(\overline{B}^{\mathrm{ord}}(\mathcal{A}), \overline{B}^{\mathrm{ord}}(\mathcal{A}))$ via the classical Deligne conjecture ($\mathsf{E}_1 + 1 = \mathsf{E}_2$). The passage from $\mathsf{E}_2$ to $\mathsf{E}_3$ for $\mathsf{E}_1$-chiral algebras requires the Drinfeld centre construction (see* [36, conj:drinfeld-center-equals-bulk]).

*Remark* 8.3 (Two distinct $\mathsf{E}_3$ structures). The paper produces two $\mathsf{E}_3$-algebra structures on the derived chiral centre of $V_k(\mathfrak{g})$ at non-critical level.

(a) *Algebraic* $\mathsf{E}_3$ (Proposition 8.2): the Hochschild cohomology of the $\mathsf{E}_2$-coalgebra $\overline{B}^{\Sigma}(V_k(\mathfrak{g}))$, via the Higher Deligne Conjecture. This requires dualisation of the coalgebra to an algebra, which uses the finiteness of $V_k(\mathfrak{g})$ (finite-type as a graded vector space in each conformal-weight stratum). No conformal vector is needed.

(b) *Topological* $\mathsf{E}_3$ (Proposition 8.23): the Sugawara Virasoro element at non-critical level topologises the holomorphic direction of the $\mathsf{E}_2$ (Warning 3.7), producing a topological $\mathsf{E}_3$-algebra on $\mathbb{R}^3 \cong X \times \mathbb{R}$. This requires the conformal vector and fails at $k = -h^\vee$.

Structure (a) is formal-algebraic; structure (b) is geometric-topological. Theorem 8.6 identifies the two as families over $(k + h^\vee)H^3(\mathfrak{g})[[k + h^\vee]]$.

**Warning 8.4** (The bar complex itself is not $\mathsf{E}_3$)**.** The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ does *not* carry an $\mathsf{E}_3$-algebra structure. The obstruction is topological: $\mathsf{E}_3$ requires $\mathrm{Conf}_k(\mathbb{R}^3)$, but a curve provides only $\mathrm{Conf}_k(\mathbb{R}^2)$. For $\widehat{\mathfrak{sl}}_2$, the obstruction space $\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$ is one-dimensional and the obstruction class is non-zero. The $\mathsf{E}_3$-algebra is the *derived centre*, not the bar complex.

8.2. **The CFG comparison.** Costello, Francis, and Gwilliam construct a filtered $\mathsf{E}_3$-algebra $\mathcal{A}^\lambda$ from BV quantisation of Chern–Simons theory on $\mathbb{R}^3$. The derived chiral centre of $V_k(\mathfrak{g})$ and the CFG algebra both deform $C^*(\mathfrak{g})$ over $(k + h^\vee)H^3(\mathfrak{g})[[k + h^\vee]]$. The question is whether these deformation families are isomorphic as $\mathsf{E}_3$-algebras.

**Theorem 8.5** (Costello–Francis–Gwilliam [12])**.** *Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra. BV quantisation of Chern–Simons theory on $\mathbb{R}^3$ with gauge algebra $\mathfrak{g}$ and invariant pairing $\lambda$ yields a filtered $\mathsf{E}_3$-algebra $\mathcal{A}^\lambda$ on the Chevalley–Eilenberg cochains $C^*(\mathfrak{g})$.*

(i) *The classical observables are $C^*(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$ with the CE differential: a commutative ($\mathsf{E}_\infty$) algebra.*

(ii) *The quantum observables $\mathcal{A}^\lambda$ are a filtered $\mathsf{E}_3$-algebra deforming $C^*(\mathfrak{g})$, with deformation space $(k + h^\vee)H^3(\mathfrak{g})[[k + h^\vee]]$.*

(iii) *For any $\mathsf{E}_3$-algebra $\mathcal{A}$, $\mathrm{Mod}_{\mathcal{A}}$ is $\mathsf{E}_2$-monoidal (braided monoidal). Filtered Koszul duality gives $\mathrm{Perf}_{C^*(\mathfrak{g})} \simeq \mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})^{\mathrm{dg}}$, so deforming $C^*(\mathfrak{g})$ as $\mathsf{E}_3$ deforms $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$ as braided monoidal: this recovers the Drinfeld–Jimbo quantum group.*

(iv) (Theorem 1.4 of [12].) *Canonical bijection between: perturbative CS quantisations, filtered* $\mathsf{E}_3$*-algebras deforming* $C^*(\mathfrak{g})$*, braided monoidal deformations of* $\mathrm{Rep}_{\mathrm{fin}}(\mathfrak{g})$*, and quasi-triangular quasi-Hopf algebras deforming* $U(\mathfrak{g})$*. All non-canonically isomorphic to* $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$*.*

**Theorem 8.6** (Deformation families of $\mathsf{E}_3$-algebras)**.** *Let* $\mathfrak{g}$ *be a simple finite-dimensional Lie algebra and* $V_k(\mathfrak{g})$ *the affine Kac–Moody vertex algebra at level* $k$*. Writing* $\lambda = k + h^\vee$ *for the departure from critical level, the derived chiral centre* $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ *of Proposition 8.2 and the CFG* $\mathsf{E}_3$*-algebra* $\mathcal{A}^\lambda$ *of Theorem 8.5 are isomorphic as formal deformation families of* $\mathsf{E}_3$*-algebras:*

$$\{Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))\}_{k+h^\vee\in\lambda\mathbb{C}[[\lambda]]} \;\simeq\; \{\mathcal{A}^\lambda\}_{\lambda\in\lambda H^3(\mathfrak{g})[[\lambda]]} \tag{8.4}$$

*over the base* $\lambda H^3(\mathfrak{g})[[\lambda]]$ *(with* $\lambda = k + h^\vee$*).*

(i) (Matching deformation spaces.) *Both families are parametrised by* $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$*: for the derived chiral centre, the deformation parameter is the departure from critical level; for the CFG algebra, it is the Chern–Simons coupling.*

(ii) (Base point mismatch.) *At the critical level* $k = -h^\vee$*, the derived chiral centre* $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{-h^\vee}(\mathfrak{g}))$ *is related to the Feigin–Frenkel centre* $\mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$ *(opers on the formal disk), not directly to* $C^*(\mathfrak{g})$*. The isomorphism is perturbative: it holds in the formal neighbourhood of critical level, order by order in* $\lambda = k + h^\vee$*.*

(iii) (Order-by-order uniqueness.) *For simple* $\mathfrak{g}$*,* $H^3(\mathfrak{g}) \cong \mathbb{C}$ *is one-dimensional, so the deformation base is one-dimensional at each order in* $\lambda$*. The isomorphism (8.4) is forced by the matching of associated graded algebras (both* $C^*(\mathfrak{g})$*), matching* $\mathsf{P}_3$ *brackets on the formal disk (Theorem 8.21*(ii)*), and the one-dimensionality of the obstruction space at each order.*

(iv) (Sugawara constraint.) *The topological enhancement (Theorem 1.3(iv)) holds at generic* $k$ *but fails at* $k = -h^\vee$*, where the Sugawara element is undefined. The perturbative identification holds order by order in* $(k + h^\vee)$*, since the Sugawara element exists at any nonzero departure from critical level.*

*Proof.* The argument is obstruction-theoretic: we construct an isomorphism of $\mathsf{E}_3$-algebras order by order in $\lambda = k + h^\vee$, using the one-dimensionality of $H^3(\mathfrak{g})$ for simple $\mathfrak{g}$ to force uniqueness at each stage.

*Step 1: setup.* Both $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ and $\mathcal{A}^\lambda$ are filtered $\mathsf{E}_3$-algebras deforming $C^*(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$ over the base $\lambda H^3(\mathfrak{g})[[\lambda]]$, with $\lambda = k + h^\vee$ (Theorem 1.3(ii) and Theorem 8.5(iv) respectively). Both filtrations are complete and exhaustive with associated graded $C^*(\mathfrak{g})$. The deformation theory of $\mathsf{E}_3$-algebras with fixed associated graded $C^*(\mathfrak{g})$ is controlled by the $\mathsf{E}_3$-deformation complex, whose relevant cohomology group is $H^3(\mathfrak{g})$: the space of $\mathsf{E}_3$-deformations of $C^*(\mathfrak{g})$ at each order $p \geq 1$ in $\lambda$, modulo those at

order $< p$, is classified by $H^3(\mathfrak{g})$ (Lurie [24], Costello–Francis–Gwilliam [12], Theorem 1.4).

*Step 2: induction on order.* We construct the isomorphism $\varphi\colon Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) \xrightarrow{\sim} \mathcal{A}^\lambda$ as a compatible family of isomorphisms $\varphi_p$ modulo $\lambda^{p+1}$, by induction on $p$.

*Base case ($p = 0$):* both associated graded algebras are $C^*(\mathfrak{g})$ (Theorem 8.21(i)), so $\varphi_0 = \mathrm{id}_{C^*(\mathfrak{g})}$.

*Inductive step:* suppose $\varphi_p$ is an $\mathsf{E}_3$-isomorphism modulo $\lambda^{p+1}$. The obstruction to extending $\varphi_p$ to an isomorphism modulo $\lambda^{p+2}$ lies in the space of $\mathsf{E}_3$-deformations of $C^*(\mathfrak{g})$ at order $p+1$, which is $H^3(\mathfrak{g})$. For simple $\mathfrak{g}$, $H^3(\mathfrak{g}) = \mathbb{C}$ is one-dimensional, generated by the Cartan three-form $\omega_3(X, Y, Z) = (X, [Y, Z])$. Two $\mathsf{E}_3$-deformations of $C^*(\mathfrak{g})$ that agree modulo $\lambda^{p+1}$ therefore differ at order $p+1$ by a scalar multiple of $\omega_3$.

It remains to show that both constructions select the *same* scalar at each order. On the formal disk $D = \operatorname{Spec}\mathbb{C}[[z]]$, the restriction functor $\Gamma(D, -)$ is symmetric monoidal and preserves $\mathsf{E}_3$-structures (Theorem 8.21, Step 1 of the proof). By Theorem 8.21(ii), the chiral $\mathsf{P}_3$ bracket restricted to $D$ recovers the CFG $\mathsf{P}_3$ bracket at every order. Since the $\mathsf{P}_3$ bracket determines the $\mathsf{E}_3$-deformation class (the bracket on generators $\mathfrak{g}^*[-1] \otimes \mathfrak{g}^*[-1] \to \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$ encodes the full deformation datum for biderivations, by the argument of Lemma 8.18, Step 2), and the space of $\mathfrak{g}$-equivariant corrections at each order is one-dimensional (spanned by the Killing-form term $k\,(a, b)\,\partial$), the two constructions agree at order $p+1$. The isomorphism $\varphi_{p+1}$ extends $\varphi_p$.

*Step 3: passage to the limit.* The compatible family $\{\varphi_p\}_{p\geq 0}$ determines an isomorphism $\varphi = \varprojlim_p \varphi_p$ of filtered $\mathsf{E}_3$-algebras over $\lambda\, H^3(\mathfrak{g})[[\lambda]]$, completing the proof of (8.4). Part (iv) is immediate: the topological enhancement (Theorem 1.3(iv)) requires the Sugawara element, which exists for $\lambda \neq 0$ and is compatible with the isomorphism $\varphi$ order by order (the Sugawara element at order $p$ depends only on the data modulo $\lambda^{p+1}$, where the two families already agree). □

8.3. **Explicit $\mathsf{E}_3$ operations for $V_k(\mathfrak{sl}_2)$.** The abstract $\mathsf{E}_3$ structure of Proposition 8.2 acquires concrete content for $\mathfrak{g} = \mathfrak{sl}_2$: the five-dimensional derived chiral centre $\mathbb{C} \oplus \mathfrak{sl}_2[-1] \oplus \mathbb{C}[-2]$ carries an explicit cup product, Gerstenhaber bracket, $\mathsf{P}_3$ bracket, and BV operator, all computable from the OPE.

**The underlying graded vector space.** By Proposition 8.2 and the chiral Hochschild computation of Example 9.2, the derived chiral centre of $V_k(\mathfrak{sl}_2)$ at generic level $k \neq -2$ is the graded vector space

$$Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2)) \;=\; \underbrace{\mathbb{C}}_{=\mathrm{HH}^0} \;\oplus\; \underbrace{\mathfrak{sl}_2[-1]}_{=\mathrm{HH}^1} \;\oplus\; \underbrace{\mathbb{C}[-2]}_{=\mathrm{HH}^2}, \tag{8.5}$$

with total dimension $1 + 3 + 1 = 5$. Write $\mathbf{1}$ for the generator of $\mathrm{HH}^0 = \mathbb{C}$ (the vacuum), $\{e, f, h\}$ for the standard basis of $\mathrm{HH}^1 = \mathfrak{sl}_2$ (outer derivations:

infinitesimal level-preserving deformations parametrised by $\mathfrak{g}$ itself), and $\eta$ for the generator of $\mathrm{HH}^2 = \mathbb{C}$ (the central charge deformation cocycle). The $\mathfrak{sl}_2$-module structure is: $\mathrm{HH}^0$ trivial, $\mathrm{HH}^1$ adjoint, $\mathrm{HH}^2$ trivial.

**Conventions.** The Killing form normalisation is $(e, f) = (f, e) = 1$, $(h, h) = 2$, all other pairings zero. Structure constants: $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$. The inverse Casimir is $\Omega = e \otimes f + f \otimes e + \frac{1}{2}\, h \otimes h$, and $h^\vee = 2$. The KZ coupling constant is $h_{\mathrm{KZ}} = 1/(k + h^\vee) = 1/(k + 2)$.

*Remark* 8.7 (Convention for $\hbar$). Throughout this paper, $\hbar$ denotes the *KZ coupling constant*

$$\hbar \;=\; h_{\mathrm{KZ}} \;=\; \frac{1}{k + h^\vee}, \tag{8.6}$$

the normalisation in the KZ connection $\nabla = d + \hbar \sum \Omega_{ij}\, dz_{ij}$. It vanishes as $k \to \infty$ (weak-coupling limit) and diverges at the critical level $k = -h^\vee$. In the $\mathfrak{sl}_2$ computations, $h^\vee = 2$, so $\hbar = 1/(k + 2)$. The *departure from critical level* is the inverse quantity $k + h^\vee = 1/\hbar$; we always write this explicitly as $k + h^\vee$ rather than introducing a separate symbol. Critical level corresponds to $k + h^\vee = 0$ (equivalently, $\hbar \to \infty$); weak coupling corresponds to $k + h^\vee \to \infty$ (equivalently, $\hbar \to 0$). For the quantum group parameter: $q = e^{2\pi i \hbar}$.

An $\mathsf{E}_3$-algebra carries three levels of operations:

- Level 0 ($\mathsf{E}_\infty$): a graded-commutative product $\mu\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q}$.
- Level 1 ($\mathsf{E}_2$ beyond $\mathsf{E}_1$): the Gerstenhaber bracket $[-,-]\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q-1}$ of degree $-1$, derived from the fundamental class of $S^1 = \mathrm{Conf}_2(\mathbb{R}^2)/\mathbb{R}^+$.
- Level 2 ($\mathsf{E}_3$ beyond $\mathsf{E}_2$): the $\mathsf{P}_3$ bracket $\{-,-\}\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q-2}$ of degree $-2$, derived from the fundamental class of $S^2 = \mathrm{Conf}_2(\mathbb{R}^3)/\mathbb{R}^+$.

We compute each in turn.

**Proposition 8.8** (Explicit $\mathsf{E}_3$ operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$)**.** *Let $k \neq -2$. The $\mathsf{E}_3$-algebra structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$ is determined by the following data.*

(I) Cup product *($\mathsf{E}_\infty$ level).*

(a) $\mathbf{1}$ *is the unit:* $\mathbf{1} \cdot f = f \cdot \mathbf{1} = f$ *for all* $f$.
(b) $\mu(X, Y) = 0$ *for all* $X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2$.
(c) $\mu(X, \eta) = 0$ *for all* $X \in \mathrm{HH}^1$, *and* $\eta \cdot \eta = 0$.

(II) Gerstenhaber bracket *($\mathsf{E}_2$ level, degree $-1$).*

$$[\mathbf{1}, -] = 0, \qquad [X, Y] = 0 \quad \textit{for all } X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2,$$
$$[X, \eta] = 0 \quad \textit{for all } X \in \mathrm{HH}^1, \qquad [\eta, \eta] = 0.$$

(III) The $\mathsf{P}_3$ bracket *($\mathsf{E}_3$ level, degree $-2$).*

$$\{X,Y\} = h_{\mathrm{KZ}} \cdot (X,Y) = \frac{1}{k+h^\vee}\,(X,Y) \quad \textit{for } X,Y \in \mathrm{HH}^1, \tag{8.7}$$

$$\{X,\eta\} = 0 \quad \textit{for } X \in \mathrm{HH}^1, \tag{8.8}$$

$$\{\mathbf{1},-\} = 0, \tag{8.9}$$

$$\{\eta,\eta\} = 0, \tag{8.10}$$

*where $(X,Y)$ denotes the normalised Killing form and $h_{\mathrm{KZ}} = 1/(k+h^\vee) = 1/(k+2)$ is the KZ coupling constant (Remark 8.7).*

*Proof.* We establish each level by $\mathfrak{sl}_2$-equivariance, degree counting, and the BV relation.

*Cup product.* The cup product $\mu\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q}$ is graded-commutative: $\mu(a,b) = (-1)^{pq}\,\mu(b,a)$. For $X,Y \in \mathrm{HH}^1$, this gives $\mu(X,Y) = -\mu(Y,X)$: the restriction to $\mathrm{HH}^1 \otimes \mathrm{HH}^1 \to \mathrm{HH}^2 = \mathbb{C}$ is an $\mathfrak{sl}_2$-invariant antisymmetric bilinear form on the adjoint representation valued in the trivial representation. Since $\Lambda^2(\mathfrak{sl}_2)^{\mathfrak{sl}_2} \cong \mathrm{Hom}_{\mathfrak{sl}_2}(\Lambda^2(\mathrm{ad}),\mathbb{C}) = 0$ (the exterior square of the adjoint decomposes as $\Lambda^2(\mathrm{ad}) \cong \mathrm{ad}$, which has no trivial summand), the cup product $\mu(X,Y) = 0$ for all $X,Y \in \mathrm{HH}^1$.

The products $\mu(\mathrm{HH}^1,\mathrm{HH}^2) \subset \mathrm{HH}^3 = 0$ and $\mu(\mathrm{HH}^2,\mathrm{HH}^2) \subset \mathrm{HH}^4 = 0$ vanish for degree reasons.

*BV operator.* The framed $\mathsf{E}_2$ structure on $\overline{B}^\Sigma(V_k(\mathfrak{sl}_2))$ (from genus-1 sewing) equips the derived centre with a BV operator $\Delta\colon \mathrm{HH}^n \to \mathrm{HH}^{n-1}$ satisfying $\Delta^2 = 0$ and the BV relation

$$[f,g] = \Delta(f\cdot g) - (\Delta f)\cdot g - (-1)^{|f|}\, f\cdot(\Delta g). \tag{8.11}$$

The component $\Delta\colon \mathrm{HH}^1 \to \mathrm{HH}^0$ is an $\mathfrak{sl}_2$-equivariant linear map from the adjoint representation to the trivial representation. Since the adjoint is irreducible and non-trivial, $\Delta|_{\mathrm{HH}^1} = 0$.

The component $\Delta\colon \mathrm{HH}^2 \to \mathrm{HH}^1$ is an $\mathfrak{sl}_2$-equivariant linear map from the trivial to the adjoint. Again, $\Delta|_{\mathrm{HH}^2} = 0$.

All other components vanish trivially ($\Delta|_{\mathrm{HH}^0}$ lands in $\mathrm{HH}^{-1} = 0$; $\Delta|_{\mathrm{HH}^n} = 0$ for $n \geq 3$). Therefore $\Delta = 0$ on the entire derived centre.

*Gerstenhaber bracket.* With $\Delta = 0$ and $\mu|_{\mathrm{HH}^1\otimes\mathrm{HH}^1} = 0$, the BV relation (8.11) gives: for $X,Y \in \mathrm{HH}^1$,

$$[X,Y] = \Delta(\mu(X,Y)) - (\Delta X)\cdot Y + X\cdot(\Delta Y) = \Delta(0) - 0 + 0 = 0.$$

For $X \in \mathrm{HH}^1$, $\eta \in \mathrm{HH}^2$: $[X,\eta] = \Delta(\mu(X,\eta)) - (\Delta X)\cdot\eta - (-1)^1\, X\cdot(\Delta\eta) = 0 - 0 + 0 = 0$ (since $\mu(X,\eta) \in \mathrm{HH}^3 = 0$, $\Delta X = 0$, $\Delta\eta = 0$). Similarly $[\eta,\eta] \in \mathrm{HH}^3 = 0$.

The vanishing of the Gerstenhaber bracket on the derived centre is a structural consequence of the $\mathfrak{sl}_2$-equivariance constraints: the adjoint representation provides no equivariant maps to or from the trivial representation. This is specific to the *derived centre* $\mathrm{HH}^*(\overline{B}^\Sigma,\overline{B}^\Sigma)$; the Gerstenhaber bracket

on the full Hochschild cochain complex $C^*(\overline{B}^\Sigma, \overline{B}^\Sigma)$ is non-trivial (it carries the commutator of derivations at the cochain level).

$\mathsf{P}_3$ *bracket.* The degree-$(-2)$ bracket $\{-,-\}\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q-2}$ satisfies the graded symmetry

$$\{a,b\} = -(-1)^{(|a|-2)(|b|-2)}\,\{b,a\} \tag{8.12}$$

of a degree-$(-2)$ Lie bracket, together with the Leibniz rule $\{a, b\cdot c\} = \{a,b\}\cdot c + (-1)^{(|a|-2)|b|}\, b\cdot\{a,c\}$.

*Step 1:* $\{\mathbf{1},-\} = 0$. The Leibniz rule applied to $\mathbf{1} = \mathbf{1}\cdot\mathbf{1}$ gives $\{\mathbf{1}, f\} = 2\,\{\mathbf{1}, f\}$ for all $f$; hence $\{\mathbf{1}, f\} = 0$.

*Step 2:* $\{X,Y\} = h_{\mathrm{KZ}}\,(X,Y)$. For $X, Y \in \mathrm{HH}^1$, $\{X,Y\} \in \mathrm{HH}^0 = \mathbb{C}$. The symmetry (8.12) at $|X| = |Y| = 1$ gives $\{X,Y\} = -(-1)^{(-1)(-1)}\,\{Y,X\} = -(-1)^1\,\{Y,X\} = \{Y,X\}$: the bracket is *symmetric* on $\mathrm{HH}^1 \otimes \mathrm{HH}^1$. By $\mathfrak{sl}_2$-equivariance, $\{X,Y\}$ is an invariant symmetric bilinear form on the adjoint representation valued in $\mathbb{C}$. The space of such forms is one-dimensional, spanned by the Killing form $(X,Y)$. Hence $\{X,Y\} = \alpha\cdot(X,Y)$ for a scalar $\alpha$ depending on $k$.

The scalar $\alpha$ is determined by the $r$-matrix residue, independently of any comparison with the CFG construction. The $\mathsf{P}_3$ bracket on the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) = \mathrm{HH}^*(\overline{B}^\Sigma(V_k(\mathfrak{g})), \overline{B}^\Sigma(V_k(\mathfrak{g})))$ is the $\mathsf{E}_2$-Hochschild cohomology operation derived from the fundamental class of $S^2 \subset \mathrm{Conf}_2(\mathbb{R}^3)$. At degree 2, this operation is the residue of the classical $r$-matrix paired with the fundamental class of $S^1 \subset \mathrm{Conf}_2(\mathbb{C})$. In the KZ normalisation (see (9.10) and the conventions of §8.10):

$$r(z) = \frac{\Omega}{(k+h^\vee)\,z}, \qquad \mathrm{Res}_{z=0}\big[r(z)\big] = \frac{\Omega}{k+h^\vee}. \tag{8.13}$$

The residue acts on $X\otimes Y \in \mathfrak{g}\otimes\mathfrak{g}$ by contraction with the inverse Casimir: $\Omega(X\otimes Y) = (X,Y)$ (the normalised Killing form). The $\mathsf{P}_3$ bracket on $\mathrm{HH}^1 \cong \mathfrak{g}$ is therefore

$$\{X,Y\} = \mathrm{Res}_{z=0}\big[r(z)(X\otimes Y)\big] = \frac{(X,Y)}{k+h^\vee} = h_{\mathrm{KZ}}\cdot(X,Y), \tag{8.14}$$

giving $\alpha = h_{\mathrm{KZ}} = 1/(k+h^\vee)$. For $\mathfrak{sl}_2$, $h^\vee = 2$ and $\alpha = 1/(k+2)$.

Verification: at $k\to\infty$, $h_{\mathrm{KZ}}\to 0$ and the bracket vanishes (the weak-coupling limit is the abelian $\mathsf{E}_3$). At the critical level $k = -h^\vee$, $h_{\mathrm{KZ}}$ diverges (the bracket degenerates, reflecting the Feigin–Frenkel singularity). Both limits are consistent with the behaviour of the KZ connection.

*Step 3:* $\{X,\eta\} = 0$. The bracket $\{X,\eta\} \in \mathrm{HH}^1 = \mathfrak{sl}_2$ for $X \in \mathrm{HH}^1$, $\eta \in \mathrm{HH}^2$ is an $\mathfrak{sl}_2$-equivariant map $\mathrm{ad}\otimes\mathbb{C} \to \mathrm{ad}$, hence $\{X,\eta\} = \beta\, X$ for some scalar $\beta$. The $\mathsf{P}_3$ Jacobi identity applied to $X, Y \in \mathrm{HH}^1$ and $\eta \in \mathrm{HH}^2$, combined with $\{X,Y\} = h_{\mathrm{KZ}}\,(X,Y) \in \mathrm{HH}^0$, gives $\{\{X,\eta\},\eta\} = 0$ (since $\{\{X,\eta\},\eta\} \in \mathrm{HH}^{-1} = 0$) and $\{\eta,\{X,\eta\}\} \in \mathrm{HH}^{-1} = 0$, while $\{X,\{\eta,\eta\}\} = \{X,0\} = 0$. The Jacobi identity is thus satisfied for all $\beta$; the constraint comes instead from the self-consistency of the $\mathsf{E}_3$ deformation. At the cohomological

level, one applies the iterated Jacobi identity to $X, Y \in \mathrm{HH}^1$ and $\eta$:

$$\begin{aligned}\{\{X,Y\},\eta\} &= \{X,\{Y,\eta\}\} - (-1)^{(|X|-2)(|Y|-2)}\,\{Y,\{X,\eta\}\}\\ &= \{X,\beta Y\} - (-1)^1\,\{Y,\beta X\}\\ &= \beta\, h_{\mathrm{KZ}}\,(X,Y) + \beta\, h_{\mathrm{KZ}}\,(Y,X)\\ &= 2\beta\, h_{\mathrm{KZ}}\,(X,Y).\end{aligned}$$

The left side: $\{\{X,Y\},\eta\} = \{h_{\mathrm{KZ}}(X,Y)\,\mathbf{1},\eta\} = h_{\mathrm{KZ}}(X,Y)\,\{\mathbf{1},\eta\} = 0$. Hence $2\beta\, h_{\mathrm{KZ}}\,(X,Y) = 0$ for all $X, Y$. Since $(e,f) = 1 \neq 0$ and $h_{\mathrm{KZ}} \neq 0$ at generic $k$, we conclude $\beta = 0$.

*Step 4:* $\{\eta,\eta\} = 0$. The bracket $\{\eta,\eta\} \in \mathrm{HH}^2 = \mathbb{C}$ and the symmetry (8.12) at $|\eta| = 2$ gives $\{\eta,\eta\} = -(-1)^{0\cdot 0}\,\{\eta,\eta\} = -\{\eta,\eta\}$, hence $\{\eta,\eta\} = 0$. □

*Remark* 8.9 (The Killing form as the $\mathsf{E}_3$ shadow of the Cartan 3-form). The bracket (8.7) has a transparent topological origin. The $\mathsf{E}_3$-deformation space is $(k+h^\vee)\,H^3(\mathfrak{sl}_2)[[k+h^\vee]] \cong (k+h^\vee)\,\mathbb{C}[[k+h^\vee]]$, one-dimensional at each order in $(k+h^\vee)$ (Theorem 8.5(iv)). The generator of $H^3(\mathfrak{sl}_2)$ is the Cartan 3-form

$$\omega_3(X,Y,Z) = (X,[Y,Z]), \tag{8.15}$$

the unique (up to scalar) $\mathfrak{sl}_2$-invariant element of $\Lambda^3(\mathfrak{sl}_2^*)$. The degree-$(-2)$ bracket $\{X,Y\}$ is the $\mathsf{E}_3$-operation induced by this deformation class scaled by $\hbar = 1/(k+h^\vee)$:

$$\{X,Y\} = \hbar\cdot(X,Y).$$

The mechanism is *not* literal contraction of the antisymmetric 3-form $\omega_3$: the $\mathsf{P}_3$ bracket is symmetric on $\mathrm{HH}^1$ (by the graded symmetry (8.12) at $|X| = |Y| = 1$), while any contraction of $\omega_3$ in two slots is antisymmetric. The connection is indirect: the one-dimensional deformation space $H^3(\mathfrak{sl}_2)$, generated by $\omega_3$, parametrises $\mathsf{E}_3$-deformations of $C^*(\mathfrak{g})$ (Theorem 8.5(iv)); on cohomology, $\mathfrak{sl}_2$-equivariance forces the induced $\mathsf{P}_3$ bracket on $\mathrm{HH}^1 \cong \mathfrak{sl}_2$ to be proportional to the Killing form $(X,Y)$, the unique invariant symmetric bilinear form. The coefficient $h_{\mathrm{KZ}} = 1/(k+h^\vee)$ is fixed by the $r$-matrix residue (equation (8.13)), not by comparison with CFG. The $\mathsf{P}_3$ bracket on $Z^{\mathrm{der}}_{\mathrm{ch}}$ is the binary shadow of the ternary Cartan deformation class, with the Killing form appearing through equivariance rather than through direct contraction.

8.4. **Quantum corrections from the vertex $R$-matrix: $\mathsf{E}_3$ operations for the EK quantum VOA.** The classical $V_k(\mathfrak{sl}_2)$ has trivial braiding $S(z) = \mathrm{id}$; the Etingof–Kazhdan quantum VOA $V_{\mathrm{EK}}$ of Example 6.13 has $S(z) = \mathrm{id} + h_{\mathrm{KZ}}\,\Omega/z + O(z^{-2})$. Does the non-trivial $R$-matrix change the $\mathsf{E}_3$-algebra structure on the derived chiral centre? On cohomology, no: the $\mathsf{P}_3$ bracket coincides with the classical answer (Proposition 8.8). On cochains, yes: the quantum corrections encode the braided monoidal structure of $\mathrm{Rep}(Y_\hbar(\mathfrak{sl}_2))$.

**The underlying graded vector space.** The EK quantum VOA is a flat formal deformation of $V_k(\mathfrak{sl}_2)$ over $\mathbb{C}[[h_{\mathrm{KZ}}]]$, with $h_{\mathrm{KZ}} = 1/(k+2)$. Since the chiral Hochschild cohomology is computed by a bounded complex of finite-dimensional $\mathfrak{sl}_2$-modules, the deformation does not change the underlying graded vector space:

$$Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}}) \;=\; \mathbb{C}[[h_{\mathrm{KZ}}]] \;\oplus\; \mathfrak{sl}_2[[h_{\mathrm{KZ}}]][-1] \;\oplus\; \mathbb{C}[[h_{\mathrm{KZ}}]][-2], \tag{8.16}$$

with total rank 5 over $\mathbb{C}[[h_{\mathrm{KZ}}]]$. The generators $\mathbf{1} \in \mathrm{HH}^0$, $\{e, f, h\} \in \mathrm{HH}^1$, and $\eta \in \mathrm{HH}^2$ are defined as in Proposition 8.8, now taken over the base ring $\mathbb{C}[[h_{\mathrm{KZ}}]]$.

**Equivariance under the quantum $R$-matrix.** The Yang $R$-matrix $R(u) = u\,I + \hbar\,P$ on $V \otimes V$ is $\mathfrak{sl}_2$-equivariant: $[R(u),\, \rho(X) \otimes I + I \otimes \rho(X)] = 0$ for all $X \in \mathfrak{sl}_2$. The full vertex $R$-matrix $S(z)$ on $V_{\mathrm{EK}}^{\otimes 2}$ (equation (6.30)) is likewise $\mathfrak{sl}_2$-equivariant, since it is constructed from $\mathcal{R}(z)$, the spectral universal $R$-matrix of the Yangian $Y_\hbar(\mathfrak{sl}_2)$, which commutes with the $\mathfrak{sl}_2$-action by the defining property of the Drinfeld coproduct.

This equivariance is the key structural constraint: all three levels of the $\mathsf{E}_3$ operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}})$ are $\mathfrak{sl}_2$-equivariant maps between representations. The equivariance arguments of Proposition 8.8 therefore apply verbatim to the quantum case.

**Proposition 8.10** ($\mathsf{E}_3$ operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}})$)**.** *Let $V_{\mathrm{EK}}$ be the Etingof–Kazhdan quantum vertex algebra for $\mathfrak{sl}_2$ at KZ coupling $h_{\mathrm{KZ}} = 1/(k+2) \neq 0$ (Example 6.13). The $\mathsf{E}_3$-algebra structure on the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_{\mathrm{EK}})$ coincides with the classical $\mathsf{E}_3$ structure of Proposition 8.8: the cup product and Gerstenhaber bracket vanish, and the $\mathsf{P}_3$ bracket is*

$$\{X, Y\}_q \;=\; h_{\mathrm{KZ}} \cdot (X, Y) \qquad \text{for } X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2, \tag{8.17}$$

*with all other brackets zero. In particular, the vertex $R$-matrix $S(z)$ does not renormalise the $\mathsf{P}_3$ bracket on cohomology.*

*Proof.* The argument has four ingredients: equivariance, one-dimensionality of the obstruction space, exactness of the rational classical $r$-matrix, and a $\mathrm{GRT}_1$ consistency check.

*Step 1: cup product and BV operator.* The cup product $\mu\colon \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q}$ is a graded-commutative product on $\mathfrak{sl}_2$-modules over $\mathbb{C}[[h_{\mathrm{KZ}}]]$. For $X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2[[h_{\mathrm{KZ}}]]$, the product $\mu(X, Y) \in \mathrm{HH}^2 = \mathbb{C}[[h_{\mathrm{KZ}}]]$ is an $\mathfrak{sl}_2$-equivariant antisymmetric bilinear form on the adjoint representation: $\Lambda^2(\mathrm{ad})^{\mathfrak{sl}_2} = 0$, so $\mu(X, Y) = 0$ over $\mathbb{C}[[h_{\mathrm{KZ}}]]$.

The BV operator $\Delta\colon \mathrm{HH}^n \to \mathrm{HH}^{n-1}$ is $\mathfrak{sl}_2$-equivariant. The components $\Delta|_{\mathrm{HH}^1}\colon \mathrm{ad} \to \mathbb{C}$ and $\Delta|_{\mathrm{HH}^2}\colon \mathbb{C} \to \mathrm{ad}$ vanish by Schur's lemma. Hence $\Delta = 0$ on the derived centre, and the BV relation gives $[X, Y] = 0$ for all $X, Y \in \mathrm{HH}^1$, just as in the classical case.

*Step 2: the $\mathsf{P}_3$ bracket is proportional to the Killing form.* For $X, Y \in \mathrm{HH}^1$, the bracket $\{X, Y\}_q \in \mathrm{HH}^0 = \mathbb{C}[[h_{\mathrm{KZ}}]]$ is an $\mathfrak{sl}_2$-equivariant symmetric

bilinear form on the adjoint representation: $\mathrm{Sym}^2(\mathrm{ad})^{\mathfrak{sl}_2} \cong \mathbb{C}$. Hence $\{X,Y\}_q = f(h_{\mathrm{KZ}}) \cdot (X,Y)$ for some $f(h_{\mathrm{KZ}}) \in \mathbb{C}[[h_{\mathrm{KZ}}]]$.

The remaining brackets ($\{X,\eta\}_q$, $\{\eta,\eta\}_q$, $\{\mathbf{1},-\}_q$) vanish by the same equivariance and Jacobi arguments as in Proposition 8.8 (the argument for $\beta = 0$ uses $h_{\mathrm{KZ}} \neq 0$, $(e,f) = 1 \neq 0$, and $\{\mathbf{1},-\} = 0$; all remain valid over $\mathbb{C}[[h_{\mathrm{KZ}}]]$).

*Step 3: the scalar $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$ (from exactness of the rational $r$-matrix).* The $\mathsf{P}_3$ bracket on cohomology is computed by the $d$-log residue of the classical $r$-matrix (Construction 8.14, equation (8.26)): for $X, Y \in \mathrm{HH}^1 \cong \mathfrak{g}$, the bracket $\{X,Y\}_q$ is determined by the coefficient of the simple pole in $r(z)$.

For the EK quantum vertex algebra, the spectral universal $R$-matrix $\mathcal{R}(z)$ of $Y_{h_{\mathrm{KZ}}}(\mathfrak{sl}_2)$ evaluates on the fundamental representation to the Yang $R$-matrix

$$R(z) = z\,I + h_{\mathrm{KZ}}\,P \;\in\; \mathrm{End}(V \otimes V)[z], \tag{8.18}$$

which is *polynomial of degree one* in the spectral parameter $z$ (equation (6.32)). The classical $r$-matrix extracted from $R(z)$ is

$$r(z) = \frac{h_{\mathrm{KZ}}\,\Omega}{z}, \tag{8.19}$$

with $\Omega = \sum_a t^a \otimes t_a$ the Casimir element dual to the Killing form. This is *exact*: the rational (Yang) $r$-matrix has a simple pole and no higher-order terms in $z^{-1}$. The higher-order corrections appearing in the vertex $R$-matrix expansion (6.33) (the $z^{-2}$, $z^{-3}$, ... terms) contribute to cochain-level operations but do not modify the classical $r$-matrix, which is determined by the $O(h_{\mathrm{KZ}})$ coefficient of $S(z)$ at the $z^{-1}$ term.

The $\mathsf{P}_3$ bracket on cohomology is the residue

$$\{X,Y\}_q = \mathrm{Res}_{z=0}\; r(z)(X \otimes Y)\,dz = \mathrm{Res}_{z=0}\; \frac{h_{\mathrm{KZ}}\,\Omega(X \otimes Y)}{z}\,dz = h_{\mathrm{KZ}}\,(X,Y), \tag{8.20}$$

where the last equality uses $\Omega(X \otimes Y) = (X,Y)$ (the Killing form pairing through the Casimir). Since $r(z)$ has *only* a simple pole (no $z^{-k}$ terms for $k \geq 2$), this residue receives no corrections at any order in $h_{\mathrm{KZ}}$. Hence $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$.

The essential point is the distinction between rational, trigonometric, and elliptic $R$-matrices. The Yang $R$-matrix $R(z) = zI + h_{\mathrm{KZ}}P$ is the unique rational solution of the Yang–Baxter equation (up to gauge equivalence); it is polynomial of degree one in $z$. A trigonometric $R$-matrix (e.g., the XXZ solution $R(z) \sim \sinh(z + h_{\mathrm{KZ}}P)/\sinh(z)$, associated to $U_q(\widehat{\mathfrak{sl}}_2)$ rather than $Y_{h_{\mathrm{KZ}}}(\mathfrak{sl}_2)$) would have poles at $z \in 2\pi i\mathbb{Z}$, producing a Laurent expansion with infinitely many negative powers of $z$ and potentially renormalising the $\mathsf{P}_3$ bracket. For the EK quantum vertex algebra, which quantises affine Kac–Moody on the formal disk, the rational solution governs, and the $\mathsf{P}_3$ bracket is protected by the polynomial structure of $R(z)$.

*Step 4:* $\mathrm{GRT}_1$ *consistency check.* The preceding argument determines $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$ from the explicit form of the Yang $R$-matrix; the $\mathrm{GRT}_1$-action provides an independent consistency check. The group $\mathrm{GRT}_1$ acts on the $\mathsf{E}_3$-operad maps by conjugation via the Drinfeld associator $\Phi_{\mathrm{KZ}}$. On the one-dimensional deformation space $h_{\mathrm{KZ}}\,\mathbb{C}[[h_{\mathrm{KZ}}]]$ (since $H^3(\mathfrak{sl}_2) \cong \mathbb{C}$), this action is by rescaling. At the cohomological level, the conjugation shifts the $\mathsf{P}_3$ bracket by Hochschild coboundaries, which vanish on $\mathrm{HH}^* = H^*(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$. The $\mathsf{P}_3$ bracket is therefore $\mathrm{GRT}_1$-invariant on cohomology, consistent with the scalar $f(h_{\mathrm{KZ}}) = h_{\mathrm{KZ}}$ being independent of the choice of associator. □

*Remark* 8.11 (Cochain-level corrections and bulk rigidity). At the *cochain* level, the $\mathsf{P}_3$ bracket on $C^1(\overline{B}^{\Sigma}, \overline{B}^{\Sigma})$ acquires quantum corrections $\{D_X, D_Y\}^{\mathrm{ch}} = h_{\mathrm{KZ}}(X,Y)\,\mathrm{id} + h_{\mathrm{KZ}}^2\,\Gamma_2 + \cdots$, where each $\Gamma_n$ is $d_{\mathrm{Hoch}}$-exact. These corrections encode the braided monoidal structure on $\mathrm{Rep}(Y_\hbar(\mathfrak{sl}_2))$ (where $\hbar = 1/(k+h^\vee)$ per Remark 8.7); they are invisible on cohomology but recover the Drinfeld–Jimbo quantum group at the cochain level. The cohomological $\mathsf{E}_3$ operations (cup product, Gerstenhaber bracket, $\mathsf{P}_3$ bracket, BV operator) are identical for the classical $V_k(\mathfrak{sl}_2)$ and the EK quantum $V_{\mathrm{EK}}$. The sole non-trivial cohomological $\mathsf{E}_3$ invariant is $h_{\mathrm{KZ}} = 1/(k+2)$: the perturbative bulk observables of Chern–Simons are insensitive to the choice of associator.

*Remark* 8.12 (Why the quantum $R$-matrix does not renormalise the $\mathsf{P}_3$ bracket). Four mechanisms, in decreasing order of directness:

(i) (Rational $r$-matrix exactness.) The Yang $R$-matrix $R(z) = zI + h_{\mathrm{KZ}}P$ is polynomial degree one in $z$, so the classical $r$-matrix $r(z) = h_{\mathrm{KZ}}\,\Omega/z$ has a simple pole and no higher-order $z$-corrections. The $\mathsf{P}_3$ bracket is the residue of $r(z)$, hence $h_{\mathrm{KZ}}\,(X,Y)$ exactly.

(ii) (Equivariance.) $\mathfrak{sl}_2$-equivariance constrains $\{X,Y\}_q$ to be proportional to the Killing form $(X,Y)$, reducing the problem to a single scalar $f(h_{\mathrm{KZ}})$.

(iii) ($\mathrm{GRT}_1$-invariance.) $\dim H^3(\mathfrak{sl}_2) = 1$ makes the deformation space a formal line; the Drinfeld associator shifts the $\mathsf{P}_3$ bracket by Hochschild coboundaries, which vanish on cohomology.

(iv) (Chern–Simons perturbative finiteness.) The $\mathsf{P}_3$ bracket is the tree-level cubic vertex; pure Chern–Simons has no higher-loop corrections.

For simple $\mathfrak{g}$ with $\dim H^3 = 1$, mechanisms (i)–(iii) all apply and the rigidity persists. For non-simple $\mathfrak{g}$ with $\dim H^3 > 1$, mechanism (ii) still constrains the bracket to the space of invariant symmetric forms, but the higher-dimensional deformation space could allow parameter mixing under the $\mathrm{GRT}_1$-action; mechanism (i) remains valid whenever the governing $R$-matrix is rational.

Both the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ and the CFG $\mathsf{E}_3$-algebra $\mathcal{A}^\lambda$ deform $C^*(\mathfrak{g})$, but through different mechanisms. The chiral bar complex produces a filtered $\mathsf{E}_3$-chiral algebra $\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g})$: a factorisation $\mathcal{D}$-module on

$\mathrm{Ran}(X)$ whose global sections on the formal disk recover the CFG $\mathsf{E}_3$-algebra $\mathcal{A}^\lambda$.

8.5. **The chiral $\mathsf{P}_3$ bracket from the PVA $\lambda$-bracket.** The $\mathsf{E}_3$ structure on the derived chiral centre is abstract; it comes from Higher Deligne applied to the $\mathsf{E}_2$-coalgebra $\overline{B}^\Sigma(\mathcal{A})$. To make it explicit, one needs a concrete bracket. The Poisson vertex algebra (PVA) $\lambda$-bracket on $V_k(\mathfrak{g})$ provides exactly this: a degree-$(-2)$ Lie bracket on the chiral Chevalley–Eilenberg complex, yielding a chiral $\mathsf{P}_3$ structure in the category of $\mathcal{D}$-modules on $X$.

**Definition 8.13** (The chiral Chevalley–Eilenberg complex)**.** Let $\mathfrak{g}$ be a simple Lie algebra with Killing form $(\cdot,\cdot)$ and dual Coxeter number $h^\vee$. The *chiral Chevalley–Eilenberg complex* is the curved chiral coalgebra

$$\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k) \;:=\; \big(\mathrm{Sym}^c(\mathfrak{g}^*[-1])\otimes\omega_X,\; d_{\mathrm{CE}},\; m_0=\tfrac{k+h^\vee}{2h^\vee}\,\kappa\big), \tag{8.21}$$

where $\omega_X$ denotes the dualising sheaf of $X$, $d_{\mathrm{CE}}$ is the chiral Chevalley–Eilenberg differential encoding the Lie bracket of $\mathfrak{g}$, and the curvature $m_0$ is the image of the level under the bar construction (the notation $\kappa$ here denotes the Koszul invariant, not the Killing form). At critical level $k=-h^\vee$ (i.e. $k+h^\vee=0$), the curvature vanishes and $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_{-h^\vee}) = (\mathrm{Sym}^c(\mathfrak{g}^*[-1])\otimes\omega_X,\, d_{\mathrm{CE}})$ is an uncurved commutative $(\mathsf{E}_\infty)$ factorisation coalgebra on $X$.

**Construction 8.14** (The chiral $\mathsf{P}_3$ bracket)**.** The PVA $\lambda$-bracket on $V_k(\mathfrak{g})$ is (see [34], §5.11):

$$\{J^a{}_\lambda\, J^b\} \;=\; f^{ab}{}_c\, J^c \;+\; k\,(a,b)\,\lambda, \tag{8.22}$$

where $f^{ab}{}_c$ are the structure constants of $\mathfrak{g}$ and $(a,b)$ is the Killing form.

At the level of the chiral Chevalley–Eilenberg complex, the $\lambda$-bracket induces a degree-$(-2)$ bracket

$$\{-,-\}^{\mathrm{ch}}\colon \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)\otimes\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k) \;\longrightarrow\; \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k), \tag{8.23}$$

defined on generators $\phi_a=(J^a)^*\in\mathfrak{g}^*[-1]$ by the two components of the PVA bracket:

(i) (Lie bracket term.) The structure constant term $f^{ab}{}_c\,J^c$ contributes the Chevalley–Eilenberg differential $d_{\mathrm{CE}}$ at the algebra level. On the dual (coalgebra) side, it induces a coderivation. This is the classical $\mathsf{P}_3$ bracket of the uncurved complex $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_0)$: the degree-$(-2)$ Lie bracket

$$\{\phi_a,\phi_b\}^{\mathrm{ch}}_{\mathrm{Lie}} \;=\; f^{ab}{}_c\,\phi_c. \tag{8.24}$$

This bracket is the coinvariant shadow of the ordered bar $R$-matrix at the critical level.

(ii) (Cocycle term.) The level-dependent term $k\,(a,b)\,\lambda$ contributes a degree-$(-2)$ symmetric pairing valued in $\mathcal{O}_X$:

$$\{\phi_a,\phi_b\}^{\mathrm{ch}}_{\mathrm{cocycle}} \;=\; k\,(a,b)\cdot\partial, \tag{8.25}$$

where $\partial$ denotes the $\mathcal{D}$-module action (the spectral parameter $\lambda$ of the PVA bracket becomes the derivation $\partial_z$ in the $\mathcal{D}$-module structure). This term is the cocycle that controls the deformation away from the commutative ($\mathsf{E}_\infty$) factorisation structure.

The total chiral $\mathsf{P}_3$ bracket is

$$\{\phi_a, \phi_b\}^{\mathrm{ch}} \;=\; f^{ab}{}_c\, \phi_c \;+\; k\,(a,b)\,\partial. \tag{8.26}$$

**Proposition 8.15** (The chiral $\mathsf{P}_3$ structure)**.** *The bracket* (8.26) *equips the chiral Chevalley–Eilenberg complex* $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$ *with the structure of a* $\mathsf{P}_3$*-algebra object in* $\mathcal{D}$-$\mathrm{mod}(X)$*. Concretely:*

(i) *The bracket is a degree-*$(-2)$ *Lie bracket satisfying graded antisymmetry and the Jacobi identity. The Jacobi identity follows from the Jacobi identity of the PVA* $\lambda$*-bracket, which in turn encodes the Jacobi identity of the Lie algebra* $\mathfrak{g}$ *together with the* $\mathfrak{g}$*-invariance of the Killing form.*

(ii) *The bracket is compatible with the commutative product on* $\mathrm{Sym}^c(\mathfrak{g}^*[-1])$ *via the Leibniz rule:*

$$\{a, b\cdot c\}^{\mathrm{ch}} = \{a,b\}^{\mathrm{ch}}\cdot c + (-1)^{(|a|-2)|b|}\, b\cdot\{a,c\}^{\mathrm{ch}}.$$

(iii) *The bracket is* $\mathcal{D}$*-linear in the sense that it respects the flat connection on* $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$ *inherited from the* $\mathcal{D}$*-module structure: the chiral* $\mathsf{P}_3$ *bracket is a morphism of* $\mathcal{D}$*-modules, not merely of* $\mathcal{O}_X$*-modules.*

*Proof.* The PVA $\lambda$-bracket on $V_k(\mathfrak{g})$ satisfies the PVA axioms (sesquilinearity, Leibniz, skew-symmetry, Jacobi); these are the $\lambda$-bracket translations of the classical master equation for the Poisson structure. The identification of the $\lambda$-bracket with a degree-$(-2)$ Lie bracket on the shifted dual follows the general correspondence between PVA brackets and shifted Poisson structures (Beilinson–Drinfeld [4], §3.8): a PVA $\lambda$-bracket of conformal weight 1 (where $\lambda$ has weight 1) corresponds to a $\mathsf{P}_3$ bracket (degree $-2 = -\dim_{\mathbb{C}} X - 1$, where $\dim_{\mathbb{C}} X = 1$) on the $\mathcal{D}$-module of chiral cochains.

The $\mathcal{D}$-linearity follows because the PVA $\lambda$-bracket is a $\mathcal{D}$-module morphism by construction: the derivation $\partial$ acts via $\lambda \mapsto \lambda$ in the $\lambda$-bracket formalism, and this action is compatible with the Leibniz rule by the sesquilinearity axiom of the PVA. □

8.6. **The filtered $\mathsf{E}_3$-chiral algebra.** The chiral $\mathsf{P}_3$ bracket and the chiral Chevalley–Eilenberg complex assemble into a single object: a filtered $\mathsf{E}_3$-chiral algebra $\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g})$, a factorisation $\mathcal{D}$-module on $\mathrm{Ran}(X)$ whose global sections on the formal disk recover the CFG $\mathsf{E}_3$-algebra.

**Definition 8.16** (The filtered $\mathsf{E}_3$-chiral algebra)**.** The *chiral* $\mathsf{E}_3$*-algebra*

$$\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g}) \;:=\; \big(\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k),\ \{-,-\}^{\mathrm{ch}},\ \mathcal{D}\text{-module structure}\big) \tag{8.27}$$

is the chiral Chevalley–Eilenberg complex (Definition 8.13) equipped with the chiral $\mathsf{P}_3$ bracket (Construction 8.14) and the $\mathcal{D}$-module structure from

the factorisation algebra on $X$. It carries a $(k+h^\vee)$-adic filtration $F^\bullet \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ defined by:

$$F^p \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g}) \;=\; (k+h^\vee)^p \cdot \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g}). \tag{8.28}$$

**Theorem 8.17** (Structure of the chiral $\mathsf{E}_3$-algebra)**.** *Let $\mathfrak{g}$ be a simple Lie algebra. The chiral $\mathsf{E}_3$-algebra $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ satisfies:*

(i) (Associated graded.) *The associated graded with respect to the $(k+h^\vee)$-adic filtration is the uncurved chiral Chevalley–Eilenberg complex at the critical level:*

$$\mathrm{gr}_F\, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g}) \;\simeq\; \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_{-h^\vee}) \;=\; \big(\mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_X,\; d_{\mathrm{CE}}\big), \tag{8.29}$$

*which is an $\mathsf{E}_\infty$-chiral algebra (commutative factorisation $\mathcal{D}$-module on $\mathrm{Ran}(X)$).*

(ii) (The $\mathsf{E}_3$ structure.) *The chiral $\mathsf{P}_3$ bracket (Proposition 8.15) deforms the $\mathsf{E}_\infty$ structure to a filtered $\mathsf{E}_3$-chiral algebra: $\mathsf{E}_2$ from the curve geometry (Warning 3.7) $+1$ from the chiral $\mathsf{P}_3$ bracket $= \mathsf{E}_3$.*

(iii) (Curvature and the deformation parameter.) *The curvature $m_0 = \frac{k+h^\vee}{2h^\vee}\,\kappa$ of the chiral Chevalley–Eilenberg complex is the obstruction to the $\mathsf{E}_\infty$ structure: at $k=-h^\vee$, the curvature vanishes and the algebra is commutative; at $k \neq -h^\vee$, the $\mathsf{P}_3$ bracket and the curvature together deform the commutative structure to a filtered $\mathsf{E}_3$-chiral algebra. The deformation parameter is $(k+h^\vee) \in (k+h^\vee)\, H^3(\mathfrak{g})[[k+h^\vee]]$, matching the deformation space of Theorem 1.3*(ii)*.*

(iv) (Factorisation property.) *The $\mathsf{E}_3$-chiral algebra $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ is a factorisation algebra on $X$ in the sense that the restriction to disjoint opens $U_1 \sqcup U_2 \hookrightarrow X$ induces*

$$\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})\big|_{U_1 \sqcup U_2} \;\simeq\; \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})\big|_{U_1} \otimes \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})\big|_{U_2}, \tag{8.30}$$

*where the tensor product is in the derived category of $\mathcal{D}$-modules. The $\mathsf{E}_3$ structure is compatible with the factorisation isomorphism.*

The following lemma supplies the key compatibility between the BV operator $\Delta$ and the chiral $\mathsf{P}_3$ bracket used in the proof of part (ii). The "independent geometric data" heuristic (genus-1 sewing vs. genus-0 collision) motivates the result, but the proof requires a precise argument.

**Lemma 8.18** (Commutativity of the BV operator and the chiral $\mathsf{P}_3$ bracket)**.** *Let $\mathfrak{g}$ be a simple Lie algebra and $k \neq -h^\vee$. The BV operator $\Delta$ on the derived chiral centre $Z_{\mathrm{ch}}^{\mathrm{der}}(V_k(\mathfrak{g}))$ (from the framed $\mathsf{E}_2$ structure, genus-1 sewing) and the chiral $\mathsf{P}_3$ bracket $\{-,-\}^{\mathrm{ch}}$ (Construction 8.14, genus-0 collision residue) satisfy*

$$[\Delta, \{a,-\}^{\mathrm{ch}}] = 0 \quad \textit{for all } a \in \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k). \tag{8.31}$$

*Proof.* The abstract $\mathsf{E}_3$-algebra structure on the derived chiral centre $Z_{\mathrm{ch}}^{\mathrm{der}}(V_k(\mathfrak{g}))$ is provided by Proposition 8.2(ii) via the Higher Deligne Conjecture applied to the $\mathsf{E}_2$-coalgebra $\overline{B}^{\Sigma}(V_k(\mathfrak{g}))$. This abstract $\mathsf{E}_3$ structure includes, in particular, an abstract BV operator $\Delta_{\mathrm{HDC}}$ (from the framed $\mathsf{E}_2$ structure) and

an abstract $\mathsf{P}_3$ bracket $\{-,-\}_{\mathrm{HDC}}$ (the degree-$(-2)$ Lie bracket from the $S^2$ fundamental class of $\mathrm{Conf}_2(\mathbb{R}^3)/\mathbb{R}^+$). These satisfy all $\mathsf{E}_3$ compatibility conditions, including $[\Delta_{\mathrm{HDC}}, \{a,-\}_{\mathrm{HDC}}] = 0$, by the operadic structure of $\mathsf{E}_3$.

It remains to identify the explicit chiral $\mathsf{P}_3$ bracket $\{-,-\}^{\mathrm{ch}}$ from Construction 8.14 with the abstract bracket $\{-,-\}_{\mathrm{HDC}}$. We argue in three steps.

*Step 1: matching on the associated graded.* At $k = -h^\vee$ (the associated graded of the $(k+h^\vee)$-adic filtration), the chiral Chevalley–Eilenberg complex becomes the uncurved commutative coalgebra $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_0)$, which is an $\mathsf{E}_\infty$-chiral algebra. Both $\{-,-\}^{\mathrm{ch}}$ and $\{-,-\}_{\mathrm{HDC}}$ reduce to the classical $\mathsf{P}_3$ bracket on $\mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_X$ induced by the Lie bracket of $\mathfrak{g}$: the bracket $\{\phi_a, \phi_b\} = f^{ab}{}_c\, \phi_c$ on generators, extended as a biderivation. This classical bracket is uniquely determined by the $\mathfrak{g}$-equivariance and degree constraints on $\mathrm{Sym}^c(\mathfrak{g}^*[-1])$: the space of degree-$(-2)$ Lie brackets on $\mathrm{Sym}^c(\mathfrak{g}^*[-1])$ that are biderivations of the commutative product and $\mathfrak{g}$-equivariant is one-dimensional (generated by the bracket induced by $\mathfrak{g}$ itself). Therefore $\{-,-\}^{\mathrm{ch}}$ and $\{-,-\}_{\mathrm{HDC}}$ agree on $\mathrm{gr}_F$.

*Step 2: matching order by order.* The $(k+h^\vee)$-adic filtration on $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$ induces a filtration on the space of degree-$(-2)$ Lie biderivations. At each order $p \geq 1$ in $(k+h^\vee)$, the bracket is determined by its value on generators $\phi_a \otimes \phi_b \in \mathfrak{g}^*[-1] \otimes \mathfrak{g}^*[-1]$ (since both brackets are biderivations of $\mu$). On generators, the space of $\mathfrak{g}$-equivariant $\mathcal{D}$-linear pairings $\mathfrak{g}^*[-1] \otimes \mathfrak{g}^*[-1] \to \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$ of degree $-2$ is spanned by $f^{ab}{}_c\, \phi_c$ and $k\,(a,b)\,\partial$ (the structure constants and the Killing form, the only $\mathfrak{g}$-invariant tensors of the appropriate type for simple $\mathfrak{g}$). Step 1 fixes the structure-constant coefficient $f^{ab}{}_c\, \phi_c$ at order 0. At each subsequent order $p \geq 1$, the *correction* to the bracket on generators lies in the subspace spanned by the Killing-form term $k\,(a,b)\,\partial$ alone (since the structure-constant component is already determined by the lower orders and the biderivation property). For simple $\mathfrak{g}$, this correction space is one-dimensional: $\mathrm{Hom}_{\mathfrak{g}}(\mathrm{Sym}^2\mathfrak{g}, \mathbb{C}) \cong \mathbb{C}$ (the Killing form is the unique $\mathfrak{g}$-invariant symmetric bilinear form). Both $\{-,-\}^{\mathrm{ch}}$ and $\{-,-\}_{\mathrm{HDC}}$ are $\mathfrak{g}$-equivariant $\mathcal{D}$-linear biderivations with the same order-0 value (Step 1), so their corrections at each order $p$ coincide by the one-dimensionality of the Killing-form component. Therefore the two brackets agree at each order.

*Step 3: conclusion.* Since $\{-,-\}^{\mathrm{ch}} = \{-,-\}_{\mathrm{HDC}}$ as formal power series in $(k+h^\vee)$, and $[\Delta_{\mathrm{HDC}}, \{a,-\}_{\mathrm{HDC}}] = 0$ holds by the $\mathsf{E}_3$ operadic structure, the commutativity (8.31) follows. □

*Remark* 8.19 (Chain-level status of Lemma 8.18). The proof reduces the commutativity to the identification of the explicit chiral $\mathsf{P}_3$ bracket with the abstract HDC bracket. This identification is proved order by order in $(k+h^\vee)$ using the one-dimensionality of the deformation space, which holds for simple $\mathfrak{g}$. For non-simple $\mathfrak{g}$, the deformation space $H^3(\mathfrak{g})[[k+h^\vee]]$ may

be higher-dimensional, and the order-by-order uniqueness argument requires a more refined comparison.

The "independent geometric data" intuition (that $\Delta$ comes from genus-1 sewing strata of $\overline{\mathcal{M}}_{1,n}$ while $\{-,-\}^{\mathrm{ch}}$ comes from genus-0 collision strata of $\overline{\mathcal{M}}_{0,n}$) captures the operadic reason: in the modular envelope of the $\mathsf{E}_3$ operad, genus-1 and genus-0 operations compose without interference because they correspond to disjoint cells in the modular operad. Making this precise at the chain level in the chiral (as opposed to topological) setting is what the identification argument above accomplishes for simple $\mathfrak{g}$.

*Proof of Theorem 8.17.* Part (i) is immediate from the filtration: the $(k+h^\vee)$-adic filtration is exhaustive, and at $k=-h^\vee$ the curvature and the cocycle term of the $\mathsf{P}_3$ bracket both vanish, leaving the uncurved commutative chiral coalgebra $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_{-h^\vee})$.

Part (ii) requires three things: that the chiral $\mathsf{P}_3$ bracket is a degree-$(-2)$ Lie bracket (Jacobi), that it satisfies the Leibniz rule with respect to the commutative product (Poisson compatibility), and that it is compatible with the $\mathsf{E}_2$ structure from the curve geometry, so that the combined data assembles into an $\mathsf{E}_3$-algebra. We prove each in turn.

*Step 1: the $\mathsf{P}_3$ Jacobi identity from the PVA Jacobi.* The PVA $\lambda$-bracket on $V_k(\mathfrak{g})$ satisfies the $(-1)$-shifted Jacobi identity (see [35], Appendix A, Proposition A.5):

$$\{a\,{}_\lambda\,\{b\,{}_\mu\,c\}\} - (-1)^{(|a|+1)(|b|+1)}\,\{b\,{}_\mu\,\{a\,{}_\lambda\,c\}\} = \{\{a\,{}_\lambda\,b\}\,{}_{\lambda+\mu}\,c\}. \tag{8.32}$$

The chiral $\mathsf{P}_3$ bracket $\{-,-\}^{\mathrm{ch}}$ on $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$ is obtained from the $\lambda$-bracket by specialising $\lambda\mapsto\partial$ (the $\mathcal{D}$-module derivation) and passing to the chiral Chevalley–Eilenberg complex. On generators $\phi_a,\phi_b,\phi_c\in\mathfrak{g}^*[-1]$ (cohomological degree 1, shifted bracket-degree $|\phi_a|-2=-1$), the degree-$(-2)$ Jacobi identity is:

$$\{\phi_a,\{\phi_b,\phi_c\}^{\mathrm{ch}}\}^{\mathrm{ch}} + \text{graded cyclic} = 0. \tag{8.33}$$

To derive this from (8.32), set $\lambda=\mu=0$ (the constant-coefficient specialisation that extracts the $\mathsf{P}_3$ bracket from the $\lambda$-bracket). The left side of (8.32) becomes $\{\phi_a,\{\phi_b,\phi_c\}^{\mathrm{ch}}\}^{\mathrm{ch}} - (-1)^{(|\phi_a|+1)(|\phi_b|+1)}\,\{\phi_b,\{\phi_a,\phi_c\}^{\mathrm{ch}}\}^{\mathrm{ch}}$, and the right side becomes $\{\{\phi_a,\phi_b\}^{\mathrm{ch}},\phi_c\}^{\mathrm{ch}}$ (at $\lambda+\mu=0$). The PVA $\lambda$-bracket has degree $-1$ (it is $(-1)$-shifted); the chiral $\mathsf{P}_3$ bracket has degree $-2$, accounting for the additional degree shift from the $\mathcal{D}$-module derivation $\partial$ (conformal weight 1). The Koszul sign $(-1)^{(|\phi_a|+1)(|\phi_b|+1)}$ in the PVA identity translates to $(-1)^{(|\phi_a|-2)(|\phi_b|-2)}$ in the $\mathsf{P}_3$ identity after incorporating the degree shift $|-|\mapsto|-|-2$ appropriate to a degree-$(-2)$ bracket. On $\mathfrak{g}^*[-1]$ where $|\phi_a|=1$, both signs evaluate to $(-1)^{(-1)(-1)}=-1$. Rearranging gives (8.33).

For general elements of $\mathrm{Sym}^c(\mathfrak{g}^*[-1])$ (not just generators), the Jacobi identity extends by the Leibniz rule (Step 2 below): the bracket is a biderivation of the free commutative algebra, so a trilinear identity that holds on generators holds on all elements.

*Step 2: the Leibniz rule from PVA sesquilinearity.* The PVA right Leibniz rule (see [35], Appendix A, Proposition A.6) states:

$$\{a\,{}_\lambda\,(b\cdot c)\} = \{a\,{}_\lambda\,b\}\cdot c + (-1)^{(|a|+1)|b|}\,b\cdot\{a\,{}_\lambda\,c\} + \int_0^\lambda \{\{a\,{}_\mu\,b\}\,{}_\lambda\,c\}\,d\mu.$$

At $\lambda = 0$, the integral term vanishes (the integration range collapses to a point), leaving:

$$\{a, b\cdot c\}^{\mathrm{ch}} = \{a,b\}^{\mathrm{ch}}\cdot c + (-1)^{(|a|+1)|b|}\,b\cdot\{a,c\}^{\mathrm{ch}}.$$

Adjusting the Koszul sign from the $(-1)$-shifted convention to the degree-$(-2)$ convention: $(|a|+1)|b|$ becomes $(|a|-2)|b|$ after the degree shift, recovering the $\mathsf{P}_3$ Leibniz rule

$$\{a, b\cdot c\}^{\mathrm{ch}} = \{a,b\}^{\mathrm{ch}}\cdot c + (-1)^{(|a|-2)|b|}\,b\cdot\{a,c\}^{\mathrm{ch}},$$

as stated in Proposition 8.15(ii). This identifies the chiral $\mathsf{P}_3$ bracket as a derivation of the commutative product $\mu$ on $\mathrm{Sym}^c(\mathfrak{g}^*[-1])\otimes\omega_X$: a commutative algebra equipped with a compatible degree-$(-2)$ Lie bracket is a $\mathsf{P}_3$-algebra.

*Step 3: compatibility with* $\mathsf{E}_2$ *and the* $\mathsf{E}_3$ *assembly.* The $\mathsf{E}_2$-coalgebra structure on $\overline{B}^{\Sigma}$ (Warning 3.7) and its framed variant (from genus-1 sewing) equip the derived centre with a BV operator $\Delta$ satisfying $[f,g] = \Delta(f\cdot g) - (\Delta f)\cdot g - (-1)^{|f|}\,f\cdot(\Delta g)$ (the BV relation (8.11)). The $\mathsf{P}_3$ bracket of degree $-2$ supplements this $\mathsf{E}_2$ structure with one additional degree of operadic freedom: the $\mathsf{P}_3$ bracket (degree $-2 = -(3-1)$) corresponds to the fundamental class of $S^2 = \mathrm{Conf}_2(\mathbb{R}^3)/\mathbb{R}^+$, the extra generator needed for $\mathsf{E}_3$ beyond $\mathsf{E}_2$.

The compatibility condition is that the $\mathsf{P}_3$ bracket must be a derivation of the Gerstenhaber bracket $[-,-]$ of degree $-1$ from the $\mathsf{E}_2$ structure. Since the Gerstenhaber bracket is determined by $\Delta$ and $\mu$ via the BV relation, it suffices to show that $\{-,-\}^{\mathrm{ch}}$ commutes with $\Delta$ and is a derivation of $\mu$ (already proved in Step 2). The commutativity $[\Delta, \{a,-\}^{\mathrm{ch}}] = 0$ follows from Lemma 8.18. Given this, for any $a,b,c\in \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k)$:

$$\begin{aligned}\{a,[b,c]\}^{\mathrm{ch}} &= \{a, \Delta(b\cdot c) - (\Delta b)\cdot c - (-1)^{|b|}\,b\cdot(\Delta c)\}^{\mathrm{ch}}\\ &= \Delta\{a, b\cdot c\}^{\mathrm{ch}} - \{a,(\Delta b)\cdot c\}^{\mathrm{ch}} - (-1)^{|b|}\,\{a, b\cdot(\Delta c)\}^{\mathrm{ch}},\end{aligned}$$

where the first term uses $[\Delta,\{a,-\}^{\mathrm{ch}}] = 0$. Expanding each term by the Leibniz rule of Step 2 and collecting via the BV relation yields

$$\{a,[b,c]\}^{\mathrm{ch}} = [\{a,b\}^{\mathrm{ch}}, c] + (-1)^{(|a|-2)|b|}\,[b,\{a,c\}^{\mathrm{ch}}]\,:$$

the $\mathsf{P}_3$ bracket is a derivation of the Gerstenhaber bracket.

This is the mixed Leibniz rule that characterises $\mathsf{E}_3 = \mathsf{E}_2 + \mathsf{P}_3$: the $\mathsf{P}_3$ bracket acts as a derivation of the $\mathsf{E}_2$ Gerstenhaber bracket. By the recognition principle for $\mathsf{E}_n$-algebras (Lurie [24], Theorem 5.1.2.2; Francis [21], Theorem 4.1): a commutative algebra with a compatible $\mathsf{E}_2$ structure and a degree-$(-2)$ Lie bracket satisfying the mutual Leibniz rules (Jacobi, Poisson

compatibility with the product, and derivation of the Gerstenhaber bracket) is an $\mathsf{E}_3$-algebra. The three verifications above establish all the required relations.

Part (iii) combines two observations. The curvature $m_0 = \frac{k+h^\vee}{2h^\vee}\,\kappa$ is the bar-complex encoding of the level departure (the identification $\mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_k) \simeq B^{\mathrm{ch}}(V_k(\mathfrak{g}))$). The cocycle term $k\,(a,b)\,\partial$ of the $\mathsf{P}_3$ bracket is linear in $k$ and hence linear in $(k+h^\vee)$. Both vanish at $k=-h^\vee$, so the deformation is controlled by $(k+h^\vee)$. The identification of the deformation space as $(k+h^\vee)H^3(\mathfrak{g})[[k+h^\vee]]$ then follows from the computation of Theorem 1.3(ii): $H^3(\mathfrak{g})$ is one-dimensional for simple $\mathfrak{g}$, and the Cartan 3-form $\omega_3(X,Y,Z) = (X,[Y,Z])$ is the generator.

Part (iv) is the factorisation property of the chiral bar complex. The chiral Chevalley–Eilenberg complex, being a chiral coalgebra, satisfies the factorisation axiom by construction (Beilinson–Drinfeld [4], Chapter 3). The $\mathsf{P}_3$ bracket, being a $\mathcal{D}$-module morphism (Proposition 8.15(iii)), is compatible with the factorisation isomorphisms. □

8.7. **The $\mathcal{D}$-module structure and the KZ connection.** The chiral $\mathsf{E}_3$-algebra $\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g})$ is not an abstract $\mathsf{E}_3$-algebra: it is an $\mathsf{E}_3$-algebra object in the category of $\mathcal{D}$-modules on $X$, or equivalently on $\mathrm{Ran}(X)$. This $\mathcal{D}$-module structure is the content that distinguishes the chiral construction from the topological CFG construction.

**Proposition 8.20** (The $\mathcal{D}$-module structure)**.** *The chiral $\mathsf{E}_3$-algebra $\mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{g})$ carries a flat connection*

$$\nabla^{\mathrm{ch}} \;=\; d \;+\; \sum_{i<j} \frac{k\,\Omega_{ij}}{z_i - z_j}\,dz_{ij} \tag{8.34}$$

*on $\mathrm{Conf}_n(X)$, where $\Omega_{ij}$ is the Casimir element acting on the $i$-th and $j$-th tensor factors. This is the Knizhnik–Zamolodchikov connection.*

(i) *The flatness $(\nabla^{\mathrm{ch}})^2 = 0$ follows from the classical Yang–Baxter equation for the $r$-matrix $r(z) = k\,\Omega/z$ (trace-form convention; at $k=0$, $r=0$): the curvature of $\nabla^{\mathrm{ch}}$ is proportional to the LHS of the CYBE, which vanishes.*

(ii) *The $\mathsf{E}_3$-algebra operations are horizontal with respect to $\nabla^{\mathrm{ch}}$: the cup product, the Gerstenhaber bracket, and the $\mathsf{P}_3$ bracket are all flat sections of the corresponding* Hom*-bundles. This is the $\mathcal{D}$-linearity of Proposition 8.15*(iii) *in the multi-point form.*

(iii) *The monodromy representation of $\nabla^{\mathrm{ch}}$ on $\pi_1(\mathrm{Conf}_n(X))$ gives the braid group action. For $X=\mathbb{C}$, the monodromy of the KZ connection is computed by the Drinfeld associator $\Phi_{\mathrm{KZ}}$ and recovers the braided monoidal structure on $\mathrm{Rep}(U_h(\mathfrak{g}))$.*

*Proof.* The KZ connection on the $n$-point conformal blocks of $V_k(\mathfrak{g})$ is classical (Knizhnik–Zamolodchikov, 1984). The presentation here identifies this

connection as the $\mathcal{D}$-module structure on the chiral Chevalley–Eilenberg complex, viewed as a factorisation algebra on $\mathrm{Ran}(X)$.

For (i): the curvature of $\nabla^{\mathrm{ch}}$ on $\mathrm{Conf}_n(X)$ is $[\nabla_i, \nabla_j] = k^2\,[\Omega_{ij}/(z_i - z_j),\, \Omega_{ik}/(z_i - z_k) + \Omega_{jk}/(z_j - z_k)]\, dz_{ij} \wedge dz_{ik}$ plus cyclic permutations. The vanishing of this expression is equivalent to the classical Yang–Baxter equation $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$ for $r_{ij} = k\,\Omega_{ij}/(z_i - z_j)$.

For (ii): the $\mathsf{E}_3$ operations are defined in terms of the PVA $\lambda$-bracket, which is itself a $\mathcal{D}$-module morphism. The cup product is a $\mathcal{D}$-module map by the commutative algebra structure on the factorisation algebra. The $\mathsf{P}_3$ bracket involves the derivation $\partial$ (from the $\lambda$ in the PVA bracket), which is the $\mathcal{D}$-module action; its compatibility with the connection follows from the PVA sesquilinearity axiom.

For (iii): this is the Drinfeld–Kohno theorem (Drinfeld [17], Kohno, Kassel): the monodromy of the KZ connection on $\mathrm{Conf}_n(\mathbb{C})$ factors through the pure braid group $P_n$ and gives, via the Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\hat{\mathfrak{t}}_3)$, a braided monoidal structure on $\mathrm{Rep}(U_\hbar(\mathfrak{g}))$. □

8.8. **Comparison with CFG: formal disk global sections.** The chiral $\mathsf{E}_3$-algebra $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ lives on $\mathrm{Ran}(X)$ and carries the KZ connection; the CFG algebra $\mathcal{A}^\lambda$ lives on $\mathbb{R}^3$ and is purely topological. The formal disk $D = \mathrm{Spec}\,\mathbb{C}[[z]]$ is the bridge: restricting to $D$ trivializes the $\mathcal{D}$-module structure and recovers the CFG algebra as the fiber.

**Theorem 8.21** (Formal disk restriction recovers CFG)**.** *Let $D = \mathrm{Spec}\,\mathbb{C}[[z]]$ be the formal disk in $X$. There is a natural map of filtered $\mathsf{E}_3$-algebras*

$$\Gamma(D,\, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})) \;\longrightarrow\; \mathcal{A}^\lambda \tag{8.35}$$

*from the global sections of the chiral $\mathsf{E}_3$-algebra on the formal disk to the CFG $\mathsf{E}_3$-algebra of Theorem 8.5, where the CFG coupling $\lambda$ equals the departure from critical level $k + h^\vee$.*

(i) (Associated graded.) *At the level of associated graded algebras,*

$$\mathrm{gr}\,\Gamma(D,\, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})) \;\simeq\; C^*(\mathfrak{g}) \;=\; \mathrm{Sym}(\mathfrak{g}^\vee[-1]), \tag{8.36}$$

*the Chevalley–Eilenberg cochains of $\mathfrak{g}$: both the chiral construction restricted to the formal disk and the CFG construction have the same classical limit.*

(ii) (Matching of $\mathsf{P}_3$ brackets.) *The restriction of the chiral $\mathsf{P}_3$ bracket (8.26) to the formal disk recovers the CFG $\mathsf{P}_3$ bracket. On the formal disk, the derivation $\partial = \partial_z$ acts on $\mathbb{C}[[z]]$; the global sections $\Gamma(D, \mathfrak{g}^*[-1] \otimes \omega_D) \cong \mathfrak{g}^*[-1][[z]]\, dz$ produce, after extracting the residue at $z = 0$, the finite-dimensional $\mathsf{P}_3$ bracket*

$$\{\phi_a, \phi_b\}_{\mathrm{CFG}} \;=\; f^{ab}{}_c\, \phi_c \;+\; k\,(a, b) \cdot \delta_0,$$

*where $\delta_0$ is the delta-function at the origin, the formal-disk avatar of $\partial$. Under the identification $\Gamma(D, \omega_D) \cong \mathbb{C}\, dz$ (taking the constant mode), this recovers the CFG bracket on $C^*(\mathfrak{g})$.*

(iii) (Matching of deformation spaces.) *Both* $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})\big|_D$ *and* $\mathcal{A}^\lambda$ *are deformations of* $C^*(\mathfrak{g})$ *over the base* $(k+h^\vee)\,H^3(\mathfrak{g})[[k+h^\vee]]$*, with the same one-dimensional fibre at each order in* $(k+h^\vee)$ *for simple* $\mathfrak{g}$*. The map* (8.35) *intertwines the* $(k+h^\vee)$*-adic filtrations.*

*Proof.* The argument has two steps: construct the $\mathsf{E}_3$-algebra structure on $\Gamma(D, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g}))$ by symmetric monoidal transport, then identify it with the CFG algebra $\mathcal{A}^\lambda$.

**Step 1: symmetric monoidal trivialization on $D$.** The formal disk $D = \operatorname{Spec}\mathbb{C}[[z]]$ is contractible: every $\mathcal{D}$-module on $D$ (that is, a $\mathbb{C}[[z]]$-module with flat connection) is isomorphic to its fibre at the origin, since every flat connection on $D$ is trivializable ($\pi_1(D^n) = 0$ for all $n$, so there is no monodromy obstruction). The global sections functor

$$\Gamma(D, -)\colon \mathcal{D}\text{-mod}(D) \xrightarrow{\sim} \mathrm{Vect} \tag{8.37}$$

is an equivalence of categories. This equivalence is symmetric monoidal: the tensor product of $\mathcal{D}$-modules on $D$ (the !-tensor product, which on the contractible formal disk coincides with the $*$-tensor product and with the ordinary tensor product of the underlying vector spaces) corresponds under (8.37) to the tensor product in Vect.

An equivalence of symmetric monoidal categories preserves all algebraic structures definable in terms of the monoidal structure. In particular, it preserves $\mathsf{E}_3$-algebra structures: if $\mathcal{B}$ is an $\mathsf{E}_3$-algebra in $\mathcal{D}$-mod$(D)$, then $\Gamma(D, \mathcal{B})$ is an $\mathsf{E}_3$-algebra in Vect, with structure maps obtained by applying $\Gamma(D, -)$ to those of $\mathcal{B}$.

Applying this to $\mathcal{B} = \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})\big|_D$: the restriction of the chiral $\mathsf{E}_3$-algebra to $D$ is an $\mathsf{E}_3$-algebra in $\mathcal{D}$-mod$(D)$ (by restriction of the $\mathsf{E}_3$-structure on $X$), and the equivalence (8.37) endows $\Gamma(D, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g}))$ with an $\mathsf{E}_3$-algebra structure in Vect. This $\mathsf{E}_3$-structure is functorial and carries the $(k+h^\vee)$-adic filtration inherited from the chiral side.

**Step 2: identification with $\mathcal{A}^\lambda$.** It remains to identify $\Gamma(D, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g}))$ with the CFG algebra $\mathcal{A}^\lambda$ as filtered $\mathsf{E}_3$-algebras, where $\lambda = k + h^\vee$. Both are filtered $\mathsf{E}_3$-algebras deforming the same classical limit over the same base, and the identification proceeds through three structural invariants.

Part (i): on $D$, the $\mathcal{D}$-module structure trivializes, so the chiral Chevalley–Eilenberg complex reduces to the ordinary one. The associated graded of the chiral $\mathsf{E}_3$-algebra on $D$ is

$$\operatorname{gr}\Gamma(D,\, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})) \;=\; \Gamma(D,\, \operatorname{gr}\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})) \;=\; \Gamma(D,\, \mathrm{Sym}^c(\mathfrak{g}^*[-1]) \otimes \omega_D).$$

On the formal disk, $\omega_D \cong \mathcal{O}_D\, dz$; the equivalence (8.37) extracts the constant mode in the $z$-expansion, giving $\mathrm{Sym}(\mathfrak{g}^\vee[-1]) = C^*(\mathfrak{g})$, the Chevalley–Eilenberg cochains of $\mathfrak{g}$ as a commutative algebra. This is the associated graded of $\mathcal{A}^\lambda$ (Theorem 8.5(i)).

Part (ii): the chiral $\mathsf{P}_3$ bracket on $X$ (equation (8.26)) is defined via the PVA $\lambda$-bracket. On the formal disk, the spectral parameter $\lambda$ becomes

the formal variable of the completed enveloping algebra: the translation generator $\partial_z$ acts on $\mathbb{C}[[z]]$, and taking global sections via (8.37) extracts the residue at $z = 0$. The resulting bracket on $\Gamma(D, \mathfrak{g}^*[-1] \otimes \omega_D) \cong \mathfrak{g}^*[-1][[z]]\, dz$ is

$$\{\phi_a, \phi_b\}_{\mathrm{CFG}} \;=\; f^{ab}{}_c\, \phi_c \;+\; k\,(a,b) \cdot \delta_0,$$

where $f^{ab}{}_c$ are the structure constants of $\mathfrak{g}$, $(a, b)$ is the Killing form, and $\delta_0$ is the delta-function at the origin (the formal-disk avatar of the derivation $\partial$). Under the identification $\Gamma(D, \omega_D) \cong \mathbb{C}\, dz$, this is the CFG $\mathsf{P}_3$ bracket on $C^*(\mathfrak{g})$ (Theorem 8.5(ii)).

Part (iii): the deformation theory of $\mathsf{E}_3$-algebras with fixed associated graded $C^*(\mathfrak{g})$ is controlled by the $\mathsf{E}_3$-Hochschild cohomology, which for $C^*(\mathfrak{g})$ with $\mathfrak{g}$ simple is $(k + h^\vee)\, H^3(\mathfrak{g})[[k + h^\vee]]$, one-dimensional at each order in $(k+h^\vee)$. Both $\Gamma(D, \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g}))$ and $\mathcal{A}^\lambda$ are deformations of $C^*(\mathfrak{g})$ over this base (Theorem 1.3(ii) and Theorem 8.5(iv) respectively). Since the deformation space is one-dimensional at each order, and both deformations have matching associated graded (Part (i)) and matching $\mathsf{P}_3$ bracket (Part (ii)), they are isomorphic as filtered $\mathsf{E}_3$-algebras. The $(k + h^\vee)$-adic filtration on the chiral side maps to the $\lambda$-adic filtration on the CFG side under $\lambda = k + h^\vee$. □

*Remark* 8.22 (The chain of spaces). The restriction passes through $\mathrm{pt} \leftarrow D \to \mathbb{A}^1 \to \mathbb{P}^1 \to X$, with each step adding geometric content: the formal disk provides the derivation $\partial$ (cocycle term $k(a, b)\partial$); $\mathbb{A}^1$ adds Arnold relations and braid monodromy; $\mathbb{P}^1$ and higher-genus $X$ add compactness and modular curvature. None of these restrictions is an identity; each step loses geometric information.

8.9. **The Khan–Zeng topological enhancement.** The chiral $\mathsf{E}_3$-algebra is holomorphic: it depends on the complex structure of $X$ through the KZ connection. At non-critical level $k \neq -h^\vee$, the Sugawara Virasoro element provides a homotopy between the holomorphic and topological directions, upgrading the $\mathsf{E}_3$ structure from holomorphic to topological. At the critical level, the Sugawara element is undefined and the enhancement fails.

**Proposition 8.23** (Topological enhancement via Sugawara)**.** *At generic non-critical level $k \neq -h^\vee$, the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ possesses a Sugawara Virasoro element*

$$T_{\mathrm{Sug}} \;=\; \frac{1}{2(k + h^\vee)} \sum_{a=1}^{\dim \mathfrak{g}} {:}J^a J_a{:}, \tag{8.38}$$

*which generates an inner action of the Virasoro algebra at central charge $c = k\, \dim(\mathfrak{g})/(k + h^\vee)$.*

*By Khan–Zeng [28], the Virasoro element upgrades the holomorphic-topological symmetry of the $3d$ gauge theory on $\Sigma \times \mathbb{R}_t$ to fully topological symmetry on $\Sigma \times \mathbb{R}_t$. At the level of the chiral $\mathsf{E}_3$-algebra, this means:*

(i) (Holomorphic-topological.) *The chiral* $\mathsf{E}_3$*-algebra* $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ *lives on* $X \times \mathbb{R}$*, where* $X$ *carries holomorphic structure and* $\mathbb{R}$ *carries topological structure. The* $\mathsf{E}_2$ *from the curve geometry is holomorphic; the* $+1$ *from the* $\mathsf{P}_3$ *bracket is topological (from the* $\mathbb{R}$ *factor).*

(ii) (Full topological.) *The Sugawara element provides an identification of the holomorphic direction in* $X$ *with a topological direction: the stress-energy tensor* $T_{\mathrm{Sug}}$ *generates translations along* $X$*, making them homotopically trivial. After this identification, the* $\mathsf{E}_3$*-structure becomes* topological $\mathsf{E}_3$*: an* $\mathsf{E}_3$*-algebra in the category of locally constant sheaves on* $\mathbb{R}^3 \cong X \times \mathbb{R}$ *(identifying* $X \cong \mathbb{C} \cong \mathbb{R}^2$*).*

(iii) (Failure at critical level.) *At* $k = -h^\vee$*, the Sugawara element* (8.38) *is undefined (the denominator* $2(k + h^\vee)$ *vanishes). The critical-level chiral* $\mathsf{E}_3$*-algebra* $\mathrm{CE}_{-h^\vee}^{\mathrm{ch}}(\mathfrak{g})$ *is genuinely holomorphic, not topological: the* $\mathsf{E}_2$*-coalgebra on* $\overline{B}^{\Sigma}(\mathcal{A})$ *is topological (Warning 3.7), but* $\mathcal{A}$ *itself retains essential dependence on the complex structure of* $X$*. Without the Sugawara element, there is no homotopy between the holomorphic and topological directions, so the full structure is not topological* $\mathsf{E}_3$*. Perturbatively, in the formal neighbourhood of critical level, the topological enhancement holds order-by-order in* $(k + h^\vee)$ *(since* $T_{\mathrm{Sug}}$ *exists at any nonzero departure from critical level).*

*Proof.* Part (i) is the HT structure of the 3d gauge theory whose boundary chiral algebra is $V_k(\mathfrak{g})$ (Costello–Gwilliam [13]). The factorisation algebra of the HT theory lives on $\Sigma \times \mathbb{R}$ with holomorphic dependence on $\Sigma$ and topological dependence on $\mathbb{R}$; the chiral $\mathsf{E}_3$-algebra is the algebraic encoding of this factorisation structure.

Part (ii). Khan–Zeng [28] prove the following at the classical level: an inner Virasoro element in a Poisson vertex algebra with non-degenerate central charge provides a homotopy between the holomorphic and topological directions, upgrading the HT theory to a fully topological theory. We extend this to the quantum vertex algebra $V_k(\mathfrak{g})$. The Sugawara construction (Frenkel–Ben-Zvi [20], Chapter 15) produces $T_{\mathrm{Sug}} \in V_k(\mathfrak{g})$ satisfying the exact Virasoro OPE

$$T_{\mathrm{Sug}}(z)\, T_{\mathrm{Sug}}(w) \;\sim\; \frac{c/2}{(z-w)^4} + \frac{2\, T_{\mathrm{Sug}}(w)}{(z-w)^2} + \frac{\partial T_{\mathrm{Sug}}(w)}{z-w}$$

at central charge $c = k\,\dim(\mathfrak{g})/(k + h^\vee)$, and the Sugawara OPE with the current generators

$$T_{\mathrm{Sug}}(z)\, J^a(w) \;\sim\; \frac{J^a(w)}{(z-w)^2} + \frac{\partial J^a(w)}{z-w}.$$

The second relation gives $L_{-1} \cdot J^a = \partial J^a$: the holomorphic translation operator $L_{-1} = (T_{\mathrm{Sug}})_{(1)}$ acts as $\partial$ on all generators. These are identities in the quantum vertex algebra, not formal-deformation statements; the normal ordering in (8.38) absorbs all quantum corrections. The BRST operator $Q$ of the 3d gauge theory satisfies $[Q, b] = L_{-1}$ where $b$ is the antighost zero-mode

(this is the standard BRST relation for the Sugawara Virasoro; see [20], §15.4), so $L_{-1}$ is BRST-exact. The Khan–Zeng argument [28] then applies verbatim: BRST-exactness of holomorphic translation is the datum required to upgrade the HT symmetry to full topological symmetry. At generic $k$, the central charge $c = k\,\dim(\mathfrak{g})/(k+h^\vee) \neq 0$, so the hypothesis is satisfied.

Part (iii): at $k = -h^\vee$, the Feigin–Frenkel centre $Z(V_{-h^\vee}(\mathfrak{g})) = \mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$ (Feigin–Frenkel) is a commutative algebra with non-trivial dependence on the complex structure of the formal disk (through the notion of $\mathfrak{g}^\vee$-oper). The absence of the Sugawara element means no homotopy between holomorphic and topological directions exists, and the chiral $\mathsf{E}_3$-algebra is genuinely holomorphic. The perturbative statement follows because the Sugawara element exists as a formal power series in $(k+h^\vee)$ for $k \neq -h^\vee$. □

8.10. **Example: $\mathfrak{sl}_2$ chiral $\mathsf{E}_3$-algebra.** The general construction specialises cleanly to $\mathfrak{g} = \mathfrak{sl}_2$: the chiral $\mathsf{P}_3$ bracket is determined by six pairings on the generators $\phi_e, \phi_f, \phi_h$, each consisting of a Lie bracket term (from the structure constants) and a cocycle term (proportional to the level $k$, vanishing at $k = 0$). Restricting to the formal disk and extracting the zero mode recovers the CFG bracket of Proposition 8.8.

**Example 8.24** (Explicit chiral $\mathsf{P}_3$ bracket for $\mathfrak{sl}_2$)**.** For $\mathfrak{g} = \mathfrak{sl}_2$ with basis $\{e, f, h\}$, Killing form $(e, f) = (f, e) = 1$, $(h, h) = 2$, structure constants $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$, and $h^\vee = 2$, the chiral $\mathsf{P}_3$ bracket (Construction 8.14) on the generators $\phi_e, \phi_f, \phi_h \in \mathfrak{sl}_2^*[-1]$ is:

$$\{\phi_e, \phi_f\}^{\mathrm{ch}} = \phi_h + k\,\partial, \tag{8.39}$$

$$\{\phi_h, \phi_e\}^{\mathrm{ch}} = 2\,\phi_e, \tag{8.40}$$

$$\{\phi_h, \phi_f\}^{\mathrm{ch}} = -2\,\phi_f, \tag{8.41}$$

$$\{\phi_h, \phi_h\}^{\mathrm{ch}} = 2k\,\partial, \tag{8.42}$$

$$\{\phi_e, \phi_e\}^{\mathrm{ch}} = 0, \tag{8.43}$$

$$\{\phi_f, \phi_f\}^{\mathrm{ch}} = 0. \tag{8.44}$$

The Lie bracket terms ($\phi_h$ in (8.39), $2\phi_e$ in (8.40), $-2\phi_f$ in (8.41)) persist at $k = 0$; the cocycle terms ($k\,\partial$ in (8.39) and (8.42)) vanish at $k = 0$ (AP126 check: at $k = 0$, the level-dependent terms vanish, and the bracket reduces to the Chevalley–Eilenberg coderivation).

At the critical level $k = -2$ (i.e. $k+h^\vee = 0$), the cocycle terms become $-2\,\partial$, and the chiral $\mathsf{E}_3$-algebra degenerates: the curvature $m_0 = 0$ (uncurved), and the commutative factorisation structure is undeformed.

Taking formal disk global sections and extracting the zero mode recovers the CFG $\mathsf{P}_3$ bracket of Proposition 8.8: $\{\phi_a, \phi_b\}_{\mathrm{CFG}} = f^{ab}{}_c\,\phi_c + k\,(a, b)\,\delta_0$, which at the cohomological level gives $\{X, Y\} = h_{\mathrm{KZ}}\,(X, Y)$ for $X, Y \in \mathrm{HH}^1 = \mathfrak{sl}_2$ (equation (8.7)), with $h_{\mathrm{KZ}} = 1/(k+2)$.

## 9. Examples

The four shadow classes $G$, $L$, $C$, $M$ produce qualitatively different ordered chiral homology theories. Class $G$ (Heisenberg) has regular singularities, finite shadow tower, and trivial kernel ker(av). Class $L$ (affine Kac–Moody) has regular singularities but a non-trivial $R$-matrix kernel. Class $C$ ($\beta\gamma$ system) has regular singularities with nilpotent monodromy (the $r$-matrix is a constant nilpotent after $d\log$-absorption; pole order 0). Class $M$ (Virasoro) has irregular singularities of unbounded rank and an infinite shadow tower.

**Example 9.1** (Heisenberg: class $G$, $r = 2$; explicit ordered chiral Hochschild cohomology)**.** The Heisenberg algebra $\mathcal{H}_k$ has a single generating field $J(z)$ of conformal weight 1, OPE $J(z)J(w) \sim k/(z-w)^2$, and $r$-matrix $r(z) = k/z$ (trace-form convention; at $k = 0$, $r = 0$). The Koszul invariant is $\kappa = k$. The algebra is $\mathsf{E}_\infty$-chiral (symmetric OPE; no simple pole) and class $G$ (shadow depth $r = 2$: $S_4 = 0$, so $\Delta = 8\kappa S_4 = 0$). By the formality bridge (Theorem 7.5), ordered and symmetric chiral homologies agree. We compute the ordered chiral Hochschild cohomology $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k)$ on the formal disk $D$ and on $\mathbb{P}^1$ explicitly, degree by degree.

**Setup.** The ordered chiral Hochschild cochain complex (Definition 8.1) is

$$\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) = \bigoplus_{n\geq 0}\Big(\Omega^*\big(\overline{\mathrm{FM}}^{\mathrm{ord}}_n(\mathbb{C})\big)\otimes \mathrm{Hom}(\mathcal{H}_k^{\otimes n}, \mathcal{H}_k),\ d_{\mathrm{dR}}+d_{\mathrm{Hoch}}+\nabla_{\mathrm{KZ}}\Big),$$

where $d_{\mathrm{Hoch}}$ is the Hochschild differential (encoding OPE residues at boundary strata) and $\nabla_{\mathrm{KZ}}$ is the KZ connection. The bar complex is $B(\mathcal{H}_k) = T^c(s^{-1}\bar{\mathcal{H}}_k)$ (desuspended augmentation ideal; $|s^{-1}v| = |v| - 1$), with deconcatenation coproduct.

**Degree 0: the vacuum.** The chiral centre $Z(\mathcal{H}_k) = \{a : a_{(n)}b = 0 \text{ for all } b, n \geq 0\}$ consists of the scalars only (since $J_{(1)}J = k \neq 0$ but $J_{(0)}J = 0$).

$$\mathrm{HH}^0 = \mathbb{C}. \tag{9.1}$$

**Degree 1: outer derivations.** A chiral derivation $D$ satisfies $D(J) = c \cdot \mathbf{1}$ for $c \in \mathbb{C}$ (the Leibniz rule and the absence of fields with regular OPE against $J$ force this). Since $J_{(0)}J = 0$ (no simple pole), all inner derivations vanish: $\mathrm{Inn}(\mathcal{H}_k) = 0$.

$$\mathrm{HH}^1 = \mathbb{C}, \tag{9.2}$$

generated by the translation $D(J) = \mathbf{1}$. The classical Hochschild $\mathrm{HH}^1$ of the Weyl algebra is 0 (every derivation is inner); the discrepancy arises because chiral inner derivations use $a_{(0)}$, not $[a, -]$.

**Degree 2: level deformation and the KZ connection.** Setting $w = z_1 - z_2$, the KZ connection is

$$\nabla_{\mathrm{KZ}} = d + k\, d\log w, \tag{9.3}$$

with solution $\Phi(w) = w^k$ and scalar monodromy

$$M = e^{2\pi i k}. \tag{9.4}$$

For generic $k \notin \mathbb{Z}$, the twisted de Rham cohomology of $\mathbb{C}^\times$ vanishes. The level deformation cocycle $\phi_k(J \otimes J) = k \cdot \mathbf{1}$ represents

$$[\phi_k] \in \mathrm{HH}^2 = \mathbb{C}, \tag{9.5}$$

the unique deformation $J(z)J(w) \sim (k+\epsilon)/(z-w)^2$.

**Degree $n \geq 3$: vanishing.** The Heisenberg is chirally Koszul with resolution of length 2, so the cochain complex at degree $n \geq 3$ is acyclic:

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) = 0 \quad \text{for } n \geq 3. \tag{9.6}$$

**Summary on $D$.** The ordered chiral Hochschild cohomology on the formal disk is

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) = \begin{cases} \mathbb{C} & n = 0 \text{ (vacuum: centre} = \mathbb{C} \cdot \mathbf{1}), \\ \mathbb{C} & n = 1 \text{ (outer derivation: } D(J) = \mathbf{1}), \\ \mathbb{C} & n = 2 \text{ (level deformation: } k \mapsto k + \epsilon), \\ 0 & n \geq 3 \text{ (Koszul resolution of length 2)}, \end{cases} \tag{9.7}$$

with Poincaré polynomial $P_{\mathcal{H}_k}(t) = 1 + t + t^2$. The total dimension is 3, and the cohomological amplitude is $[0, 2]$, consistent with Theorem H. The derived chiral centre is $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k) \cong \mathbb{C} \oplus \mathbb{C}[-1] \oplus \mathbb{C}[-2]$ as a graded vector space.

**The computation on $\mathbb{P}^1$.** The computation on $\mathbb{P}^1$ gives the same result:

$$\mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch},\mathbb{P}^1}(\mathcal{H}_k, \mathcal{H}_k) \cong \mathrm{HH}^{n,\mathrm{ord}}_{\mathrm{ch},D}(\mathcal{H}_k, \mathcal{H}_k) \quad \text{for all } n. \tag{9.8}$$

The factorisation $\mathcal{D}$-module is insensitive to the compactification for chirally Koszul algebras: the Koszul resolution has length 2, so only degrees 0, 1, 2 contribute, and at these degrees the global geometry of $\mathbb{P}^1$ introduces no new cohomology beyond what the formal disk sees.

**The $\mathsf{E}_3$ structure.** The derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k)$ is an $\mathsf{E}_3$-algebra by Proposition 8.2. The obstruction space $\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^\Sigma(\mathcal{H}_k), \overline{B}^\Sigma(\mathcal{H}_k)) \cong \Lambda^3(\mathbb{C}^1) = 0$ vanishes, so the $\mathsf{E}_3$-algebra structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k) \cong \mathbb{C} \oplus \mathbb{C}[-1] \oplus \mathbb{C}[-2]$ is the unique (trivial) one. The 3d theory is abelian Chern–Simons at level $k$; the triviality reflects the absence of perturbative quantum corrections in the abelian theory.

**Comparison with topological Hochschild.** By the formality bridge (Theorem 7.5), $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\mathcal{H}_k, \mathcal{H}_k) \simeq \mathrm{HH}^{*,\mathrm{top}}(\mathcal{H}_k, \mathcal{H}_k)$: no information is lost in the passage from holomorphic (ordered, $z$-dependent) to topological ($S^1$, $z$-independent).

**Example 9.2** (Affine Kac–Moody: class $L$, $r = 3$; explicit computation for $\widehat{\mathfrak{sl}}_2$)**.** We carry out the ordered chiral Hochschild computation for $V_k(\mathfrak{sl}_2)$ on the formal disk, the simplest non-abelian Kac–Moody algebra.

**Data.** $\mathfrak{g} = \mathfrak{sl}_2$, $h^\vee = 2$, $\dim(\mathfrak{g}) = 3$. Generators $\{e, f, h\}$ with brackets $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$. Normalised invariant form: $(e, f) = (f, e) = 1$, $(h, h) = 2$. Inverse Casimir: $\Omega = e \otimes f + f \otimes e + \frac{1}{2} h \otimes h$. The Koszul invariant is

$$\kappa = 3(k + 2)/4. \tag{9.9}$$

The $r$-matrix in trace-form convention is $r(z) = k\Omega/z$ (at $k = 0$, $r = 0$); in KZ normalisation, $r(z) = \Omega/((k + 2)z)$. Class $L$ (shadow depth $r = 3$): $\ell_1, \ell_2, \ell_3$ non-trivial, $\ell_k = 0$ for $k \geq 4$.

**Degree 2: KZ conformal blocks.** The ordered chiral Hochschild at degree 2 is the de Rham cohomology of $\mathrm{Conf}_2(\mathbb{C})$ with coefficients in the KZ local system. The KZ equation at degree 2, setting $z = z_1 - z_2$, is

$$\frac{\partial \Phi}{\partial z} = \frac{\Omega}{(k + 2)\, z}\, \Phi. \tag{9.10}$$

Since $V_k(\mathfrak{sl}_2)$ is $\mathsf{E}_\infty$-chiral, the coefficients live in $V_k(\mathfrak{sl}_2)^{\otimes 2}$, which decomposes under $\mathfrak{sl}_2$ by Clebsch–Gordan. For the adjoint representation acting on itself ($j = 1$): $V_1 \otimes V_1 = V_0 \oplus V_1 \oplus V_2$. The Casimir eigenvalue of $\Omega$ on the isospin-$J$ component is

$$\lambda_J = \frac{C_2(J) - C_2(1) - C_2(1)}{2} = \frac{J(J + 1) - 4}{2}, \tag{9.11}$$

where $C_2(j) = j(j + 1)$ in the normalisation $(\theta|\theta) = 2$. The eigenvalues are:

$$\lambda_0 = -2, \quad \lambda_1 = -1, \quad \lambda_2 = 1.$$

On each isospin component, the KZ equation (9.10) has the explicit solution $\Phi_J(z) = z^{\lambda_J/(k+2)}$. The monodromy around $z = 0$ (the diagonal $\Delta_{12} \subset X^2$) is

$$M_J = \exp\bigl(2\pi i\, \lambda_J/(k + 2)\bigr).$$

The de Rham cohomology $H^*_{\mathrm{dR}}(\mathrm{Conf}_2(\mathbb{C}), \nabla_{\mathrm{KZ}})$ decomposes as:

- $H^0 = \ker(M - \mathrm{id})$: the subspace on which the monodromy is trivial. At generic $k$, this is zero (no isospin channel has $\lambda_J/(k + 2) \in \mathbb{Z}$).
- $H^1 = \mathrm{coker}(M - \mathrm{id})$: the space of KZ conformal blocks at degree 2. At generic $k$, $H^1 \cong V_0 \oplus V_1 \oplus V_2$ (three-dimensional over the isospin decomposition: one block per channel).

The averaging map $\mathrm{av}_2 \colon H^1_{\mathrm{ord}} \to H^1_\Sigma$ projects each conformal block to its $\Sigma_2$-coinvariant. Under the flip $z \mapsto -z$, the monodromy on $V_J$ picks up a sign $(-1)^J$ (from the permutation of the two adjoint insertions); at generic $k$, $\mathrm{av}_2$ is surjective with $\ker(\mathrm{av}_2)$ carrying the odd-$J$ blocks. The scalar $\kappa = 3(k + 2)/4$ (equation (9.9)) is the complete information surviving the averaging.

**Degree 3: the Drinfeld associator.** At degree 3, the KZ system on $\mathrm{Conf}_3(\mathbb{C})$ is

$$d\Phi = \frac{1}{k+2}\left(\frac{\Omega_{12}}{z_1-z_2}\,d(z_1-z_2) + \frac{\Omega_{13}}{z_1-z_3}\,d(z_1-z_3) + \frac{\Omega_{23}}{z_2-z_3}\,d(z_2-z_3)\right)\Phi. \tag{9.12}$$

The fundamental group $\pi_1(\mathrm{Conf}_3(\mathbb{C})) = PB_3$ (the pure braid group on three strands) has generators $\sigma_{12}^2, \sigma_{23}^2$ (Dehn twists around the diagonals). The KZ monodromy gives a representation $PB_3 \to \mathrm{GL}(V_k(\mathfrak{g})^{\otimes 3})$. The Drinfeld associator $\Phi_{\mathrm{KZ}} \in \exp(\widehat{\mathrm{Lie}}(t_{12}, t_{23}))$ is the regularised holonomy along the path from the boundary stratum $|z_1 - z_2| \ll |z_2 - z_3|$ to the stratum $|z_2 - z_3| \ll |z_1 - z_2|$ in the real locus of $\overline{\mathrm{FM}}_3^{\mathrm{ord}}(\mathbb{C})$.

The kernel of the averaging map $\mathrm{av}_3$ is the space of ordered data invisible to $\overline{B}^{\Sigma}$. For $\mathfrak{g} = \mathfrak{sl}_2$, this kernel is one-dimensional: $\ker(\mathrm{av}_3) \cong \Lambda^3(\mathbb{C}^3) = \mathbb{C}$, carrying the Drinfeld associator. The associator $\Phi_{\mathrm{KZ}}$ is the unique (up to $\mathrm{GRT}_1$-action) non-trivial element of this kernel.

For general $\mathfrak{g}$, the ordered data at degree 3 is controlled by $\Lambda^3(\mathfrak{g}) = \Lambda^3(\mathbb{C}^{\dim \mathfrak{g}})$, with $\dim \Lambda^3(\mathfrak{g}) = \binom{\dim \mathfrak{g}}{3}$. At $\mathfrak{g} = \mathfrak{sl}_2$: $\binom{3}{3} = 1$.

**$\mathsf{E}_3$ obstruction.** The $\mathsf{E}_2$-coalgebra $\overline{B}^{\Sigma}(V_k(\mathfrak{sl}_2))$ (Warning 3.7) has obstruction to $\mathsf{E}_3$-promotion in

$$\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}, \overline{B}^{\Sigma}) \;\cong\; H^3(\mathfrak{sl}_2) \;\cong\; \Lambda^3(\mathfrak{sl}_2^*)^{\mathfrak{sl}_2} \;=\; \mathbb{C}, \tag{9.13}$$

which is one-dimensional and carries the non-trivial obstruction class (Warning 8.4). The $\mathsf{E}_3$-algebra is the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$, not $\overline{B}^{\Sigma}$ itself.

**Identification with CFG.** The $\mathsf{E}_3$-deformation space is $(k+2)H^3(\mathfrak{sl}_2)[[k+2]] \cong (k+2)\mathbb{C}[[k+2]]$: one-dimensional at each order in $(k+2)$. For $\mathfrak{sl}_2$, $H^3(\mathfrak{sl}_2) = \mathbb{C}$ (generated by the three-form $\omega_3 = (e^* \wedge f^* \wedge h^*) \circ [\cdot,\cdot]$, the Cartan form). The deformation parameter is the departure from critical level $k+2 = k+h^\vee$. By Theorem 8.21, the formal disk restriction $\Gamma(D, \mathrm{CE}^{\mathrm{ch}}_k(\mathfrak{sl}_2))$ recovers the CFG filtered $\mathsf{E}_3$-algebra $C^*_{k+2}(\mathfrak{sl}_2)$ of perturbative Chern–Simons quantisations; the full identification of the deformation families is Theorem 8.6. At the associative level, the Drinfeld–Kohno theorem identifies the KZ monodromy with the $R$-matrix of $U_q(\mathfrak{sl}_2)$ at $q = \exp(2\pi i/(k+2))$: the chiral quantum group equivalence (Theorem 6.5) lifts this from monodromy representations to the algebra.

**Summary of ordered chiral Hochschild for $\widehat{\mathfrak{sl}}_2$.**

| Degree $n$ | Ordered datum | Symmetric shadow |
|---|---|---|
| 1 | $V_k(\mathfrak{sl}_2)$ | $V_k(\mathfrak{sl}_2)$ |
| 2 | KZ local system; conformal blocks $V_0 \oplus V_1 \oplus V_2$ | scalar $\kappa = 3(k+2)/4$ |
| 3 | KZ associator $\Phi_{\mathrm{KZ}}$; $\ker(\mathrm{av}_3) = \mathbb{C}$ | trivial (associator lost) |
| $\geq 4$ | higher KZ holonomy | trivial |

The KZ connection (4.6) has simple poles: regular singularities. The 3d theory is Chern–Simons gauge theory with gauge group $G$.

**Example 9.3** ($\beta\gamma$ system: class $C$, $r = 4$; explicit ordered chiral Hochschild cohomology)**.** The $\beta\gamma$ system at conformal weight $\lambda$ has two generating fields $\beta(z)$ of weight $\lambda$ and $\gamma(z)$ of weight $1-\lambda$, with OPE

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}, \qquad \gamma(z)\beta(w) \sim \frac{1}{z-w}, \qquad \beta(z)\beta(w) \sim 0, \qquad \gamma(z)\gamma(w) \sim 0.$$

The central charge is $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and the Koszul invariant is $\kappa = c_{\beta\gamma}/2 = 6\lambda^2 - 6\lambda + 1$. (Verification: at $\lambda = 1/2$, $c_{\beta\gamma} = -1$; at $\lambda = 2$, $c_{\beta\gamma} = 26$.) The algebra is $\mathsf{E}_\infty$-chiral (the OPE is symmetric after clearing the simple pole by locality) and class $C$ (shadow depth $r = 4$: the quartic contact shadow is nontrivial, but the quintic obstruction vanishes by rank-one abelian rigidity).

By the formality bridge (Theorem 7.5), ordered and symmetric chiral homologies agree. The distinctive feature of this example is that the KZ connection has *unipotent* monodromy (rather than scalar or semisimple), and the nilpotency $\Theta^2 = 0$ makes the parallel transport computable in closed form at every degree. The $\beta\gamma$ system is the *hydrogen atom* for ordered chiral homology with unipotent local systems: all KZ solutions are explicit polynomials in logarithms, with no hypergeometric functions and no Drinfeld associator.

**The collision residue and $R$-matrix.** The generating space is $V = \mathbb{C}\beta \oplus \mathbb{C}\gamma$ (two-dimensional). The OPE has a simple pole between $\beta$ and $\gamma$. After $d$-log absorption (pole order $1 - 1 = 0$), the collision residue is a *constant* nilpotent endomorphism

$$\Theta \in \mathrm{End}(V \otimes V), \qquad \Theta(\beta\otimes\gamma) = \mathbf{1}, \quad \Theta(\gamma\otimes\beta) = \mathbf{1}, \quad \Theta(\beta\otimes\beta) = 0, \quad \Theta(\gamma\otimes\gamma) = 0, \tag{9.14}$$

where $\mathbf{1}$ denotes the vacuum in the full algebra $A$ (not in $V \otimes V$). The map $\Theta$ sends $V \otimes V \to A$, with image in the one-dimensional vacuum subspace $\mathbb{C} \cdot \mathbf{1} \subset A$. The key algebraic fact: $\Theta^2 = 0$ (nilpotent of order 2), because the image of $\Theta$ is the vacuum subspace, and $\Theta$ annihilates the vacuum. Concretely: $\Theta$ has rank 1 on the four-dimensional space $V^{\otimes 2}$ (both $\beta \otimes \gamma$ and $\gamma \otimes \beta$ map to $\mathbf{1}$, hence the image is one-dimensional), with $\ker(\Theta) \supset \mathbb{C}(\beta \otimes \beta) \oplus \mathbb{C}(\gamma \otimes \gamma)$ (the "same-type" tensors, on which the OPE is regular) and $\mathrm{im}(\Theta) = \mathbb{C} \cdot \mathbf{1}$.

The $R$-matrix is the monodromy $R = \exp(2\pi i\, \Theta)$. Since $\Theta^2 = 0$, the exponential terminates:

$$R_{\beta\gamma} = I + 2\pi i\, \Theta. \tag{9.15}$$

This termination at the linear term is the defining feature of class $C$. For affine Kac–Moody (class $L$), the $R$-matrix is an infinite series in the spectral parameter; for $\beta\gamma$, nilpotency truncates the series.

**The KZ connection.** At degree $n$, the KZ connection on $\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{C})$ is

$$\nabla_n^{\beta\gamma} = d + \sum_{1\le i<j\le n} \Theta_{ij}\,\omega_{ij}, \qquad \omega_{ij} = d\log(z_i - z_j), \tag{9.16}$$

where $\Theta_{ij} \in \mathrm{End}(V^{\otimes n})$ acts as $\Theta$ on the $i$-th and $j$-th tensor factors and as the identity elsewhere. The connection is logarithmic: regular singularities along every diagonal $\Delta_{ij}$. This is the analogue of the KZ connection for affine KM, with the simplification that $\Theta_{ij}^2 = 0$.

Flatness: the condition $(\nabla_n^{\beta\gamma})^2 = 0$ reduces to the classical Yang–Baxter equation for $\Theta$:

$$[\Theta_{12}, \Theta_{13}] + [\Theta_{12}, \Theta_{23}] + [\Theta_{13}, \Theta_{23}] = 0 \quad \text{in } \mathrm{End}(V^{\otimes 3}). \tag{9.17}$$

This holds because $\Theta$ is the collision residue of a vertex algebra OPE. Each commutator $[\Theta_{ij}, \Theta_{jk}]$ produces a "triple collision" term (contraction at all three sites via the shared index $j$), and the three such terms cancel by the cyclic symmetry of the Arnold relation.

**Explicit KZ solutions via nilpotency.** Since $\Theta^2 = 0$, the parallel transport is polynomial in logarithms. At degree 2:

$$\Phi(w) = \exp(\Theta_{12}\log w) = I + \Theta_{12}\,\log w, \tag{9.18}$$

with monodromy $R = I + 2\pi i\,\Theta_{12}$. At degree 3, the solution terminates at degree two in logarithms (triple contractions vanish on the two-dimensional $V$), and the Drinfeld associator is trivial. Since $\Theta^2 = 0$, the iterated-integral corrections from non-commutativity of the connection vanish: the Picard series $\Phi = I + \sum_{i<j}\Theta_{ij}\log(z_i - z_j) + \sum_{(i<j),(k<l)}\Theta_{ij}\Theta_{kl}\log(z_i - z_j)\log(z_k - z_l) + \cdots$ reduces to ordinary products of logarithms (rather than Chen iterated integrals), because the ordering ambiguity in the iterated integral $\int \Theta_{ij}\omega_{ij} \circ \Theta_{kl}\omega_{kl}$ versus $\int \Theta_{kl}\omega_{kl} \circ \Theta_{ij}\omega_{ij}$ differs by the commutator $[\Theta_{ij}, \Theta_{kl}]$, which vanishes when $\Theta^2 = 0$ (the image of each $\Theta_{ij}$ is the vacuum, annihilated by every $\Theta_{kl}$).

**The chiral Hochschild computation.** The $\beta\gamma$ system is chirally Koszul with Koszul resolution of length 2 (generating space $V = \mathbb{C}\beta \oplus \mathbb{C}\gamma$, relation space $R = \mathbb{C}\cdot(\beta_{(0)}\gamma - \mathbf{1})$), so only degrees 0, 1, 2 contribute.

**Degree 0: the vacuum.** Since $\beta_{(0)}\gamma = \mathbf{1} \neq 0$, neither $\beta$ nor $\gamma$ is central; only the scalars survive.

$$\mathrm{HH}^0 = \mathbb{C}. \tag{9.19}$$

**Degree 1: outer derivations.** The simple-pole OPE $\beta_{(0)}\gamma = \mathbf{1}$ makes $a_{(0)}$ surjective on the generating space, so every chiral derivation is inner ($\beta \mapsto \beta + \epsilon$ is generated by $(\epsilon\gamma)_{(0)}$).

$$\mathrm{HH}^1 = 0. \tag{9.20}$$

This inverts the Heisenberg pattern: a double pole makes inner derivations trivial ($\mathrm{HH}^1 = \mathbb{C}$); a simple pole makes all derivations inner ($\mathrm{HH}^1 = 0$). The dichotomy is a shadow of the Koszul duality $(\beta\gamma)^! \simeq bc$.

**Degree 2: the unipotent KZ connection.** The KZ connection (9.16) at degree 2 is

$$\nabla_{\mathrm{KZ}}^{\beta\gamma} = d + \Theta_{12}\,\omega_{12}, \qquad \omega_{12} = d\log(z_1 - z_2). \tag{9.21}$$

The solution is $\Phi(w) = I + \Theta_{12}\log w$ (equation (9.18)), with *unipotent* monodromy

$$M = R_{\beta\gamma} = I + 2\pi i\,\Theta_{12}, \qquad (M - I)^2 = 0. \tag{9.22}$$

For a unipotent local system ($N = M - I$, $N^2 = 0$), the de Rham cohomology is $H^0 = \ker(N)$, $H^1 = \operatorname{coker}(N)$. Since $\Theta_{12}$ has rank 1 on $V^{\otimes 2} \cong \mathbb{C}^4$ (both $\beta\otimes\gamma$ and $\gamma\otimes\beta$ map to $\mathbf{1}$, so the image is one-dimensional; $\beta\otimes\beta$ and $\gamma\otimes\gamma$ are killed), $\dim\ker(N) = 3$ and $\dim\operatorname{coker}(N) = 3$. The monodromy is scalar for the Heisenberg ($e^{2\pi i k}$, generic), semisimple for affine KM ($e^{2\pi i\lambda_J/(k+2)}$, channel-dependent), and unipotent for $\beta\gamma$ (always nontrivial flat sections).

The unique deformation cocycle $\phi(\beta\otimes\gamma) = \epsilon\cdot\mathbf{1}$ represents

$$[\phi] \in \mathrm{HH}^2 = \mathbb{C}, \tag{9.23}$$

corresponding to $\beta(z)\gamma(w) \sim (1+\epsilon)/(z-w)$.

**Degree $n \geq 3$: vanishing.** By the Koszul resolution of length 2:

$$\mathrm{HH}_{\mathrm{ch}}^{n,\mathrm{ord}}(\beta\gamma,\beta\gamma) = 0 \quad \text{for } n \geq 3. \tag{9.24}$$

**Summary on $D$.** The ordered chiral Hochschild cohomology on the formal disk is

$$\mathrm{HH}_{\mathrm{ch}}^{n,\mathrm{ord}}(\beta\gamma,\beta\gamma) = \begin{cases} \mathbb{C} & n = 0 \text{ (vacuum: centre} = \mathbb{C}\cdot\mathbf{1}), \\ 0 & n = 1 \text{ (all derivations inner: simple-pole OPE)}, \\ \mathbb{C} & n = 2 \text{ (OPE residue deformation: } 1/(z-w) \mapsto (1+\epsilon)/(z-w)), \\ 0 & n \geq 3 \text{ (Koszul resolution of length 2)}, \end{cases} \tag{9.25}$$

with Poincaré polynomial $P_{\beta\gamma}(t) = 1 + t^2$. The total dimension is 2, and the cohomological amplitude is $[0,2]$, consistent with Theorem H. The derived chiral centre is $Z_{\mathrm{ch}}^{\mathrm{der}}(\beta\gamma) \cong \mathbb{C} \oplus \mathbb{C}[-2]$ as a graded vector space.

*Remark* 9.4 (Poincaré polynomials and Koszul duality). $P_{\mathcal{H}_k}(t) = 1 + t + t^2$ versus $P_{\beta\gamma}(t) = 1 + t^2$: the missing $t$-term reflects $\mathrm{HH}^1 = 0$. By Koszul duality $(\beta\gamma)^! \simeq bc$, the $bc$ ghost system has $P_{bc}(t) = 1 + t^2$ as well ($\mathrm{HH}^n(\beta\gamma) \cong \mathrm{HH}^{2-n}(bc)^*$).

**The computation on $\mathbb{P}^1$.** The computation on $\mathbb{P}^1$ gives the same result (as for the Heisenberg, the factorisation $\mathcal{D}$-module is insensitive to the compactification for chirally Koszul algebras):

$$\mathrm{HH}_{\mathrm{ch},\mathbb{P}^1}^{n,\mathrm{ord}}(\beta\gamma,\beta\gamma) \cong \mathrm{HH}_{\mathrm{ch},D}^{n,\mathrm{ord}}(\beta\gamma,\beta\gamma) \quad \text{for all } n. \tag{9.26}$$

**Koszul duality and complementarity.** The Koszul dual is the $bc$ ghost system at weight $\lambda$, with $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$, satisfying the complementarity

$$\kappa(\beta\gamma) + \kappa(bc) = 0. \tag{9.27}$$

(Verification: $c_{\beta\gamma}(2) = 26$, $c_{bc}(2) = -26$, sum $= 0$.) The complementarity $K = 0$ places $\beta\gamma$ in the same family as the Heisenberg and lattice algebras: the Koszul conductor vanishes.

**The $\mathsf{E}_3$ structure.** As for the Heisenberg, the $\mathsf{E}_3$ obstruction space $\mathrm{HH}^3_{\mathsf{E}_2}(\overline{B}^{\Sigma}(\beta\gamma), \overline{B}^{\Sigma}(\beta\gamma)) \cong \Lambda^3(\mathbb{C}^2) = 0$ vanishes for dimensional reasons, and the $\mathsf{E}_3$-algebra structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\beta\gamma) \cong \mathbb{C} \oplus \mathbb{C}[-2]$ is the unique (trivial) one.

**Comparison with topological Hochschild.** By the formality bridge (Theorem 7.5), $\mathrm{HH}^{*,\mathrm{ord}}_{\mathrm{ch}}(\beta\gamma, \beta\gamma) \simeq \mathrm{HH}^{*,\mathrm{top}}(\beta\gamma, \beta\gamma)$. The transferred $A_\infty$-structure carries nontrivial operations $m_3^{\mathrm{tr}} \neq 0$ and $m_4^{\mathrm{tr}} \neq 0$ (shadow depth $r = 4$): the formality bridge preserves the chain complex but does not kill the higher operations. The information lost in passing from ordered to topological is the explicit $z$-dependence of the flat sections: $\Phi(w) = I + \Theta \log w$ is replaced by the abstract homotopy transfer data.

**Example 9.5** (Virasoro: class $M$, $r = \infty$; the irregular $\mathcal{D}$-module and Stokes phenomenon)**.** The Virasoro algebra $\mathrm{Vir}_c$ has a single generating field $T(z)$ of conformal weight 2, OPE

$$T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} + \frac{2\,T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w},$$

and $r$-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ (quartic OPE pole, $d$-log absorption gives cubic $r$-matrix pole). The Koszul invariant is $\kappa = c/2$. The algebra is $\mathsf{E}_\infty$-chiral (symmetric OPE: the commutator $[T(z), T(w)]$ is commutative after clearing poles) and class $M$ (shadow depth $r = \infty$: all higher shadow operations $\ell_k \neq 0$ in the symmetric convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$).

**Formality bridge and $\mathsf{E}_\infty$ descent.** By the formality bridge (Theorem 7.5), ordered and symmetric chiral homologies agree; $\ker(\mathrm{av}) = 0$ at each degree. The infinite shadow tower of class $M$ is a *symmetric* phenomenon. The content genuinely new to class $M$ is the *singularity type* of the factorisation $\mathcal{D}$-module: irregular, with Stokes data that has no analogue in class $G/L/C$.

**The irregular KZ connection at degree 2.** At degree 2, setting $z = z_1 - z_2$, the KZ equation (Proposition 4.6(ii)) reads

$$\frac{\partial \Phi}{\partial z} = \left(\frac{c/2}{z^3} + \frac{2T}{z}\right)\Phi. \tag{9.28}$$

This has a pole of order 3 in $z$: the singular point $z = 0$ is *irregular* of Poincaré rank 2 (pole order minus one). The classical Riemann–Hilbert correspondence of Deligne does not apply; one needs the irregular Riemann–Hilbert correspondence of D'Agnolo–Kashiwara [16].

For comparison, the affine Kac–Moody KZ equation at degree 2 (Example 9.2) is $\partial\Phi/\partial z = (\Omega/((k+h^\vee)z))\,\Phi$, a *simple* pole. Its solution is the power function $\Phi(z) = z^{\Omega/(k+h^\vee)}$: purely algebraic monodromy, no exponential corrections, no Stokes data. The difference between class $L$ and class $M$ is the difference between regular and irregular singularities.

**Formal solution and the exponential factor.** On the primary sector, $T$ acts by the eigenvalue $h$ (conformal weight of the field being transported). Equation (9.28) restricts to the scalar ODE

$$\frac{d\varphi}{dz} = \left(\frac{c/2}{z^3} + \frac{2h}{z}\right)\varphi, \tag{9.29}$$

which integrates exactly:

$$\varphi(z) = z^{2h}\,\exp\!\left(-\frac{c}{4z^2}\right). \tag{9.30}$$

The factor $z^{2h}$ is the monodromy piece: it contributes $\exp(4\pi i h)$ upon circling $z = 0$, matching the conformal spin. The factor $\exp(-c/(4z^2))$ is the *irregular* piece: it has an essential singularity at $z = 0$, invisible to monodromy, carrying exponentially small corrections in different angular sectors of the $z$-plane.

In the Hukuhara–Turrittin–Levelt normal form [45, 41], the formal fundamental matrix is $\Phi^{\mathrm{form}}(z) = \widehat{F}(z)\,z^L\,\exp(-c/(4z^2)\,\mathrm{Id})$, where $\widehat{F}(z) \in \mathrm{GL}(V)[[z^{-1}]]$ is a Gevrey-1/2 formal gauge transformation.

**Stokes phenomenon.** The exponential $\exp(-c/(4z^2))$ produces four Stokes rays at $\theta = 0, \pi/2, \pi, 3\pi/2$ (matching Poincaré rank 2: $2 \times 2 = 4$ rays). Across each ray, the true solution jumps by a Stokes multiplier $\mathfrak{S}$:

$$\Phi_+^{\mathrm{true}}(z) = \Phi_-^{\mathrm{true}}(z) \cdot (\mathrm{Id} + \mathfrak{S}). \tag{9.31}$$

The flat sections are classified by a *Stokes-filtered local system*, not a monodromy representation. This contrasts with class $G/L/C$, where regular singularities produce no Stokes data.

**Wall-crossing at $c = -22/5$.** The Stokes multipliers depend on the central charge $c$. At $c = -22/5$ (the $(2,5)$ minimal model, the Yang–Lee edge singularity), two Stokes rays coalesce in a wall-crossing: the connection resonates, and the Stokes multipliers undergo a discontinuous change. In the language of the shadow obstruction tower [34], the shadow invariant $S_4 = 10/(c(5c+22))$ has a pole at $c = -22/5$, signalling the coalescence.

**De Rham cohomology of the irregular $\mathcal{D}$-module.** The factorisation $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}(\mathrm{Vir}_c)$ is holonomic on $X^n$ for any smooth proper curve $X$ (Proposition 4.6(i)). By Mebkhout's theorem [40], holonomic $\mathcal{D}$-modules on proper varieties have finite-dimensional de Rham cohomology regardless of regularity. The finite-dimensionality of $H^*_{\mathrm{dR}}(X^n, \mathcal{F}_n^{\mathrm{ord}})$ is therefore unconditional: the irregular singularities of class $M$ do not obstruct finiteness.

The de Rham cohomology computes the flat sections of the connection with coefficients in the *Stokes-filtered local system*, not the naive local system (which is the wrong coefficient system for irregular connections). For $X = \mathbb{P}^1$ at degree $n = 2$: the irregular de Rham space $H^*_{\mathrm{dR}}(\mathrm{Conf}_2(\mathbb{P}^1), \nabla^{\mathrm{Vir}}_{\mathrm{KZ}})$ is finite-dimensional, with the dimension depending on $c$ through the Stokes multipliers.

**Shadow tower and the gravitational partition function.** The class $M$ shadow obstruction tower is infinite:

$$\begin{aligned}
S_2 &= c/2 = \kappa,\\
S_3 &= 2 \quad (c\text{-independent}),\\
S_4 &= 10/(c(5c+22)),\\
S_5 &= -48/(c^2(5c+22)),\\
S_6 &= 80(45c+193)/(3\,c^3(5c+22)^2),
\end{aligned}$$

with all $S_r$ belonging to $\mathbb{Q}(c)$ and having poles only at $c = 0$ and $c = -22/5$. The spectral discriminant is $\Delta = 8\kappa S_4 = 40/(5c+22) \neq 0$: the tower does not terminate, confirming class $M$.

The scalar gravitational partition function on the scalar lane is

$$Z^{\mathrm{scal}}_{\mathrm{grav}}(\mathrm{Vir}_c;\, t) = \frac{c}{2} \cdot \frac{\sqrt{t}/2}{\sin(\sqrt{t}/2)} \qquad \text{(UNIFORM-WEIGHT)}, \tag{9.32}$$

where $t$ is the genus-expansion formal variable, converging for $|t| < 4\pi^2$. This is $\kappa \cdot (\sqrt{t}/2)/\sin(\sqrt{t}/2)$ evaluated at $\kappa = c/2$. At $c = 26$ (the bosonic string), $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = 13 + (-13) = 0$: the scalar gravitational partition function vanishes, and the non-scalar corrections from the shadow tower become the leading contribution.

The full gravitational partition function receives corrections from all degrees of the shadow tower. For class $M$, infinitely many degrees contribute at each genus $g \geq 2$. The leading non-scalar correction involves the quartic contact invariant $S_4 = 10/(c(5c+22))$, which encodes the quartic gravitational counterterm in the 3d theory.

**Koszul duality and complementarity.** The Koszul dual is $\mathrm{Vir}_{26-c}$, with $\kappa + \kappa' = c/2 + (26-c)/2 = 13$ (self-dual at $c = 13$; Virasoro complementarity $K = 13$, distinct from the vanishing $K = 0$ of KM/Heis/lattice). Under $c \mapsto 26 - c$:

$$\Delta(c) + \Delta(26-c) = \frac{40}{5c+22} + \frac{40}{152-5c} = \frac{6960}{(5c+22)(152-5c)}. \tag{9.33}$$

The shadow growth rates are complementary: $\rho(c) \cdot \rho(26-c)$ is *not* constant, but both vanish in the limit $c \to \infty$.

**The expected 3d gravitational theory.** The expected 3d theory dual to $\mathrm{Vir}_c$ is holomorphic-topological gravity in the Beltrami parametrisation [13].

The irregular singularity of the factorisation $\mathcal{D}$-module is the algebraic counterpart of the unbounded spin tower in the gravitational theory: the $T(z)$ OPE generates fields of arbitrarily high spin through repeated collisions, producing the infinite shadow tower $S_2, S_3, S_4, \ldots$. In the 3d language, each $S_r$ is a perturbative gravitational counterterm at $r$-point contact. The essential singularity $\exp(-c/(4z^2))$ in the flat sections of the KZ connection is the chiral-algebraic incarnation of the non-perturbative gravitational instanton corrections.

**Summary.**

| **Feature** | **Class $L$ (KM)** | **Class $M$ (Vir)** |
|---|---|---|
| $r$-matrix pole | 1 (simple) | 3 (cubic) |
| Singularity type | regular (Fuchsian) | irregular (Poincaré rk. 2) |
| Flat sections | $z^{\Omega/(k+h^\vee)}$ | $z^{2h}\exp(-c/(4z^2))$ |
| Monodromy data | representation of $\pi_1$ | Stokes-filtered local system |
| RH correspondence | Deligne (classical) | D'Agnolo–Kashiwara |
| Stokes rays | 0 | 4 |
| Shadow tower | terminates ($r=3$) | infinite ($r=\infty$) |
| 3d theory | CS gauge theory | HT gravity |

## 10. Ordered chiral homology on elliptic curves: $\int_{E_\tau}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2)$

The preceding sections develop ordered chiral homology at genus 0: the base curve is $\mathbb{P}^1$ (or the formal disk $D$), the bar propagator is $d\log(z_1-z_2)$, and the KZ connection governs the flat transport. The five explicit examples (§9) treat chiral algebras from classes $G$, $L$, $C$, $M$ and the genuinely $\mathsf{E}_1$ Yangian on $D^\times$.

The purpose of this section is to construct the ordered chiral homology of the Yangian $Y_\hbar(\mathfrak{sl}_2)$ on an elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$. This is a genuinely new invariant: (i) the base curve has genus 1, so the bar propagator acquires the elliptic function $\wp_1 = \theta_1'/\theta_1$ in place of $1/z$; (ii) the algebra is genuinely $\mathsf{E}_1$, so the ordered theory differs from the symmetric; (iii) the KZ connection is replaced by the KZB connection, and the $B$-cycle monodromy produces the quantum group parameter $q = e^{2\pi i\hbar}$.

The construction proceeds degree by degree.

### 10.1. Setup: the ordered chiral chain complex on $E_\tau$.

Fix $\tau \in \mathbb{H} = \{\mathrm{Im}(\tau) > 0\}$ and let $E_\tau = \mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$ with coordinate $z$. The ordered configuration space at degree $n$ is

$$\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau) = \{(z_1,\ldots,z_n) \in E_\tau^n : z_i \neq z_j \text{ for } i \neq j\}. \tag{10.1}$$

No ordering on $E_\tau$ itself is needed: "ordered" means the labelling of points is retained (no $\Sigma_n$-quotient).

The bar propagator on $E_\tau$ is

$$\eta_{ij}^{(1)} = d\log E(z_i,z_j) = \wp_1(z_i-z_j,\tau)\,d(z_i-z_j), \tag{10.2}$$

where $E(z,w)$ is the prime form on $E_\tau$ and $\wp_1(z,\tau) = \theta_1'(z|\tau)/\theta_1(z|\tau)$ is the logarithmic derivative of the Jacobi theta function. This is weight 1 (AP27: the bar propagator is always weight 1).

The ordered chiral chain complex at degree $n$ on $E_\tau$ is

$$\mathcal{C}_n^{\mathrm{ord}}(E_\tau,\mathcal{A}) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(E_\tau)\big)\otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{total}}\Big) \tag{10.3}$$

with

$$d_{\mathrm{total}} = d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZB}}. \tag{10.4}$$

The decisive change from genus 0 to genus 1: the KZ connection $\nabla_{\mathrm{KZ}} = d - \sum r_{ij}\,dz_{ij}$ is replaced by the KZB connection

$$\nabla_{\mathrm{KZB}} = d - \frac{1}{k+h^\vee}\sum_{i<j}\Omega_{ij}\,\wp_1(z_{ij},\tau)\,dz_{ij} - \frac{1}{2\pi i(k+h^\vee)}\sum_{i<j}\wp(z_{ij},\tau)\,\Omega_{ij}\,d\tau, \tag{10.5}$$

where $\hbar = 1/(k+h^\vee) = 1/(k+2)$ for $\mathfrak{sl}_2$ (the Yangian convention $\hbar = h_{\mathrm{KZ}}$; see Remark 8.7 for the relationship to the departure-from-critical convention used in §8). The $d\tau$ term is absent at genus 0; it encodes the curvature $\kappa(\mathcal{A})\cdot\omega_1$ from the Hodge class $\omega_1 = c_1(\lambda)$ on $\overline{\mathcal{M}}_{1,n}$.

**Definition 10.1** (Ordered chiral homology on $E_\tau$)**.** The *ordered chiral homology of $\mathcal{A}$ on $E_\tau$* is

$$\int_{E_\tau}^{\mathrm{ord}} \mathcal{A} \ :=\ R\Gamma_{\mathrm{dR}}\big(\mathrm{Ran}^{\mathrm{ord}}(E_\tau),\ \mathcal{F}^{\mathrm{ord}}(\mathcal{A})\big)\ =\ \bigoplus_{n\geq 0} H^*\big(\mathcal{C}_n^{\mathrm{ord}}(E_\tau,\mathcal{A}),\ d_{\mathrm{total}}\big). \tag{10.6}$$

10.2. **Degree 0: the center.** At degree 0 there are no marked points and no configuration space; the ordered chiral homology reduces to the chiral center of $\mathcal{A}$.

**Proposition 10.2** (Degree-0 ordered chiral homology on $E_\tau$)**.** *For $\mathcal{A} = Y_\hbar(\mathfrak{sl}_2)$:*

$$H^*\big(\mathcal{C}_0^{\mathrm{ord}}(E_\tau, Y_\hbar(\mathfrak{sl}_2))\big) = Z(Y_\hbar(\mathfrak{sl}_2)) = \mathbb{C}[\mathrm{qdet}\ T(u)]. \tag{10.7}$$

*The quantum determinant* $\mathrm{qdet}\ T(u) = A(u)\,D(u-\hbar) - B(u)\,C(u-\hbar)$ *generates the center as a polynomial algebra (Molev). The averaging map* $\mathrm{av}_0$ *is an isomorphism (there is nothing to symmetrise at degree* 0*), so* $\ker(\mathrm{av}_0) = 0$.

*Remark* 10.3. The center is independent of the base curve: it depends only on the algebra $\mathcal{A}$ and not on $E_\tau$. This reflects the degree-0 factorisation $\mathcal{D}$-module being a skyscraper at the empty configuration.

10.3. **Degree 1: de Rham cohomology with Yangian coefficients.** At degree 1, $\mathrm{Conf}_1^{\mathrm{ord}}(E_\tau) = E_\tau$ and the ordered chiral chain complex is

$$\mathcal{C}_1^{\mathrm{ord}}(E_\tau, Y_\hbar) = \Big(\Omega^*(E_\tau)\otimes s^{-1}\bar{Y}_\hbar,\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}}\Big), \tag{10.8}$$

where $\bar{Y}_\hbar = \ker(\varepsilon)$ is the augmentation ideal of $Y_\hbar(\mathfrak{sl}_2)$ and $|s^{-1}v| = |v| - 1$. At degree 1 the KZB connection is trivial (no pairs $i<j$), so $\nabla_{\mathrm{KZB}} = d$,

absorbed into $d_{\mathrm{dR}}$. The bar differential $d_{\mathrm{bar}}$ acts on each factor of $s^{-1}\bar{Y}_\hbar$ via the $A_\infty$ structure maps $m_k$ of the chiral algebra.

The de Rham complex of $E_\tau$ has cohomology

$$H^0(E_\tau) = \mathbb{C}, \qquad H^1(E_\tau) = \mathbb{C}^2, \qquad H^2(E_\tau) = \mathbb{C}, \tag{10.9}$$

generated respectively by 1; the $A$-cycle form $dz$ and the $B$-cycle form $d\bar{z}/\mathrm{Im}(\tau)$; and the volume form $dz \wedge d\bar{z}/(2i\,\mathrm{Im}(\tau))$. By the Künneth theorem:

$$H^*\big(\mathcal{C}_1^{\mathrm{ord}}\big) = H^*(E_\tau) \otimes H^*(s^{-1}\bar{Y}_\hbar, d_{\mathrm{bar}}). \tag{10.10}$$

The bar cohomology $H^*(s^{-1}\bar{Y}_\hbar, d_{\mathrm{bar}})$ is the degree-1 bar cohomology of the chiral algebra. For $Y_\hbar(\mathfrak{sl}_2)$ viewed through its RTT presentation, the bar differential at degree 1 is the internal differential of the Yangian (the differential inherited from the dg structure of the ordered bar complex of $V_k(\mathfrak{sl}_2)$ at the first page of the bar spectral sequence). Since $Y_\hbar(\mathfrak{sl}_2)$ is the Koszul dual of $V_k(\mathfrak{sl}_2)$ and the affine algebra is Koszul (class $L$), the bar cohomology at degree 1 is concentrated in degree 1:

$$H^*(s^{-1}\bar{Y}_\hbar, d_{\mathrm{bar}}) = s^{-1}\mathfrak{sl}_2 \ \text{ (concentrated in cohomological degree 1).} \tag{10.11}$$

This is the desuspended Lie algebra: the three generators $\mathcal{E}^{(0)}, \mathcal{F}^{(0)}, H^{(0)}$ in mode zero, shifted by $s^{-1}$.

**Proposition 10.4** (Degree-1 ordered chiral homology on $E_\tau$)**.**

$$H^*\big(\mathcal{C}_1^{\mathrm{ord}}(E_\tau, Y_\hbar(\mathfrak{sl}_2))\big) \cong H^*(E_\tau) \otimes s^{-1}\mathfrak{sl}_2 \cong (s^{-1}\mathfrak{sl}_2)^{\oplus 4}, \tag{10.12}$$

*with the four copies sitting in cohomological degrees* $1, 2, 2, 3$ (*from the tensor product of* $H^{0,1,1,2}(E_\tau)$ *with the degree-*1 *bar cohomology* $s^{-1}\mathfrak{sl}_2$). *The total dimension is* $4 \cdot 3 = 12$.

*Remark* 10.5. At degree 1, $\Sigma_1 = \{e\}$, so the averaging map is the identity and $\ker(\mathrm{av}_1) = 0$. The ordered and symmetric theories agree at degree 1: the non-trivial ordered data enters only at degree $\geq 2$.

*Remark* 10.6 (Comparison with genus 0). At genus 0, $H^*(D^\times)$ contributes only in degree 0 (and degree 1 from the circle at infinity), so the degree-1 cohomology is 3-dimensional. The passage to genus 1 multiplies by $\dim H^*(E_\tau) = 4$: the new $B$-cycle produces the extra copy.

10.4. **Degree 2: the elliptic KZB local system.** At degree 2, the configuration space is $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau) = E_\tau^2 \setminus \Delta$, and the KZB connection acts non-trivially. The ordered chiral chain complex is

$$\mathcal{C}_2^{\mathrm{ord}}(E_\tau, Y_\hbar) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_2^{\mathrm{ord}}(E_\tau)\big) \otimes (s^{-1}\bar{Y}_\hbar)^{\otimes 2},\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZB}}\Big). \tag{10.13}$$

The key new ingredient is the elliptic collision residue: the degree-2 classical $r$-matrix on $E_\tau$ is the Belavin elliptic $r$-matrix

$$r^{\mathrm{ell}}(z,\tau) = k \cdot r^{\mathrm{Belavin}}(z,\tau) = k\Big(\frac{\wp_1(z,\tau)}{2}\, H \otimes H + \phi_+(z,\tau)\, E \otimes F + \phi_-(z,\tau)\, F \otimes E\Big), \tag{10.14}$$

where $\phi_\pm(z,\tau) = \theta_1'(0|\tau)\,\theta_1(z \pm \frac{1}{2}|\tau)/(\theta_1(z|\tau)\,\theta_1(\pm\frac{1}{2}|\tau))$ are Belavin's $\phi$-functions. At $k=0$, $r^{\mathrm{ell}}=0$ (the algebra becomes abelian). The degeneration $\tau \to i\infty$ sends $\wp_1(z,\tau) \to \pi\cot(\pi z) \to 1/z + O(z)$ and $\phi_\pm \to 1/z + O(1)$, recovering $r^{\mathrm{rat}}(z) = k\,\Omega/z$.

The KZB connection at degree 2 on the relative coordinate $w = z_1 - z_2$ is (Bernard [5]; Calaque–Enriquez–Etingof [6])

$$\nabla^{(2)}_{\mathrm{KZB}} = d - \frac{1}{k+h^\vee}\,\Omega\,\wp_1(w,\tau)\,dw - \frac{1}{2\pi i(k+h^\vee)}\,\wp(w,\tau)\,\Omega\,d\tau. \tag{10.15}$$

The flat sections satisfy

$$\frac{\partial\Phi}{\partial w} = \frac{1}{k+h^\vee}\,\wp_1(w,\tau)\,\Omega\,\Phi, \tag{10.16}$$

$$\frac{\partial\Phi}{\partial \tau} = \frac{1}{2\pi i(k+h^\vee)}\,\wp(w,\tau)\,\Omega\,\Phi. \tag{10.17}$$

The flatness of the KZB system follows from the heat equation for theta functions: $\partial_\tau\,\wp_1(z,\tau) = \frac{1}{2\pi i}\,\partial_z\,\wp(z,\tau)$.

**Decomposition by $\mathfrak{sl}_2$-isospin.** On the fundamental representation $V = \mathbb{C}^2$, the Casimir decomposes as before:

$$V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V, \qquad \Omega\big|_{\mathrm{Sym}^2} = +\tfrac{1}{2}, \quad \Omega\big|_{\bigwedge^2} = -\tfrac{3}{2}.$$

The KZB system decomposes into scalar equations on each isospin channel. On the symmetric channel:

$$\frac{\partial\Phi_{\mathrm{sym}}}{\partial w} = \frac{1}{2(k+h^\vee)}\,\wp_1(w,\tau)\,\Phi_{\mathrm{sym}}. \tag{10.18}$$

**Monodromy.** The function $\wp_1(w,\tau) = \theta_1'(w|\tau)/\theta_1(w|\tau)$ is periodic under $w \mapsto w+1$ and quasi-periodic under $w \mapsto w+\tau$ with increment $-2\pi i$. The puncture monodromy (small loop around $w=0$) captures the residue:

$$M_\gamma = \exp\bigl(2\pi i\,\hbar\,\Omega\bigr) \quad \text{where } \hbar = \frac{1}{k+h^\vee}. \tag{10.19}$$

On each channel: $M_\gamma^{\mathrm{sym}} = e^{\pi i/(k+2)}$, $M_\gamma^{\mathrm{alt}} = e^{-3\pi i/(k+2)}$.

The $B$-cycle monodromy ($w \mapsto w+\tau$) is the genuinely new genus-1 datum. The quasi-periodicity $\wp_1(w+\tau) = \wp_1(w) - 2\pi i$ produces:

$$M_B = \exp\bigl(-2\pi i\,\hbar\,\Omega\bigr) = M_\gamma^{-1}. \tag{10.20}$$

*Remark* 10.7 (Dynamical parameter). The $B$-cycle monodromy (10.20) is the leading-order result from the quasi-periodicity of $\wp_1$. The full genus-1 monodromy representation involves the *dynamical parameter* $\lambda \in \mathfrak{h}^*$ (Felder [32]; Felder–Varchenko [33]): the $B$-cycle monodromy operator depends on the basepoint through a shift $\lambda \mapsto \lambda - h^{(i)}$, producing the dynamical Yang–Baxter equation rather than the ordinary YBE (Etingof–Varchenko [19]; see also [34, Thm 5.6]). The gauge-invariant content is the constraint $M_\gamma \cdot [M_A, M_B] = \mathrm{id}$ from $\pi_1(E_\tau \setminus \{0\})$.

On each isospin channel:

$$M_B^{\mathrm{sym}} = e^{-\pi i/(k+2)}, \qquad M_B^{\mathrm{alt}} = e^{3\pi i/(k+2)}. \tag{10.21}$$

**Proposition 10.8** (Degree-2 ordered chiral homology on $E_\tau$)**.** *The degree-2 ordered chiral homology $H^*(\mathcal{C}_2^{\mathrm{ord}}(E_\tau, Y_\hbar))$ is the de Rham cohomology of the KZB local system on $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau)$ with coefficients in $(s^{-1}\bar{Y}_\hbar)^{\otimes 2}$. On evaluation modules $V_a \otimes V_b$:*

(i) *The KZB local system has puncture monodromy $M_\gamma = q^\Omega$ and $B$-cycle monodromy $M_B = q^{-\Omega}$, where $q = e^{2\pi i\hbar}$.*

(ii) *The de Rham cohomology of the KZB local system on $E_\tau \setminus \{0\}$ (equivalently, $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau)$ in the relative coordinate $w = z_1 - z_2$) is:*

$$H^p_{\mathrm{dR}}(E_\tau\backslash\{0\}, \nabla_{\mathrm{KZB}}) = \begin{cases} \ker(M_\gamma - I) \cap \ker(M_B - I) = \ker(q^\Omega - I) & p = 0, \\ (\textit{see (10.23) below}) & p = 1. \end{cases} \tag{10.22}$$

(iii) *For generic $\hbar$ (equivalently, generic level $k$), $q^\Omega$ has distinct eigenvalues on each isospin channel, and $H^0_{\mathrm{dR}} = 0$: there are no global flat sections.*

(iv) *The $H^1$ is computed by the Euler characteristic of the local system. By the Riemann–Hilbert correspondence and the Euler characteristic formula for a rank-$d^2$ local system on a genus-1 curve with one puncture:*

$$\dim H^1_{\mathrm{dR}}(E_\tau \setminus \{0\}, \nabla_{\mathrm{KZB}}) = d^2 \cdot (2 \cdot 1 - 2 + 1) = d^2, \tag{10.23}$$

*where $d = \dim V = 2$, so $\dim H^1 = 4$.*

*Proof.* The Euler characteristic of a local system of rank $r$ on a surface of genus $g$ with $s$ punctures is $\chi = r(2 - 2g - s)$. For $g = 1$, $s = 1$, $r = d^2 = 4$: $\chi = 4(2 - 2 - 1) = -4$. Since $H^0 = 0$ for generic $\hbar$ (the monodromy has no invariant vectors) and $H^2 = 0$ (the puncture forces vanishing), we have $\dim H^1 = 4$. The four-dimensional space decomposes as $3 + 1$ along the isospin channels $\mathrm{Sym}^2 V$ (dimension 3) and $\bigwedge^2 V$ (dimension 1). □

*Remark* 10.9 (The symmetric shadow). The averaging map $\mathrm{av}_2$ at degree 2 sends the ordered de Rham cohomology to $\Sigma_2$-coinvariants. The $\Sigma_2$-action is the composition of the geometric swap $w \mapsto -w$ with the $R$-matrix. On the symmetric channel, the $R$-matrix eigenvalue is $+\hbar/2 + O(\hbar^2)$; on the antisymmetric channel, $-3\hbar/2 + O(\hbar^2)$. The scalar surviving $\mathrm{av}_2$ is $\kappa = 3(k+2)/4$ (equation (9.9)).

The kernel $\ker(\mathrm{av}_2)$ at genus 1 is richer than at genus 0: the $B$-cycle monodromy contributes an additional antisymmetric class (from the $e^{3\pi i/(k+2)}$ eigenvalue of $M_B$ on $\bigwedge^2 V$) that is invisible to the symmetric theory.

10.5. **Degree $n$: elliptic braid group monodromy.** At degree $n$, the fundamental group of $\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)$ is the *elliptic braid group* $\mathrm{EB}_n$, which is an extension

$$1 \to PB_n \to \mathrm{EB}_n \to \pi_1(E_\tau)^n \to 1, \tag{10.24}$$

where $PB_n$ is the pure braid group on $n$ strands and $\pi_1(E_\tau) \cong \mathbb{Z}^2$ contributes the $A$- and $B$-cycle monodromies for each point.

The KZB monodromy representation at degree $n$ is

$$\rho_n^{\mathrm{KZB}} \colon \mathrm{EB}_n \to \mathrm{GL}(V^{\otimes n}). \tag{10.25}$$

The ordered chiral homology at degree $n$ is the de Rham cohomology of the KZB local system on the Fulton–MacPherson compactification:

$$\int_{E_\tau,n}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2) = H^*_{\mathrm{dR}}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(E_\tau),\ \nabla^{(n)}_{\mathrm{KZB}}\big) \otimes (s^{-1}\bar{Y}_\hbar)^{\otimes n}. \tag{10.26}$$

**Proposition 10.10** (Euler characteristic at low degrees)**.** *The twisted Euler characteristics of the KZB local system on* $\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)$*, with fibre* $V^{\otimes n}$ $(V = \mathbb{C}^2)$ *for generic* $\hbar$*, are:*

$$n = 0: \quad \chi = \infty \ \ (\mathbb{C}[\mathrm{qdet}] \textit{ is infinite-dimensional}), \tag{10.27}$$

$$n = 1: \quad \chi = 0 \ \ (\textit{trivial local system on } E_\tau,\ \chi(E_\tau) = 0), \tag{10.28}$$

$$n = 2: \quad \chi = -4 \ \ (\textit{rank } 4 \textit{ on } E_\tau \setminus \{0\}). \tag{10.29}$$

*Proof.* At $n = 1$: $\mathrm{Conf}_1(E_\tau) = E_\tau$ and the KZB connection has no pairwise terms (there are no pairs $i < j$), so the local system is trivial of rank $d = 2$. The twisted Euler characteristic is $\chi = d \cdot \chi(E_\tau) = 2 \cdot 0 = 0$, consistent with $\dim H^0 = 2$, $\dim H^1 = 4$, $\dim H^2 = 2$ from the Künneth computation: $\chi = 2 - 4 + 2 = 0$.

At $n = 2$: the relative coordinate $w = z_1 - z_2$ identifies $\mathrm{Conf}_2^{\mathrm{ord}}(E_\tau)$ (modulo the $E_\tau$-translation in the centre-of-mass) with $E_\tau \setminus \{0\}$, a genus-1 surface with one puncture. The KZB local system has rank $d^2 = 4$. The Euler characteristic of a rank-$r$ local system on a surface of genus $g$ with $s$ punctures is $\chi = r(2 - 2g - s)$: here $\chi = 4(2 - 2 - 1) = -4$. For generic $\hbar$, the monodromy has no invariant vectors and $H^0 = 0$; the puncture forces $H^2 = 0$; so $\dim H^1 = 4$. This is consistent with Proposition 10.8. □

*Remark* 10.11 (Higher degrees). For $n \geq 3$, $\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)$ is a smooth quasi-projective variety of complex dimension $n$. Its topological Euler characteristic vanishes: $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)) = 0$ for all $n \geq 1$ (the product formula $\chi = \prod_{j=0}^{n-1}(\chi(E_\tau) - j) = 0$ since $\chi(E_\tau) = 0$). The twisted Euler characteristic of the KZB local system is also zero: $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau),\ \mathcal{L}_{\mathrm{KZB}}) = 0$ for $n \geq 3$. This does *not* mean the de Rham cohomology vanishes; it means the dimensions of the cohomology groups cancel in alternating sum. The individual Betti numbers with KZB coefficients require explicit computation of the elliptic braid group monodromy at each degree.

10.6. **The total ordered chiral homology and the averaging kernel.** Assembling all degrees, the total ordered chiral homology is

$$\int_{E_\tau}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2) = \bigoplus_{n \geq 0} \int_{E_\tau,n}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2). \tag{10.30}$$

At each degree, the averaging map $\mathrm{av}_n\colon \int^{\mathrm{ord}}_{E_\tau,n} \to (\int^{\mathrm{ord}}_{E_\tau,n})_{\Sigma_n}$ projects to the symmetric shadow. The kernel $\ker(\mathrm{av}) = \bigoplus_n \ker(\mathrm{av}_n)$ carries the genuinely $\mathsf{E}_1$ data that is invisible to the symmetric theory.

**Theorem 10.12** (Structure of $\int^{\mathrm{ord}}_{E_\tau} Y_\hbar(\mathfrak{sl}_2)$)**.** *For generic $\hbar = 1/(k+2)$ with $k \notin \{-2\} \cup \mathbb{Z}_{\leq 0}$:*

(i) (Degree decomposition.) *The degree-by-degree structure is:*

| $n$ | $\int^{\mathrm{ord}}_{E_\tau,n} Y_\hbar$ | $\ker(\mathrm{av}_n)$ | *Shadow* |
|---|---|---|---|
| 0 | $\mathbb{C}[\mathrm{qdet}\ T(u)]$ | 0 | $\mathbb{C}[\mathrm{qdet}]$ |
| 1 | $(s^{-1}\mathfrak{sl}_2)^{\oplus 4}$, *dim* 12 | 0 | $(s^{-1}\mathfrak{sl}_2)^{\oplus 4}$ |
| 2 | *KZB flat sections, dim* 4 | $\bigwedge^2 V \oplus$ *B-cycle class* | $\kappa = 3(k{+}2)/4$ |
| 3 | *elliptic braid monodromy* | $\supseteq \Lambda^3(\mathfrak{g})$ | *traces* |
| $n$ | $\mathrm{EB}_n$*-monodromy of KZB* | *matrix-valued* | *scalar traces* |

(ii) (Genus-1 enhancement of the kernel.) *The kernel* $\ker(\mathrm{av}_2)$ *at genus* 1 *is strictly larger than the kernel at genus* 0*: the B-cycle monodromy* $q^\Omega$ *introduces new antisymmetric classes in the* $\bigwedge^2 V$ *channel that have no genus-*0 *analogue. The Drinfeld associator at degree* 3 *is deformed by the KZB associator* $\Phi_{\mathrm{KZB}}$*, which is a genus-*1 *deformation of* $\Phi_{\mathrm{KZ}}$*.*

(iii) (Degeneration.) *In the limit $\tau \to i\infty$ (the B-cycle collapses and $E_\tau$ degenerates to $\mathbb{C}^\times$), the KZB connection degenerates to the KZ connection, the B-cycle monodromy becomes trivial, and*

$$\lim_{\tau\to i\infty} \int^{\mathrm{ord}}_{E_\tau} Y_\hbar(\mathfrak{sl}_2) = \int^{\mathrm{ord}}_{D^\times} Y_\hbar(\mathfrak{sl}_2), \tag{10.31}$$

*recovering the genus-*0 *ordered chiral homology of §9.*

(iv) (Modular dependence.) *The ordered chiral homology varies with $\tau$ as a section of a vector bundle over $\mathbb{H}$ with flat connection $\nabla_\tau = \partial_\tau - \frac{\hbar}{2\pi i}\sum_{i<j} \wp(z_{ij},\tau)\,\Omega_{ij}$. The $\mathrm{SL}_2(\mathbb{Z})$-action on $\tau$ acts on the ordered chiral homology through the modular transformation of the KZB connection. The kernel* $\ker(\mathrm{av})$ *transforms as a non-trivial representation of the modular group: the B-cycle monodromy is exchanged with the A-cycle monodromy under $\tau \mapsto -1/\tau$: the pair $(M_\gamma, M_B) = (q^\Omega, q^{-\Omega})$ is interchanged. The parameter $q = e^{2\pi i\hbar}$ itself is $\tau$-independent; the modular group acts on the monodromy representation, not on $q$.*

*Proof sketch.* Parts (i)–(ii) follow from the degree-by-degree computation of §§10.2–10.5.

Part (iii): the Belavin $r$-matrix degenerates to the rational $r$-matrix as $\tau \to i\infty$ ($\wp_1(z,\tau) = \theta_1'(z|\tau)/\theta_1(z|\tau) \to 1/z$ and $\phi_\pm(z,\tau) \to 1/z$), the $B$-cycle shrinks, and the KZB flat sections degenerate to KZ flat sections. The degree-by-degree Euler characteristics are continuous under this degeneration.

Part (iv): the modular transformation of the KZB connection is the standard result of Bernard [5]: the $d\tau$ term ensures compatibility with the $\mathrm{SL}_2(\mathbb{Z})$-action, and the flat sections transform with the modular anomaly proportional to $\kappa(\mathcal{A})$ (equation (9.9)). □

*Remark* 10.13 (Comparison with the quantum group). The object $\int_{E_\tau}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2)$ computes the conformal blocks of the Yangian on $E_\tau$. By the Yangian–quantum group deformation [34, Theorem 5.5.1], the genus-1 line-operator algebra is $U_q(\mathfrak{sl}_2)$ with $q = e^{2\pi i\hbar}$. The ordered chiral homology packages the full KZB monodromy data into a single invariant: it sees both the $R$-matrix ($A$-cycle, Yangian data) and the quantum group parameter ($B$-cycle, $q$-data) simultaneously. The symmetric shadow $(\int_{E_\tau}^{\mathrm{ord}})_{\Sigma_n}$ retains only the scalar traces (characters of $U_q(\mathfrak{sl}_2)$ representations), which are the WZW conformal blocks; the kernel ker(av) carries the matrix-valued monodromy from which knot invariants on the torus are extracted.

*Remark* 10.14 (Roots of unity). At integer level $k \in \mathbb{Z}_{>0}$, the quantum group parameter is $q = e^{2\pi i/(k+2)}$ and $q^{k+2} = 1$. The KZB local system acquires finite monodromy, and the ordered chiral homology at each degree becomes a finite-dimensional representation of the mapping class group of $E_\tau$ with $n$ marked points. The $k+1$ integrable modules $V_0, \ldots, V_k$ of $U_q(\mathfrak{sl}_2)$ correspond to the truncation of the degree-by-degree decomposition: at degree $n > k$, the representation theory constrains certain components to vanish.

*Remark* 10.15 (The elliptic $R$-matrix as ordered datum). The Belavin elliptic $r$-matrix $r^{\mathrm{ell}}(z,\tau) = k \cdot r^{\mathrm{Belavin}}(z,\tau)$ is the degree-2 projection of the genus-1 Maurer–Cartan element $\Theta_{\mathcal{A}}^{(1)} \in \mathfrak{g}_{\mathcal{A}}^{\mathsf{E}_1}$. It satisfies the classical Yang–Baxter equation with spectral parameter (verified numerically in `elliptic_rmatrix_shadow.py`), and its averaging gives $\kappa = 3(k+2)/4$ (equation (9.9)), the same value as at genus 0. The agreement is forced: the Sugawara shift $\dim(\mathfrak{g})/2 = 3/2$ comes from the simple-pole self-contraction, a local computation independent of the global topology of the base curve.

10.7. **Recovery of the Verlinde formula.** At integer level $k \geq 1$, the quantum group parameter $q = e^{2\pi i/(k+2)}$ is a primitive $(k+2)$-th root of unity. The representation category of $U_q(\mathfrak{sl}_2)$ is semisimple with $k+1$ integrable modules $V_0, \ldots, V_k$, and the KZB local system has finite monodromy. The ordered chiral chain complex of $V_k(\mathfrak{sl}_2)$ on a genus-$g$ curve $\Sigma_g$ therefore computes a finite-dimensional invariant at each degree, and the symmetric coinvariants recover the space of conformal blocks (Tsuchiya–Ueno–Yamada [46]). The following proposition shows that the resulting dimension is given by the Verlinde formula [47].

**Proposition 10.16** (Verlinde formula from ordered chiral homology)**.** *Let $k \geq 1$ be a positive integer, and let $S_{jl} = \sqrt{2/(k+2)}\,\sin\big(\pi(j+1)(l+1)/(k+2)\big)$ be the modular $S$-matrix for $\widehat{\mathfrak{sl}}_2$ at level $k$, where $j, l \in \{0, 1, \ldots, k\}$.*

(i) (Genus 0.) *The ordered chiral homology at genus* 0 *with no marked points recovers the unique vacuum:*

$$\dim H^0\big(\mathcal{M}_{0,0},\, V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^2 = 1. \tag{10.32}$$

*This is the unitarity of the $S$-matrix: $\sum_l S_{jl}^2 = \delta_{j0}$ for $j = 0$ follows from the orthogonality of the sine functions.*

(ii) (Genus 1.) *The ordered chiral homology at genus* 1 *recovers the number of integrable representations:*

$$\dim H^0\big(\mathcal{M}_{1,0},\, V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^0 = k+1. \tag{10.33}$$

*By the Zhu algebra identification $A(V_k(\mathfrak{sl}_2)) \cong U(\mathfrak{sl}_2)/(e^{k+1}, f^{k+1})$, the $k+1$ simple modules of $A(V_k)$ are $V_0, \dots, V_k$, and the genus-*1 *conformal blocks are their characters. From the ordered chiral homology: Proposition* 10.2 *computes the degree-*0 *center at integrable level as $\mathbb{C}^{k+1}$ (the center of the integrable quotient of $Y_\hbar$ at the root of unity $q = e^{2\pi i/(k+2)}$).*

(iii) (General genus.) *At genus $g \geq 2$, the Verlinde formula gives:*

$$Z_g(k) := \dim H^0\big(\mathcal{M}_{g,0},\, V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^{2-2g}. \tag{10.34}$$

*This is an integer for all $g \geq 0$, $k \geq 1$. The first values are:*

| $k$ | $Z_0$ | $Z_1$ | $Z_2$ | $Z_3$ | $Z_4$ |
|---|---|---|---|---|---|
| 1 | 1 | 2 | 4 | 8 | 16 |
| 2 | 1 | 3 | 10 | 36 | 136 |
| 3 | 1 | 4 | 20 | 120 | 800 |
| 4 | 1 | 5 | 35 | 329 | 3,611 |
| 5 | 1 | 6 | 56 | 784 | 13,328 |

*At $k = 1$: $S_{00} = S_{01} = 1/\sqrt{2}$, so $Z_g(1) = 2 \cdot (1/\sqrt{2})^{2-2g} = 2^g$.*

(iv) (Handle attachment.) *The ordered bar complex factorization under non-separating sewing recovers the TQFT handle-attachment formula: each isospin channel $j$ contributes a handle operator $H_j = S_{0j}^{-2}$, so*

$$Z_{g+1} = \sum_{j=0}^{k} S_{0j}^{-2} \cdot S_{0j}^{2-2g} = \sum_{j=0}^{k} S_{0j}^{2-2(g+1)}. \tag{10.35}$$

(v) (Separating factorization.) *Under separating degeneration* $\Sigma_g \rightsquigarrow \Sigma_{g_1} \cup \Sigma_{g_2}$ *with* $g = g_1 + g_2$*, the bar complex factorization gives*

$$Z_g = \sum_{j=0}^{k} S_{0j}^{2-2g_1} \cdot S_{0j}^{2-2g_2} \cdot S_{0j}^{-2} = \sum_{j=0}^{k} S_{0j}^{2-2g}. \tag{10.36}$$

*Proof.* (i) The identity $\sum_j S_{0j}^2 = 1$ is the $(0,0)$-entry of $S \cdot S^T = I$, which follows from the orthogonality relation $\sum_{j=0}^{k} \sin(\pi(j+1)a/(k+2)) \sin(\pi(j+1)b/(k+2)) = (k+2)\delta_{ab}/2$. From the ordered chiral homology: at genus 0 with no marked points, the conformal block space is the space of vacua $\mathbb{C} \cdot |0\rangle$, which is one-dimensional.

(ii) At genus 1, $S_{0j}^0 = 1$ for all $j$, so $Z_1 = k+1$. This has three independent derivations:

- (S-matrix.) Direct computation: $\sum_{j=0}^{k} 1 = k+1$.
- (Zhu algebra.) The Zhu algebra $A(V_k(\mathfrak{sl}_2))$ has $k+1$ simple modules; each contributes a one-dimensional space of genus-1 conformal blocks (the character).
- (Ordered center.) Proposition 10.2 identifies the degree-0 center of $Y_\hbar(\mathfrak{sl}_2)$ at integrable level $k$ as $\mathbb{C}^{k+1}$ (the center of the integrable quotient). The averaging map $\mathrm{av}_0$ is an isomorphism (nothing to symmetrise at degree 0), so all $k+1$ dimensions survive.

(iii) The Verlinde formula [47] gives $Z_g = \sum_j S_{0j}^{2-2g}$ as the dimension of the space of conformal blocks on a genus-$g$ curve with no insertions. From the ordered chiral homology: the symmetric coinvariants of the ordered chain complex compute the conformal blocks via the factorization property (TUY [46]). The KZB local system at integrable level decomposes into $k+1$ channels labeled by integrable representations; each channel $j$ contributes $S_{0j}^{2-2g}$ to the partition function. The integrality of $Z_g$ follows from the identification with the rank of the Verlinde bundle over $\mathcal{M}_g$.

(iv) Handle attachment: a genus-$g$ surface with a non-separating cycle cut open gives a genus-$(g-1)$ surface with two extra marked points. The bar complex sewing map reglues these marked points, summing over the $k+1$ intermediate representations with the propagator $S_{0j}^{-2}$ (the inverse of the $j$-th eigenvalue of the sewing operator). This gives $Z_{g+1} = \sum_j H_j \cdot S_{0j}^{2-2g}$.

(v) Separating factorization: the bar complex factorization under separating degeneration (Theorem A for the ordered bar complex on $\mathrm{Ran}^{\mathrm{ord}}$) decomposes the genus-$g$ partition function into a sum over channels of the product of the two lower-genus contributions, normalized by the propagator $S_{0j}^{-2}$.

All numerical values in the table have been verified by direct computation in `verlinde_ordered_engine.py` using three independent code paths (direct $S$-matrix summation, quantum-dimension formula, and handle recursion), cross-checked against the Verlinde table of `verlinde_shadow_algebra.py` (198 tests, all passing). □

*Remark* 10.17 (Verlinde bundle vs shadow tower). The Verlinde dimension $Z_g$ and the shadow partition function $F_g = \kappa \cdot \lambda_g$ are different invariants of the same chiral algebra. $Z_g$ is the rank of the Verlinde bundle over $\mathcal{M}_g$ (an integer, from the $S$-matrix); $F_g$ is the first Chern class integrated over $\overline{\mathcal{M}}_g$ (a rational number, from $\kappa$). For $V_k(\mathfrak{sl}_2)$: $\kappa = 3(k+2)/4$ (equation (9.9)), $F_1 = \kappa/24$, and $Z_1 = k+1$. The two invariants encode complementary data: $Z_g$ counts the conformal blocks (combinatorial, from the modular tensor category), while $F_g$ measures the curvature (analytic, from the Hodge class $\omega_g = c_1(\lambda)$ on $\overline{\mathcal{M}}_g$; AP130: $\omega_g$ lives on $\overline{\mathcal{M}}_g$, not on the curve $\Sigma_g$).

*Remark* 10.18 (The ordered refinement). The Verlinde dimension $Z_g = \dim(\int_{\Sigma_g}^{\mathrm{ord}} V_k)_{\Sigma_n}$ is the dimension of the symmetric coinvariants. The full ordered chiral homology $\int_{\Sigma_g}^{\mathrm{ord}} V_k(\mathfrak{sl}_2)$ is strictly larger: its degree-$n$ component carries the full KZB monodromy representation of the elliptic braid group $\mathrm{EB}_n$ (Proposition 10.8, Theorem 10.12), from which the Verlinde dimension is recovered by taking $\Sigma_n$-coinvariants and restricting to $n = 0$. The kernel $\ker(\mathrm{av}_n)$ carries the matrix-valued monodromy data: $R$-matrices, $6j$-symbols, and (at genus $\geq 1$) dynamical quantum group data.

## 11. Ordered chiral homology at genus 2: $\int_{\Sigma_2}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2)$

The genus-1 construction of §10 replaces the rational propagator $1/z$ by the elliptic function $\wp_1 = \theta_1'/\theta_1$ and the KZ connection by the KZB connection. At genus 2, three qualitatively new features enter: (i) the prime form $E(z,w)$ on a genus-2 Riemann surface $\Sigma_2$ replaces the Jacobi theta function; (ii) the period matrix $\Omega \in \mathbb{H}_2 = \{Z \in M_2(\mathbb{C}) : Z = Z^t,\ \mathrm{Im}\, Z > 0\}$ is a $2 \times 2$ symmetric matrix, and the KZB connection coefficients involve the matrix $(\mathrm{Im}\,\Omega)^{-1}$ rather than the scalar $1/\mathrm{Im}(\tau)$; (iii) the fundamental group $\pi_1(\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2))$ is the genus-2 surface braid group, an extension of the pure braid group by $\pi_1(\Sigma_2)^n \cong (\mathbb{Z}^4)^n$ with four generators per point (two $A$-cycles, two $B$-cycles).

11.1. **Setup: the genus-2 ordered chiral chain complex.** Let $\Sigma_2$ be a smooth projective curve of genus 2 with Siegel period matrix $\Omega = \left(\begin{smallmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{12} & \Omega_{22} \end{smallmatrix}\right) \in \mathbb{H}_2$. Fix a symplectic basis $\{A_1, A_2, B_1, B_2\}$ of $H_1(\Sigma_2, \mathbb{Z})$ with intersection pairing $A_i \cdot B_j = \delta_{ij}$, and let $\omega_1, \omega_2 \in H^0(\Sigma_2, K_{\Sigma_2})$ be the normalised holomorphic differentials satisfying $\oint_{A_i} \omega_j = \delta_{ij}$, $\oint_{B_i} \omega_j = \Omega_{ij}$.

The bar propagator on $\Sigma_2$ is

$$\eta_{ij}^{(2)} = d \log E(z_i, z_j), \tag{11.1}$$

where $E(z,w)$ is the *prime form* on $\Sigma_2$: a section of $K_{\Sigma_2}^{-1/2} \boxtimes K_{\Sigma_2}^{-1/2}$ with a simple zero along the diagonal $z = w$ and no other zeros or poles. This is weight 1 (the bar propagator is always weight 1). Explicitly, in terms of the

Riemann theta function $\theta[\delta](z|\Omega)$ with odd characteristic $\delta$:

$$E(z,w) = \frac{\theta[\delta]\big(\int_w^z \vec{\omega} \,\big|\, \Omega\big)}{h_\delta(z)\, h_\delta(w)}, \tag{11.2}$$

where $\vec{\omega} = (\omega_1, \omega_2)^t$, $\int_w^z \vec{\omega} \in \mathbb{C}^2$ is the Abel–Jacobi map, and $h_\delta$ is a half-differential depending on the odd spin structure $\delta$. In the genus-1 limit ($\Omega_{12} \to 0$, $\Omega_{22} \to i\infty$), one component decouples and $E(z,w) \to \theta_1(z - w|\Omega_{11})/\theta_1'(0|\Omega_{11})$, recovering the elliptic prime form.

*Remark* 11.1 (The Szegő kernel at genus 2). The logarithmic derivative of the prime form,

$$S_2(z,w) = d_z \log E(z,w) = \partial_z \log \theta[\delta]\Big(\int_w^z \vec{\omega} \,\Big|\, \Omega\Big) \cdot \omega(z) - d_z \log h_\delta(z), \tag{11.3}$$

is the genus-2 Szegő kernel. It has a simple pole at $z = w$ with residue 1 (matching $1/(z-w)$ locally) and is quasi-periodic under the $B$-cycles:

$$S_2(z + B_\alpha, w) = S_2(z,w) - 2\pi i\, \omega_\alpha(z), \qquad \alpha = 1, 2. \tag{11.4}$$

The two-component quasi-periodicity increment $-2\pi i\, \omega_\alpha(z)$ replaces the scalar $-2\pi i\, dz$ of the genus-1 case; it is the source of the $2 \times 2$ matrix structure in the genus-2 KZB connection.

The ordered chiral chain complex at degree $n$ on $\Sigma_2$ is

$$\mathcal{C}_n^{\mathrm{ord}}(\Sigma_2, \mathcal{A}) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\Sigma_2)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes n},\ \ d_{\mathrm{total}}\Big) \tag{11.5}$$

with

$$d_{\mathrm{total}} = d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZB}}^{(g=2)}. \tag{11.6}$$

**Definition 11.2** (Ordered chiral homology on $\Sigma_2$)**.** The *ordered chiral homology of* $\mathcal{A}$ *on* $\Sigma_2$ is

$$\int_{\Sigma_2}^{\mathrm{ord}} \mathcal{A} \ := \ \bigoplus_{n \geq 0} H^*\big(\mathcal{C}_n^{\mathrm{ord}}(\Sigma_2, \mathcal{A}),\ d_{\mathrm{total}}\big). \tag{11.7}$$

**Curvature.** The fiberwise bar differential at genus 2 acquires curvature (UNIFORM-WEIGHT):

$$d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_2, \tag{11.8}$$

where $\omega_2 = c_1(\lambda) \in H^2(\overline{\mathcal{M}}_2)$ is the Hodge class on $\overline{\mathcal{M}}_2$ (a class on the *moduli space*, not a form on the curve), and $\kappa(\mathcal{A}) = 3(k+2)/4$ for $\mathcal{A} = Y_\hbar(\mathfrak{sl}_2)$. The period-corrected differential $D^{(2)} = d_{\mathrm{total}} - \kappa(\mathcal{A}) \cdot \omega_2 \cdot [\text{Hodge correction}]$ restores $(D^{(2)})^2 = 0$ at the total level.

11.2. **The genus-2 KZB connection.** At degree $n \geq 2$, the KZB connection on $\Sigma_2$ is the genus-2 analogue of (10.5) (Bernard [5], Calaque–Enriquez–Etingof [6] for the universal formulation at arbitrary genus). The connection has two components: a spatial part on $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$ and a modular part along the Siegel upper half-space $\mathbb{H}_2$.

The spatial part at degree $n$ is

$$\nabla^{(2)}_{\text{spatial}} = d - \hbar \sum_{i<j} \big(S_2(z_i, z_j)\big)\, \Omega_{ij}, \tag{11.9}$$

where $\hbar = 1/(k + h^\vee) = 1/(k+2)$ for $\mathfrak{sl}_2$, $S_2(z_i, z_j) = d_{z_i} \log E(z_i, z_j)$ is the Szegő kernel (11.3), and $\Omega_{ij}$ is the Casimir acting in positions $i, j$. This reduces to the genus-1 spatial connection $\hbar\, \Omega\, \wp_1(w, \tau)\, dw$ in the degeneration $\Sigma_2 \to E_\tau$ (one handle collapses).

The modular part is a connection along the Siegel directions $\Omega_{\alpha\beta}$ ($\alpha, \beta = 1, 2$):

$$\nabla^{(2)}_{\text{modular}} = \sum_{\alpha \leq \beta} \Big( \partial_{\Omega_{\alpha\beta}} - \frac{1}{2\pi i}\, \hbar \sum_{i<j} \partial_{z_i} \partial_{z_j} \log E(z_i, z_j) \cdot \omega_\alpha(z_i)\, \omega_\beta(z_j)\, \Omega_{ij} \Big) d\Omega_{\alpha\beta}. \tag{11.10}$$

At genus 1, the Siegel matrix reduces to the scalar $\tau = \Omega_{11}$, $\omega_1(z) = dz$, and the modular part reduces to $\partial_\tau - \frac{\hbar}{2\pi i}\, \wp(w, \tau)\, \Omega\, d\tau$ of equation (10.17). At genus 2, the modular part has *three* independent components $d\Omega_{11}$, $d\Omega_{12}$, $d\Omega_{22}$ (the three directions in $\mathbb{H}_2$). The flatness of the full KZB system follows from the genus-2 heat equation for the theta function:

$$\frac{\partial}{\partial \Omega_{\alpha\beta}} \theta[\delta](z|\Omega) = \frac{1}{2\pi i}\, \frac{\partial^2}{\partial z_\alpha\, \partial z_\beta} \theta[\delta](z|\Omega). \tag{11.11}$$

*Remark* 11.3 (The $(2 \times 2)$-matrix structure). The genus-2 KZB system makes essential use of the matrix $(\mathrm{Im}\, \Omega)^{-1}$. In the de Rham description, the harmonic projection requires the inner product on $H^1(\Sigma_2)$ given by the Hodge star, which on a genus-2 surface is encoded by the $2 \times 2$ positive-definite matrix $\mathrm{Im}\, \Omega$. Its inverse

$$(\mathrm{Im}\, \Omega)^{-1} = \frac{1}{\det(\mathrm{Im}\, \Omega)} \begin{pmatrix} \mathrm{Im}\, \Omega_{22} & -\mathrm{Im}\, \Omega_{12} \\ -\mathrm{Im}\, \Omega_{12} & \mathrm{Im}\, \Omega_{11} \end{pmatrix} \tag{11.12}$$

is a genuine $2 \times 2$ matrix (not a scalar as at genus 1 where $(\mathrm{Im}\, \Omega)^{-1} = 1/\mathrm{Im}(\tau)$). The $B$-cycle monodromy at genus 2 involves the matrix exponential $\exp(-2\pi i\, \hbar\, \Omega_{ij})$ with the Casimir acting through a $2 \times 2$ block corresponding to the two $B$-cycles, producing a *doubly-dynamical* parameter.

11.3. **Degree 0: the center.** At degree 0, there are no marked points and no configuration space; the ordered chiral homology reduces to the chiral center of $\mathcal{A}$, independent of the base curve.

**Proposition 11.4** (Degree-0 ordered chiral homology on $\Sigma_2$)**.** *For* $\mathcal{A} = Y_\hbar(\mathfrak{sl}_2)$*:*

$$H^*\big(\mathcal{C}_0^{\mathrm{ord}}(\Sigma_2, Y_\hbar(\mathfrak{sl}_2))\big) = Z(Y_\hbar(\mathfrak{sl}_2)) = \mathbb{C}[\mathrm{qdet}\ T(u)]. \tag{11.13}$$

*This is identical to degree* 0 *at genus* 0 *and genus* 1 (*Proposition* 10.2).

*Proof.* The degree-0 factorisation $\mathcal{D}$-module is a skyscraper at the empty configuration. It depends only on the algebra $\mathcal{A}$, not on the base curve. □

11.4. **Degree 1: de Rham cohomology of $\Sigma_2$ with Yangian coefficients.** At degree 1, $\mathrm{Conf}_1^{\mathrm{ord}}(\Sigma_2) = \Sigma_2$ and the KZB connection has no pairwise terms ($i < j$ is empty), so the ordered chiral chain complex is

$$\mathcal{C}_1^{\mathrm{ord}}(\Sigma_2, Y_\hbar) = \Big(\Omega^*(\Sigma_2) \otimes s^{-1}\bar{Y}_\hbar,\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}}\Big), \tag{11.14}$$

where $\bar{Y}_\hbar = \ker(\varepsilon)$ is the augmentation ideal. The de Rham complex of $\Sigma_2$ has cohomology

$$H^0(\Sigma_2) = \mathbb{C}, \qquad H^1(\Sigma_2) = \mathbb{C}^4, \qquad H^2(\Sigma_2) = \mathbb{C}, \tag{11.15}$$

generated respectively by 1; the four cycle forms $\omega_1, \omega_2$ ($A$-cycles) and $\bar{\omega}_1, \bar{\omega}_2$ ($B$-cycles, with normalisation involving $(\mathrm{Im}\,\Omega)^{-1}$); and the volume form. By the Künneth theorem, with bar cohomology $H^*(s^{-1}\bar{Y}_\hbar, d_{\mathrm{bar}}) = s^{-1}\mathfrak{sl}_2$ concentrated in degree 1 (equation (10.11)):

**Proposition 11.5** (Degree-1 ordered chiral homology on $\Sigma_2$)**.**

$$H^*\big(\mathcal{C}_1^{\mathrm{ord}}(\Sigma_2, Y_\hbar(\mathfrak{sl}_2))\big) \cong H^*(\Sigma_2) \otimes s^{-1}\mathfrak{sl}_2 \cong (s^{-1}\mathfrak{sl}_2)^{\oplus 6}, \tag{11.16}$$

*with the six copies sitting in cohomological degrees* $1, 2, 2, 2, 2, 3$ (*from the tensor product of* $H^{0,1,1,1,1,2}(\Sigma_2)$ *with the degree-*1 *bar cohomology* $s^{-1}\mathfrak{sl}_2$). *The total dimension is* $6 \cdot 3 = 18$.

*Proof.* Since the KZB connection is trivial at degree 1 (no pairs $i < j$), the complex splits as a tensor product. The bar cohomology at degree 1 is $s^{-1}\mathfrak{sl}_2$ by Koszulness of $V_k(\mathfrak{sl}_2)$ (equation (10.11)). The de Rham Betti numbers of $\Sigma_2$ are $(1, 4, 1)$, and the Künneth formula gives $6 = 1 + 4 + 1$ copies of $s^{-1}\mathfrak{sl}_2$. □

*Remark* 11.6 (Comparison with genus 0 and genus 1). At genus 0, the degree-1 cohomology is 3-dimensional (1 copy of $s^{-1}\mathfrak{sl}_2$ from $H^0(D^\times)$, with additional degree-1 contribution from the circle). At genus 1, the Betti numbers $(1, 2, 1)$ give $12 = 4 \cdot 3$. At genus 2, the Betti numbers $(1, 4, 1)$ give $18 = 6 \cdot 3$. The pattern at genus $g$ is $\dim H^*(\mathcal{C}_1^{\mathrm{ord}}) = 3(2g + 2)$ from $\dim H^*(\Sigma_g) = 2 + 2g$, reflecting the $2g$ extra 1-cycles on $\Sigma_g$.

11.5. **Degree 2: the genus-2 KZB local system.** At degree 2, the configuration space $\mathrm{Conf}_2^{\mathrm{ord}}(\Sigma_2) = \Sigma_2^2 \setminus \Delta$ carries the KZB connection with non-trivial pairwise interaction. After separating the centre-of-mass coordinate (translation on $\Sigma_2$ via the Jacobian $\mathrm{Jac}(\Sigma_2) = \mathbb{C}^2/(\mathbb{Z}^2 + \Omega\,\mathbb{Z}^2)$, which has dimension 2), the relative configuration is a punctured genus-2 surface: the KZB local system lives on $\Sigma_2 \setminus \{0\}$, a genus-2 surface with one puncture.

The ordered chiral chain complex at degree 2 is

$$\mathcal{C}_2^{\mathrm{ord}}(\Sigma_2, Y_\hbar) = \Big(\Omega^*_{\log}\big(\overline{\mathrm{FM}}_2^{\mathrm{ord}}(\Sigma_2)\big) \otimes (s^{-1}\bar{Y}_\hbar)^{\otimes 2},\ \ d_{\mathrm{dR}} + d_{\mathrm{bar}} + \nabla_{\mathrm{KZB}}^{(g=2)}\Big). \tag{11.17}$$

The collision residue is governed by the genus-2 $r$-matrix

$$r^{(g=2)}(z_i, z_j) = \hbar\, S_2(z_i, z_j)\, \Omega = \frac{1}{k+h^\vee}\, d_{z_i} \log E(z_i, z_j)\, \Omega, \tag{11.18}$$

where $\hbar = 1/(k+h^\vee) = 1/(k+2)$ and $\Omega$ is the $\mathfrak{sl}_2$ Casimir (in the KZ normalisation consistent with equation (10.5)).

**Decomposition by $\mathfrak{sl}_2$-isospin.** On $V \otimes V = \mathrm{Sym}^2 V \oplus \bigwedge^2 V$ with $V = \mathbb{C}^2$:

$$\Omega\big|_{\mathrm{Sym}^2} = +\tfrac{1}{2}, \qquad \Omega\big|_{\bigwedge^2} = -\tfrac{3}{2},$$

exactly as at genus 0 and genus 1. The KZB system decomposes into scalar equations on each isospin channel, now with genus-2 coefficients.

**Monodromy.** The fundamental group $\pi_1(\Sigma_2 \setminus \{0\})$ has presentation

$$\pi_1(\Sigma_2 \setminus \{0\}) = \big\langle a_1, b_1, a_2, b_2, \gamma \;\big|\; [a_1, b_1]\,[a_2, b_2]\,\gamma = 1 \big\rangle \tag{11.19}$$

with $a_\alpha, b_\alpha$ the $A$- and $B$-cycle generators ($\alpha = 1, 2$) and $\gamma$ the puncture loop. The KZB connection produces a monodromy representation

$$\rho^{(2)} \colon \pi_1(\Sigma_2 \setminus \{0\}) \to \mathrm{GL}(V \otimes V) \tag{11.20}$$

with monodromy matrices:

- Puncture loop: $M_\gamma = \exp(2\pi i\, \hbar\, \Omega)$ (identical to genus 0 and genus 1, capturing the local collision residue).
- $A$-cycles: $M_{a_\alpha}$ ($\alpha = 1, 2$); the $A$-cycle monodromy is trivial at leading order since $S_2(z, w)$ is periodic under $A$-cycles: $S_2(z + A_\alpha, w) = S_2(z, w)$.
- $B$-cycles: $M_{b_\alpha}$ ($\alpha = 1, 2$); the quasi-periodicity (11.4) gives a shift by $-2\pi i\, \omega_\alpha$, producing

$$M_{b_\alpha} = \exp\big(-2\pi i\, \hbar\, \Omega \cdot \omega_\alpha\big), \qquad \alpha = 1, 2. \tag{11.21}$$

  At genus 1, $\omega_1 = dz$ is a scalar and $M_B = M_\gamma^{-1}$; at genus 2, the two $B$-cycle monodromies $M_{b_1}, M_{b_2}$ are independent and involve the two holomorphic differentials $\omega_1, \omega_2$ separately.

The constraint from the fundamental group relation is

$$[M_{a_1}, M_{b_1}]\,[M_{a_2}, M_{b_2}]\, M_\gamma = I. \tag{11.22}$$

*Remark* 11.7 (Dynamical parameters at genus 2). The $B$-cycle monodromies at genus 2 depend on a *two-component* dynamical parameter $\lambda = (\lambda_1, \lambda_2) \in \mathfrak{h}^* \oplus \mathfrak{h}^*$, replacing the single dynamical parameter $\lambda \in \mathfrak{h}^*$ of the genus-1 theory (Remark 10.7). The dynamical Yang–Baxter equation is replaced by a system of two coupled equations, one for each $B$-cycle. This is the genus-2 instantiation of the Felder dynamical $R$-matrix theory [32, 33].

**Proposition 11.8** (Degree-2 ordered chiral homology on $\Sigma_2$)**.** *The degree-2 ordered chiral homology $H^*(\mathcal{C}_2^{\mathrm{ord}}(\Sigma_2, Y_\hbar))$ is the de Rham cohomology of the KZB local system on $\mathrm{Conf}_2^{\mathrm{ord}}(\Sigma_2)$ with coefficients in $(s^{-1}\bar{Y}_\hbar)^{\otimes 2}$. On evaluation modules $V_a \otimes V_b$ ($V = \mathbb{C}^2$, so the local system has rank $d^2 = 4$):*

(i) *The KZB local system on* $\Sigma_2 \setminus \{0\}$ (*a genus-2 surface with one puncture*) *has monodromy representation* (11.20) *with generators* $M_{a_1}, M_{b_1}, M_{a_2}, M_{b_2}, M_\gamma$.

(ii) *For generic* $\hbar$*, the monodromy has no invariant vectors:* $H^0_{\mathrm{dR}} = 0$.

(iii) *The Euler characteristic of a rank-r local system on a surface of genus g with s punctures is* $\chi = r(2 - 2g - s)$. *For* $g = 2$, $s = 1$, $r = d^2 = 4$*:*

$$\chi = 4 \cdot (2 - 4 - 1) = -12. \tag{11.23}$$

(iv) *Since* $H^0 = 0$ *for generic* $\hbar$ *and* $H^2 = 0$ (*the puncture forces vanishing*)*, we obtain:*

$$\dim H^1_{\mathrm{dR}}(\Sigma_2 \setminus \{0\}, \nabla_{\mathrm{KZB}}) = 12. \tag{11.24}$$

*Proof.* The surface $\Sigma_2 \setminus \{0\}$ has genus $g = 2$ and $s = 1$ punctures. The Euler characteristic of a local system of rank $r$ is $\chi = r(2-2g-s)$: $\chi = 4(2-4-1) = 4 \cdot (-3) = -12$. For generic $\hbar$ (equivalently, generic level $k$), the monodromy around the puncture is $M_\gamma = \exp(2\pi i\, \hbar\, \Omega)$ with eigenvalues $e^{\pi i/(k+2)}$ (on $\mathrm{Sym}^2 V$, dimension 3) and $e^{-3\pi i/(k+2)}$ (on $\bigwedge^2 V$, dimension 1); generically neither is 1, so $\ker(M_\gamma - I) = 0$ and $H^0 = 0$. The four $B$-cycle monodromies $M_{b_1}, M_{b_2}$ impose further non-trivial constraints, reinforcing $H^0 = 0$. Poincaré–Verdier duality on $\Sigma_2 \setminus \{0\}$ (a non-compact surface) gives $H^2 = (H^0_c)^\vee = 0$ for generic monodromy; alternatively, the puncture forces $H^2 = 0$ by a residue argument. The alternating sum gives $0 - \dim H^1 + 0 = -12$, so $\dim H^1 = 12$. The 12-dimensional space decomposes along the isospin channels: $\mathrm{Sym}^2 V$ (rank 3) contributes $3 \cdot 3 = 9$, and $\bigwedge^2 V$ (rank 1) contributes $1 \cdot 3 = 3$; total $9 + 3 = 12$. □

*Remark* 11.9 (Comparison of $H^1$ dimensions). The degree-2 $H^1$ dimension grows linearly with genus:

| $g$ | $\chi(\Sigma_g \setminus \{0\})$ | $\chi = r \cdot (2 - 2g - 1)$ | $\dim H^1$ |
|---|---|---|---|
| 0 | 1 | $4 \cdot 1 = 4$ | 4 |
| 1 | $-1$ | $4 \cdot (-1) = -4$ | 4 |
| 2 | $-3$ | $4 \cdot (-3) = -12$ | 12 |
| $g$ | $1 - 2g$ | $4(1 - 2g)$ | $4(2g - 1)$ |

The jump from $\dim H^1 = 4$ at $g = 1$ to $\dim H^1 = 12$ at $g = 2$ reflects the richer monodromy: at genus 1, the monodromy group is generated by three elements $(M_\gamma, M_a, M_b)$ with one relation; at genus 2, by five elements $(M_\gamma, M_{a_1}, M_{b_1}, M_{a_2}, M_{b_2})$ with one relation, allowing a 12-dimensional space of flat sections with prescribed singular behaviour at the puncture. The genus-0 entry $\dim H^1 = 4$ uses $\chi(\mathbb{P}^1 \setminus \{0, \infty\}) = 0$ for the once-punctured sphere (the relative coordinate lives on $\mathbb{C}^\times$), so $\chi = 0$ and the $H^1$-dimension does not follow from $\chi$ alone; it is computed directly from KZ flat sections (cf. §10.4).

*Remark* 11.10 (The symmetric shadow at genus 2). The averaging map $\mathrm{av}_2$ at degree 2 sends the 12-dimensional ordered de Rham cohomology to $\Sigma_2$-coinvariants. The scalar surviving $\mathrm{av}_2$ is $\kappa = 3(k+2)/4$ (equation (9.9)), the same value as at genus 0 and genus 1: the averaging map is a local computation (residue extraction), independent of the global topology. The kernel $\ker(\mathrm{av}_2)$ at genus 2 is richer than at genus 1: the four $B$-cycle monodromies contribute independent antisymmetric classes that are invisible to the symmetric theory.

11.6. **Degree $n$: genus-2 surface braid group monodromy.** At degree $n$, the fundamental group of $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$ is the genus-2 *surface braid group* $\mathrm{SB}_n(\Sigma_2)$, which fits into the extension

$$1 \to PB_n \to \mathrm{SB}_n(\Sigma_2) \to \pi_1(\Sigma_2)^n \to 1, \tag{11.25}$$

generalising the elliptic braid group extension (10.24) (at genus 1, $\pi_1(E_\tau) \cong \mathbb{Z}^2$ contributes two generators per point; at genus 2, $\pi_1(\Sigma_2) \cong \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1\rangle$ contributes four generators per point).

The ordered chiral homology at degree $n$ is

$$\int_{\Sigma_2,n}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2) = H^*_{\mathrm{dR}}\big(\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\Sigma_2),\ \nabla^{(n)}_{\mathrm{KZB}}\big) \otimes (s^{-1}\bar{Y}_\hbar)^{\otimes n}. \tag{11.26}$$

**Proposition 11.11** (Euler characteristic at low degrees, genus 2)**.** *The twisted Euler characteristics of the KZB local system on* $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$*, with fibre* $V^{\otimes n}$ $(V = \mathbb{C}^2)$ *for generic* $\hbar$*, are:*

$$n = 0: \quad \chi = \infty \ \ (\mathbb{C}[\mathrm{qdet}] \textit{ is infinite-dimensional}), \tag{11.27}$$

$$n = 1: \quad \chi = -4 \ \ (\textit{trivial local system of rank } 2 \textit{ on } \Sigma_2,\ \chi(\Sigma_2) = -2), \tag{11.28}$$

$$n = 2: \quad \chi = -12 \ \ (\textit{rank } 4 \textit{ on } \Sigma_2 \setminus \{0\}). \tag{11.29}$$

*For* $n \geq 3$*, the Euler characteristic of* $\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)$ *with coefficients in a rank-*$d^n$ *local system is:*

$$\chi\big(\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2),\ \mathcal{L}_{\mathrm{KZB}}\big) = d^n \cdot \prod_{j=0}^{n-1}(\chi(\Sigma_2) - j) = 2^n \cdot \prod_{j=0}^{n-1}(-2-j). \tag{11.30}$$

*In particular:* $n = 3$ *gives* $8 \cdot (-2)(-3)(-4) = -192$*;* $n = 4$ *gives* $16 \cdot (-2)(-3)(-4)(-5) = -1920$*.*

*Proof.* At $n = 1$: $\mathrm{Conf}_1(\Sigma_2) = \Sigma_2$ and the KZB connection is trivial (no pairwise terms). The local system has rank $d = 2$ and the Euler characteristic is $\chi = d \cdot \chi(\Sigma_2) = 2 \cdot (-2) = -4$. (In contrast to genus 1 where $\chi(E_\tau) = 0$ gives $\chi = 0$, the negative Euler characteristic of $\Sigma_2$ makes the twisted Euler characteristic non-zero even at degree 1.)

At $n = 2$: Proposition 11.8.

At $n \geq 3$: the product formula for the Euler characteristic of ordered configuration spaces gives $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(\Sigma_2)) = \prod_{j=0}^{n-1}(\chi(\Sigma_2) - j) = \prod_{j=0}^{n-1}(-2-$

$j$), and the multiplicativity of the Euler characteristic with respect to local systems gives (11.30). (Unlike genus 1, where $\chi(E_\tau) = 0$ kills the product at the first factor and gives $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)) = 0$ for all $n \geq 1$, here $\chi(\Sigma_2) = -2 \neq 0$ and the product is non-vanishing for all $n$.) □

11.7. **Summary: genus** 0 **vs. genus** 1 **vs. genus** 2**.**

| | Genus 0 ($\mathbb{P}^1$ or $D^\times$) | Genus 1 ($E_\tau$) | Genus 2 ($\Sigma_2$) |
|---|---|---|---|
| Base curve | $\mathbb{P}^1$ | $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ | $\Sigma_2$ |
| Period matrix | — | $\tau \in \mathbb{H}$, scalar | $\Omega \in \mathbb{H}_2$, $2 \times 2$ |
| Bar propagator | $d\log(z_1 - z_2)$ | $d\log E(z,w) = \wp_1\, dz$ | $d\log E(z,w)$ (prime form) |
| weight | 1 | 1 | 1 |
| Connection | KZ | KZB | KZB (genus 2) |
| $\pi_1$ per point | 0 | $\mathbb{Z}^2$ ($A, B$-cycle) | $\mathbb{Z}^4$ ($A_1, B_1, A_2, B_2$) |
| $B$-cycle monodromy | — | $M_B = e^{-2\pi i\,\hbar\,\Omega}$ | $M_{b_\alpha} = e^{-2\pi i\,\hbar\,\Omega\cdot\omega_\alpha}$ |
| Dynamical parameter | — | $\lambda \in \mathfrak{h}^*$, one component | $(\lambda_1, \lambda_2) \in (\mathfrak{h}^*)^2$ |
| Degree 0 | $\mathbb{C}[\mathrm{qdet}]$ | $\mathbb{C}[\mathrm{qdet}]$ | $\mathbb{C}[\mathrm{qdet}]$ |
| Degree 1 dim | 3 (on $D^\times$) | $12 = 4 \times 3$ | $18 = 6 \times 3$ |
| Degree 2: $\dim H^1$ | 4 | 4 | 12 |
| Degree 2: $\chi$ | 4 ($g = 0$, $s = 2$) | $-4$ | $-12$ |
| $\chi(\mathrm{Conf}_n)$, $n \geq 1$ | varies | 0 (all $n \geq 1$) | $\prod_{j=0}^{n-1}(-2-j) \neq 0$ |
| Curvature $d^2_{\mathrm{fib}}$ | 0 | $\kappa(\mathcal{A}) \cdot \omega_1$ | $\kappa(\mathcal{A}) \cdot \omega_2$ |
| lives on | — | $\overline{\mathcal{M}}_1$ | $\overline{\mathcal{M}}_2$ |
| Degeneration | — | $\tau \to i\infty$: $E_\tau \to \mathbb{C}^\times$ | $\Omega_{12} \to 0$, $\Omega_{22} \to i\infty$: $\Sigma_2 \to E_\tau$ |

*Remark* 11.12 (Non-vanishing Euler characteristics at genus 2)*.* The qualitative difference between genus 1 and genus 2 at degree $n \geq 3$ is striking. At genus 1, the vanishing $\chi(E_\tau) = 0$ forces $\chi(\mathrm{Conf}_n^{\mathrm{ord}}(E_\tau)) = 0$ for all $n \geq 1$, so the Euler characteristic provides no information about the individual cohomology dimensions. At genus 2, $\chi(\Sigma_2) = -2 \neq 0$ and the product formula gives non-zero (and rapidly growing) Euler characteristics: $-4, -12, -192, -1920, \ldots$ at degrees $1, 2, 3, 4, \ldots$. These provide genuine lower bounds on the de Rham cohomology dimensions of the KZB local system, making the genus-2 theory computationally richer: the alternating sum is a useful invariant at every degree.

*Remark* 11.13 (Degeneration $\Sigma_2 \to E_\tau$)*.* In the separating degeneration $\Sigma_2 \to E_\tau \cup E_{\tau'}$ (a genus-2 curve pinches along a separating cycle to two elliptic curves joined at a node), the genus-2 KZB connection splits into two independent genus-1 KZB connections. The ordered chiral homology satisfies

a factorisation constraint at this boundary stratum of $\overline{\mathcal{M}}_2$:

$$\int_{\Sigma_2}^{\mathrm{ord}} Y_\hbar \xrightarrow{\mathrm{res}} \int_{E_\tau}^{\mathrm{ord}} Y_\hbar \otimes \int_{E_{\tau'}}^{\mathrm{ord}} Y_\hbar.$$

In the non-separating degeneration $\Sigma_2 \to E_\tau$ (a single cycle pinches, reducing the genus by one), the period matrix $\Omega = \left(\begin{smallmatrix} \tau & \epsilon \\ \epsilon & i/\epsilon \end{smallmatrix}\right)$ degenerates as $\epsilon \to 0$, and the genus-2 theory reduces to the genus-1 theory of §10 with additional data from the pinched cycle. This non-separating degeneration carries genuinely genus-2 information (the off-diagonal entry $\Omega_{12}$); the separating degeneration does not.

## 12. Comparison with related work

Two existing programmes address the integration of braided algebraic structures over surfaces: Ben-Zvi–Brochier–Jordan integrate braided monoidal *categories*, while Latyntsev constructs chiral homology for factorisation *coalgebras*. Ordered chiral homology occupies a different position: it integrates chiral *algebras* at the chain level, applies uniformly to all four shadow classes, and does not require a finite ribbon category.

**Conjecture 12.1** (BBJ decategorification)**.** *Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, $k$ an admissible rational level (so that $\mathrm{Rep}_q(\mathfrak{g})$ is a finite ribbon category with $q = e^{2\pi i/(k+h^\vee)}$), $\Sigma$ a closed oriented surface, and $X$ a smooth projective curve of the same genus.*

(i) (Decategorification quasi-isomorphism.) *There is a quasi-isomorphism*

$$\mathrm{HH}_*\Big(\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})\Big) \simeq \int_X^{\mathrm{ord}} V_k(\mathfrak{g}), \tag{12.1}$$

*where the left side is the Hochschild homology of the BBJ categorical factorisation homology and the right side is the ordered chiral homology of the affine vertex algebra.*

(ii) ($\mathsf{E}_3$-intertwining.) *The quasi-isomorphism (12.1) intertwines the $\mathsf{E}_3$-structure on the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ with the $\mathsf{E}_3$-structure on the Hochschild cochains of the categorical invariant.*

(iii) (Scope: admissible rational levels.) *The comparison requires $k$ admissible rational, so that $\mathrm{Rep}_q(\mathfrak{g})$ is a finite ribbon category (the BBJ construction uses the ribbon structure and the finiteness condition). At irrational or generic $k$, the categorical side is not defined; ordered chiral homology applies without restriction.*

(iv) (Kernel.) *The kernel of the decategorification map (12.1) detects higher categorical data: objects of $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ that are invisible at the chain level (i.e. whose Hochschild homology vanishes).*

*Remark* 12.2 (Ben-Zvi–Brochier–Jordan). Conjecture 12.1 makes precise the expected relationship between Ben-Zvi, Brochier, and Jordan's factorisation homology for braided monoidal categories [2, 3] and ordered chiral homology. The two constructions operate at different levels:

- BBJ integrate braided monoidal *categories* over surfaces (categorical level).
- Ordered chiral homology integrates chiral *algebras* over curves (chain level).

For affine Kac–Moody at admissible rational level: BBJ's $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ is a category, while $\int_X^{\mathrm{ord}} V_k(\mathfrak{g})$ is a chain complex. Part (i) of the conjecture asserts that Hochschild homology of the former recovers the latter.

For class $M$ (Virasoro at generic $c$): the BBJ framework requires a finite ribbon category, which is not available for generic central charge. Ordered chiral homology applies without restriction to class $M$, providing an integration theory where the categorical approach does not (currently) reach.

*Remark* 12.3 (Latyntsev). Latyntsev [31] constructs chiral homology for factorisation coalgebras on curves, providing an algebraic counterpart to the analytic construction of this paper. The relationship between the two approaches at the chain level, particularly for irregular singularities (class $M$), deserves further investigation.

*Remark* 12.4 (Wilson lines by ambient dimension). The type of Wilson line, the form of the $R$-matrix, and the resulting Yang–Baxter equation are controlled by the ambient space dimension. The following table summarizes the principal cases.

| Ambient | $R$-matrix | YBE | Algebra | Integrable system |
|---|---|---|---|---|
| $\mathbb{R}^3$ | constant $R$ | quantum $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$ | $U_q(\mathfrak{g})$ | Jones polynomial |
| $\mathbb{R}\times\mathbb{C}$ | $r(z)$, 1 spectral | CYBE $[r_{12}, r_{13}] + \mathrm{cyc} = 0$ | $Y_\hbar(\mathfrak{g})$ | Gaudin |
| $\mathbb{R}^2\times\mathbb{C}$ | $R(u)$, spectral | quantum spectral YBE | $U_q(L\mathfrak{g})$ | XXZ |
| $\mathbb{C}^2$ | $r(z,w)$, 2 params | 2d CYBE | $\mathfrak{g}[u,v]$ | Hitchin on surface |
| $\mathbb{R}\times\mathbb{C}^2$ | $R(u,v)$, 2 spectral | quantum + dynamical | $U_{q,t}(\widehat{\widehat{\mathfrak{g}}})$ | eCM + Hitchin |
| $\mathbb{C}^3$ (CY) | $r + \mathbb{Z}\times\mathbb{Z}$ grading | 6d YBE | $\mathsf{E}_1$-chiral QG | DT/BPS |

This paper works in the $\mathbb{R}\times\mathbb{C}$ setting (Costello's 4d Chern–Simons [14, 15, 12, 13]), producing the Yangian $Y_\hbar(\mathfrak{g})$ as the line-operator algebra. The extensions to $\mathbb{R}^2\times\mathbb{C}$ (quantum loop algebras), $\mathbb{C}^2$ (double current algebras), $\mathbb{R}\times\mathbb{C}^2$ (quantum toroidal algebras), and $\mathbb{C}^3$ (6d holomorphic Chern–Simons defects) are frontier directions; ordered chiral homology provides the natural integration theory for the $\mathbb{R}\times\mathbb{C}$ column.

## 13. Conclusion and open problems

This paper defines ordered chiral homology $\int_X^{\mathrm{ord}} \mathcal{A}$ for $\mathsf{E}_1$-chiral algebras on algebraic curves, constructs the chiral $\mathsf{E}_3$-algebra $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ as the holomorphic refinement of the Costello–Francis–Gwilliam perturbative Chern–Simons algebra, and establishes the chiral quantum group equivalence (Theorem 6.5)

relating vertex $R$-matrices, $A_\infty$-structures, and chiral coproducts as three faces of a single ordered bar complex. The chiral Yangian datum (Definition 6.7) packages this equivalence as a single axiomatised object, and Conjecture 2.4 predicts the full structure of the ordered derived chiral centre: infinite-dimensional, carrying an $\mathsf{E}_2$-algebra structure via the classical Deligne conjecture, and recovering $\mathfrak{g}$ from the degree-1 Gerstenhaber bracket. The representation-theoretic kernel formula $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ (Proposition 2.6) provides the quantitative underpinning: the ordered data grows exponentially while the symmetric shadow grows polynomially. Five explicit computations across the shadow classification (Heisenberg: class $G$; affine Kac–Moody: class $L$; $\beta\gamma$: class $C$; Virasoro: class $M$; Yangian: genuinely $\mathsf{E}_1$) verify the formality bridge, the regular/irregular dichotomy, and the Stokes phenomena that distinguish gauge theory from gravity at the level of the KZ $\mathcal{D}$-module. Section 10 extends the construction to genus 1, computing $\int_{E_\tau}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2)$ degree by degree: the Belavin elliptic $r$-matrix replaces the rational $r$-matrix, the KZB connection replaces the KZ connection, and the $B$-cycle monodromy produces the quantum group parameter $q = e^{2\pi i\hbar}$. Section 11 extends to genus 2, where the $2 \times 2$ Siegel period matrix, the prime form, and the genus-2 surface braid group produce a theory with non-vanishing Euler characteristics at every degree ($\chi = -12$ at degree 2, compared with $-4$ at genus 1).

The following problems remain open.

(a) (The $\mathsf{E}_3$ identification for affine Kac–Moody: resolved.) Theorem 8.6 proves that the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ and the CFG algebra $\mathcal{A}^\lambda$ are isomorphic as formal deformation families of $\mathsf{E}_3$-algebras over $(k + h^\vee)H^3(\mathfrak{g})[[k + h^\vee]]$, for simple $\mathfrak{g}$. The proof is obstruction-theoretic: the one-dimensionality of $H^3(\mathfrak{g})$ for simple $\mathfrak{g}$ forces the two deformations to agree at each order in $(k + h^\vee)$, given the matching of associated graded algebras and $\mathsf{P}_3$ brackets on the formal disk (Theorem 8.21). The remaining question is the extension to non-simple $\mathfrak{g}$, where $H^3(\mathfrak{g})$ may be higher-dimensional and the order-by-order uniqueness argument requires additional invariants to pin down the deformation class.

(b) (The ordered derived chiral centre of the chiral Yangian.) Conjecture 2.4 predicts that the ordered derived chiral centre of $Y_\hbar(\mathfrak{sl}_2)^{\mathrm{ch}}$ is infinite-dimensional and carries an $\mathsf{E}_2$-algebra structure (but not $\mathsf{E}_3$, since the symmetric bar does not exist for genuinely $\mathsf{E}_1$-chiral algebras). Proposition 2.6 proves the representation-theoretic kernel formula $\dim\ker(\mathrm{av}_n) = d^n - \binom{n+d-1}{d-1}$ for any $d$-dimensional representation of any simple Lie algebra; the open content is the passage from this bound to the actual cohomology of the ordered bar complex at each degree. In particular: does the degree-1 Gerstenhaber bracket on the ordered chiral centre recover $\mathfrak{g}$ for all simple Lie algebras, or is the identification specific to $\mathfrak{sl}_2$?

(c) (Holomorphic Chern–Simons invariants of curves.) The chiral $\mathsf{E}_3$-algebra $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ on a smooth proper curve $X$ of genus $g$ should produce, via the higher-genus ordered chiral homology and factorisation on $\overline{\mathcal{M}}_{g,n}$, invariants of $(X, \mathfrak{g}, k)$ that refine Reshetikhin–Turaev invariants in the same way that perturbative Chern–Simons refines Witten's partition function. Concretely: does the genus-$g$ ordered chiral homology $\int_X^{\mathrm{ord}} \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ recover the perturbative Chern–Simons partition function $Z_{\mathrm{CS}}(X, \mathfrak{g}, k)$ as its $\Sigma_n$-coinvariant shadow, and what invariants live in the kernel of the averaging map av?

(d) (The $\mathcal{W}_{1+\infty}$ chiral quantum group: remaining questions.) Theorem 6.15 establishes that $\mathcal{W}_{1+\infty}[\Psi]$ carries a chiral quantum group datum with OPE compatibility at all spins, proved by the coderivation property on the Koszul-locus bar complex and independently by the JKL vertex bialgebra theorem on the CoHA. A constant coproduct does not exist for $c \neq 0$ (conformal anomaly obstruction). Theorem 6.22 extends this to $\mathcal{W}_N$ at all finite $N \geq 2$, where the $R$-matrix is Yang's matrix $R(u) = u\, I + \Psi\, P$ and the RTT is non-trivial. The remaining questions (Remark 6.19) are: (i) the vertex-algebraic antipode (does $S(T(u)) = T(u)^{-1}$ on the Yangian lift to a vertex Hopf algebra structure on $\mathcal{W}_{1+\infty}$?); (ii) the Stokes multipliers of the irregular KZ connection for $\mathcal{W}_N$ (Poincaré rank $N-1$), and whether they encode the higher-spin BPS spectrum.

(e) (Higher-genus ordered chiral homology.) Section 5 defines ordered chiral homology at genus 0, Section 10 constructs the genus-1 theory $\int_{E_\tau}^{\mathrm{ord}} Y_\hbar(\mathfrak{sl}_2)$ explicitly, and Section 11 extends the construction to genus 2, where the period matrix is a $2 \times 2$ symmetric matrix, the KZB connection acquires three modular components, and the Euler characteristic $\chi = -12$ at degree 2 is three times the genus-1 value. At genus $g \geq 3$, the curvature obstruction $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ (UNIFORM-WEIGHT), where $\omega_g = c_1(\lambda)$ is the Hodge class on $\overline{\mathcal{M}}_g$, requires a period-corrected differential whose coefficients involve the $(g \times g)$-matrix $(\operatorname{Im}\Omega)^{-1}$ of the Siegel period matrix. Does the ordered chiral chain complex at genus $g \geq 3$ define a flat connection over $\overline{\mathcal{M}}_g$ compatible with the Mumford–Morita class $\kappa_1$, and does the resulting system of conformal blocks satisfy factorisation at the boundary divisors of $\overline{\mathcal{M}}_g$?

(f) (Trigonometric and elliptic $R$-matrices beyond $\mathfrak{sl}_2$.) Section 10 treats the Belavin elliptic $R$-matrix for $\mathfrak{sl}_2$ on $E_\tau$. The trigonometric $R$-matrix $R^{\mathrm{trig}}(u)$ (quantum affine algebras) and the elliptic Belavin $R$-matrix for $\mathfrak{sl}_N$ ($N \geq 3$) produce richer ordered chiral homology with qualitatively different behaviour: higher-order Stokes phenomena, theta-function monodromy, and flat sections valued in elliptic hypergeometric integrals. Does the regular/irregular dichotomy of Theorem 1.1 extend to a trichotomy (regular, irregular-rational,

irregular-elliptic) correlated with the three types of quantum groups? For $\mathfrak{sl}_N$, what is the degree-2 de Rham cohomology of the KZB local system of rank $N^2$ on $E_\tau \setminus \{0\}$, and how does the isospin decomposition into irreducibles of $V^{\otimes 2} = \mathrm{Sym}^2 V \oplus \bigwedge^2 V$ generalise?

(g) (Categorification.) Ben-Zvi–Brochier–Jordan [2, 3] integrate braided monoidal categories over surfaces; ordered chiral homology integrates chiral algebras over curves. Conjecture 12.1 makes the expected bridge precise: the decategorification quasi-isomorphism (12.1) identifies the Hochschild homology of the categorical invariant $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ with the ordered chiral homology $\int_X^{\mathrm{ord}} V_k(\mathfrak{g})$. Is there a direct functor realising this comparison that intertwines the $\mathsf{E}_3$-structures (part (ii)), and what is the structure of the kernel (part (iv)): which objects of $\int_\Sigma \mathrm{Rep}_q(\mathfrak{g})$ are invisible at the chain level?

(h) (Refinements of the topological enhancement.) Khan–Zeng [28] prove the topological enhancement at the classical (PVA) level; Proposition 8.23 extends this to the full quantum vertex algebra $V_k(\mathfrak{g})$ at non-critical level, using the exactness of the Sugawara element in $V_k(\mathfrak{g})$. Two refinements remain open: (a) does the topological $\mathsf{E}_3$-structure on $\mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ extend to a topological $\mathsf{E}_3$-structure on the higher-genus ordered chiral homology $\int_X^{\mathrm{ord}} \mathrm{CE}_k^{\mathrm{ch}}(\mathfrak{g})$ for $g \geq 2$, and (b) at the critical level $k = -h^\vee$, where the Sugawara element is undefined, is the resulting holomorphic $\mathsf{E}_3$-algebra distinguished from the topological one by a computable invariant (a secondary characteristic class of the chiral $\mathsf{E}_3$-algebra)?

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